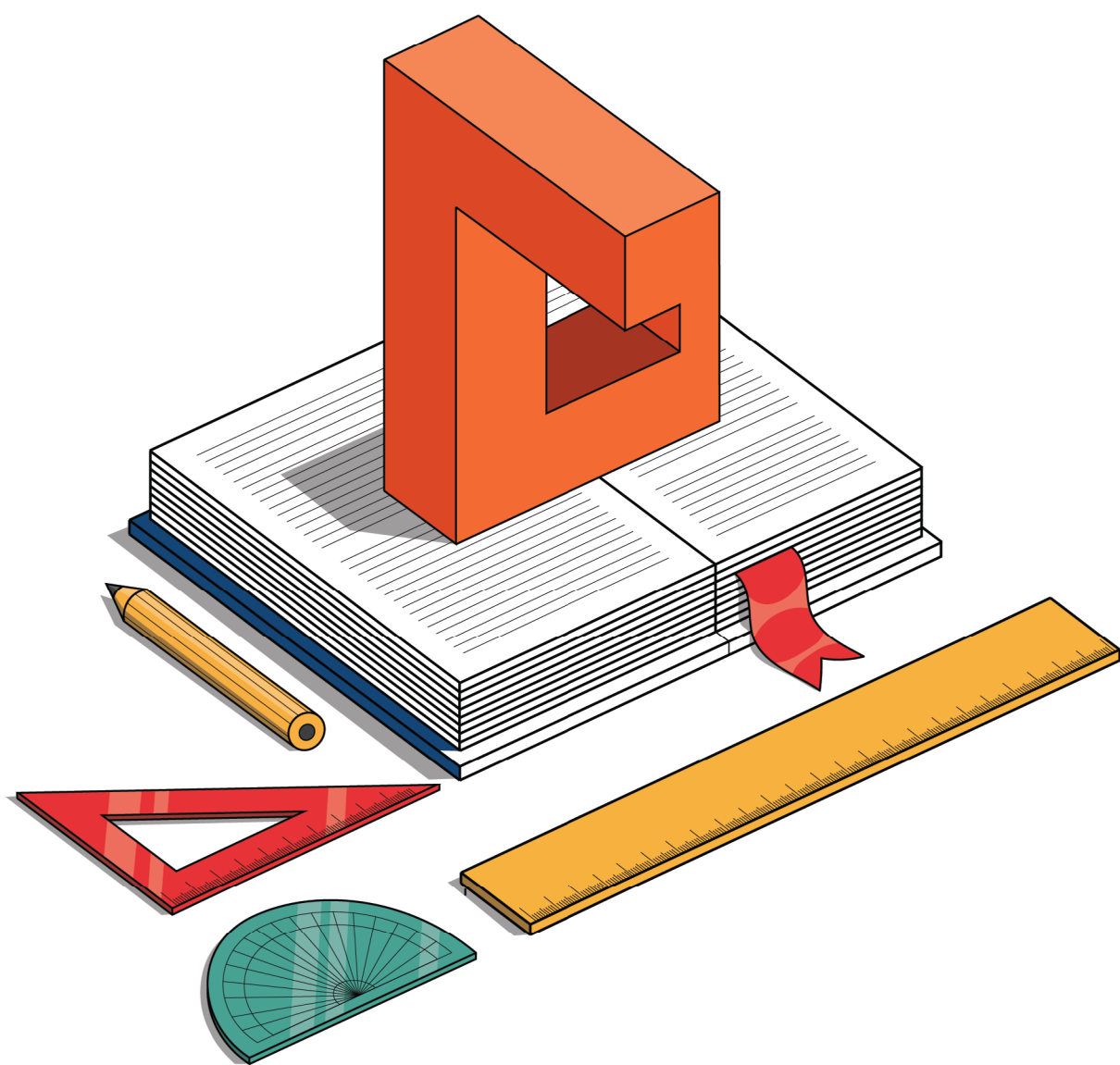


Vedantu's Tatva Practice Book

Mathematics

Volume 3

Class 12 **JEE**



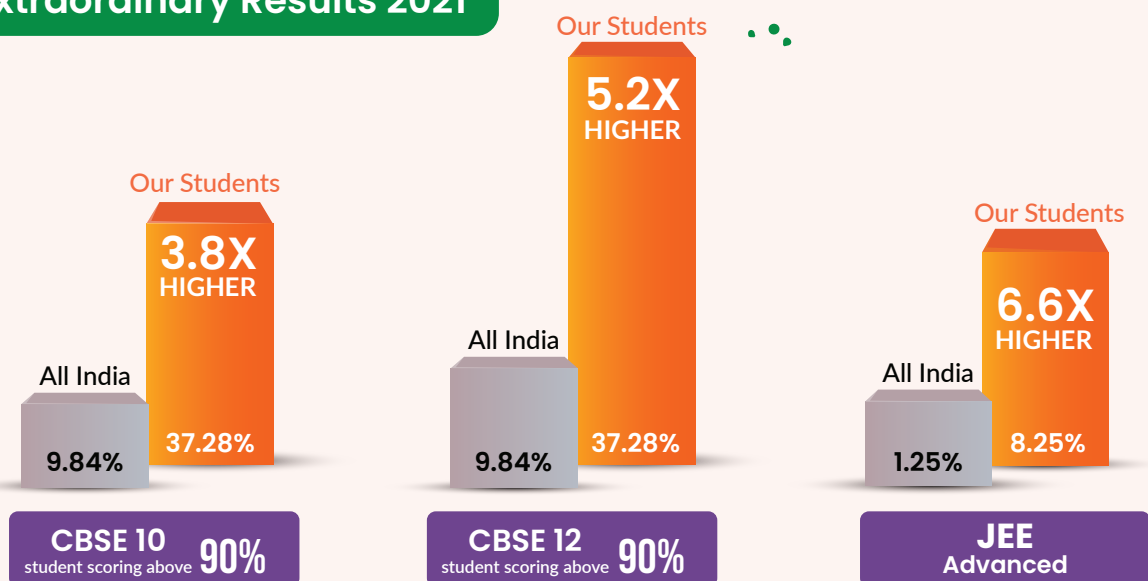
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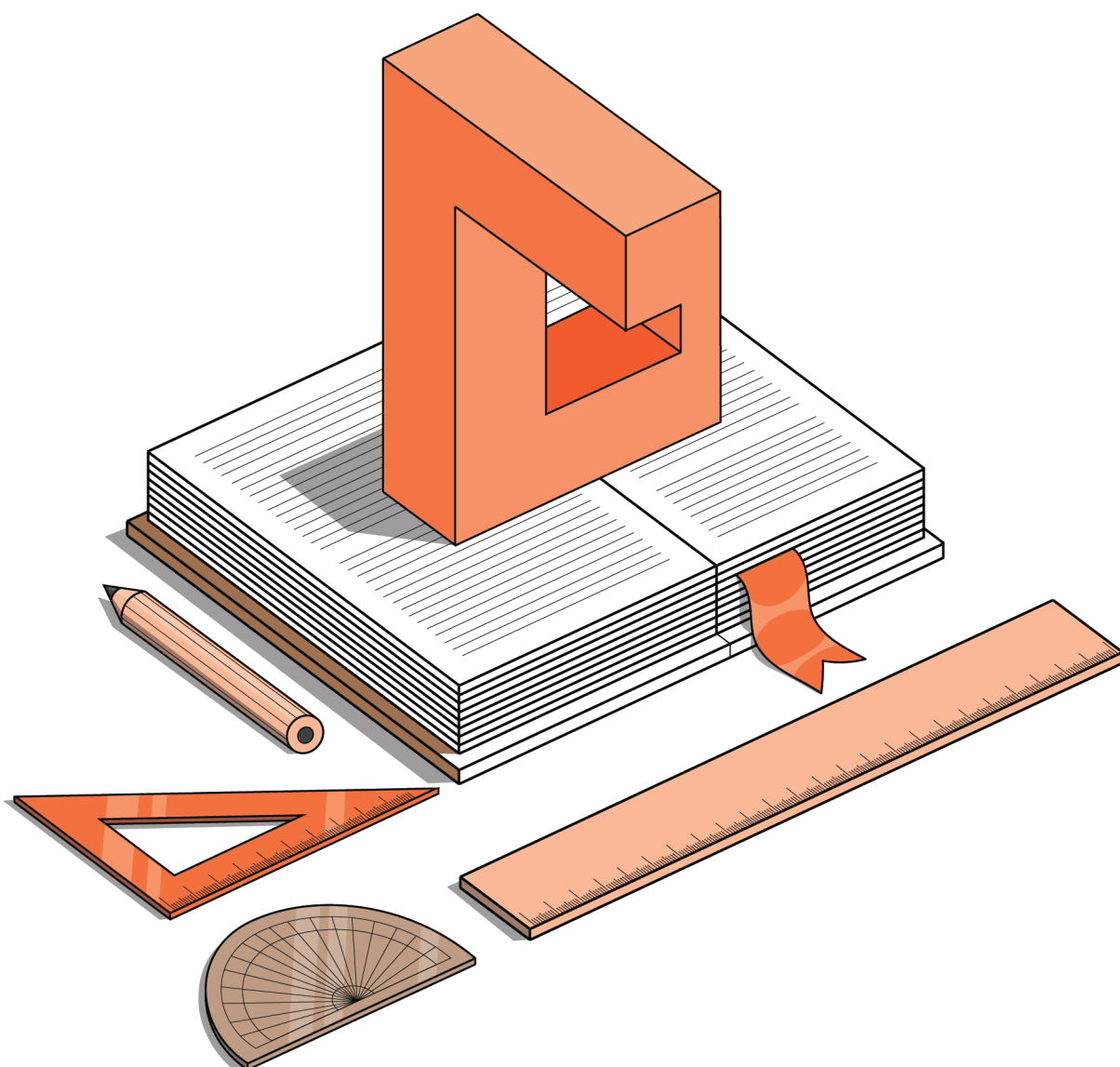
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How to use your Tatva Practice Book

4. Fundamental Principle of Counting :

If an event can occur in 'm' different ways following which another event can occur in 'n' different ways following which another event can occur in 'p' different ways then



SCAN CODE
Permutations

1

Scan the QR Code in each chapter's theory section to view micro concept videos related to the chapter, on the Vedantu app.

2

Solve all types of exercise questions based on the latest JEE pattern.

Exercise - 2:

1. If every element of a third order determinant of value Δ is multiplied by 5, then the value of new determinant is:

(JEE 2019)

- (a) Δ (b) 5Δ
(c) 25Δ (d) 125Δ

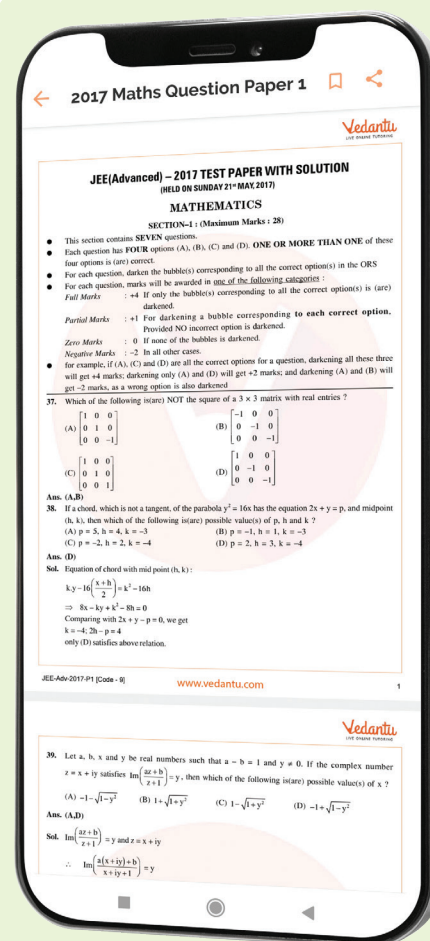
Answer Key

CHAPTER-1: DETERMINANTS
Exercise-1: Basic Objective



3

Scan the QR code in the Answers Section to view detailed solutions for all exercise questions.



For extra exam preparation content, visit the Vedantu app. You can find previous years' JEE papers with solutions and detailed analysis.

Founder's Message

Dear Student,

I am delighted to present to you a Ready Reckoner and an amazing book to guide you for your exams-'TATVA'. Tatva—which means the 'Core' is fully aligned with the culture, the mission, and the vision of Vedantu and therefore it gives me immense pleasure and joy to share this book with you. We at Vedantu have always believed in revolutionizing the teaching and learning process and always speedily progressed in the direction of bringing superior quality education and content to your table. Tatva is a step forward in this direction. This book is your guide, your practice guru, and your companion in moving towards your dreams. The book is a result of the consistent effort, diligence, and research by our experienced team of subject experts and teachers.

This book has been customized with curated content to suit the needs of JEE aspirants like you and guide you on the right path to cracking JEE and optimizing your efficiency. Tatva is a comprehensive amalgamation of important concepts, theories, derivations, definitions, solved examples, concept videos, practice questions, and important questions. We have ensured that high-quality content and the right ingredients are in place in this booklet to help you climb up the success ladder.

A few guiding points to optimally use Tatva with a planned approach:

Tatva equips you with Theory, Concept Videos, and Solved examples to help you revise concepts, mark your notes, walk you through the entire summary, and eventually makes you capable of clearing all your conceptual doubts all by yourself.

We suggest revision of theory followed by practice of solved examples.

Practice relevant questions daily after finishing Vedantu lectures and session assignments. We believe that a daily dose of Tatva will keep all your exam blues at bay.

Use the Tatva booklet to mark notes so that it always comes in handy for last-minute revision sessions before your exams. Notes should include key points of theory, solved examples, and some questions which you couldn't solve in the first attempt.

Exercise 1 and Exercise 2 of JEE Tatva deal with basic questions and those which can be asked or already asked in JEE Main. Similarly, Exercise 3 and Exercise 4 deal with JEE Advanced level questions. We recommend you solve basic JEE Main questions before moving to JEE Advanced level questions.

Before wrapping up, the practice mantra: "Don't practice until you get it right. Practice until you can't get it wrong."

We strongly believe in you and your capabilities. So believe in yourself because success is only one step away. Wishing that your talent shines bright. All the very best!

Anand Prakash

Founder and Academic Head, Vedantu



Anand Prakash Sir has been a pioneer in producing Top Ranks in JEE/NEET and Olympiads. He has personally taught and mentored AIR 1, 6, 7 (JEE Advanced), AIR-1, 7, 9(AIIMS), and thousands of more students who have successfully cleared these competitive exams in the last few years.

Credits

*“Happiness lies in the joy of achievement
and the thrill of creative effort.”*

—Franklin D. Roosevelt

Tatva is the brainchild of a group of creative Vedans who have strived tirelessly to weave success stories for you. We extend our heartfelt gratitude to the superb team of Vedans who give wings to the vision of Vedantu, starting with our leaders who have been guiding and encouraging us at every step of the way:

Vamsi Krishna Sir, Anand Prakash Sir and Pulkit Jain Sir

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Sahil Bhatia, Sudhanshu Jain, Shubam Gupta, Ajay Mittal, Arshad Shahid, Jaideep Sontakke

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Harish Rao, Neha Surana, Charubak Chakrabarti, Prashant Palande

Mathematics Team

We truly appreciate all the Master Teachers of Vedantu whose relentless efforts helped us translate this vision into reality. Our heartfelt gratitude to our creative content developers and the typesetting team, who have put in their hard work, insight, and eagerness to nurture and execute Tatva into ‘your ready handbook’ and bring a positive learning experience to you.

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The journey of bringing Tatva to life, from an idea to the book you are holding, would not have been possible without the extensive support of our diligent Operations Team, our amazing Academic Team, our dedicated team of Teachers, and our talented Tech Team.

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VECTORS & 3-DIMENSIONAL GEOMETRY

VECTORS & 3-DIMENSIONAL GEOMETRY



VECTORS

1. VECTORS & THEIR TYPES

Vector quantities are specified by definite magnitudes and definite directions. A vector is generally represented by a directed line segment, say $\overrightarrow{AB}$. A is called the **initial point** and B is called the **terminal point**. The magnitude of vector $\overrightarrow{AB}$ is expressed by $|\overrightarrow{AB}|$. $\overrightarrow{AB}$ may also be represented by $\vec{a}$ and its magnitude by $|\vec{a}|$.

1.1 Zero Vector

A vector of zero magnitude is a zero vector i.e. which has the same initial & terminal point, is called a **Zero Vector**. It is denoted by $\vec{0}$. The direction of zero vector is indeterminate.

1.2 Unit Vector

A vector of unit magnitude in direction of a vector $\vec{a}$ is called unit vector along $\vec{a}$ and is denoted by $\hat{a}$. Symbolically $\hat{a} = \frac{\vec{a}}{|\vec{a}|}$.

1.3 Equal Vector

Two vectors are said to be equal if they have the same magnitude, direction & represent the same physical quantity.

1.4 Collinear Vector

Two vectors are said to be collinear if their directed line segments are parallel disregards to their direction. Collinear vectors are also called **Parallel Vectors**. If they have the same direction they are named as like vectors otherwise unlike vectors.

Symbolically, two non-zero vectors $\vec{a}$ and $\vec{b}$ are collinear if and only, if $\vec{a} = K\vec{b}$, where $K \in \mathbb{R} - \{0\}$.

1.5 Coplanar Vector

A given number of vectors are called coplanar if their line segments are all parallel to the same plane. Note that **“Two Vectors Are Always Coplanar”**.

1.6 Position Vector of A Point

Let O be a fixed origin, then the position vector (pv) of a point P is the vector $\overrightarrow{OP}$. If $\vec{a}$ and $\vec{b}$ are position vectors of two points A and B, then, $\overrightarrow{AB} = \vec{b} - \vec{a}$ = pv of B – pv of A.

1.7 Section Formula

If $\vec{a}$ and $\vec{b}$ are the position vectors to two points A and B then the p.v. of a point which divides AB in the ratio m : n is given by :

$$\vec{r} = \frac{n\vec{a} + m\vec{b}}{m + n} \text{ . Note p.v. of mid point of AB} = \frac{\vec{a} + \vec{b}}{2}$$

2. ALGEBRA OF VECTORS

2.1 Addition of vectors

If two vectors $\vec{a}$ & $\vec{b}$ are represented by $\overrightarrow{OA}$ & $\overrightarrow{OB}$, then their sum $\vec{a} + \vec{b}$ is a vector represented by $\overrightarrow{OC}$, where OC is the diagonal of the parallelogram OACB.

- $\vec{a} + \vec{b} = \vec{b} + \vec{a}$ (commutative)
- $(\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c})$ (associative)
- $\vec{a} + \vec{0} = \vec{a} = \vec{0} + \vec{a}$
- $\vec{a} + (-\vec{a}) = \vec{0} = (-\vec{a}) + \vec{a}$

2.2 Multiplication of a Vector by a scalar

If $\vec{a}$ is a vector & m is a scalar, then $m\vec{a}$ is vector parallel to $\vec{a}$ whose modulus is |m| times that of $\vec{a}$. If $m > 0$ then $m\vec{a}$ and $\vec{a}$ have same direction and if $m < 0$, then they have opposite



directions. This multiplication is called **Scalar Multiplication**.

If $\vec{a}$ & $\vec{b}$ are vectors & m, n are scalars, then :

$$m(\vec{a}) = (\vec{a})m = m\vec{a}$$

$$m(n\vec{a}) = n(m\vec{a}) = (mn)\vec{a}$$

$$(m+n)\vec{a} = m\vec{a} + n\vec{a}$$

$$m(\vec{a} + \vec{b}) = m\vec{a} + m\vec{b}$$

2.3 Subtraction of Vectors

$\vec{a} - \vec{b}$ is defined as addition of vectors $\vec{a}$ and $(-\vec{b})$.

3. TEST OF COLLINEARITY

Three points A, B, C with position vectors $\vec{a}, \vec{b}, \vec{c}$ respectively are collinear, if & only if there exist scalar x, y, z not all zero simultaneously such that; $x\vec{a} + y\vec{b} + z\vec{c} = \vec{0}$, where $x + y + z = 0$

4. TEST OF COPLANARITY

Four points A, B, C, D with position vectors $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ respectively are coplanar if and only if there exist scalars x, y, z, w not all zero simultaneously such that $x\vec{a} + y\vec{b} + z\vec{c} + w\vec{d} = \vec{0}$ where, $x + y + z + w = 0$

5. SCALAR PRODUCT OF TWO VECTORS

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta \quad (0 \leq \theta \leq \pi)$$

note that if θ is acute then $\vec{a} \cdot \vec{b} > 0$ & if θ is obtuse then

$$\vec{a} \cdot \vec{b} < 0$$

$$\vec{a} \cdot \vec{a} = |\vec{a}|^2 = a^2$$

$$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a} \quad (\text{commutative})$$

$$\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c} \quad (\text{distributive})$$

$$\vec{a} \cdot \vec{b} = 0 \Leftrightarrow \vec{a} \perp \vec{b} \quad (\vec{a} \neq 0, \vec{b} \neq 0)$$

$$(m\vec{a}) \cdot \vec{b} = \vec{a} \cdot (m\vec{b}) = m(\vec{a} \cdot \vec{b}), \text{ where } m \text{ is scalar.}$$

$$\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1;$$

$$\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$$

$$\bullet \quad \text{projection of } \vec{a} \text{ on } \vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$$

$$\bullet \quad \text{the angle } \phi \text{ between } \vec{a} \text{ \& } \vec{b} \text{ is given by } \cos \phi = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$$

$$0 \leq \phi \leq \pi$$

$$\bullet \quad \text{if } \vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k} \text{ and } \vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k} \text{ then}$$

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

$$\bullet \quad |\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

$$\bullet \quad |\vec{b}| = \sqrt{b_1^2 + b_2^2 + b_3^2}$$

NOTES :

$$(i) \quad \text{Maximum value of } \vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}|$$

$$(ii) \quad \text{Minimum values of } \vec{a} \cdot \vec{b} = -|\vec{a}| |\vec{b}|$$

(iii) Any vector $\vec{a}$ can be written as,

$$\vec{a} = (\vec{a} \cdot \hat{i})\hat{i} + (\vec{a} \cdot \hat{j})\hat{j} + (\vec{a} \cdot \hat{k})\hat{k}.$$

(iv) A vector in the direction of the bisector of the angle

$$\text{between two vectors } \vec{a} \text{ \& } \vec{b} \text{ is } \frac{\vec{a}}{|\vec{a}|} + \frac{\vec{b}}{|\vec{b}|}.$$

Hence bisector of the angle between the two vectors

$$\vec{a} \text{ \& } \vec{b} \text{ is } \lambda(\hat{a} + \hat{b}), \text{ where } \lambda \in \mathbb{R}^+.$$

Bisector of the exterior angle between $\vec{a}$ & $\vec{b}$ is $\lambda(\hat{a} - \hat{b})$

$$\lambda \in \mathbb{R} - \{0\}.$$



6. VECTOR PRODUCT OF TWO VECTORS

- If $\vec{a}$ & $\vec{b}$ are two vectors & θ is the angle between them then $\vec{a} \times \vec{b} = |\vec{a}| |\vec{b}| \sin \theta \hat{n}$, where $\hat{n}$ is the unit vector perpendicular to both $\vec{a}$ & $\vec{b}$ such that $\vec{a}$, $\vec{b}$ & $\hat{n}$ forms a right handed screw system.
- Geometrically $|\vec{a} \times \vec{b}|$ equals area of the parallelogram whose two adjacent sides are represented by $\vec{a}$ & $\vec{b}$
- $\vec{a} \times \vec{b} = \vec{0} \Leftrightarrow \vec{a}$ and $\vec{b}$ are parallel (collinear) (provided $\vec{a} \neq \vec{0}, \vec{b} \neq \vec{0}$) i.e. $\vec{a} = K\vec{b}$, where K is scalar.
- $\vec{a} \times \vec{b} \neq \vec{b} \times \vec{a}$ (not commutative)
- $(m\vec{a}) \times \vec{b} = \vec{a} \times (m\vec{b}) = m(\vec{a} \times \vec{b})$ where m is scalar
- $\vec{a} \times (\vec{b} + \vec{c}) = (\vec{a} \times \vec{b}) + (\vec{a} \times \vec{c})$ (distributive)
- $\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = 0$
- $\hat{i} \times \hat{j} = \hat{k}, \hat{j} \times \hat{k} = \hat{i}, \hat{k} \times \hat{i} = \hat{j}$
- If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ then

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

- Unit vector perpendicular to the plane of $\vec{a}$ & $\vec{b}$ is
- $$\hat{n} = \pm \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$$
- A vector of magnitude 'r' & perpendicular to the plane of $\vec{a}$ and $\vec{b}$ is $\pm \frac{r(\vec{a} \times \vec{b})}{|\vec{a} \times \vec{b}|}$

- If θ is the angle between $\vec{a}$ & $\vec{b}$ then $\sin \theta = \frac{|\vec{a} \times \vec{b}|}{|\vec{a}| |\vec{b}|}$

- If $\vec{a}, \vec{b}$ & $\vec{c}$ are the pv's of 3 points A, B and C then the vector area of triangle ABC = $\frac{1}{2} [\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}]$.

The point A, B & C are collinear if $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a} = \vec{0}$

- **Area of any quadrilateral** whose diagonal vectors are $\vec{d}_1$ & $\vec{d}_2$ is given by $\frac{1}{2} |\vec{d}_1 \times \vec{d}_2|$

- **Lagranges Identity** : for any two vector $\vec{a}$ & $\vec{b}$;

$$(\vec{a} \times \vec{b})^2 = |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} \\ \vec{a} \cdot \vec{b} & \vec{b} \cdot \vec{b} \end{vmatrix}$$

7. SCALAR TRIPLE PRODUCT

- The scalar triple product of three vectors $\vec{a}, \vec{b}$ & $\vec{c}$ is defined as :

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = |\vec{a}| |\vec{b}| |\vec{c}| \sin \theta \cos \phi$$

where θ is the angle between $\vec{c}$ & $\vec{b}$ & ϕ is angle between $\vec{a}$ and $\vec{b} \times \vec{c}$

It is also written as $[\vec{a} \vec{b} \vec{c}]$, spelled as **box product**.

- Scalar triple product geometrically represents the volume of the parallelepiped whose three coterminous edges are represented by $\vec{a}, \vec{b}$ & $\vec{c}$ i.e. $V = [\vec{a} \vec{b} \vec{c}]$

- In a scalar triple product the position of dot & cross can be interchanged i.e.

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \cdot \vec{c} \text{ Also } [\vec{a} \vec{b} \vec{c}] = [\vec{b} \vec{c} \vec{a}] = [\vec{c} \vec{a} \vec{b}]$$

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = -\vec{a} \cdot (\vec{c} \times \vec{b}) \text{ i.e. } [\vec{a} \vec{b} \vec{c}] = -[\vec{a} \vec{c} \vec{b}]$$

- If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$; $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ and

$$\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k} \text{ then } [\vec{a} \vec{b} \vec{c}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$



In general, if $\vec{a} = a_1\vec{\ell} + a_2\vec{m} + a_3\vec{n}$; $\vec{b} = b_1\vec{\ell} + b_2\vec{m} + b_3\vec{n}$ and

$$\vec{c} = c_1\vec{\ell} + c_2\vec{m} + c_3\vec{n} \text{ then } [\vec{a} \vec{b} \vec{c}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} [\vec{\ell} \vec{m} \vec{n}];$$

where $\vec{\ell}$, $\vec{m}$ & $\vec{n}$ are non-coplanar vectors.

- $\vec{a}, \vec{b}, \vec{c}$ are coplanar $\Leftrightarrow [\vec{a} \vec{b} \vec{c}] = 0$.
- Scalar product of three vectors, two of which are equal or parallel is 0.

NOTES:

- If $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar then $[\vec{a} \vec{b} \vec{c}] > 0$ for right handed system & $[\vec{a} \vec{b} \vec{c}] < 0$ for left handed system.
- $\begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \end{bmatrix} = 1$
- $[K \vec{a} \vec{b} \vec{c}] = K [\vec{a} \vec{b} \vec{c}]$
- $[(\vec{a} + \vec{b}) \vec{c} \vec{d}] = [\vec{a} \vec{c} \vec{d}] + [\vec{b} \vec{c} \vec{d}]$
- **The volume of the tetrahedron OABC** with O as origin & the pv's of A, B and C being $\vec{a}, \vec{b}$ & $\vec{c}$ respectively is given by $V = \frac{1}{6} [\vec{a} \vec{b} \vec{c}]$
- The position vector of the centroid of a tetrahedron if the pv's of its angular vertices are $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are given by $\frac{1}{4} [\vec{a} + \vec{b} + \vec{c} + \vec{d}]$.

Note that this is also the point of concurrency of the lines joining the vertices to the centroids of the opposite faces and is also called the centre of the tetrahedron. In case the tetrahedron is regular it is equidistant from the vertices and the four faces of the tetrahedron.

- **Remember that:** $[(\vec{a} - \vec{b})(\vec{b} - \vec{c})(\vec{c} - \vec{a})] = 0$ &

$$[(\vec{a} + \vec{b})(\vec{b} + \vec{c})(\vec{c} + \vec{a})] = 2 [\vec{a} \vec{b} \vec{c}]$$

8. VECTOR TRIPLE PRODUCT

Let $\vec{a}, \vec{b}, \vec{c}$ be any three vectors, then the expression $\vec{a} \times (\vec{b} \times \vec{c})$ is vector & is called vector triple product.

Geometrical Interpretation of $\vec{a} \times (\vec{b} \times \vec{c})$

Consider the expression $\vec{a} \times (\vec{b} \times \vec{c})$ which itself is a vector. Since it is a cross product of two vectors $\vec{a}$ and $(\vec{b} \times \vec{c})$. Now $\vec{a} \times (\vec{b} \times \vec{c})$ is a vector perpendicular to the plane containing $\vec{a}$ and $(\vec{b} \times \vec{c})$ but $\vec{b} \times \vec{c}$ is a vector perpendicular to the plane containing $\vec{b}$ & $\vec{c}$, therefore $\vec{a} \times (\vec{b} \times \vec{c})$ is a vector lying in the plane of $\vec{b}$ & $\vec{c}$ and perpendicular to $\vec{a}$. Hence we can express $\vec{a} \times (\vec{b} \times \vec{c})$ in terms of $\vec{b}$ & $\vec{c}$ i.e. $\vec{a} \times (\vec{b} \times \vec{c}) = x\vec{b} + y\vec{c}$ where x and y are scalars.

- $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$
- $(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a}$
- $(\vec{a} \times \vec{b}) \times \vec{c} \neq \vec{a} \times (\vec{b} \times \vec{c})$

9. LINEAR COMBINATIONS

Given a finite set of vectors $\vec{a}, \vec{b}, \vec{c}, \dots$ then the vector $\vec{r} = x\vec{a} + y\vec{b} + z\vec{c} + \dots$ is called a linear combination of $\vec{a}, \vec{b}, \vec{c}, \dots$ for any $x, y, z, \dots \in \mathbb{R}$. We have the following results:

- If $\vec{a}, \vec{b}$ are non zero, non-collinear vectors then $x\vec{a} + y\vec{b} = x'\vec{a} + y'\vec{b} \Rightarrow x = x'; y = y'$
- Fundamental Theorem in plane:** Let $\vec{a}, \vec{b}$ be non-zero, non-collinear vectors. Then any vector $\vec{r}$ coplanar with $\vec{a}, \vec{b}$ can be expressed uniquely as a linear combination of $\vec{a}, \vec{b}$ i.e. There exist some unique $x, y \in \mathbb{R}$ such that $x\vec{a} + y\vec{b} = \vec{r}$
- If $\vec{a}, \vec{b}, \vec{c}$ are non-zero, non-coplanar vectors then : $x\vec{a} + y\vec{b} + z\vec{c} = x'\vec{a} + y'\vec{b} + z'\vec{c} \Rightarrow x = x', y = y', z = z'$



- (d) **Fundamental Theorem in Space :** Let $\vec{a}, \vec{b}, \vec{c}$ be non-zero, non-coplanar vectors in space. Then any vector $\vec{r}$ can be uniquely expressed as a linear combination of $\vec{a}, \vec{b}, \vec{c}$ i.e. There exist some unique $x, y, z \in \mathbb{R}$ such that $x\vec{a} + y\vec{b} + z\vec{c} = \vec{r}$.
- (e) If $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n$ are n non zero vectors & $k_1, k_2, \dots, k_n$ are n scalars & if the linear combination $k_1 \vec{x}_1 + k_2 \vec{x}_2 + \dots + k_n \vec{x}_n = 0 \Rightarrow k_1 = 0, k_2 = 0, \dots, k_n = 0$ then we say that vectors $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n$ are **Linearly Independent Vectors**.
- (f) If $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n$ are not Linearly Independent then they are said to be Linearly Dependent vectors i.e. if $k_1 \vec{x}_1 + k_2 \vec{x}_2 + \dots + k_n \vec{x}_n = 0$ & if there exists at least one $k_i \neq 0$ then $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n$ are said to be **Linearly Dependent**.

NOTES :

- If $\vec{a} = 3\hat{i} + 2\hat{j} + 5\hat{k}$ then $\vec{a}$ is expressed as a Linear Combination of vectors $\hat{i}, \hat{j}, \hat{k}$. Also $\vec{a}, \hat{i}, \hat{j}, \hat{k}$ form a linearly dependent set of vectors. In general, **every set of four vectors is a linearly dependent system**.
- $\hat{i}, \hat{j}, \hat{k}$ are Linearly Independent set of vectors. For $K_1\hat{i} + K_2\hat{j} + K_3\hat{k} = 0 \Rightarrow K_1 = 0 = K_2 = K_3$
- Two vectors $\vec{a}$ & $\vec{b}$ are linearly dependent $\Rightarrow \vec{a}$ is parallel to $\vec{b}$ i.e. $\vec{a} \times \vec{b} = 0 \Rightarrow$ linear dependence of $\vec{a}$ & $\vec{b}$. Conversely if $\vec{a} \times \vec{b} \neq 0$ then $\vec{a}$ & $\vec{b}$ are linearly independent.
- **If three vectors $\vec{a}, \vec{b}, \vec{c}$ are linearly dependent, then they are coplanar i.e. $[\vec{a}, \vec{b}, \vec{c}] = 0$ conversely, if $[\vec{a}, \vec{b}, \vec{c}] \neq 0$, then the vectors are linearly independent.**

10. RECIPROCAL SYSTEM OF VECTORS

If $\vec{a}, \vec{b}, \vec{c}$ & $\vec{a}', \vec{b}', \vec{c}'$ are two sets of non-coplanar vectors such that $\vec{a} \cdot \vec{a}' = \vec{b} \cdot \vec{b}' = \vec{c} \cdot \vec{c}' = 1$ then the two systems are called **Reciprocal System of vectors**.

$$a' = \frac{\vec{b} \times \vec{c}}{[\vec{a} \vec{b} \vec{c}]}; b' = \frac{\vec{c} \times \vec{a}}{[\vec{a} \vec{b} \vec{c}]}; c' = \frac{\vec{a} \times \vec{b}}{[\vec{a} \vec{b} \vec{c}]}$$

3 - DIMENSIONAL GEOMETRY**1. INTRODUCTION**

There are infinite number of points in space. We want to identify each and every point of space with the help of three mutually perpendicular coordinate axes OX, OY and OZ.

1.1 Axes

Three mutually perpendicular lines OX, OY, OZ are considered as three axes.

1.2 Coordinate planes

Plane formed with the help of x and y axes is known as x-y plane similarly plane formed with y and z axes is known as y-z plane and plane formed with z and x axis z-x plane.

1.3 Coordinate of a Point

Consider any point P on the space and drop a perpendicular from that point to x-y plane; then the algebraic length of this perpendicular is considered as z-coordinate; and from the foot of the perpendicular drop perpendiculars to x and y axes the algebraic length of these perpendiculars are considered as y and x coordinates respectively.

1.4 Vector Representation of a Point in Space

If the coordinates of a point P in space are (x, y, z) then the position vector of the point P with respect to the same origin is $x\hat{i} + y\hat{j} + z\hat{k}$.

2. DISTANCE FORMULA

Distance between any two points (x_1, y_1, z_1) and (x_2, y_2, z_2) is given as $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$



Vector method

We know that if position vector of two points A and B are given as $\vec{OA}$ and $\vec{OB}$ then

$$|AB| = |\vec{OB} - \vec{OA}|$$

$$\Rightarrow |AB| = (x_2\mathbf{i} + y_2\mathbf{j} + z_2\mathbf{k}) - (x_1\mathbf{i} + y_1\mathbf{j} + z_1\mathbf{k})$$

$$\Rightarrow |AB| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

2.1 Distance of a Point P From Coordinate Axes

Let PA, PB and PC be the distances of the point P(x, y, z) from the coordinate axes OX, OY and OZ respectively then

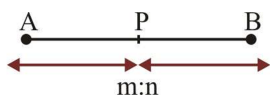
$$PA = \sqrt{y^2 + z^2}, \quad PB = \sqrt{z^2 + x^2}, \quad PC = \sqrt{x^2 + y^2}$$

3. SECTION FORMULA

(i) Internal Division

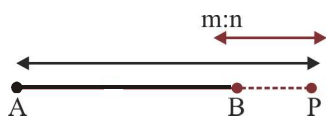
If point P divides the distance between the points A (x_1, y_1, z_1) and B (x_2, y_2, z_2) in the ratio of m : n (internally) then the coordinates of P are given by

$$\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}, \frac{mz_2 + nz_1}{m+n} \right)$$



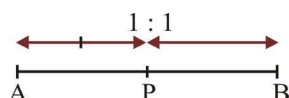
(ii) External division

$$\left(\frac{mx_2 - nx_1}{m-n}, \frac{my_2 - ny_1}{m-n}, \frac{mz_2 - nz_1}{m-n} \right)$$



(iii) Mid point

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right)$$

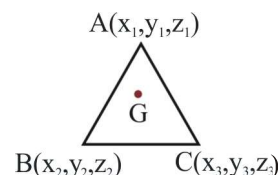


NOTES :

All these formulae are very much similar to two dimensional coordinate geometry.

3.1 Centroid of a Triangle

$$G \equiv \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3} \right)$$



3.2. In - Centre of Triangle ABC

$$\left(\frac{ax_1 + bx_2 + cx_3}{a+b+c}, \frac{ay_1 + by_2 + cy_3}{a+b+c}, \frac{az_1 + bz_2 + cz_3}{a+b+c} \right)$$

Where $|AB| = a, |BC| = b, |CA| = c$

3.3 Centroid of a Tetrahedron

If A (x_1, y_1, z_1), B (x_2, y_2, z_2), C (x_3, y_3, z_3), and D (x_4, y_4, z_4) are the vertices of a tetrahedron then the coordinates of its centroid (G) are given by

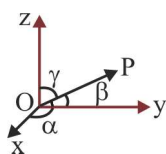
$$\left(\frac{\sum x_i}{4}, \frac{\sum y_i}{4}, \frac{\sum z_i}{4} \right)$$

4. RELATION BETWEEN TWO LINES

Two lines in the space may be coplanar and may be non-coplanar. Non-coplanar lines are called skew lines if they never intersect each other. Two parallel lines are also non intersecting lines but they are coplanar. Whether two lines are intersecting or non intersecting, the angle between them can be obtained.

5. DIRECTION COSINES AND DIRECTION RATIOS

- (i) **Direction cosines :** Let α, β, γ be the angles which a directed line makes with the positive directions of the axes of x, y and z respectively, then $\cos \alpha, \cos \beta, \cos \gamma$ are called the direction cosines of the line. The direction cosines are usually denoted by (l, m, n).



Thus $l = \cos \alpha$, $m = \cos \beta$, $n = \cos \gamma$.

- (ii) If l , m , n , be the direction cosines of a lines, then

$$l^2 + m^2 + n^2 = 1$$

- (iii) **Direction ratios** : Let a , b , c be proportional to the direction cosines, l , m , n , then a , b , c are called the direction ratios.

If a , b , c are the direction ratio of any line L then $a\hat{i} + b\hat{j} + c\hat{k}$ will be a vector parallel to the line L .

If l , m , n are direction cosines of line L then $l\hat{i} + m\hat{j} + n\hat{k}$ is a unit vector parallel to the line L .

- (iv) If l , m , n be the direction cosines and a , b , c be the direction ratios of a vector, then

$$\left(l = \frac{a}{\sqrt{a^2 + b^2 + c^2}}, m = \frac{b}{\sqrt{a^2 + b^2 + c^2}}, n = \frac{c}{\sqrt{a^2 + b^2 + c^2}} \right)$$

$$\text{or } \left(l = \frac{-a}{\sqrt{a^2 + b^2 + c^2}}, m = \frac{-b}{\sqrt{a^2 + b^2 + c^2}}, n = \frac{-c}{\sqrt{a^2 + b^2 + c^2}} \right)$$

- (v) If $OP = r$, when O is the origin and the direction cosines of OP are l , m , n then the coordinates of P are (lr, mr, nr) .

If direction cosines of the line AB are l , m , n , $|AB| = r$, and the coordinates of A are (x_1, y_1, z_1) then the coordinates of B are given as $(x_1 + rl, y_1 + rm, z_1 + rn)$

- (vi) If the coordinates of P and Q are (x_1, y_1, z_1) and (x_2, y_2, z_2) then the direction ratios of line PQ are $a = x_2 - x_1$, $b = y_2 - y_1$ and $c = z_2 - z_1$ and the direction cosines of line

$$PQ \text{ are } l = \frac{x_2 - x_1}{|PQ|}, m = \frac{y_2 - y_1}{|PQ|} \text{ and } n = \frac{z_2 - z_1}{|PQ|}$$

- (vii) **Direction cosines of axes** : Since the positive x -axis makes angles 0° , 90° , 90° with axes of x , y and z respectively, therefore

Direction cosines of x -axis are $(1, 0, 0)$

Direction cosines of y -axis are $(0, 1, 0)$

Direction cosines of z -axis are $(0, 0, 1)$

6. ANGLE BETWEEN TWO LINES

If two lines having direction ratios a_1, b_1, c_1 and a_2, b_2, c_2 respectively then we can consider two vectors parallel to the lines as $a_1\hat{i} + b_1\hat{j} + c_1\hat{k}$ and $a_2\hat{i} + b_2\hat{j} + c_2\hat{k}$ and the angle between them can be given as

$$\cos \theta = \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

- (i) The lines will be perpendicular if $a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$

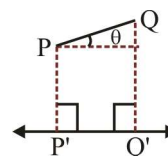
- (ii) The lines will be parallel if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

- (iii) Two parallel lines have same direction cosines i.e. $l_1 = l_2$, $m_1 = m_2$, $n_1 = n_2$

7. PROJECTION OF A LINE SEGMENT ON A LINE

- (i) If the coordinates of P and Q are (x_1, y_1, z_1) and (x_2, y_2, z_2) then the projection of the line segments PQ on a line having direction cosines l , m , n is

$$|l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)|$$



- (ii) **Vector form** : projection of a vector $\vec{a}$ on another vector $\vec{b}$

$$\text{is } \vec{a} \cdot \hat{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} \text{ In the above case we can consider } \vec{PQ}$$

as $(x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$ in place of $\vec{a}$ and $l\hat{i} + m\hat{j} + n\hat{k}$ in place of $\vec{b}$.

- (iii) $l|\vec{r}|$, $m|\vec{r}|$, and $n|\vec{r}|$ are the projection of $\vec{r}$ in x , y and z axes.

- (iv) $\vec{r} = |\vec{r}|(l\hat{i} + m\hat{j} + n\hat{k})$



8. EQUATION OF A LINE

- (i) The equation of a line passing through the point (x_1, y_1, z_1) and having direction ratios a, b, c is

$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c} = r$. This form is called symmetric form. A general point on the line is given by $(x_1 + ar, y_1 + br, z_1 + cr)$.

- (ii) Vector equation : Vector equation of a straight line passing through a fixed point with position vector $\vec{a}$ and parallel to a given vector $\vec{b}$ is $\vec{r} = \vec{a} + \lambda \vec{b}$ where λ is a scalar.

- (iii) The equation of the line passing through the points (x_1, y_1, z_1) and (x_2, y_2, z_2) is

$$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$$

- (iv) Vector equation of a straight line passing through two points with position vectors $\vec{a}$ and $\vec{b}$ is $\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$.

- (v) Reduction of cartesian form of equation of a line to vector form and vice versa

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$$

$$\Leftrightarrow \vec{r} = (x_1\hat{i} + y_1\hat{j} + z_1\hat{k}) + \lambda(a\hat{i} + b\hat{j} + c\hat{k}).$$

NOTES :

Straight lines parallel to co-ordinate axes :

	Straight lines	Equation
(i)	Through origin	$y = mx, z = nx$
(ii)	x-axis	$y = 0, z = 0$
(iii)	y-axis	$x = 0, z = 0$
(iv)	z-axis	$x = 0, y = 0$
(v)	Parallel to x-axis	$y = p, z = q$
(vi)	Parallel to y-axis	$x = h, z = q$
(vii)	Parallel to z-axis	$x = h, y = p$

9. SKEW LINES

- (i) The straight lines which are not parallel and non-coplanar i.e. non-intersecting are called skew lines.

$$\text{If } \Delta = \begin{vmatrix} \alpha' - \alpha & \beta' - \beta & \gamma' - \gamma \\ \ell & m & n \\ \ell' & m' & n' \end{vmatrix} \neq 0, \text{ then the lines are skew.}$$

- (ii) Vector Form : For lines $\vec{a}_1 + \lambda \vec{b}_1$ and $\vec{a}_2 + \lambda \vec{b}_2$ to be skew $(\vec{b}_1 \times \vec{b}_2) \cdot (\vec{a}_2 - \vec{a}_1) \neq 0$ or $[\vec{b}_1 \vec{b}_2 (\vec{a}_2 - \vec{a}_1)] \neq 0$.

- (iii) Shortest distance between the two parallel lines

$$\vec{r} = \vec{a}_1 + \lambda \vec{b} \text{ and } \vec{r} = \vec{a}_2 + \mu \vec{b} \text{ is } d = \frac{|(\vec{a}_2 - \vec{a}_1) \times \vec{b}|}{|\vec{b}|}.$$

10. PLANE

If a line joining any two points on a surface lies completely on it then the surface is a plane.

OR

If a line joining any two points on a surface is perpendicular to some fixed straight line, then the surface is called a plane. This fixed line is called the normal to the plane.

10.1 Equation of a Plane

- (i) Normal form of the equation of a plane is $lx + my + nz = p$ where l, m, n are the direction cosines of the normal to the plane and p is the distance of the plane from the origin.
- (ii) General form : $ax + by + cz + d = 0$ is the equation of a plane where a, b, c are the direction ratios of the normal to the plane.
- (iii) The equation of a plane passing through the point (x_1, y_1, z_1) is given by $a(x-x_1) + b(y-y_1) + c(z-z_1) = 0$ where a, b, c are the direction ratios of the normal to the plane.
- (iv) Plane through three points : The equation of the plane through three non-collinear points $(x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3)$ is

$$(x_2, y_2, z_2), (x_3, y_3, z_3) \text{ is } \begin{vmatrix} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{vmatrix} = 0$$



- (v) Intercept Form : The equation of a plane cutting intercepts a, b, c on the axes is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$
- (vi) Vector form : The equation of a plane passing through a point having position vector $\vec{a}$ and normal to vector $\vec{n}$ is $(\vec{r} - \vec{a}) \cdot \vec{n} = 0$ or $\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$

NOTES :

- (a) Vector equation of a plane normal to unit vector $\hat{n}$ and at a distance d from the origin is $\vec{r} \cdot \hat{n} = d$
- (b) Planes parallel to the coordinate planes
- (i) Equation of yz -plane is $x = 0$
- (ii) Equation of xz -plane is $y = 0$
- (iii) Equation of xy -plane is $z = 0$
- (c) Planes parallel to the axes :
- If $a = 0$, the plane is parallel to x -axis i.e. equation of the plane parallel to the x -axis is $by + cz + d = 0$.
Similarly, equation of planes parallel to y -axis and parallel to z -axis are $ax + cz + d = 0$ and $ax + by + d = 0$ respectively.
- (d) Plane through origin : Equation of plane passing through origin is $ax + by + cz = 0$.
- (e) Transformation of the equation of a plane to the normal form : To reduce any equation $ax + by + cz - d = 0$ to the normal form, first write the constant term on the right hand side and make it positive, then divided each term by $\sqrt{a^2 + b^2 + c^2}$, where a, b, c are coefficients of x, y and z respectively e.g.

$$\frac{ax}{\pm\sqrt{a^2 + b^2 + c^2}} + \frac{by}{\pm\sqrt{a^2 + b^2 + c^2}} + \frac{cz}{\pm\sqrt{a^2 + b^2 + c^2}} = \frac{d}{\pm\sqrt{a^2 + b^2 + c^2}}$$

Where (+) sign is to be taken if $d > 0$ and (-) sign is to be taken if $d < 0$.

- (f) Any plane parallel to the given plane $ax + by + cz + d = 0$ is $ax + by + cz + \lambda = 0$. Distance between two parallel planes $ax + by + cz + d_1 = 0$ and $ax + by + cz + d_2 = 0$ is given as $\frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}}$

- (g) Equation of a plane passing through a given point and parallel to the given vectors : The equation of a plane passing through a point having position vector $\vec{a}$ and parallel to $\vec{b}$ and $\vec{c}$ is $\vec{r} = \vec{a} + \lambda \vec{b} + \mu \vec{c}$ (parametric form where λ and μ are scalars).
or $\vec{r} \cdot (\vec{b} \times \vec{c}) = \vec{a} \cdot (\vec{b} \times \vec{c})$ (non-parametric form)

11 ANGLE BETWEEN TWO PLANES

- (i) Consider two planes $ax + by + cz + d = 0$ and $a'x + b'y + c'z + d' = 0$. Angle between these planes is the angle between their normals. Since direction ratios of their normals are (a, b, c) and (a', b', c') respectively, hence the angle θ between them is given by

$$\cos \theta = \frac{aa' + bb' + cc'}{\sqrt{a^2 + b^2 + c^2} \sqrt{a'^2 + b'^2 + c'^2}}$$

Planes are perpendicular if $aa' + bb' + cc' = 0$ and planes are

$$\text{parallel if } \frac{a}{a'} = \frac{b}{b'} = \frac{c}{c'}$$

- (ii) The angle θ between the planes $\vec{r} \cdot \vec{n}_1 = d_1$, $\vec{r} \cdot \vec{n}_2 = d_2$ is given by $\cos \theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1| |\vec{n}_2|}$. Planes are perpendicular if $\vec{n}_1 \cdot \vec{n}_2 = 0$ and Planes are parallel if $\vec{n}_1 = \lambda \vec{n}_2$.

12. A PLANE AND A POINT

- (i) Distance of the point (x', y', z') from the plane $ax + by + cz + d = 0$ is given by $\frac{|ax' + by' + cz' + d|}{\sqrt{a^2 + b^2 + c^2}}$
- (ii) The length of the perpendicular from a point having position vector $\vec{a}$ to a plane $\vec{r} \cdot \vec{n} = d$ is given by $p = \frac{|\vec{a} \cdot \vec{n} - d|}{|\vec{n}|}$.

13. ANGLE BISECTORS

- (i) The equations of the planes bisecting the angle between two given planes $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ are



$$\frac{a_1x + b_1y + c_1z + d_1}{\sqrt{a_1^2 + b_1^2 + c_1^2}} = \pm \frac{a_2x + b_2y + c_2z + d_2}{\sqrt{a_2^2 + b_2^2 + c_2^2}}$$

- (ii) **Equation of bisector of the angle containing origin** : First make both the constant terms positive. Then the positive

$$\text{sign in } \frac{a_1x + b_1y + c_1z + d_1}{\sqrt{a_1^2 + b_1^2 + c_1^2}} = \pm \frac{a_2x + b_2y + c_2z + d_2}{\sqrt{a_2^2 + b_2^2 + c_2^2}}$$

gives the bisector of the angle which contains the origin.

- (iii) **Bisector of acute/obtuse angle** : First make both the constant terms positive. Then

$$a_1a_2 + b_1b_2 + c_1c_2 > 0$$

⇒ origin lies in obtuse angle

$$a_1a_2 + b_1b_2 + c_1c_2 < 0$$

⇒ origin lies in acute angle

14. FAMILY OF PLANES

- (i) The equation of any plane passing through the line of intersection of non-parallel planes $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ is given by $a_1x + b_1y + c_1z + d_1 + \lambda(a_2x + b_2y + c_2z + d_2) = 0$
- (ii) The equation of plane passing through the intersection of the planes $\vec{r} \cdot \vec{n}_1 = d_1$ and $\vec{r} \cdot \vec{n}_2 = d_2$ is $\vec{r} \cdot (\vec{n}_1 + \lambda\vec{n}_2) = d_1 + \lambda d_2$ where λ is an arbitrary scalar
- (iii) **Plane through a given line** : Equation of any plane through the line in symmetrical form.

$$\frac{x-x_1}{\ell} = \frac{y-y_1}{m} = \frac{z-z_1}{n} \text{ is } A(x-x_1) + B(y-y_1) +$$

$$C(z-z_1) = 0 \text{ where } A\ell + Bm + Cn = 0$$

NOTES :

A straight line in space is characterised by the intersection of two planes which are not parallel and therefore, the equation of a straight line is a solution of the system constituted by the equations of the two planes $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$. This form is also known as non-symmetrical form.

15. ANGLE BETWEEN A PLANE AND A LINE

- (i) If θ is the angle between the line $\frac{x-x_1}{\ell} = \frac{y-y_1}{m} = \frac{z-z_1}{n}$ and the plane $ax + by + cz + d = 0$, then

$$\sin \theta = \left[\frac{a\ell + bm + cn}{\sqrt{(a^2 + b^2 + c^2)} \sqrt{\ell^2 + m^2 + n^2}} \right]$$

- (ii) Vector form : If θ is the angle between a line $\vec{r} = (\vec{a} + \lambda\vec{b})$

$$\text{and } \vec{r} \cdot \vec{n} = d \text{ then } \sin \theta = \left[\frac{\vec{b} \cdot \vec{n}}{|\vec{b}| |\vec{n}|} \right].$$

- (iii) Condition for perpendicularity $\frac{\ell}{a} = \frac{m}{b} = \frac{n}{c}$, $\vec{b} \times \vec{n} = 0$

- (iv) Condition for parallelism $a\ell + bm + cn = 0$, $\vec{b} \cdot \vec{n} = 0$

16. CONDITION FOR A LINE TO LIE IN A PLANE

- (i) Cartesian form : Line $\frac{x-x_1}{\ell} = \frac{y-y_1}{m} = \frac{z-z_1}{n}$ would lie in a plane $ax + by + cz + d = 0$ if $ax_1 + by_1 + cz_1 + d = 0$ and $a\ell + bm + cn = 0$.
- (ii) Vector form : Line $\vec{r} = \vec{a} + \lambda\vec{b}$ would lie in the plane $\vec{r} \cdot \vec{n} = d$ if $\vec{b} \cdot \vec{n} = 0$ and $\vec{a} \cdot \vec{n} = d$

17. COPLANER LINES

- (i) If the given lines are $\frac{x-\alpha}{\ell} = \frac{y-\beta}{m} = \frac{z-\gamma}{n}$ and

$$\frac{x-\alpha'}{\ell'} = \frac{y-\beta'}{m'} = \frac{z-\gamma'}{n'}, \text{ then the condition for}$$

$$\text{intersection/coplanarity is } \begin{vmatrix} \alpha-\alpha' & \beta-\beta' & \gamma-\gamma' \\ \ell & m & n \\ \ell' & m' & n' \end{vmatrix} = 0 \text{ and}$$

the plane containing the above two lines is

$$\begin{vmatrix} x-\alpha & y-\beta & z-\gamma \\ \ell & m & n \\ \ell' & m' & n' \end{vmatrix} = 0$$



- (ii) Condition of coplanarity if both the lines are in general asymmetric form :-

$$ax + by + cz + d = 0 = a'x + b'y + c'z + d' \text{ and}$$

$$\alpha x + \beta y + \gamma z + \delta = 0 = \alpha'x + \beta'y + \gamma'z + \delta'$$

They are coplanar if
$$\begin{vmatrix} a & b & c & d \\ a' & b' & c' & d' \\ \alpha & \beta & \gamma & \delta \\ \alpha' & \beta' & \gamma' & \delta' \end{vmatrix} = 0$$

18. COPLANARITY OF FOUR POINTS

If the points $A(x_1, y_1, z_1)$, $B(x_2, y_2, z_2)$, $C(x_3, y_3, z_3)$ and $D(x_4, y_4, z_4)$ are coplanar then

$$\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \\ x_4 - x_1 & y_4 - y_1 & z_4 - z_1 \end{vmatrix} = 0$$

Similarly, in vector method the points $A(\vec{r}_1)$, $B(\vec{r}_2)$, $C(\vec{r}_3)$ and $D(\vec{r}_4)$ are coplanar if there exists 4 scalars a, b, c and d such that

$$a + b + c + d = 0 \text{ and } a\vec{r}_1 + b\vec{r}_2 + c\vec{r}_3 + d\vec{r}_4 = 0$$

19. SIDES OF A PLANE

- (i) A plane divides the three dimensional space in two parts. Two points $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ are on the same side of the plane $ax + by + cz + d = 0$ if $ax_1 + by_1 + cz_1 + d$ and $ax_2 + by_2 + cz_2 + d$ are both positive or both negative and are on opposite sides of plane if these values are opposite in signs.

- (ii) A plane $ax + by + cz + d = 0$ divides the line segment joining (x_1, y_1, z_1) and (x_2, y_2, z_2) in the ratio

$$\left(-\frac{ax_1 + by_1 + cz_1 + d}{ax_2 + by_2 + cz_2 + d} \right)$$

- (iii) The xy -plane divides the line segment joining the point (x_1, y_1, z_1) and (x_2, y_2, z_2) in the ratio $-\frac{z_1}{z_2}$. Similarly

yz -plane in the ratio of $-\frac{x_1}{x_2}$ and zx -plane in the ratio

$$\text{of } -\frac{y_1}{y_2}$$

20. FOOT OF PERPENDICULAR AND IMAGE OF A POINT W.R.T. A PLANE

Let $P(x_1, y_1, z_1)$ be a given point and $ax + by + cz + d = 0$ be a given plane. Let (x', y', z') be the image point. Then,

$$\frac{x' - x_1}{a} = \frac{y' - y_1}{b} = \frac{z' - z_1}{c} = \frac{-2(ax_1 + by_1 + cz_1 + d)}{a^2 + b^2 + c^2}$$

And if (x_2, y_2, z_2) is foot of perpendicular from point P on given plane, then

$$\frac{x_2 - x_1}{a} = \frac{y_2 - y_1}{b} = \frac{z_2 - z_1}{c} = \frac{-(ax_1 + by_1 + cz_1 + d)}{a^2 + b^2 + c^2}$$



SOLVED EXAMPLES

Example – 1

Show that the point $P(\vec{a} + 2\vec{b} + \vec{c})$, $Q(\vec{a} - \vec{b} - \vec{c})$, $R(3\vec{a} + \vec{b} + 2\vec{c})$ and $S(5\vec{a} + 3\vec{b} + 5\vec{c})$ are coplanar given that $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar.

Sol. To show that P, Q, R, S are coplanar, we will show that $\overrightarrow{PQ}, \overrightarrow{PR}, \overrightarrow{PS}$ are coplanar.

$$\overrightarrow{PQ} = -3\vec{b} - 2\vec{c}$$

$$\overrightarrow{PR} = 2\vec{a} - \vec{b} + \vec{c}$$

$$\overrightarrow{PS} = 4\vec{a} + \vec{b} + 4\vec{c}$$

$$\text{Let } \overrightarrow{PQ} = \lambda \overrightarrow{PR} + \mu \overrightarrow{PS}$$

$$\Rightarrow -3\vec{b} - 2\vec{c} = \lambda (2\vec{a} - \vec{b} + \vec{c}) + \mu (4\vec{a} + \vec{b} + 4\vec{c})$$

$$\Rightarrow -3\vec{b} - 2\vec{c} = (2\lambda + 4\mu)\vec{a} + (-\lambda + \mu)\vec{b} + (\lambda + 4\mu)\vec{c}$$

As the vectors, $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar, we can equate their coefficients.

$$\Rightarrow 0 = 2\lambda + 4\mu$$

$$\Rightarrow -3 = -\lambda + \mu$$

$$\Rightarrow -2 = \lambda + 4\mu$$

$\lambda = 2, \mu = -1$ is the unique solution for the above system of equations.

$$\Rightarrow \overrightarrow{PQ} = 2\overrightarrow{PR} - \overrightarrow{PS}$$

$\overrightarrow{PQ}, \overrightarrow{PR}, \overrightarrow{PS}$ are coplanar because $\overrightarrow{PQ}$ is a linear combination of $\overrightarrow{PR}$ and $\overrightarrow{PS}$.

$\Rightarrow$ the points P, Q, R, S are also coplanar.

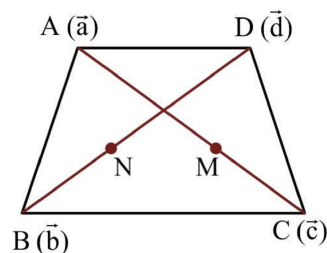
Example – 2

Prove that the segment joining mid-points of the diagonals of a trapezium is parallel to the parallel sides of the trapezium and is equal to half the difference of their lengths.

Sol. Let ABCD be the given trapezium and M, N be the mid points of the diagonals AC and BD.

Let $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ be the position vectors of A, B, C, D respectively.

using section formula, mid points of AC and BD are :



$$\vec{m} = \frac{\vec{a} + \vec{c}}{2}, \quad \vec{n} = \frac{\vec{b} + \vec{d}}{2}$$

$$\Rightarrow \overrightarrow{NM} = \vec{m} - \vec{n} = \frac{(\vec{a} + \vec{c}) - (\vec{b} + \vec{d})}{2}$$

$$\Rightarrow \overrightarrow{NM} = \left(\frac{\vec{c} - \vec{b}}{2} \right) - \left(\frac{\vec{d} - \vec{a}}{2} \right)$$

$$\Rightarrow \overrightarrow{NM} = 1/2 (\overrightarrow{BC} - \overrightarrow{AD})$$

$$\text{Let } \overrightarrow{BC} = k(\overrightarrow{AD})$$

$$\Rightarrow \overrightarrow{NM} = 1/2 (k - 1) \overrightarrow{AD}$$

$$NM \parallel AD \quad \text{and} \quad NM = 1/2 (k - 1) AD$$

$$\Rightarrow NM = \frac{k(AD) - AD}{2} = \frac{BC - AD}{2}$$

$\Rightarrow$ NM is parallel to AD (and BC) and is half the difference of BC and AD.

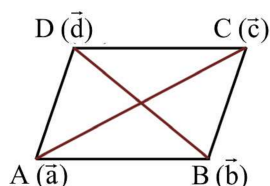


Example – 3

Show that the diagonals of a parallelogram bisect each other.

Sol. Let $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ be the position vectors of a vertices of a parallelogram ABCD.

$AB = DC$ and $AB \parallel DC$
(because ABCD is a parallelogram)



$$\Rightarrow \overline{AB} = \overline{DC}$$

$$\Rightarrow \vec{b} - \vec{a} = \vec{c} - \vec{d}$$

$$\Rightarrow (\vec{b} + \vec{d}) / 2 = (\vec{a} + \vec{c}) / 2$$

$$\Rightarrow \text{P.V of mid point of BD} = \text{P.V of mid point of AC}$$

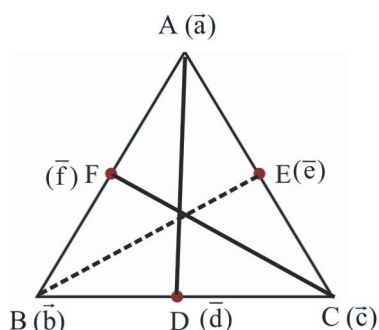
$$\Rightarrow \text{mid points of BD and AC coincide. Hence AC and BD bisect each other.}$$

Example – 4

Show that the medians of the triangle are concurrent and the point of concurrence divides each median in the ratio 2 : 1.

Sol. Let $\vec{a}, \vec{b}, \vec{c}$ be the position vectors of the vertices of a triangle ABC.

Let D, E, F be the mid-points of sides with P.V $\vec{d}, \vec{e}, \vec{f}$ as shown.



$$\Rightarrow \vec{d} = (\vec{b} + \vec{c}) / 2 \quad \Rightarrow \quad 2\vec{d} = \vec{b} + \vec{c}$$

$$\Rightarrow \vec{e} = (\vec{c} + \vec{a}) / 2 \quad \Rightarrow \quad 2\vec{e} = \vec{c} + \vec{a}$$

$$\Rightarrow \vec{f} = (\vec{a} + \vec{b}) / 2 \quad \Rightarrow \quad 2\vec{f} = \vec{a} + \vec{b}$$

Now try to make the RHS of each equation equal.

$$\Rightarrow 2\vec{d} + \vec{a} = \vec{a} + \vec{b} + \vec{c}$$

$$\Rightarrow 2\vec{e} + \vec{b} = \vec{b} + \vec{c} + \vec{a}$$

$$\Rightarrow 2\vec{f} + \vec{c} = \vec{c} + \vec{a} + \vec{b}$$

$$\Rightarrow 2\vec{d} + \vec{a} = 2\vec{e} + \vec{b} = 2\vec{f} + \vec{c} = \vec{a} + \vec{b} + \vec{c}$$

Note that the sum of scalar coefficients of vectors is equal to 3 in each expression. We divide each term by 3.

$$\Rightarrow \frac{2\vec{d} + \vec{a}}{3} = \frac{2\vec{e} + \vec{b}}{3} = \frac{2\vec{f} + \vec{c}}{3} = \frac{\vec{a} + \vec{b} + \vec{c}}{3}$$

$$\Rightarrow \frac{2\vec{d} + \vec{a}}{2+1} = \frac{2\vec{e} + \vec{b}}{2+1} = \frac{2\vec{f} + \vec{c}}{2+1} = \frac{\vec{a} + \vec{b} + \vec{c}}{2+1}$$

$\Rightarrow$ the point G $[(\vec{a} + \vec{b} + \vec{c}) / 3]$ divides AD, BE and CF each internally in ratio 2 : 1. Hence G is the common point of intersection of all medians.

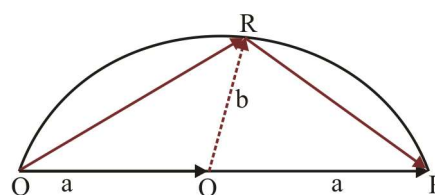
$\Rightarrow$ medians are concurrent and centroid G divides each median in 2 : 1.

$$\text{Centroid G} \equiv \left(\frac{\vec{a} + \vec{b} + \vec{c}}{3} \right)$$

Example – 5

Show that the angle in semi-circle is a right angle.

Sol. Let O be the centre and r be the radius of the semi-circle.



$$\text{Let } \overline{OP} = \overline{OQ} = \vec{a} \text{ and } \overline{OR} = \vec{b}$$

$$\Rightarrow \overline{QR} = \vec{a} + \vec{b} \text{ and } \overline{RP} = \vec{a} - \vec{b}$$

$$\text{Now } \overline{QR} \cdot \overline{RP} = (\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b})$$

$$= a^2 - b^2$$

$$= a^2 - b^2 = 0$$

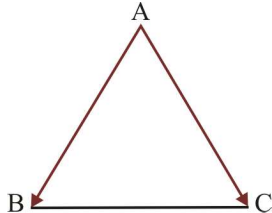
because $a = b = \text{radius of the semi-circle.}$

$$\therefore \overline{QR} \cdot \overline{RP} = 0 \Rightarrow \theta = 90^\circ$$

Example – 6

The vertices of a triangle are A (2, 3, 0), B (–3, 2, 1) and C (4, –1, 0). Find the area of the triangle ABC and unit vector normal to the plane of this triangle.

Sol. Area of $\triangle ABC = \frac{1}{2} |\overrightarrow{AB} \times \overrightarrow{AC}|$



$$\overrightarrow{AB} = (-3 - 2)\hat{i} + (2 - 3)\hat{j} + (1 - 0)\hat{k}$$

$$\Rightarrow \overrightarrow{AB} = -5\hat{i} - \hat{j} + \hat{k}$$

$$\text{and } \overrightarrow{AC} = 2\hat{i} - 4\hat{j} + 0\hat{k}$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -5 & -1 & 1 \\ 2 & -4 & 0 \end{vmatrix} = 4\hat{i} + 2\hat{j} + 22\hat{k}$$

$$\Rightarrow \text{area of } \triangle ABC = \frac{1}{2} \sqrt{16 + 4 + 484} = \sqrt{126} \text{ sq. units}$$

and unit vector normal to the plane of this triangle

$$= \frac{\overrightarrow{AB} \times \overrightarrow{AC}}{|\overrightarrow{AB} \times \overrightarrow{AC}|} = \frac{4\hat{i} + 2\hat{j} + 22\hat{k}}{2\sqrt{126}}$$

$$= \frac{2\hat{i} + \hat{j} + 11\hat{k}}{\sqrt{126}}$$

Example – 7

A line makes angles α , β , γ and δ with the diagonals of a cube. Prove that :

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = 4/3$$

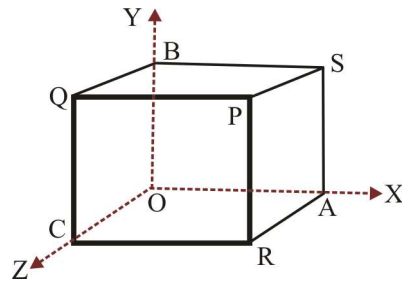
Sol. Let the origin O be one of the vertices of the cube and OA, OB, OC be the edges through O along the axes so that :

$$\overrightarrow{OA} = a\hat{i}, \quad \overrightarrow{OB} = a\hat{j}, \quad \overrightarrow{OC} = a\hat{k}$$

where a is the length of the edge of the cube

Let P, Q, R, S be the other vertices of the

cube opposite to O, A, B, C respectively.



Hence the diagonals of the cube are OP, AQ, BR, & CS.

$$\overrightarrow{OP} = a\hat{i} + a\hat{j} + a\hat{k}$$

$$\overrightarrow{AQ} = a\hat{i} + a\hat{j} + a\hat{k}$$

$$\overrightarrow{BR} = a\hat{i} - a\hat{j} + a\hat{k}$$

$$\overrightarrow{CS} = a\hat{i} + a\hat{j} - a\hat{k}$$

If $\hat{n} = x\hat{i} + y\hat{j} + z\hat{k}$ is the unit vector along the line which makes the angles α , β , γ & δ with diagonals,

$$\cos \alpha = \frac{\hat{n} \cdot \overrightarrow{OP}}{|\overrightarrow{OP}| |\hat{n}|} = \frac{ax + ay + az}{a\sqrt{3}} = \frac{x + y + z}{\sqrt{3}}$$

$$\cos \beta = \frac{-x + y + z}{\sqrt{3}}; \quad \cos \gamma = \frac{x - y + z}{\sqrt{3}}$$

$$\cos \delta = \frac{x + y - z}{\sqrt{3}}$$

$$\Rightarrow \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta$$

$$= \frac{1}{3} [(x + y + z)^2 + (-x + y + z)^2 + (x - y + z)^2 + (x + y - z)^2]$$

$$= \frac{1}{3} 4(x^2 + y^2 + z^2) = 4/3 \quad [\because x^2 + y^2 + z^2 = 1]$$

Example – 8

Show that : $\hat{i} \times (\vec{a} \times \hat{i}) + \hat{j} \times (\vec{a} \times \hat{j}) + \hat{k} \times (\vec{a} \times \hat{k}) = 2\vec{a}$.

Sol. L.H.S. = $[(\hat{i} \cdot \hat{i})\vec{a} - (\hat{i} \cdot \vec{a})\hat{i}] + [(\hat{j} \cdot \hat{j})\vec{a} - (\hat{j} \cdot \vec{a})\hat{j}] + [(\hat{k} \cdot \hat{k})\vec{a} - (\hat{k} \cdot \vec{a})\hat{k}]$

$$= \vec{a} + \vec{a} + \vec{a} - [(\hat{i} \cdot \vec{a})\hat{i} + (\hat{j} \cdot \vec{a})\hat{j} + (\hat{k} \cdot \vec{a})\hat{k}]$$

$$= 3\vec{a} - [a_x\hat{i} + a_y\hat{j} + a_z\hat{k}] = 3\vec{a} - \vec{a} = 2\vec{a} = \text{R.H.S.}$$

Note : It is useful to remember that x-component of $\vec{a} = a_x\hat{i}$ etc



Example – 9

Find a vector of magnitude 5 units coplanar with vectors $3\hat{i} - \hat{j} - \hat{k}$ and $\hat{i} + \hat{j} - 2\hat{k}$ and perpendicular to the vector $2\hat{i} + 2\hat{j} + \hat{k}$.

Sol. Let $\vec{a} = 3\hat{i} - \hat{j} - \hat{k}$

$$\vec{b} = \hat{i} + \hat{j} - 2\hat{k}$$

and $\vec{c} = 2\hat{i} + 2\hat{j} + \hat{k}$

A vector coplanar with $\vec{a}$ and $\vec{b}$ and perpendicular to $\vec{c}$ can be taken as

$$\vec{r} = \ell \vec{c} \times (\vec{a} \times \vec{b}) \quad \text{where } \ell \text{ is a scalar}$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & -1 \\ 1 & 1 & -2 \end{vmatrix} = 3\hat{i} + 5\hat{j} + 4\hat{k}$$

$$\vec{c} \times (\vec{a} \times \vec{b}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 2 & 1 \\ 3 & 5 & 4 \end{vmatrix} = 3\hat{i} - 5\hat{j} + 4\hat{k}$$

$$\Rightarrow \vec{r} = \ell (3\hat{i} - 5\hat{j} + 4\hat{k})$$

$$|\vec{r}| = |\ell| \sqrt{9+25+16} = 5$$

$$\ell = \pm \frac{5}{5\sqrt{2}} = \pm \frac{1}{\sqrt{2}}$$

the required vector is $\vec{r} = \pm \frac{1}{\sqrt{2}} (3\hat{i} - 5\hat{j} + 4\hat{k})$

Example – 10

Show that the lines $\vec{r} = 3\hat{i} - \hat{j} + \hat{k} + \lambda (\hat{i} + \hat{j} + \hat{k})$ and $\vec{r} = 2\hat{i} + 2\hat{j} - 2\hat{k} + \mu (\hat{i} - \hat{j} + 2\hat{k})$ are intersecting and hence find their point of intersection.

Sol. Let $\vec{p}$ be the position vector of their point of intersection.

$$\Rightarrow \vec{p} = 3\hat{i} - \hat{j} + \hat{k} + \lambda (\hat{i} + \hat{j} + \hat{k}) = 2\hat{i} + 2\hat{j} - 2\hat{k} + \mu (\hat{i} - \hat{j} + 2\hat{k})$$

$$\Rightarrow (3 + \lambda) \hat{i} + (\lambda - 1) \hat{j} + (\lambda + 1) \hat{k} = (\mu + 2) \hat{i} + (2 - \mu) \hat{j} + (2\mu - 2) \hat{k}$$

$$\Rightarrow 3 + \lambda = \mu + 2 \quad \dots(i)$$

$$\Rightarrow \lambda - 1 = 2 - \mu \quad \dots(ii)$$

$$\Rightarrow \lambda + 1 = 2\mu - 2 \quad \dots(iii)$$

The lines are intersecting if these equations are consistent.

from (i) and (ii), we get

$$\lambda = 1, \mu = 2$$

Substituting these values in (iii), we get

$$1 + 1 = 2(2) - 2$$

$$\Rightarrow 2 = 2$$

$$\Rightarrow \lambda = 1, \mu = 2$$

satisfied (iii) also

Hence lines are intersecting and the point of intersection is :

$$\vec{p} = 3\hat{i} - \hat{j} + \hat{k} + \lambda (\hat{i} + \hat{j} + \hat{k})$$

$$= 3\hat{i} - \hat{j} + \hat{k} + (\hat{i} + \hat{j} + \hat{k})$$

$$= 4\hat{i} + 2\hat{k}$$

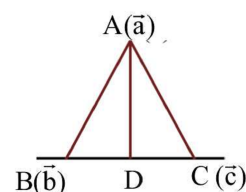
$$\Rightarrow \text{the coordinates of this point are } (4, 0, 2).$$

Example – 11

The vertices of a triangle ABC are A (1, 0, 2), B (-2, 1, 3) and C (2, -1, 1). Find the equation of the line BC, the foot of the perpendicular from A to BC and the length of the perpendicular.

Sol. A vector parallel to BC is

$$\vec{BC} = \vec{c} - \vec{b} = 4\hat{i} - 2\hat{j} - 2\hat{k}$$



$$\Rightarrow \text{the equation of BC is : } \vec{r} = \vec{b} + t(\vec{c} - \vec{b})$$

$$\Rightarrow \vec{r} = -2\hat{i} + \hat{j} + 3\hat{k} + t(4\hat{i} - 2\hat{j} - 2\hat{k})$$

Let position vector of D be



$$\vec{d} = -2\hat{i} + \hat{j} + 3\hat{k} + t(4\hat{i} - 2\hat{j} - 2\hat{k})$$

because D lies on line BC.

Now $\overline{AD} \perp \overline{BC}$

$$\Rightarrow \overline{AD} \cdot \overline{BC} = 0$$

$$\Rightarrow (\vec{d} - \vec{a}) \cdot (\vec{c} - \vec{b}) = 0$$

$$\Rightarrow [-3\hat{i} + \hat{j} + \hat{k} + t(4\hat{i} - 2\hat{j} - 2\hat{k})] \cdot (4\hat{i} - 2\hat{j} - 2\hat{k}) = 0$$

$$\Rightarrow (4t - 3)4 + (1 - 2t)(-2) + (1 - 2t)(-2) = 0$$

$$\Rightarrow 24t - 16 = 0$$

$$\Rightarrow t = 2/3$$

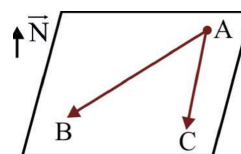
$$\Rightarrow \vec{d} = -2\hat{i} + \hat{j} + 3\hat{k} + (2/3)(4\hat{i} - 2\hat{j} - 2\hat{k})$$

$$\vec{d} = \frac{2}{3}\hat{i} - \frac{1}{3}\hat{j} + \frac{5}{3}\hat{k}$$

$$\Rightarrow D \equiv \left(\frac{2}{3}, -\frac{1}{3}, \frac{5}{3}\right)$$

$$\overline{AD} = \vec{d} - \vec{a} = \left(\frac{2}{3} - 1\right)\hat{i} - \frac{1}{3}\hat{j} + \left(\frac{5}{3} - 2\right)\hat{k}$$

$$= -\frac{1}{3}\hat{i} - \frac{1}{3}\hat{j} - \frac{1}{3}\hat{k} \quad AD = |\overline{AD}| = \frac{\sqrt{3}}{3} = \frac{1}{\sqrt{3}} \text{ units}$$



$\Rightarrow$ the equation of the plane is

$$(\vec{r} - \vec{a}) \cdot \vec{N} = 0 \text{ where } \vec{a} = 2\hat{i} + \hat{j} + 3\hat{k}$$

$$\Rightarrow \vec{r} \cdot \vec{N} = \vec{a} \cdot \vec{N}$$

$$\Rightarrow \vec{r} \cdot (-3\hat{i} - 8\hat{j} - \hat{k}) = -6 - 8 - 3$$

$$\Rightarrow \vec{r} \cdot (-3\hat{i} - 8\hat{j} - \hat{k}) = -17$$

The given line is $\vec{r} = \hat{i} - \hat{j} + \hat{k} + \lambda(2\hat{i} + \hat{k})$

To find the point of intersection, we solve these equations simultaneously.

$$\Rightarrow [\hat{i} - \hat{j} + \hat{k} + \lambda(2\hat{i} + \hat{k})] \cdot (-3\hat{i} - 8\hat{j} - \hat{k}) = -17$$

$$\Rightarrow (2\lambda + 1)(-3) + 8 + (\lambda + 1)(-1) = -17$$

$$\Rightarrow \lambda = 3$$

$\Rightarrow$ the point of intersection is

$$\vec{r} = \hat{i} - \hat{j} + \hat{k} + 3(2\hat{i} + \hat{k})$$

$$\Rightarrow \vec{r} = 7\hat{i} - \hat{j} + 4\hat{k}$$

$\Rightarrow$ coordinates are (7, -1, 4)

Example - 12

Find the equation of the plane passing through the points A (2, 1, 3), B (-1, 2, 4) and C (0, 2, 1). Hence find the coordinate of the point of intersection of the plane ABC and the line $\vec{r} = \hat{i} - \hat{j} + \hat{k} + \lambda(2\hat{i} + \hat{k})$.

Sol. Let $\vec{N}$ be a vector perpendicular to the plane of ΔABC .

$$\Rightarrow \vec{N} = \overline{AB} \times \overline{AC}$$

$$= (-3\hat{i} + \hat{j} + \hat{k}) \times (-2\hat{i} + \hat{j} - 2\hat{k})$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -3 & 1 & 1 \\ -2 & 1 & -2 \end{vmatrix}$$

$$= -3\hat{i} - 8\hat{j} - \hat{k}$$

Example - 13

From the point A (1, 2, 0), perpendicular is drawn to the plane $\vec{r} \cdot (3\hat{i} - \hat{j} + \hat{k}) = 2$ meeting it at the point P. Find the coordinates of point P and the distance AP.

Sol. Let us first find the equation of line AP. As AP is normal to the plane, the vector $\vec{N} = 3\hat{i} - \hat{j} + \hat{k}$ is parallel to AP.

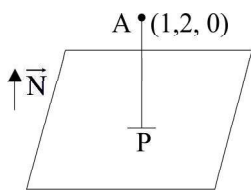
$$\Rightarrow \text{equation of AP is } \vec{r} = \hat{i} + 2\hat{j} + t(3\hat{i} - \hat{j} + \hat{k})$$

Now we solve equations of AP and plane to get point P.

$$\Rightarrow [\hat{i} + 2\hat{j} + t(3\hat{i} - \hat{j} + \hat{k})] \cdot (3\hat{i} - \hat{j} + \hat{k}) = 2$$

$$\Rightarrow (3t + 1)3 + (2 - t)(-1) + t = 2$$

$$\Rightarrow t = 1/11$$



⇒ point P is $\vec{r} = \hat{i} + 2\hat{j} + \frac{1}{11}(3\hat{i} - \hat{j} + \hat{k})$

$$\vec{r} = \frac{14}{11}\hat{i} + \frac{21}{11}\hat{j} + \frac{1}{11}\hat{k}$$

$$\Rightarrow P \equiv \left(\frac{14}{11}, \frac{21}{11}, \frac{1}{11}\right)$$

$$AP = \sqrt{\left(\frac{14}{11} - 1\right)^2 + \left(\frac{21}{11} - 2\right)^2 + \left(\frac{1}{11} - 0\right)^2}$$

$$= \frac{1}{\sqrt{11}}$$

Example – 14

The position vectors of the points P and Q are $5\hat{i} + 7\hat{j} - 2\hat{k}$ and $-3\hat{i} + 3\hat{j} + 6\hat{k}$ respectively. The vector $\vec{A} = 3\hat{i} - \hat{j} + \hat{k}$ passes through the point P and the vector $\vec{B} = -3\hat{i} + 2\hat{j} + 4\hat{k}$ passes through the point Q. A third vector $2\hat{i} + 7\hat{j} - 5\hat{k}$ intersects vectors $\vec{A}$ and $\vec{B}$. Find the position vectors of the points of intersection.

Sol. Equation of line AP $\equiv \vec{r} = 5\hat{i} + 7\hat{j} - 2\hat{k} + \lambda_1(3\hat{i} - \hat{j} + \hat{k})$

Equation of line BQ $\equiv$

$$\vec{r} = -3\hat{i} + 3\hat{j} + 6\hat{k} + \lambda_2(-3\hat{i} + 2\hat{j} + 4\hat{k})$$

Since Point D lies on AP, its position vector can be taken as : $\vec{d} = 5\hat{i} + 7\hat{j} - 2\hat{k} + \lambda_1(3\hat{i} - \hat{j} + \hat{k})$

A vector parallel to line CD is $2\hat{i} + 7\hat{j} - 5\hat{k}$

Equation of line CD $\equiv$

$$\vec{r} = 5\hat{i} + 7\hat{j} - 2\hat{k} + \lambda_1(3\hat{i} - \hat{j} + \hat{k}) + \lambda_2(2\hat{i} + 7\hat{j} - 5\hat{k})$$

solve equation of line BQ with equation of line CD to get point of intersection C.

Solve BQ and CD to get :

$$5\hat{i} + 7\hat{j} - 2\hat{k} + \lambda_1(3\hat{i} - \hat{j} + \hat{k}) + \lambda_2(2\hat{i} + 7\hat{j} - 5\hat{k})$$

$$= -3\hat{i} + 3\hat{j} + 6\hat{k} + \lambda_3(-3\hat{i} + 2\hat{j} + 4\hat{k})$$

Equating the coefficients of i, j and k, we get

$$5 + 3\lambda_1 + 2\lambda_2 = -3(1 + \lambda_3) \quad \dots(1)$$

$$7 - \lambda_1 + 7\lambda_2 = 3 + 2\lambda_3 \quad \dots(2)$$

$$-2 + \lambda_1 - 5\lambda_2 = 6 + 4\lambda_3 \quad \dots(3)$$

Solve equations (1), (2) and (3) to get :

$$\lambda_2 = -1, \lambda_3 = -1 \quad \text{and} \quad \lambda_1 = -1$$

$$\Rightarrow D \equiv (2, 8, -3) \quad \text{and} \quad C \equiv (0, 1, 2)$$

Example – 15

The volume of the parallelopiped whose sides are given by $\vec{OA} = 2\hat{i} - 3\hat{j}$, $\vec{OB} = \hat{i} + \hat{j} - \hat{k}$, $\vec{OC} = 3\hat{i} - \hat{k}$, is :

- (a) $\frac{4}{13}$ (b) 4
- (c) $\frac{2}{7}$ (d) none of these

Ans. (b)

Sol. The volume of parallelopiped $= \begin{vmatrix} \vec{a} & \vec{b} & \vec{c} \end{vmatrix} = \begin{vmatrix} 2 & -3 & 0 \\ 1 & 1 & -1 \\ 3 & 0 & -1 \end{vmatrix}$

$$= 2(-1) + 3(-1 + 3) = -2 + 6 = 4$$

Example – 16

The points with position vectors

$60\hat{i} + 3\hat{j}$, $40\hat{i} - 8\hat{j}$, $a\hat{i} - 52\hat{j}$ are collinear, if:

- (a) $a = -40$ (b) $a = 40$
- (c) $a = 20$ (d) none of these

Ans. (a)

Sol. if $\vec{AB} = -20\hat{i} - 11\hat{j}$

and $\vec{AC} = (a - 60)\hat{i} - 55\hat{j}$, then the three points A, B and C will be collinear if

$$\frac{a - 60}{-20} = \frac{-55}{-11} \Rightarrow a = -40$$



Example – 17

A vector $\vec{a}$ has components $2p$ and 1 with respect to a rectangular cartesian system. This system is rotated through a certain angle about the origin in the counter clockwise sense. If, with respect to the new system, $\vec{a}$ has components $p+1$ and 1 , then :

(a) $p=0$ (b) $p=1$ or $p=-\frac{1}{3}$

(c) $p=1$ or $p=\frac{1}{3}$ (d) $p=1$ or $p=-1$

Ans. (b)

Sol. Here, $\vec{a} = (2p)\hat{i} + \hat{j}$, when a system is rotated, the new component of $\vec{a}$ are $(p+1)$ and 1 .

$$\text{i.e. } \vec{b} = (p+1)\hat{i} + \hat{j} \Rightarrow |\vec{a}|^2 = |\vec{b}|^2$$

$$\text{or } 4p^2 + 1 = (p+1)^2 + 1 \Rightarrow 4p^2 = p^2 + 2p + 1$$

$$\Rightarrow 3p^2 - 2p - 1 = 0 \Rightarrow (3p+1)(p-1) = 0$$

$$\Rightarrow p = 1, -\frac{1}{3}$$

Example – 18

Let a, b, c be distinct non-negative numbers. If the vectors $a\hat{i} + a\hat{j} + c\hat{k}$, $\hat{i} + \hat{k}$ and $c\hat{i} + c\hat{j} + b\hat{k}$ lie in a plane, then c is:

- (a) the Arithmetic Mean of a and b
- (b) the Geometric Mean of a and b
- (c) the Harmonic Mean of a and b
- (d) equal to zero

Ans. (b)

Sol. Since, three vectors are coplanar.

$$\begin{vmatrix} a & a & c \\ 1 & 0 & 1 \\ c & c & b \end{vmatrix} = 0$$

$$\text{Applying } C_1 \rightarrow C_1 - C_2 \begin{vmatrix} 0 & a & c \\ 1 & 0 & 1 \\ 0 & c & b \end{vmatrix} = 0$$

$$\Rightarrow -1(ab - c^2) = 0 \Rightarrow ab = c^2$$

Example – 19

If $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar unit vectors such that

$\vec{a} \times (\vec{b} \times \vec{c}) = \frac{(\vec{b} + \vec{c})}{\sqrt{2}}$, then the angle between $\vec{a}$ and $\vec{b}$ is :

(a) $\frac{3\pi}{4}$ (b) $\frac{\pi}{4}$

(c) $\frac{\pi}{2}$ (d) π

Ans. (a)

Sol. Since, $\vec{a} \times (\vec{b} \times \vec{c}) = \frac{(\vec{b} + \vec{c})}{\sqrt{2}}$

$$\Rightarrow (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} = \frac{1}{\sqrt{2}}\vec{b} + \frac{1}{\sqrt{2}}\vec{c}$$

On equating the coefficient of $\vec{c}$, we get

$$\vec{a} \cdot \vec{b} = -\frac{1}{\sqrt{2}} \Rightarrow |\vec{a}||\vec{b}|\cos\theta = -\frac{1}{\sqrt{2}}$$

$$\therefore \cos\theta = -\frac{1}{\sqrt{2}} \Rightarrow \theta = \frac{3\pi}{4}$$

Example – 20

Let $\vec{u}, \vec{v}$ and $\vec{w}$ be vectors such that $\vec{u} + \vec{v} + \vec{w} = \vec{0}$. If

$|\vec{u}| = 3, |\vec{v}| = 4$ and $|\vec{w}| = 5$, then $\vec{u} \cdot \vec{v} + \vec{v} \cdot \vec{w} + \vec{w} \cdot \vec{u}$ is :

(a) 47 (b) -25

(c) 0 (d) 25

Ans. (b)

Sol. Since, $\vec{u} + \vec{v} + \vec{w} = \vec{0}$

$$\Rightarrow |\vec{u} + \vec{v} + \vec{w}|^2 = 0$$

$$\Rightarrow |\vec{u}|^2 + |\vec{v}|^2 + |\vec{w}|^2 + 2(\vec{u} \cdot \vec{v} + \vec{v} \cdot \vec{w} + \vec{w} \cdot \vec{u}) = 0$$

$$\Rightarrow 9 + 16 + 25 + 2(\vec{u} \cdot \vec{v} + \vec{v} \cdot \vec{w} + \vec{w} \cdot \vec{u}) = 0$$

$$\Rightarrow \vec{u} \cdot \vec{v} + \vec{v} \cdot \vec{w} + \vec{w} \cdot \vec{u} = -25$$



Example – 21

If $\vec{a}, \vec{b}$ and $\vec{c}$ are three non-coplanar vectors, then

$(\vec{a} + \vec{b} + \vec{c}) \cdot [(\vec{a} + \vec{b}) \times (\vec{a} + \vec{c})]$ equals :

- (a) 0 (b) $[\vec{a} \vec{b} \vec{c}]$
(c) $2 \cdot [\vec{a} \vec{b} \vec{c}]$ (d) $-[\vec{a} \vec{b} \vec{c}]$

Ans. (d)

Sol. $(\vec{a} + \vec{b} + \vec{c}) \cdot [(\vec{a} + \vec{b}) \times (\vec{a} + \vec{c})]$
 $= (\vec{a} + \vec{b} + \vec{c}) \cdot [\vec{a} \times \vec{a} + \vec{a} \times \vec{c} + \vec{b} \times \vec{a} + \vec{b} \times \vec{c}]$
 $= \{\vec{a} \cdot (\vec{a} \times \vec{c}) + \vec{a} \cdot (\vec{b} \times \vec{a}) + \vec{a} \cdot (\vec{b} \times \vec{c})\}$
 $+ \{\vec{b} \cdot (\vec{a} \times \vec{c}) + \vec{b} \cdot (\vec{b} \times \vec{a}) + \vec{b} \cdot (\vec{b} \times \vec{c})\}$
 $+ \{\vec{c} \cdot (\vec{a} \times \vec{c}) + \vec{c} \cdot (\vec{b} \times \vec{a}) + \vec{c} \cdot (\vec{b} \times \vec{c})\}$
 $\Rightarrow (\vec{a} \vec{b} \vec{c}) + (\vec{b} \vec{a} \vec{c}) + (\vec{c} \vec{b} \vec{a}) \Rightarrow \therefore (\vec{a} \vec{b} \vec{c})$

Example – 22

If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = 4\hat{i} + 3\hat{j} + 4\hat{k}$ and $\vec{c} = \hat{i} + \alpha\hat{j} + \beta\hat{k}$ are linearly dependent vectors and $|\vec{c}| = \sqrt{3}$, then :

- (a) $\alpha = 1, \beta = -1$ (b) $\alpha = 1, \beta = \pm 1$
(c) $\alpha = -1, \beta = \pm 1$ (d) $\alpha = \pm 1, \beta = 1$

Ans. (d)

Sol. Since, $\vec{a}, \vec{b}, \vec{c}$ are linearly dependent vectors.

$$\Rightarrow [\vec{a} \vec{b} \vec{c}] = 0$$

$$\Rightarrow \begin{vmatrix} 1 & 1 & 1 \\ 4 & 3 & 4 \\ 1 & \alpha & \beta \end{vmatrix} = 0$$

Applying, $C_1 \rightarrow C_2 - C_1, C_3 \rightarrow C_3 - C_1$,

$$\begin{vmatrix} 1 & 0 & 0 \\ 4 & -1 & 0 \\ 1 & \alpha - 1 & \beta - 1 \end{vmatrix} = 0 \Rightarrow -(\beta - 1) = 0 \Rightarrow \beta = 1$$

Also, $|\vec{c}| = \sqrt{3}$ [given]

$$\Rightarrow 1 + \alpha^2 + \beta^2 = 3 \quad [\text{given, } c = \hat{i} + \alpha\hat{j} + \beta\hat{k}]$$

$$\Rightarrow 1 + \alpha^2 + 1 = 3 \Rightarrow \alpha^2 = 1 \Rightarrow \alpha = \pm 1$$

Example – 23

Let $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{b} = \hat{i} + \hat{j}$. If $\vec{c}$ is a vector such that $\vec{a} \cdot \vec{c} = |\vec{c}|$, $|\vec{c} - \vec{a}| = 2\sqrt{2}$ and the angle between $(\vec{a} \times \vec{b})$ and $\vec{c}$ is 30° , then $[(\vec{a} \times \vec{b}) \times \vec{c}]$ is equal to :

- (a) $\frac{2}{3}$ (b) $\frac{3}{2}$
(c) 2 (d) 3

Ans. (b)

Sol. In this equation, vector $\vec{c}$ is not given therefore,

we cannot apply the formulae of $\vec{a} \times \vec{b} \times \vec{c}$

(vector triple product).

$$[(\vec{a} \times \vec{b}) \times \vec{c}] = |\vec{a} \times \vec{b}| |\vec{c}| \sin 30^\circ \dots (i)$$

$$|\vec{a} \times \vec{b}| = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -2 \\ 1 & 1 & 0 \end{vmatrix} = 2\hat{i} - 2\hat{j} + \hat{k}$$

$$\Rightarrow |\vec{a} \times \vec{b}| = \sqrt{2^2 + (-2)^2 + 1} = \sqrt{4 + 4 + 1} = \sqrt{9} = 3$$

Since, $|\vec{c} - \vec{a}| = 2\sqrt{2}$ [given]

$$\Rightarrow |\vec{c} - \vec{a}|^2 = 8$$

$$\Rightarrow (\vec{c} - \vec{a}) \cdot (\vec{c} - \vec{a}) = 8$$

$$\Rightarrow \vec{c} \cdot \vec{c} - \vec{c} \cdot \vec{a} - \vec{a} \cdot \vec{c} + \vec{a} \cdot \vec{a} = 8$$

$$\Rightarrow |\vec{c}|^2 + |\vec{a}|^2 - 2\vec{a} \cdot \vec{c} = 8$$

$$\Rightarrow |\vec{c}|^2 + 9 - 2|\vec{c}| = 8$$



$$\Rightarrow |\vec{c}|^2 - 2|\vec{c}| + 1 = 0$$

$$\Rightarrow (|\vec{c}| - 1)^2 = 0 \Rightarrow |\vec{c}| = 1$$

$$\text{From Eq.(1), } |(\vec{a} \times \vec{b}) \times \vec{c}| = (3)(1) \cdot \left(\frac{1}{2}\right) = \frac{3}{2}$$

$$\Rightarrow \vec{c} = p(-3\hat{j} + 3\hat{k}) \Rightarrow |\vec{c}| = p\sqrt{(-3)^2 + 3^2}$$

$$\Rightarrow |\vec{c}|^2 = p^2 (\sqrt{18})^2 \Rightarrow |\vec{c}| = p^2 \cdot 18$$

$$\Rightarrow 1 = p^2 \cdot 18 \left[\because |\vec{c}| = 1 \right]$$

$$\Rightarrow p^2 = \frac{1}{18} \Rightarrow p = \pm \frac{1}{3\sqrt{2}}$$

$$\therefore \vec{c} = \pm \frac{1(-\hat{j} + \hat{k})}{\sqrt{2}}$$

Example - 24

Let $\vec{a} = 2\hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$ and a unit vector $\vec{c}$ be coplanar. If $\vec{c}$ is perpendicular to $\vec{a}$, then $\vec{c}$ is equal to :

(a) $\frac{1}{\sqrt{2}}(-\hat{j} + \hat{k})$

(b) $\frac{1}{\sqrt{3}}(-\hat{i} - \hat{j} - \hat{k})$

(c) $\frac{1}{\sqrt{5}}(\hat{i} - 2\hat{j})$

(d) $\frac{1}{\sqrt{5}}(\hat{i} - \hat{j} - \hat{k})$

Ans. (a)

Sol. It is given that $\vec{c}$ is coplanar with $\vec{a}$ and $\vec{b}$, we take

$$\vec{c} = p\vec{a} + q\vec{b} \dots (i)$$

Where, p and q are scalars.

$$\text{Since, } \vec{c} \perp \vec{a} \Rightarrow \vec{c} \cdot \vec{a} = 0$$

Taking dot product of $\vec{a}$ in Eq.(i), we get

$$\vec{c} \cdot \vec{a} = p\vec{a} \cdot \vec{a} + q\vec{b} \cdot \vec{a} \Rightarrow 0 = p|\vec{a}|^2 + q|\vec{b} \cdot \vec{a}|$$

$$\left[\begin{array}{l} \because \vec{a} = 2\hat{i} + \hat{j} + \hat{k} \\ |\vec{a}| = \sqrt{2^2 + 1 + 1} = \sqrt{6} \\ \vec{a} \cdot \vec{b} = (2\hat{i} + \hat{j} + \hat{k}) \cdot (\hat{i} + 2\hat{j} - \hat{k}) \\ = 2 + 2 - 1 = 3 \end{array} \right]$$

$$\Rightarrow 0 = p \cdot 6 + q \cdot 3 \Rightarrow q = -2p$$

On putting in Eq.(i), we get

$$\vec{c} = p\vec{a} + b(-2p)$$

$$\Rightarrow \vec{c} = p\vec{a} - 2p\vec{b} \Rightarrow \vec{c} = p(\vec{a} - 2\vec{b})$$

$$\Rightarrow \vec{c} = p[(2\hat{i} + \hat{j} + \hat{k}) - 2(\hat{i} + 2\hat{j} - \hat{k})]$$

Example - 25

Let $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$ and $\vec{c} = \hat{i} + \hat{j} - 2\hat{k}$ be three vectors. A vector in the plane of $\vec{b}$ and $\vec{c}$, whose projection on $\vec{a}$ is of magnitude $\sqrt{2/3}$, is

(a) $2\hat{i} + 3\hat{j} - 3\hat{k}$

(b) $2\hat{i} + 3\hat{j} + 3\hat{k}$

(c) $-2\hat{i} - \hat{j} + 5\hat{k}$

(d) $2\hat{i} + \hat{j} + 5\hat{k}$

Ans. (a,c)

Sol. Given vectors are $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$ and

$$\vec{c} = \hat{i} + \hat{j} - 2\hat{k}$$

Any vector $\vec{r}$ in the plane of $\vec{b}$ and $\vec{c}$ is

$$\vec{r} = \vec{b} + t(\vec{c}) = \hat{i} + 2\hat{j} - \hat{k} + t(\hat{i} + \hat{j} - 2\hat{k})$$

$$= (1+t)\hat{i} + (2+t)\hat{j} - (1+2t)\hat{k} \dots (i)$$

Since, projection $\vec{r}$ on $\vec{a}$ is $\sqrt{\frac{2}{3}}$.

$$\therefore \frac{\vec{r} \cdot \vec{a}}{|\vec{a}|} = \sqrt{\frac{2}{3}}$$

$$\Rightarrow \left| \frac{2(1+t) - (2+t) - (1+2t)}{\sqrt{6}} \right| = \sqrt{\frac{2}{3}}$$

$$\Rightarrow |-(1+t)| = 2 \Rightarrow t = 1 \text{ or } -3$$



On putting $t=1, -3$ in Eq.(i) respectively, we get

$$\hat{r} = 2\hat{i} + 3\hat{j} - 3\hat{k}$$

$$\text{or } \hat{r} = -2\hat{i} - \hat{j} + 5\hat{k}$$

Example – 26

Which of the following expressions are meaningful questions?

(a) $\vec{u} \cdot (\vec{v} \times \vec{w})$ (b) $(\vec{u} \cdot \vec{v}) \cdot \vec{w}$

(c) $(\vec{u} \cdot \vec{v}) \vec{w}$ (d) $\vec{u} \times (\vec{v} \cdot \vec{w})$

Ans. (a,c)

Sol. (a) $\vec{u} \cdot (\vec{v} \times \vec{w})$ is a meaningful operation.

Therefore, (a) is the answer.

(b) $\vec{u} \cdot (\vec{v} \cdot \vec{w})$ is not meaningful, since $\vec{v} \cdot \vec{w}$ is a scalar quantity and for dot product both quantities should be vector.

Therefore, (b) is not the answer.

(c) $(\vec{u} \cdot \vec{v}) \vec{w}$ is meaningful, since it is a simple multiplication of vector and scalar quantity.

Therefore, (c) is the answer.

(d) $\vec{u} \times (\vec{v} \cdot \vec{w})$ is not meaningful, since $\vec{v} \cdot \vec{w}$ is a scalar quantity and for cross product, both quantities should be vector.

Therefore (d) is not the answer.

Example – 27

Let $\vec{a}$ and $\vec{b}$ be two non-collinear unit vectors. If

$$\vec{u} = \vec{a} - (\vec{a} \cdot \vec{b}) \vec{b} \text{ and } \vec{v} = \vec{a} \times \vec{b}, \text{ then } |\vec{v}| \text{ is}$$

(a) $|\vec{u}|$ (b) $|\vec{u}| + |\vec{u} \cdot \vec{a}|$

(c) $|\vec{u}| + |\vec{u} \cdot \vec{b}|$ (d) $|\vec{u}| + \vec{u} \cdot (\vec{a} + \vec{b})$

Ans. (a,c)

Sol. Let θ be the angle between $\vec{a}$ and $\vec{b}$. Since, $\vec{a}$ and $\vec{b}$ are non-collinear vectors, then $\theta \neq 0$ and $\theta \neq \pi$.

$$\text{We have, } \vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

$$= \cos \theta \quad [\because |\vec{a}| = 1, |\vec{b}| = 1, \text{ given}]$$

$$\text{Now, } \vec{u} = \vec{a} - (\vec{a} \cdot \vec{b}) \vec{b} \Rightarrow |\vec{u}| = |\vec{a} - (\vec{a} \cdot \vec{b}) \vec{b}|$$

$$\Rightarrow |\vec{u}|^2 = |\vec{a} - (\vec{a} \cdot \vec{b}) \vec{b}|^2$$

$$\Rightarrow |\vec{u}|^2 = |\vec{a} - \cos \theta \vec{b}|^2$$

$$\Rightarrow |\vec{u}|^2 = |\vec{a}|^2 + \cos^2 \theta |\vec{b}|^2 - 2 \cos \theta (\vec{a} \cdot \vec{b})$$

$$\Rightarrow |\vec{u}|^2 = 1 + \cos^2 \theta - 2 \cos^2 \theta$$

$$\Rightarrow |\vec{u}|^2 = 1 - \cos^2 \theta \Rightarrow |\vec{u}|^2 = \sin^2 \theta$$

$$\text{Also, } \vec{v} = \vec{a} \times \vec{b} \quad [\text{given}]$$

$$\Rightarrow |\vec{v}|^2 = |\vec{a} \times \vec{b}|^2 \Rightarrow |\vec{v}|^2 = |\vec{a}|^2 |\vec{b}|^2 \sin^2 \theta$$

$$\Rightarrow |\vec{v}|^2 = \sin^2 \theta$$

$$\therefore |\vec{u}|^2 = |\vec{v}|^2$$

$$\text{Now, } \vec{u} \cdot \vec{a} = [\vec{a} - (\vec{a} \cdot \vec{b}) \vec{b}] \cdot \vec{a} = \vec{a} \cdot \vec{a} - (\vec{a} \cdot \vec{b}) (\vec{b} \cdot \vec{a})$$

$$= |\vec{a}|^2 - \cos^2 \theta = 1 - \cos^2 \theta = \sin^2 \theta$$

$$\therefore |\vec{u}| + |\vec{u} \cdot \vec{a}| = \sin \theta + \sin^2 \theta \neq |\vec{v}|$$

$$\vec{u} \cdot \vec{b} = [\vec{a} - (\vec{a} \cdot \vec{b}) \vec{b}] \cdot \vec{b}$$

$$= \vec{a} \cdot \vec{b} - (\vec{a} \cdot \vec{b}) (\vec{b} \cdot \vec{b}) = \vec{a} \cdot \vec{b} - \vec{a} \cdot \vec{b} |\vec{b}|^2$$

$$= \vec{a} \cdot \vec{b} - \vec{a} \cdot \vec{b} = 0 \quad \dots(i)$$

$$\therefore |\vec{u}| + |\vec{u} \cdot \vec{b}| = |\vec{u}| + 0 = |\vec{u}| = |\vec{v}|$$

$$\text{Also, } \vec{u} \cdot (\vec{a} + \vec{b}) = \vec{u} \cdot \vec{a} + \vec{u} \cdot \vec{b} = \vec{u} \cdot \vec{a}$$

$$\Rightarrow |\vec{u}| + \vec{u} \cdot (\vec{a} + \vec{b}) = |\vec{u}| + \vec{u} \cdot \vec{a} \neq |\vec{v}|$$



Example – 28

Let $\vec{A}, \vec{B}, \vec{C}$ be three vectors of lengths 3, 4, 5 respectively.

Let $\vec{A}$ be perpendicular to $\vec{B} + \vec{C}$, $\vec{B}$ to $\vec{C} + \vec{A}$ and $\vec{C}$ to $\vec{A} + \vec{B}$. Then the length of vector $\vec{A} + \vec{B} + \vec{C}$ is...

Ans. $(5\sqrt{2})$

Sol. Given, $|\vec{A}| = 3, |\vec{B}| = 4, |\vec{C}| = 5$

Since, $\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{B} \cdot (\vec{C} + \vec{A}) = \vec{C} \cdot (\vec{A} + \vec{B}) = 0 \dots (i)$

$$\therefore |\vec{A} + \vec{B} + \vec{C}|^2 = |\vec{A}|^2 + |\vec{B}|^2 + |\vec{C}|^2 + 2(\vec{A} \cdot \vec{B} + \vec{B} \cdot \vec{C} + \vec{C} \cdot \vec{A})$$

$$= 9 + 16 + 25 + 0$$

$$\left[\text{from Eq. (i)} \vec{A} \cdot \vec{B} + \vec{B} \cdot \vec{C} + \vec{C} \cdot \vec{A} = 0 \right]$$

$$\therefore |\vec{A} + \vec{B} + \vec{C}|^2 = 50$$

$$\therefore |\vec{A} + \vec{B} + \vec{C}| = 5\sqrt{2}$$

Example – 29

A, B, C and D are four points in a plane with position vectors $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ respectively such that

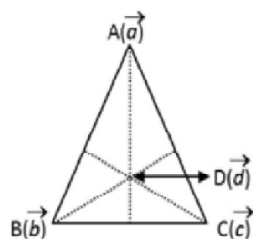
$(\vec{a} - \vec{d}) \cdot (\vec{b} - \vec{c}) = (\vec{b} - \vec{d}) \cdot (\vec{c} - \vec{a}) = 0$. The point D, then, is the.... of the triangle ABC.

Ans. (Orthocentre)

Sol. As, $(\vec{a} - \vec{d}) \cdot (\vec{b} - \vec{c}) = (\vec{b} - \vec{d}) \cdot (\vec{c} - \vec{a}) = 0$

$$\Rightarrow AD \perp BC \text{ and } BD \perp CA$$

which clearly represents from figure that D is orthocentre of $\triangle ABC$



Example – 30

If $\vec{A}, \vec{B}, \vec{C}$ are three non-coplanar vectors, then

$$\frac{\vec{A} \cdot (\vec{B} \times \vec{C})}{(\vec{C} \times \vec{A}) \cdot \vec{B}} + \frac{\vec{B} \cdot (\vec{A} \times \vec{C})}{\vec{C} \cdot (\vec{A} \times \vec{B})} = \dots$$

Ans. (0)

Sol.
$$\frac{\vec{A} \cdot (\vec{B} \times \vec{C})}{(\vec{C} \times \vec{A}) \cdot \vec{B}} + \frac{\vec{B} \cdot (\vec{A} \times \vec{C})}{\vec{C} \cdot (\vec{A} \times \vec{B})}$$

$$= \frac{(\vec{A} \vec{B} \vec{C})}{(\vec{C} \vec{A} \vec{B})} + \frac{(\vec{B} \vec{A} \vec{C})}{(\vec{C} \vec{A} \vec{B})} = \frac{(\vec{A} \vec{B} \vec{C}) - (\vec{A} \vec{B} \vec{C})}{(\vec{C} \vec{A} \vec{B})} = 0$$

Example – 31

If $\vec{A} = (1, 1, 1)$, $\vec{C} = (0, 1, -1)$ are given vectors, then a vector $\vec{B}$ satisfying the equations $\vec{A} \times \vec{B} = \vec{C}$ and $\vec{A} \cdot \vec{B} = 3$ is

Ans. $\Rightarrow \frac{5}{2}\hat{i} + \frac{2}{3}\hat{j} + \frac{2}{3}\hat{k}$

Sol. Let $\vec{B} = x\hat{i} + y\hat{j} + z\hat{k}$

Given $\vec{A} = \hat{i} + \hat{j} + \hat{k}, \vec{C} = \hat{i} - \hat{j}$

Also, given $\vec{A} \times \vec{B} = \vec{C}$

$$\Rightarrow (z - y)\hat{i} - (x - z)\hat{j} + (y - x)\hat{k} = \hat{i} - \hat{j}$$

$$\Rightarrow z - y = 0, x - z = 1, y - x = -1$$

Also, $\vec{A} \cdot \vec{B} = 3 \Rightarrow x + y + z = 3$

On solving above equations, we get

$$x = \frac{5}{3}, y = z = \frac{2}{3}$$

$$\Rightarrow \frac{5}{2}\hat{i} + \frac{2}{3}\hat{j} + \frac{2}{3}\hat{k}$$



Example – 32

If the vectors $a\hat{i} + \hat{j} + \hat{k}$, $\hat{i} + b\hat{j} + \hat{k}$ and $\hat{i} + \hat{j} + c\hat{k}$ ($a \neq b \neq c \neq 1$) are coplanar, then the value of

$$\frac{1}{(1-a)} + \frac{1}{(1-b)} + \frac{1}{(1-c)} = \dots$$

Ans. (1)

Sol. Since, vectors are coplanar

$$\therefore \begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} = 0$$

Applying $R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$

$$\begin{vmatrix} a & 1 & 1 \\ 1-a & b-1 & 0 \\ 1-a & 0 & c-1 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} a & 1 & 1 \\ \frac{1}{(1-a)} & \frac{1}{(1-b)} & \frac{1}{(1-c)} \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{vmatrix} = 0$$

$$\therefore \frac{a}{1-a}(1) - \frac{1}{1-b}(-1) + \frac{1}{1-c}(1) = 0$$

$$\therefore \frac{a}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} = 0$$

$$\therefore -1 + \frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} = 0$$

$$\therefore \frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} = 1$$

Example – 33

Let $\vec{b} = 4\hat{i} + 3\hat{j}$ and $\vec{c}$ be two vectors perpendicular to each other in the xy-plane. All vectors in the same plane having projections 1 and 2 along $\vec{b}$ and $\vec{c}$, respectively, are given by

Ans. $(2\hat{i} - \hat{j})$

Sol. Let $\vec{c} = a\hat{i} + b\hat{j}$

Since, $\vec{b}$ and $\vec{c}$ are perpendiculars to each other.

$$\text{Then, } \vec{b} \cdot \vec{c} = 0 \Rightarrow (4\hat{i} + 3\hat{j}) \cdot (a\hat{i} + b\hat{j}) = 0$$

$$\Rightarrow 4a + 3b = 0 \Rightarrow a : b = 3 : -4$$

$$\therefore \vec{c} = \lambda(3\hat{i} + 4\hat{j}), \text{ where } \lambda \text{ is constant to ratio.}$$

Let the required vectors be $\vec{a} = p\hat{i} + q\hat{j}$

$$\text{Projection } \vec{a} \text{ on } \vec{b} \text{ is } \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$$

$$\therefore 1 = \frac{4p + 3q}{5} \Rightarrow 4p + 3q = 5 \dots (i)$$

$$\text{Also, projection of } \vec{a} \text{ on } \vec{c} \text{ is } \frac{\vec{a} \cdot \vec{c}}{|\vec{c}|}$$

$$\Rightarrow 2 = \frac{3\lambda p - 4\lambda q}{5\lambda} \Rightarrow 3p - 4q = 10$$

On solving above equations, we get $p=2, q=-1$

$$\therefore \vec{c} = 2\hat{i} - \hat{j}$$



Example – 34

A unit vector coplanar with $\hat{i} + \hat{j} + 2\hat{k}$ and $\hat{i} + 2\hat{j} + \hat{k}$ and perpendicular to $\hat{i} + \hat{j} + \hat{k}$ is.....

Ans. $\frac{1}{\sqrt{2}} \cdot (-\hat{j} + \hat{k})$

Sol. Any vector coplanar with $\hat{i} + \hat{j} + 2\hat{k}$ and $\hat{i} + 2\hat{j} + \hat{k}$ is given by

$$\vec{a} = x(\hat{i} + \hat{j} + 2\hat{k}) + y(\hat{i} + 2\hat{j} + \hat{k})$$

$$= (x + y)\hat{i} + (x + 2y)\hat{j} + (2x + y)\hat{k}$$

This vector is perpendicular to $\hat{i} + \hat{j} + \hat{k}$, if

$$(x + y)1 + (x + 2y)1 + (2x + y)1 = 0$$

$$\Rightarrow 4x + 4y = 0 \Rightarrow -x = y$$

$$\therefore \vec{a} = -x\hat{j} + x\hat{k} = x(-\hat{j} + \hat{k}) \Rightarrow |\vec{a}| = \sqrt{2}|x|$$

Hence, the required unit vector is

$$\frac{1}{\sqrt{2}} \cdot (-\hat{j} + \hat{k})$$

Example – 35

A non-zero vector $\vec{a}$ is parallel to the line of intersection of the plane determined by the vectors $\hat{i}, \hat{i} + \hat{j}$ and the plane determined by the vectors $\hat{i} - \hat{j}, \hat{i} + \hat{k}$. The angle between $\vec{a}$ and the vector $\hat{i} - 2\hat{j} + 2\hat{k}$ is.....

Ans. $\frac{\pi}{4}$ or $\frac{3\pi}{4}$

Sol. Equation of plane containing $\hat{i}$ and $\hat{i} + \hat{j}$ is

$$\left[(\vec{r} - \hat{i}) \quad \hat{i} \quad (\hat{i} + \hat{j}) \right] = 0$$

$$\Rightarrow (\vec{r} - \hat{i}) \cdot [\hat{i} \times (\hat{i} + \hat{j})] = 0$$

$$\Rightarrow \left\{ (x\hat{i} + y\hat{j} + z\hat{k}) - \hat{i} \right\} \cdot [\hat{i} \times \hat{i} + \hat{i} \times \hat{j}] = 0$$

$$\Rightarrow \left\{ (x-1)\hat{i} + y\hat{j} + z\hat{k} \right\} \cdot [\hat{k}] = 0$$

$$\Rightarrow (x-1)\hat{i} \cdot \hat{k} + y\hat{j} \cdot \hat{k} + z\hat{k} \cdot \hat{k} = 0$$

$$\Rightarrow z = 0 \quad \dots(i)$$

Equation of plane containing $\hat{i} - \hat{j}$ and $\hat{i} + \hat{k}$ is

$$\left[(\vec{r} - (\hat{i} - \hat{j})) (\hat{i} - \hat{j}) (\hat{i} + \hat{k}) \right] = 0$$

$$\Rightarrow (\vec{r} - \hat{i} + \hat{j}) \cdot [(\hat{i} - \hat{j}) \times (\hat{i} + \hat{k})] = 0$$

$$\Rightarrow \left\{ (x\hat{i} + y\hat{j} + z\hat{k}) - (\hat{i} - \hat{j}) \right\} \cdot [\hat{i} \times \hat{i} + \hat{i} \times \hat{k} - \hat{j} \times \hat{i} - \hat{j} \times \hat{k}] = 0$$

$$\Rightarrow \left\{ (x-1)\hat{i} + (y+1)\hat{j} + z\hat{k} \right\} \cdot [-\hat{j} + \hat{k} - \hat{i}] = 0$$

$$\Rightarrow -(x-1) - (y+1) + z = 0 \quad \dots(ii)$$

$$\text{Let } \vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$$

Since, $\vec{a}$ is parallel to Eqs.(i) and (ii), we get

$$a_3 = 0$$

$$\text{and } a_1 + a_2 - a_3 = 0 \Rightarrow a_1 = -a_2, a_3 = 0$$

Thus, a vector in the direction $\vec{a}$ is $\hat{i} - \hat{j}$

If θ is the angle between $\vec{a}$ and $\hat{i} - 2\hat{j} + 2\hat{k}$, then

$$\cos \theta = \pm \frac{(1)(1) + (-1)(-2)}{\sqrt{1+1}\sqrt{1+4+4}} = \pm \frac{3}{\sqrt{2} \cdot 3}$$

$$\Rightarrow \cos \theta = \pm \frac{1}{\sqrt{2}} \Rightarrow \theta = \frac{\pi}{4} \text{ or } \frac{3\pi}{4}$$



Example – 36

Let $\overrightarrow{OA} = \vec{a}$, $\overrightarrow{OB} = 10\vec{a} + 2\vec{b}$, and $\overrightarrow{OC} = \vec{b}$, where O, A and C are non-collinear points. Let p denote the area of the quadrilateral OABC, and let q denote the area of the parallelogram with OA and OC as adjacent sides. If $p = kq$, then $k = \dots\dots\dots$

Ans. (6)

Sol. Since, $q = \text{area of parallelogram with } \overrightarrow{OA} \text{ and } \overrightarrow{OC} \text{ as adjacent sides}$

$$= |\overrightarrow{OA} \times \overrightarrow{OC}| = |\vec{a} \times \vec{b}|$$

and $p = \text{area of quadrilateral OABC}$

$$= \frac{1}{2} |\overrightarrow{OA} \times \overrightarrow{OB}| + \frac{1}{2} |\overrightarrow{OB} \times \overrightarrow{OC}|$$

$$= \frac{1}{2} |\vec{a} \times (10\vec{a} + 2\vec{b})| + \frac{1}{2} |(10\vec{a} + 2\vec{b}) \times \vec{b}|$$

$$= |\vec{a} \times \vec{b}| + 5|\vec{a} \times \vec{b}| = 6|\vec{a} \times \vec{b}|$$

$$\therefore p = 6q$$

$$\Rightarrow k = 6$$

Example – 37

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three vectors having magnitudes 1, 1 and 2 respectively. If $\vec{a} \times (\vec{a} \times \vec{c}) + \vec{b} = \vec{0}$, then the acute angle between $\vec{a}$ and $\vec{c}$ is

Ans. $\frac{\pi}{6}$

Sol. Given, $\vec{a} \times (\vec{a} \times \vec{c}) + \vec{b} = \vec{0}$

$$\Rightarrow (\vec{a} \cdot \vec{c})\vec{a} - (\vec{a} \cdot \vec{a})\vec{c} + \vec{b} = \vec{0}$$

$$\Rightarrow (2\cos\theta)\vec{a} - \vec{c} + \vec{b} = \vec{0}$$

$$\Rightarrow (2\cos\theta\vec{a} - \vec{c})^2 = (-\vec{b})^2$$

$$\Rightarrow 4\cos^2\theta \cdot |\vec{a}|^2 + |\vec{c}|^2 - 2 \cdot 2\cos\theta |\vec{a}| \cdot |\vec{c}| = |\vec{b}|^2$$

$$\Rightarrow 4\cos^2\theta + 4 - 8\cos\theta = 1$$

$$\Rightarrow 4\cos^2\theta = 3$$

$$\Rightarrow \cos\theta = \pm \frac{\sqrt{3}}{2}$$

$$\text{For } \theta \text{ to be acute, } \cos\theta = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \theta = \frac{\pi}{6}$$

Example – 38

Find all the values of λ such that $x, y, z \neq (0, 0, 0)$ and $(\hat{i} + \hat{j} + 3\hat{k})x + (3\hat{i} - 3\hat{j} + \hat{k})y + (-4\hat{i} + 5\hat{j})z = \lambda(\hat{i}x + \hat{j}y + \hat{k}z)$ where $\hat{i}, \hat{j}, \hat{k}$ are unit vectors along the coordinate axes.

Ans. (0, -1)

Sol. Since, $(\vec{i} + \vec{j} + 3\vec{k})x + (3\vec{i} - 3\vec{j} + \vec{k})y + (-4\vec{i} + 5\vec{j})z = \lambda(\vec{i}x + \vec{j}y + \vec{k}z)$

$$= \lambda(\vec{i}x + \vec{j}y + \vec{k}z)$$

$$\Rightarrow x + 3y - 4z = \lambda x, x - 3y + 5z = \lambda y, 3x + y + 0z = \lambda z$$

$$\Rightarrow (1 - \lambda)x + 3y - 4z = 0, x - (3 + \lambda)y + 5z = 0 \text{ and}$$

$$3x + y - \lambda z = 0$$

Since, $(x, y, z) \neq (0, 0, 0)$

$\therefore$ Non-trivial solution.

$$\Rightarrow \Delta = 0$$

$$\Rightarrow \begin{vmatrix} 1-\lambda & 3 & -4 \\ 1 & -(3+\lambda) & 5 \\ 3 & 1 & -\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda)(3\lambda + \lambda^2 - 5) - 3(-\lambda - 15)$$

$$-4(1 + 9 + 3\lambda) = 0$$

$$\Rightarrow \lambda(\lambda + 1)^2 = 0$$

$$\therefore \lambda = 0, -1$$



Example – 39

The position vectors of the points A, B, C and D are $3\hat{i}-2\hat{j}-\hat{k}$, $2\hat{i}+3\hat{j}-4\hat{k}$, $-\hat{i}+\hat{j}+2\hat{k}$ and $4\hat{i}+5\hat{j}+\lambda\hat{k}$, respectively. If the points A, B, C and D lie on a plane, find the value of λ .

Ans. $-\frac{146}{17}$

Sol. Here, $\overrightarrow{AB} = -\hat{i} + 5\hat{j} - 3\hat{k}$

$$\overrightarrow{AC} = -4\hat{i} + 3\hat{j} + 3\hat{k} \text{ and } \overrightarrow{AD} = \hat{i} + 7\hat{j} + (\lambda + 1)\hat{k}$$

We know that A, B, C, D lie in a plane if $\overrightarrow{AB}$, $\overrightarrow{AC}$, $\overrightarrow{AD}$ are coplanar i.e.

$$\begin{vmatrix} \overrightarrow{AB} & \overrightarrow{AC} & \overrightarrow{AD} \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} -1 & 5 & -3 \\ -4 & 3 & 3 \\ 1 & 7 & \lambda + 1 \end{vmatrix} = 0$$

$$\Rightarrow -(3\lambda + 3 - 21) - 5(-4\lambda - 4 - 3) - 3(-28 - 3) = 0$$

$$\Rightarrow 17\lambda + 146 = 0$$

$$\therefore \lambda = \frac{-146}{17}$$

Example – 40

If A, B, C, D are any four points in space, prove that

$$|\overrightarrow{AB} \times \overrightarrow{CD} + \overrightarrow{BC} \times \overrightarrow{AD} + \overrightarrow{CA} \times \overrightarrow{BD}| = 4 \text{ (area of } \Delta ABC).$$

Sol. Let the position vectors of points A, B, C, D be $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ respectively.

$$\text{Then, } \overrightarrow{AB} = \vec{b} - \vec{a}, \overrightarrow{BC} = \vec{c} - \vec{b}, \overrightarrow{AD} = \vec{d} - \vec{a},$$

$$\overrightarrow{BD} = \vec{d} - \vec{b}, \overrightarrow{CA} = \vec{a} - \vec{c}, \overrightarrow{CD} = \vec{d} - \vec{c}$$

$$\text{Now, } |\overrightarrow{AB} \times \overrightarrow{CD} + \overrightarrow{BC} \times \overrightarrow{AD} + \overrightarrow{CA} \times \overrightarrow{BD}|$$

$$= |(\vec{b} - \vec{a}) \times (\vec{d} - \vec{c}) + (\vec{c} - \vec{b}) \times (\vec{d} - \vec{a}) + (\vec{a} - \vec{c}) \times (\vec{d} - \vec{b})|$$

$$= |\vec{b} \times \vec{d} - \vec{a} \times \vec{d} - \vec{b} \times \vec{c} + \vec{a} \times \vec{c} + \vec{c} \times \vec{d} - \vec{c} \times \vec{a} - \vec{b} \times \vec{d}$$

$$+ \vec{b} \times \vec{a} + \vec{a} \times \vec{d} - \vec{a} \times \vec{b} - \vec{c} \times \vec{d} + \vec{c} \times \vec{b}|$$

$$= 2|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}| \dots (i)$$

Also, area of ΔABC

$$= \frac{1}{2} |\overrightarrow{AB} \times \overrightarrow{AC}| = \frac{1}{2} |(\vec{b} - \vec{a}) \times (\vec{c} - \vec{a})|$$

$$= \frac{1}{2} |\vec{b} \times \vec{c} - \vec{b} \times \vec{a} - \vec{a} \times \vec{c} + \vec{a} \times \vec{a}|$$

$$= \frac{1}{2} |\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}| \dots (ii)$$

From Eqs. (i) and (ii)

$$|\overrightarrow{AB} \times \overrightarrow{CD} + \overrightarrow{BC} \times \overrightarrow{AD} + \overrightarrow{CA} \times \overrightarrow{BD}| = 2(2 \text{ area of } \Delta ABC)$$

$$= 4(\text{area of } \Delta ABC)$$

Example – 41

If vectors $\vec{a}, \vec{b}, \vec{c}$ are coplanar, show that

$$\begin{vmatrix} \vec{a} & \vec{b} & \vec{c} \\ \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \end{vmatrix} = 0$$

Sol. Given that, $\vec{a}, \vec{b}, \vec{c}$ are coplanar vectors.

$\therefore$ There exists scalars x, y, z not all zero, such that

$$x\vec{a} + y\vec{b} + z\vec{c} = 0 \dots (i)$$

Taking dot with $\vec{a}$ and $\vec{b}$ respectively, we get

$$x(\vec{a} \cdot \vec{a}) + y(\vec{a} \cdot \vec{b}) + z(\vec{a} \cdot \vec{c}) = 0 \dots (ii)$$

$$\text{and } x(\vec{a} \cdot \vec{b}) + y(\vec{b} \cdot \vec{b}) + z(\vec{b} \cdot \vec{c}) = 0 \dots (iii)$$

Since, Eqs. (i), (ii) and (iii) represent homogeneous equations with $(x, y, z) \neq (0, 0, 0)$

$\Rightarrow$ Non-trivial solutions

$$\therefore \Delta = 0$$

$$\Rightarrow \begin{vmatrix} \vec{a} & \vec{b} & \vec{c} \\ \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \end{vmatrix} = 0$$



Example – 42

Let $\vec{A} = 2\hat{i} + \hat{k}$, $\vec{B} = \hat{i} + \hat{j} + \hat{k}$, and $\vec{C} = 4\hat{i} - 3\hat{j} + 7\hat{k}$.

Determine a vector $\vec{R}$ satisfying

$$\vec{R} \times \vec{B} = \vec{C} \times \vec{B} \text{ and } \vec{R} \cdot \vec{A} = 0.$$

Ans. $(-\hat{i} - 8\hat{j} + 2\hat{k})$

Sol. Let $\vec{R} = x\hat{i} + y\hat{j} + z\hat{k}$

$$\therefore \vec{R} \times \vec{B} = \vec{C} \times \vec{B}$$

$$\Rightarrow \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ 1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -3 & 7 \\ 1 & 1 & 1 \end{vmatrix}$$

$$\Rightarrow (y-z)\hat{i} - (x-z)\hat{j} + (x-y)\hat{k} = -10\hat{i} + 3\hat{j} + 7\hat{k}$$

$$\Rightarrow y-z = -10, z-x = 3, x-y = 7$$

$$\text{and } \vec{R} \cdot \vec{A} = 0$$

$$\Rightarrow 2x + z = 0$$

On solving above equations,

$$x = -1, y = -8 \text{ and } z = 2$$

$$\therefore \vec{R} = -\hat{i} - 8\hat{j} + 2\hat{k}$$

Example – 43

Determine the value of 'c' so that for all real x, the vector $c\hat{x} - 6\hat{j} - 3\hat{k}$ and $\hat{x} + 2\hat{j} + 2cx\hat{k}$ make an obtuse angle with each other.

Ans. $\left(\frac{-4}{3}, 0\right)$

Sol. Let $\vec{a} = c\hat{x} - 6\hat{j} + 3\hat{k}$ and $\vec{b} = \hat{x} + 2\hat{j} + 2cx\hat{k}$.

Since, $\vec{a}$ and $\vec{b}$ make an obtuse angle.

$$\Rightarrow \vec{a} \cdot \vec{b} < 0 \Rightarrow cx^2 - 12 + 6cx < 0$$

$$\Rightarrow c < 0 \text{ and discriminant } < 0$$

$$\Rightarrow c < 0 \text{ and } 36c^2 - 4(-12)c < 0$$

$$\Rightarrow c < 0 \text{ and } 12c(3c + 4) < 0$$

$$\Rightarrow c < 0 \text{ and } c > -\frac{4}{3}$$

$$\therefore c \in \left(-\frac{4}{3}, 0\right)$$

Example – 44

For any two vectors $\vec{u}$ and $\vec{v}$ prove that

$$(a) |\vec{u} \cdot \vec{v}|^2 + |\vec{u} \times \vec{v}|^2 = |\vec{u}|^2 |\vec{v}|^2 \text{ and}$$

$$(b) (1 + |\vec{u}|^2)(1 + |\vec{v}|^2) = |1 - \vec{u} \cdot \vec{v}|^2 + |\vec{u} + \vec{v} + (\vec{u} \times \vec{v})|^2$$

Sol. (a) Since, $\vec{u} \cdot \vec{v} = |\vec{u}| |\vec{v}| \cos \theta$

$$\text{and } \vec{u} \times \vec{v} = |\vec{u}| |\vec{v}| \sin \theta \hat{n}$$

where, θ is the angle between $\vec{u}$ and $\vec{v}$ and $\hat{n}$ is unit vector perpendicular to the plane of $\vec{u}$ and $\vec{v}$.

$$\text{Again, } |\vec{u} \cdot \vec{v}|^2 = |\vec{u}|^2 |\vec{v}|^2 \cos^2 \theta$$

$$|\vec{u} \times \vec{v}|^2 = |\vec{u}|^2 |\vec{v}|^2 \sin^2 \theta$$

$$\therefore |\vec{u} \cdot \vec{v}|^2 + |\vec{u} \times \vec{v}|^2 = |\vec{u}|^2 |\vec{v}|^2 (\cos^2 \theta + \sin^2 \theta)$$

$$= |\vec{u}|^2 |\vec{v}|^2 \dots (i)$$

$$(b) |\vec{u} + \vec{v} + (\vec{u} \times \vec{v})|^2$$

$$= |\vec{u} + \vec{v}|^2 + |\vec{u} \times \vec{v}|^2 + 2|\vec{u} + \vec{v}| \cdot |\vec{u} \times \vec{v}|$$

$$= |\vec{u}|^2 + |\vec{v}|^2 + 2\vec{u} \cdot \vec{v} + |\vec{u} \times \vec{v}|^2 + 0$$

[$\therefore \vec{u} \times \vec{v}$ is perpendicular to the plane of $\vec{u}$ and $\vec{v}$]

$$\therefore |\vec{u} + \vec{v} + (\vec{u} \times \vec{v})|^2 = |1 - \vec{u} \cdot \vec{v}|^2$$



$$\begin{aligned}
 &= |\vec{u}|^2 + |\vec{v}|^2 + 2\vec{u} \cdot \vec{v} + |\vec{u} \times \vec{v}|^2 + 1 - 2\vec{u} \cdot \vec{v} + |\vec{u} \cdot \vec{v}|^2 \\
 &= |\vec{u}|^2 + |\vec{v}|^2 + 1 + |\vec{u}|^2 |\vec{v}|^2 \quad [\text{from Eq.(i)}] \\
 &= |\vec{u}|^2 (1 + |\vec{v}|^2) + (1 + |\vec{v}|^2) = (1 + |\vec{v}|^2) (1 + |\vec{u}|^2)
 \end{aligned}$$

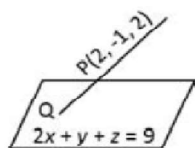
Example – 45

A line with positive direction cosines passes through the point P(2, -1, 2) and makes equal angles with the coordinate axes. The line meets the plane $2x + y + z = 9$ at point Q. The length of the line segment PQ equals

- (a) 1 (b) $\sqrt{2}$
(c) $\sqrt{3}$ (d) 2

Ans. (c)

Sol. Since, $l = m = n = \frac{1}{\sqrt{3}}$



$$\therefore \text{Equation of line is } \frac{x-2}{\frac{1}{\sqrt{3}}} = \frac{y+1}{\frac{1}{\sqrt{3}}} = \frac{z-2}{\frac{1}{\sqrt{3}}}$$

$$\Rightarrow x-2 = y+1 = z-2 = r \text{ [say]}$$

$\therefore$ Any point on the line is

$$Q \equiv (r+2, r-1, r+2)$$

$\because Q$ lies on the plane $2x + y + z = 9$

$$\therefore 2(r+2) + (r-1) + (r+2) = 9$$

$$\Rightarrow 4r + 5 = 9$$

$$\Rightarrow r = 1$$

$$\Rightarrow Q(3, 0, 3)$$

$$\therefore PQ = \sqrt{(3-2)^2 + (0+1)^2 + (3-2)^2} = \sqrt{3}$$

Example – 46

The area of the triangle whose vertices are A (1, -1, 2), B(2, 1, -1), C(3, -1, 2) is

Ans. $\sqrt{13}$ sq. unit

Sol. Area of $\triangle ABC = \frac{1}{2} |\overrightarrow{AB} \times \overrightarrow{AC}|$

$$\overrightarrow{AB} = \hat{i} + 2\hat{j} - 3\hat{k}$$

$$\text{and } \overrightarrow{AC} = 2\hat{i}$$

$$\therefore \overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -3 \\ 2 & 0 & 0 \end{vmatrix} = 2(-3\hat{j} - 2\hat{k})$$

$$\Rightarrow \text{Area of triangle} = \frac{1}{2} |\overrightarrow{AB} \times \overrightarrow{AC}|$$

$$= \frac{1}{2} \cdot 2 \cdot \sqrt{9+4}$$

$$= \sqrt{13} \text{ sq units}$$

Example – 47

The unit vector perpendicular to the plane determined by P(1, -1, 2), Q(2, 0, -1) & R(0, 2, 1) is

Ans. $\pm \frac{(2\hat{i} + \hat{j} + \hat{k})}{\sqrt{6}}$

Sol. A unit vector perpendicular to the plane determined by P, Q, R

$$= \pm \frac{(\overrightarrow{PQ}) \times (\overrightarrow{PR})}{|\overrightarrow{PQ} \times \overrightarrow{PR}|}$$

$$\therefore \text{Unit vector} = \pm \frac{(\overrightarrow{PQ}) \times (\overrightarrow{PR})}{|\overrightarrow{PQ} \times \overrightarrow{PR}|}$$

$$\overrightarrow{PQ} = \hat{i} + \hat{j} - 3\hat{k} \text{ and } \overrightarrow{PR} = -\hat{i} + 3\hat{j} - \hat{k}$$

$$\therefore \overrightarrow{PQ} \times \overrightarrow{PR} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -3 \\ -1 & 3 & -1 \end{vmatrix}$$



$$= \hat{i}(-1+9) + \hat{j}(-1-3) + \hat{k}(3+1)$$

$$= 8\hat{i} - 4\hat{j} + 4\hat{k}$$

$$\Rightarrow |\overrightarrow{PQ} \times \overrightarrow{PR}| = 4\sqrt{4+1+1} = 4\sqrt{6}$$

Hence the unit vector is

$$= \pm \frac{(2\hat{i} + \hat{j} + \hat{k})}{\sqrt{6}}$$

Example - 48

If the angle between the line $x = \frac{y-1}{2} = \frac{z-3}{\lambda}$ and the plane

$x + 2y + 3z = 4$ is $\cos^{-1}\left(\frac{\sqrt{5}}{\sqrt{14}}\right)$, then λ equals to

(a) $\frac{3}{2}$

(b) $\frac{2}{5}$

(c) $\frac{5}{3}$

(d) $\frac{2}{3}$

Ans. (d)

Sol. $\frac{x-0}{1} = \frac{y-1}{2} = \frac{z-3}{\lambda}$

$$x + 2y + 3z = 4$$

Angle between line and plane (by definition)

$$= \sin^{-1}\left(\frac{1 \cdot 1 + 2 \cdot 2 + \lambda \cdot 3}{\sqrt{1+4+9}\sqrt{1+4+\lambda^2}}\right) = \sin^{-1}\left(\frac{5+3\lambda}{\sqrt{14}\sqrt{5+\lambda^2}}\right)$$

$$\text{So, } \frac{(5+3\lambda)^2}{14(5+\lambda^2)} + \frac{5}{14} = 1 \quad (\because \sin^2 \theta + \cos^2 \theta = 1)$$

$$\Rightarrow \frac{(5+3\lambda)^2}{5+\lambda^2} + 5 = 14$$

$$\Rightarrow (5+3\lambda)^2 + 5(5+\lambda^2) = 14(5+\lambda^2)$$

$$\Rightarrow 25 + 30\lambda + 9\lambda^2 + 25 + 5\lambda^2 = 70 + 14\lambda^2$$

$$\Rightarrow 30\lambda + 50 = 70$$

$$\Rightarrow 30\lambda = 20 \quad \therefore \lambda = \frac{2}{3}$$

Example - 49

The length of the perpendicular drawn from the point

$(3, -1, 11)$ to the line $\frac{x}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ is

(a) $\sqrt{66}$

(b) $\sqrt{29}$

(c) $\sqrt{33}$

(d) $\sqrt{53}$

Ans. (d)

Sol. Let feet of perpendicular is

$$(2\alpha, 3\alpha + 2, 4\alpha + 3)$$

$\Rightarrow$ direction ratio of lines are 2, 3, 4

$$\Rightarrow 2(2\alpha - 3) + 3(3\alpha + 3) + 4(4\alpha - 8) = 0$$

$$\Rightarrow \alpha = 1$$

$\Rightarrow$ Foot of perpendicular is (2, 5, 7)

$$\Rightarrow \text{length of perpendicular} = \sqrt{1+6^2+4^2} = \sqrt{53}$$

Example - 50

An equation of a plane parallel to the plane $x - 2y + 2z - 5 = 0$ and at a unit distance from the origin is

(a) $x - 2y + 2z - 3 = 0$

(b) $x - 2y + 2z + 1 = 0$

(c) $x - 2y + 2z - 1 = 0$

(d) $x - 2y + 2z + 5 = 0$

Ans. (a)

Sol. Equation of a plane parallel to $x - 2y + 2z - 5 = 0$ and at a unit distance from origin is

$$x - 2y + 2z + k = 0$$

$$\Rightarrow \frac{|k|}{3} = 1 \Rightarrow |k| = 3$$

$$\therefore x - 2y + 2z - 3 = 0 \text{ or } x - 2y + 2z + 3 = 0$$

Example - 51

If the line $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4}$ and $\frac{x-3}{1} = \frac{y-k}{2} = \frac{z}{1}$ intersect, then k is equal to

(a) -1

(b) $\frac{2}{9}$

(c) $\frac{9}{2}$

(d) 0

Ans. (c)



Sol. $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4} = r_1$ and $\frac{x-3}{1} = \frac{y-k}{2} = \frac{z}{1} = r_2$

or $2r_1 + 1 = r_2 + 3, 3r_1 - 1 = 2r_2 + k, 4r_1 + 1 = r_2$

$\Rightarrow 2r_1 - r_2 = 2$ and $4r_1 - r_2 = -1$

$-2r_1 = 3 \Rightarrow r_1 = \frac{-3}{2}$ and $r_2 = -5$

$\therefore -\frac{9}{2} - 1 = -10 + k \Rightarrow k = 10 - \frac{11}{2} = \frac{9}{2}$

Example - 52

The image of the line $\frac{x-1}{3} = \frac{y-3}{1} = \frac{z-4}{-5}$ in the plane

$2x - y + z + 3 = 0$ is the line :

(a) $\frac{x-3}{-3} = \frac{y+5}{-1} = \frac{z-2}{5}$

(b) $\frac{x+3}{3} = \frac{y-5}{1} = \frac{z-2}{-5}$

(c) $\frac{x+3}{-3} = \frac{y-5}{-1} = \frac{z+2}{5}$

(d) $\frac{x-3}{3} = \frac{y+5}{1} = \frac{z-2}{-5}$

Ans. (b)

Sol. DR's of line = 3, 1, -5

DR's of the normal of plane = 2, -1, 1

$\Rightarrow 3(2) + 1(-1) - 5 = 0$

So, the line is parallel to the plane

Image is point (1, 3, 4) in plane

$\frac{x-1}{2} = \frac{y-3}{-1} = \frac{z-4}{1} = -\frac{2(2-3+4+3)}{6}$

$\Rightarrow \frac{x-1}{2} = \frac{y-3}{-1} = \frac{z-4}{1} = -2$

$\Rightarrow x = -3, y = 5, z = 2$

$\Rightarrow$ Equation of image of the line is

$\frac{x+3}{3} = \frac{y-5}{1} = \frac{z-2}{-5}$

Example - 53

The angle between the lines whose direction cosines satisfy the equations $l + m + n = 0$ and $l^2 = m^2 + n^2$ is :

(a) $\frac{\pi}{2}$

(b) $\frac{\pi}{3}$

(c) $\frac{\pi}{4}$

(d) $\frac{\pi}{6}$

Ans. (b)

Sol. $l + m + n = 0$ (1)

$l^2 = m^2 + n^2$ (2)

From (1) and (2)

$(m+n)^2 = m^2 + n^2$

$\Rightarrow 2mn = 0 \Rightarrow m = 0$, or $n = 0$

If $m = 0$ then from (1)

$l + n = 0 \Rightarrow l = -n \Rightarrow \frac{l}{1} = \frac{n}{-1}$

If $n = 0$ then from (1)

$l + m = 0 \Rightarrow l = -m \Rightarrow \frac{l}{1} = \frac{m}{-1}$

$\Rightarrow$ angle between the lines

$\cos \theta = \frac{(1)(1) + (-1)(0) + (-1)(0)}{\sqrt{1+0+1}\sqrt{1+0+1}}$

$= \frac{1}{2}$

$\Rightarrow \theta = \frac{\pi}{3}$



Example – 54

A line in the 3-dimensional space makes an angle θ ($0 < \theta \leq \frac{\pi}{2}$) with both the x and y axis. Then the set of all values of θ is the interval:

- (a) $\left(0, \frac{\pi}{4}\right]$ (b) $\left[\frac{\pi}{6}, \frac{\pi}{3}\right]$
 (c) $\left[\frac{\pi}{4}, \frac{\pi}{2}\right]$ (d) $\left(\frac{\pi}{3}, \frac{\pi}{2}\right]$

Ans. (c)

Sol. We know that $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$

$$\Rightarrow \cos^2 \theta + \cos^2 \theta + \cos^2 \gamma = 1$$

$$\Rightarrow \cos^2 \gamma = 1 - 2\cos^2 \theta$$

$$\because 0 \leq \cos^2 \gamma \leq 1$$

$$\Rightarrow 0 \leq 1 - 2\cos^2 \theta \leq 1$$

$$\Rightarrow -1 \leq -2\cos^2 \theta \leq 0$$

$$\Rightarrow 0 \leq 2\cos^2 \theta \leq 1$$

$$\Rightarrow 0 \leq \cos^2 \theta \leq \frac{1}{2} \Rightarrow 0 \leq \cos \theta \leq \frac{1}{\sqrt{2}}$$

$$\Rightarrow \theta \in \left[\frac{\pi}{4}, \frac{\pi}{2}\right]$$

Example – 55

The plane containing the line $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3}$ and

parallel to the line $\frac{x}{1} = \frac{y}{1} = \frac{z}{4}$ passes through the point:

- (a) (1, -2, 5) (b) (1, 0, 5)
 (c) (0, 3, -5) (d) (-1, -3, 0)

Ans. (b)

Sol. DR's of the given lines are (1,2,3) and (1,1,4)

$\Rightarrow$ DR's of the normal to the plane

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 3 \\ 1 & 1 & 4 \end{vmatrix}$$

$$= \hat{i}(5) - \hat{j}(1) + \hat{k}(-1)$$

$$= 5\hat{i} - \hat{j} - \hat{k}$$

It contains the point (1,2,3)

$\Rightarrow$ Equation of the plane is

$$5(x-1) - 1(y-2) - 1(z-3) = 0$$

$$\Rightarrow 5x - y - z = 0$$

It also passes through (1,0,5)



EXERCISE - 1 : BASIC OBJECTIVE QUESTIONS

VECTORS

Basics of Vectors

- In a regular hexagon ABCDEF, $\overrightarrow{AB} = \vec{a}$, $\overrightarrow{BC} = \vec{b}$ and $\overrightarrow{CD} = \vec{c}$. Then, $\overrightarrow{AE} =$
 - $\vec{a} + \vec{b} + \vec{c}$
 - $2\vec{a} + \vec{b} + \vec{c}$
 - $\vec{b} + \vec{c}$
 - $\vec{a} + 2\vec{b} + 2\vec{c}$
- If ABCDE is a pentagon, then $\overrightarrow{AB} + \overrightarrow{AE} + \overrightarrow{BC} + \overrightarrow{DC} + \overrightarrow{ED} + \overrightarrow{AC}$ equals
 - $3\overrightarrow{AD}$
 - $3\overrightarrow{AC}$
 - $3\overrightarrow{BE}$
 - $3\overrightarrow{CE}$
- Which of the following is unit vectors
 - $\hat{i} + \hat{j}$
 - $\frac{(\hat{i} + \hat{j} + \hat{k})}{\sqrt{2}}$
 - $\hat{i} + \hat{j} + \hat{k}$
 - $\frac{(\hat{i} + \hat{j} + \hat{k})}{\sqrt{3}}$
- If $\vec{a} = 2\hat{i} + 5\hat{j}$ and $\vec{b} = 2\hat{i} - \hat{j}$, then unit vector in the direction of $\vec{a} + \vec{b}$ is
 - $\hat{i} + \hat{j}$
 - $\sqrt{2}(\hat{i} + \hat{j})$
 - $(\hat{i} + \hat{j})/\sqrt{2}$
 - $(\hat{i} - \hat{j})/\sqrt{2}$
- For any two vector $\vec{a}$ and $\vec{b}$, correct statement is
 - $|\vec{a} - \vec{b}| = |\vec{a}| - |\vec{b}|$
 - $|\vec{a} + \vec{b}| \geq |\vec{a}| - |\vec{b}|$
 - $|\vec{a} + \vec{b}| = |\vec{a}| + |\vec{b}|$
 - $|\vec{a} - \vec{b}| \leq |\vec{a}| - |\vec{b}|$
- If $\hat{i}, \hat{j}, \hat{k}$ are position vectors of A, B, C and $\overrightarrow{AB} = \overrightarrow{CX}$, then position vector of X is
 - $-\hat{i} + \hat{j} + \hat{k}$
 - $\hat{i} - \hat{j} + \hat{k}$
 - $\hat{i} + \hat{j} - \hat{k}$
 - $\hat{i} + \hat{j} + \hat{k}$
- If the position vector of points A and B with respect to point P are respectively $\vec{a}$ and $\vec{b}$ then the position vector of middle point of AB is
 - $\frac{\vec{b} - \vec{a}}{2}$
 - $\frac{\vec{a} + \vec{b}}{2}$
 - $\frac{\vec{a} - \vec{b}}{2}$
 - none of these
- The points with position vectors $3\hat{i} - 4\hat{j} - 4\hat{k}$, $2\hat{i} - \hat{j} + \hat{k}$ and $\hat{i} - 3\hat{j} - 5\hat{k}$ form
 - an equilateral triangle
 - an isosceles triangle
 - a right angle triangle
 - none of these
- The position vector of two points P and Q are respectively $\vec{p}$ and $\vec{q}$ then the position vector of the point dividing PQ in 2 : 5 is
 - $\frac{\vec{p} + \vec{q}}{2 + 5}$
 - $\frac{5\vec{p} + 2\vec{q}}{2 + 5}$
 - $\frac{2\vec{p} + 5\vec{q}}{2 + 5}$
 - $\frac{\vec{p} - \vec{q}}{2 + 5}$
- The position vector of the vertices of triangle ABC are $\hat{i}, \hat{j}$ and $\hat{k}$ then the position vector of its orthocentre is
 - $\hat{i} + \hat{j} + \hat{k}$
 - $2(\hat{i} + \hat{j} + \hat{k})$
 - $\frac{1}{3}(\hat{i} + \hat{j} + \hat{k})$
 - $\frac{1}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})$
- If, D, E, F are mid points of sides BC, CA and AB respectively of a triangle ABC, and $\hat{i} + \hat{j}, \hat{j} + \hat{k}, \hat{k} + \hat{i}$ are p.v. of points A, B and C respectively, then p.v. of centroid of $\triangle DEF$ is
 - $\frac{\hat{i} + \hat{j} + \hat{k}}{3}$
 - $\hat{i} + \hat{j} + \hat{k}$
 - $2(\hat{i} + \hat{j} + \hat{k})$
 - $\frac{2(\hat{i} + \hat{j} + \hat{k})}{3}$



12. If $\vec{a} = 3\hat{i} - 2\hat{j} + \hat{k}$ and $\vec{b} = -\hat{i} + \hat{j} + \hat{k}$, then the unit vector parallel to $\vec{a} + \vec{b}$, is
- (a) $\frac{1}{3}(2\hat{i} - \hat{j} + 2\hat{k})$ (b) $\frac{1}{5}(2\hat{i} - \hat{j} + 2\hat{k})$
- (c) $\frac{1}{\sqrt{3}}(2\hat{i} - \hat{j} + 2\hat{k})$ (d) none of these
13. A parallelepiped is formed by planes drawn through the points (2, 3, 5) and (5, 9, 7), parallel to the co-ordinate planes. The length of a diagonal of the parallelepiped is:
- (a) 7 (b) $\sqrt{38}$
- (c) $\sqrt{155}$ (d) None of these
14. The vector $\vec{c}$, directed along the internal bisector of the angle between the vectors, $\vec{a} = 7\hat{i} - 4\hat{j} - 4\hat{k}$ and $\vec{b} = -2\hat{i} - \hat{j} + 2\hat{k}$ with $|\vec{c}| = 5\sqrt{6}$ is:
- (a) $\frac{5}{3}(\hat{i} - 7\hat{j} + 2\hat{k})$ (b) $\frac{5}{3}(5\hat{i} + 5\hat{j} + 2\hat{k})$
- (c) $\frac{5}{3}(\hat{i} + 7\hat{j} + 2\hat{k})$ (d) $\frac{5}{3}(-5\hat{i} + 5\hat{j} + 2\hat{k})$
15. The vector $\vec{a} = \alpha\hat{i} + 2\hat{j} + \beta\hat{k}$ lies in the plane of the vectors $\vec{b} = \hat{i} + \hat{j}$ and $\vec{c} = \hat{j} + \hat{k}$ and bisects the angle between $\vec{b}$ and $\vec{c}$. Then which one of the following gives possible values of α and β ?
- (a) $\alpha = 1, \beta = 1$ (b) $\alpha = 2, \beta = 2$
- (c) $\alpha = 1, \beta = 2$ (d) $\alpha = 2, \beta = 1$
17. If the vector $\vec{b}$ is collinear with the vector $\vec{a} = (2\sqrt{2}, -1, 4)$ & $|\vec{b}| = 10$, then
- (a) $\vec{a} \pm \vec{b} = 0$ (b) $\vec{a} \pm 2\vec{b} = 0$
- (c) $2\vec{a} \pm \vec{b} = 0$ (d) none
18. If $\vec{a} = 4\hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{b} = -8\hat{i} + 4\hat{j} - 6\hat{k}$ are two vectors then $\vec{a}, \vec{b}$ are
- (a) like parallel (b) unlike parallel
- (c) non-collinear (d) perpendicular
19. If $\begin{vmatrix} a & a^2 & 1+a^3 \\ b & b^2 & 1+b^3 \\ c & c^2 & 1+c^3 \end{vmatrix} = 0$ and vectors $(1, a, a^2), (1, b, b^2)$ and $(1, c, c^2)$ are non-coplanar, then the product abc equals
- (a) 2 (b) -1
- (c) 1 (d) 0

Scalar and vector products

20. If the moduli of vectors $\vec{a}$ and $\vec{b}$ are 1 and 2 respectively and $\vec{a} \cdot \vec{b} = 1$, then the angle θ between them is:
- (a) $\theta = \pi/6$ (b) $\theta = \pi/3$
- (c) $\theta = \pi/2$ (d) $\theta = 2\pi/3$
21. If the angle between $\vec{a}$ and $\vec{b}$ is θ then for $\vec{a} \cdot \vec{b} \geq 0$
- (a) $0 \leq \theta \leq \pi$ (b) $0 < \theta$ or $\theta > \pi/2$
- (c) $\pi/2 \leq \theta \leq \pi$ (d) $0 \leq \theta \leq \pi/2$
22. $(\vec{A} + \vec{B})^2 + (\vec{A} - \vec{B})^2$ equals
- (a) $2(\vec{A}^2 + \vec{B}^2)$ (b) $4\vec{A} \cdot \vec{B}$
- (c) $\vec{A}^2 + \vec{B}^2$ (d) none of these
23. If $|\vec{a}| = |\vec{b}|$, then $(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b})$ is
- (a) positive (b) negative
- (c) zero (d) none of these
24. If $\vec{a} = \hat{i} + 3\hat{j} - 2\hat{k}$ and $\vec{b} = 4\hat{i} - 2\hat{j} + 4\hat{k}$, then $(2\vec{a} + \vec{b}) \cdot (\vec{a} - 2\vec{b})$ equals
- (a) 14 (b) -14
- (c) 0 (d) none of these

Collinearity and Coplanarity Conditions

16. If position vectors of A, B, C, D are respectively $2\hat{i} + 3\hat{j} + 5\hat{k}, \hat{i} + 2\hat{j} + 3\hat{k}, -5\hat{i} + 4\hat{j} - 2\hat{k}$ and $\hat{i} + 10\hat{j} + 10\hat{k}$, then
- (a) $AB \parallel CD$ (b) $DC \parallel AD$
- (c) A, B, C are collinear (d) B, C, D are collinear



25. Angle between the vectors $2\hat{i} + 6\hat{j} + 3\hat{k}$ and $12\hat{i} - 4\hat{j} + 3\hat{k}$ is
- (a) $\cos^{-1}\left(\frac{1}{10}\right)$ (b) $\cos^{-1}\left(\frac{9}{11}\right)$
- (c) $\cos^{-1}\left(\frac{9}{91}\right)$ (d) $\cos^{-1}\left(\frac{1}{9}\right)$
26. If $\hat{i} + \hat{j} + \hat{k}$, $2\hat{i} + 5\hat{j}$, $3\hat{i} + 2\hat{j} - 3\hat{k}$ and $\hat{i} - 6\hat{j} - \hat{k}$ be p.v. of four points A, B, C and D respectively, then the angle between $\overline{AB}$ and $\overline{CD}$ is
- (a) $\pi/4$ (b) $\pi/2$
- (c) π (d) none of these
27. Projection vector of $\vec{a}$ on $\vec{b}$ is
- (a) $\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \vec{b}$ (b) $\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$
- (c) $\frac{\vec{a} \cdot \vec{b}}{|\vec{a}|}$ (d) $\frac{\vec{a} \cdot \vec{b}}{|\vec{a}|^2} \hat{b}$
28. The angle between the vectors $\vec{a} + \vec{b}$ & $\vec{a} - \vec{b}$, given $|\vec{a}| = 2$, $|\vec{b}| = 1$ and angle between $\vec{a}$ & $\vec{b}$ is $\pi/3$, is
- (a) $\tan^{-1} \frac{2}{\sqrt{3}}$ (b) $\tan^{-1} \sqrt{\frac{2}{3}}$
- (c) $\tan^{-1} \sqrt{\frac{3}{7}}$ (d) none
29. Given the vectors $\vec{a}$ & $\vec{b}$ the angle between which equals 120° . If $|\vec{a}| = 3$ & $|\vec{b}| = 4$ then the length of the vector $2\vec{a} - \frac{3}{2}\vec{b}$ is
- (a) $6\sqrt{3}$ (b) $7\sqrt{2}$
- (c) $4\sqrt{5}$ (d) none
30. $\left[\frac{\vec{a}}{|\vec{a}|^2} - \frac{\vec{b}}{|\vec{b}|^2} \right]^2 =$
- (a) $|\vec{a}|^2 - |\vec{b}|^2$ (b) $\left[\frac{\vec{a} - \vec{b}}{|\vec{a}| |\vec{b}|} \right]^2$
- (c) $\left[\frac{|\vec{a}| |\vec{a}| - |\vec{b}| |\vec{b}|}{|\vec{a}| |\vec{b}|} \right]^2$ (d) none
31. If the coordinates of the points A, B, C be $(-1, 3, 2)$, $(2, 3, 5)$ and $(3, 5, -2)$ respectively, then $\angle A =$
- (a) 0° (b) 45°
- (c) 60° (d) 90°
32. The coordinates of the points A, B, C, D are $(4, \alpha, 2)$, $(5, -3, 2)$, $(\beta, 1, 1)$ & $(3, 3, -1)$. Line AB would be perpendicular to line CD when
- (a) $\alpha = -1, \beta = -1$ (b) $\alpha = 1, \beta = 2$
- (c) $\alpha = 2, \beta = 1$ (d) $\alpha = 2, \beta = 2$
33. If $\vec{a}$ and $\vec{b}$ are vectors of equal magnitude 2 and α be the angle between them, then magnitude of $\vec{a} + \vec{b}$ will be 2 if
- (a) $\alpha = \pi/3$ (b) $\alpha = \pi/4$
- (c) $\alpha = \pi/2$ (d) $\alpha = 2\pi/3$
34. If θ be the angle between vectors $\hat{i} + 2\hat{j} + 3\hat{k}$ and $3\hat{i} + 2\hat{j} + \hat{k}$, then the value of $\sin \theta$ is
- (a) $\sqrt{6/7}$ (b) $\frac{2\sqrt{6}}{7}$
- (c) $1/7$ (d) none of these
35. Two non zero vectors $\vec{a}$ and $\vec{b}$ will be parallel, if
- (a) $\vec{a} \cdot \vec{b} = 0$ (b) $\vec{a} \times \vec{b} \neq \vec{0}$
- (c) $\vec{a} = \vec{b}$ (d) none of these
36. If $\vec{a}$ and $\vec{b}$ are two vectors, then -
- (a) $|\vec{a} \times \vec{b}| \geq |\vec{a}| |\vec{b}|$ (b) $|\vec{a} \times \vec{b}| \leq |\vec{a}| |\vec{b}|$
- (c) $|\vec{a} \times \vec{b}| > |\vec{a}| |\vec{b}|$ (d) $|\vec{a} \times \vec{b}| < |\vec{a}| |\vec{b}|$
37. If $|\vec{a} \times \vec{b}| = \vec{a} \cdot \vec{b}$ then angle between a and b is
- (a) 0° (b) 90°
- (c) 60° (d) 45°



38. The unit vector perpendicular to vectors $\hat{i} + \hat{j}$ and $\hat{j} + \hat{k}$ is
- (a) $\frac{1}{\sqrt{3}}(\hat{i} - \hat{j} + \hat{k})$ (b) $\frac{1}{\sqrt{3}}(\hat{i} + \hat{j} - \hat{k})$
- (c) $\frac{1}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})$ (d) none of these
39. If $|\vec{a} \cdot \vec{b}| = 3$ and $|\vec{a} \times \vec{b}| = 4$, then the angle between $\vec{a}$ and $\vec{b}$ is
- (a) $\cos^{-1} 3/4$ or $\pi - \cos^{-1} 3/4$
- (b) $\cos^{-1} 3/5$ or $\pi - \cos^{-1} 3/5$
- (c) $\sin^{-1} 3/5$ or $\pi - \sin^{-1} 3/5$
- (d) $\pi/4$ or $3\pi/4$
40. If $\vec{a}$, $\vec{b}$, $\vec{c}$ are any vectors then which one of the following is a **wrong statement**.
- (a) $\vec{a} + \vec{b} = \vec{b} + \vec{a}$ (b) $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$
- (c) $\vec{a} \times \vec{b} = \vec{b} \times \vec{a}$ (d) $|\vec{a} \times \vec{b}| = |\vec{b} \times \vec{a}|$
41. If for vectors $\vec{a}$ and $\vec{b}$, $\vec{a} \times \vec{b} = 0$ and $\vec{a} \cdot \vec{b} = 0$, then
- (a) $\vec{a} \parallel \vec{b}$ (b) $\vec{a} \perp \vec{b}$
- (c) $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$ (d) none of these
42. If $\vec{a} \times \vec{b} = \vec{a} \times \vec{c}$ then correct statement is
- (a) $\vec{a} = 0$
- (b) $\vec{b} = 0 = \vec{c}$
- (c) $\vec{b} = \vec{c}$
- (d) above three are not necessary
43. For any vectors $\vec{a}$, $\vec{b}$; $|\vec{a} \times \vec{b}|^2 + (\vec{a} \cdot \vec{b})^2$ is equal to
- (a) $|\vec{a}|^2 |\vec{b}|^2$ (b) $\vec{a}^2 + \vec{b}^2$
- (c) $\vec{a}^2 - \vec{b}^2$ (d) 0
44. If $\vec{a} \times \vec{b} = \vec{c} \times \vec{d}$ and $\vec{a} \times \vec{c} = \vec{b} \times \vec{d}$ and $\vec{a} \neq \vec{d}$, $\vec{b} \neq \vec{c}$, Then
- (a) $\vec{a} - \vec{d}$ is parallel to $\vec{b} - \vec{c}$.
- (b) $\vec{a} - \vec{d}$ is perpendicular to $\vec{b} - \vec{c}$.
- (c) $\vec{a} - \vec{d}$ is equal to $\vec{b} - \vec{c}$.
- (d) none of these
45. Vectors $\vec{a}$ & $\vec{b}$ make an angle $\theta = \frac{2\pi}{3}$. If $|\vec{a}| = 1$, $|\vec{b}| = 2$ then $\{(\vec{a} + 3\vec{b}) \times (3\vec{a} - \vec{b})\}^2 =$
- (a) 225 (b) 250
- (c) 275 (d) 300
46. Unit vector perpendicular to the plane of the triangle ABC with pv's $\vec{a}$, $\vec{b}$, $\vec{c}$ of the vertices A, B, C is
- (a) $\frac{(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a})}{\Delta}$ (b) $\frac{(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a})}{2\Delta}$
- (c) $\frac{(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a})}{4\Delta}$ (d) none
47. The area of the quadrilateral ABCD, where A(0, 4, 1), B(2, 3, -1), C(4, 5, 0) and D(2, 6, 2), is equal to
- (a) 9 sq. units (b) 18 sq. units
- (c) 27 sq. units (d) 81 sq. units
48. If the vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ from the sides BC, CA and AB respectively of a triangle ABC, then
- (a) $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 0$ (b) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$
- (c) $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 0$ (d) $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a} = 0$
- Scalar and vector triple product.**
49. For non-zero vectors $\vec{a}$, $\vec{b}$, $\vec{c}$, $|\vec{a} \times \vec{b} \cdot \vec{c}| = |\vec{a}| |\vec{b}| |\vec{c}|$ holds if and only if
- (a) $\vec{a} \cdot \vec{b} = 0$, $\vec{b} \cdot \vec{c} = 0$ (b) $\vec{c} \cdot \vec{a} = 0$, $\vec{a} \cdot \vec{b} = 0$
- (c) $\vec{a} \cdot \vec{c} = 0$, $\vec{b} \cdot \vec{c} = 0$ (d) $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 0$



- 50.** The volume of the parallelepiped whose sides are given by $\vec{OA} = 2\hat{i} - 3\hat{j}$, $\vec{OB} = \hat{i} + \hat{j} - \hat{k}$ and $\vec{OC} = 3\hat{i} - \hat{k}$ is :
- (a) $4/13$ (b) 4
(c) $2/7$ (d) none
- 51.** If $(\vec{a} \times \vec{b}) \times \vec{c} = \vec{a} \times (\vec{b} \times \vec{c})$, where $\vec{a}, \vec{b}, \vec{c}$ are any three vectors such that $\vec{a} \cdot \vec{b} \neq 0$, $\vec{b} \cdot \vec{c} \neq 0$, then $\vec{a}$ and $\vec{c}$ are
- (a) inclined at an angle of $\frac{\pi}{6}$ between them
(b) perpendicular
(c) parallel
(d) inclined at an angle of $\frac{\pi}{3}$ between them
- 52.** Let $\vec{a} = \hat{j} - \hat{k}$ and $\vec{c} = \hat{i} - \hat{j} - \hat{k}$. Then the vector $\vec{b}$ satisfying $\vec{a} \times \vec{b} + \vec{c} = 0$ and $\vec{a} \cdot \vec{b} = 3$, is
- (a) $-\hat{i} + \hat{j} - 2\hat{k}$ (b) $2\hat{i} - \hat{j} + 2\hat{k}$
(c) $\hat{i} - \hat{j} - 2\hat{k}$ (d) $\hat{i} + \hat{j} - 2\hat{k}$
- 53.** Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non-zero vectors such that no two of them are collinear and $(\vec{a} \times \vec{b}) \times \vec{c} = \frac{1}{3} |\vec{b}| |\vec{c}| |\vec{a}|$. If θ is the angle between vector $\vec{b}$ and $\vec{c}$, then a value of $\sin \theta$ is:
- (a) $\frac{2}{3}$ (b) $\frac{-2\sqrt{3}}{3}$
(c) $\frac{2\sqrt{2}}{3}$ (d) $\frac{-\sqrt{2}}{3}$
- 54.** Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three unit vectors such that $\vec{a} \times (\vec{b} \times \vec{c}) = \frac{\sqrt{3}}{2} (\vec{b} + \vec{c})$. If $\vec{b}$ is not parallel to $\vec{c}$, then the angle between $\vec{a}$ and $\vec{b}$ is :
- (a) $\frac{\pi}{2}$ (b) $\frac{2\pi}{3}$
(c) $\frac{5\pi}{6}$ (d) $\frac{3\pi}{4}$
- Numerical Value Type Questions**
- 55.** If G is the intersection of diagonals of a parallelogram ABCD and O is any point and $\vec{OA} + \vec{OB} + \vec{OC} + \vec{OD} = k \vec{OG}$. Then the value of k is
- 56.** If vector $2\hat{i} + 3\hat{j} - 2\hat{k}$ and $\hat{i} + 2\hat{j} + \hat{k}$ represents the adjacent sides of any parallelogram then the sum of squares of lengths of diagonals of parallelogram is
- 57.** The 2 vectors $\hat{j} + \hat{k}$ and $3\hat{i} - \hat{j} + 4\hat{k}$ represents the two sides AB and AC, respectively of a ΔABC . Then twice of square of length of the median through A is
- 58.** P is a point on the line segment joining the points (3, 2, -1) and (6, 2, -2). If x co-ordinate of P is 5, then its y co-ordinate is
- 59.** The plane XOZ divides the join of (1, -1, 5) & (2, 3, 4) in the ratio $\lambda : 1$, then 6λ is :
- 60.** If vectors $(x - 2) \hat{i} + \hat{j}$ and $(x + 1) \hat{i} + 2\hat{j}$ are collinear, then the value of x is
- 61.** If points $\hat{i} + 2\hat{k}$, $\hat{j} + \hat{k}$ and $\lambda\hat{i} + \mu\hat{j}$ are collinear, then $\lambda + \mu$ equals
- 62.** If $\vec{a}, \vec{b}, \vec{c}$ are non coplanar vectors, then the points with p.v. $\vec{a} - 2\vec{b} + 3\vec{c}$, $2\vec{a} + \lambda\vec{b} - 4\vec{c}$, $-7\vec{b} + 10\vec{c}$ will be collinear if the value of λ is
- 63.** The number of distinct real values of λ for which the vectors $-\lambda^2\hat{i} + \hat{j} + \hat{k}$, $\hat{i} - \lambda^2\hat{j} + \hat{k}$ and $\hat{i} + \hat{j} - \lambda^2\hat{k}$ are coplanar is
- 64.** If $\vec{a}$ and $\vec{b}$ are unit vectors and 60° is the angle between them, then $(2\vec{a} - 3\vec{b}) \cdot (4\vec{a} + \vec{b})$ equals
- 65.** If vectors $3\hat{i} + 2\hat{j} + 8\hat{k}$ and $2\hat{i} + x\hat{j} + \hat{k}$ are perpendicular then -x is equal to
- 66.** If vector $\vec{a} + \vec{b}$ is perpendicular to $\vec{b}$ and $2\vec{b} + \vec{a}$ is perpendicular to $\vec{a}$, then $\left(\frac{|\vec{a}|}{|\vec{b}|}\right)^2$ equals



67. If $\vec{a} = 2\hat{i} + \hat{j} + \hat{k}$, $\vec{b} = 3\hat{i} - 4\hat{j} + 2\hat{k}$ and $\vec{c} = \hat{i} - 2\hat{j} + 2\hat{k}$ then 3 times the projection of $\vec{a} + \vec{b}$ on $\vec{c}$ is

68. If the angle between two vectors $\hat{i} + \hat{k}$ and $\hat{i} - \hat{j} + a\hat{k}$ is $\pi/3$, then the non-negative value of a is

69. The number of vectors of unit length perpendicular to the vectors $\vec{a} = 2\hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = \hat{j} + \hat{k}$ is

70. Let $\vec{A}, \vec{B}, \vec{C}$ be vectors of length 3, 4 and 5 respectively. Let $\vec{A}$ be perpendicular to $\vec{B} + \vec{C}$, $\vec{B}$ to $\vec{C} + \vec{A}$ and $\vec{C}$ to $\vec{A} + \vec{B}$. Then square of length of the vector, $\vec{A} + \vec{B} + \vec{C}$ is:

71. If $\vec{a} = 2\hat{i} + \hat{j} + 3\hat{k}$, $\vec{b} = \hat{i} + 3\hat{j} + 3\hat{k}$, then $|\vec{a} \times \vec{b}|^2$ is

72. If $(\vec{a} \times \vec{b})^2 + (\vec{a} \cdot \vec{b})^2 = 144$ and $|\vec{a}| = 4$, then $|\vec{b}|$ is equal to

73. $(\hat{i} + \hat{j}) \cdot [(\hat{j} + \hat{k}) \times (\hat{k} + \hat{i})]$ equals

74. If the diagonals of a parallelogram are respectively $\vec{a} = \hat{i} + \hat{j} - 2\hat{k}$, $\vec{b} = \hat{i} - 3\hat{j} + 4\hat{k}$, then the square of the area of parallelogram is

75. Twice of the area of the parallelogram constructed on the vectors $\vec{a} = \vec{p} + 2\vec{q}$ & $\vec{b} = 2\vec{p} + \vec{q}$ where $\vec{p}$ & $\vec{q}$ are unit vectors forming an acute angle of 30° is

76. If the value of $[(\vec{a} + 2\vec{b} - \vec{c}) \cdot (\vec{a} - \vec{b}) \cdot (\vec{a} - \vec{b} - \vec{c})]$ is equal to $k [\vec{a} \cdot \vec{b} \cdot \vec{c}]$. Then the value of k is

77. Let $\vec{a}, \vec{b}, \vec{c}$ be three non-coplanar vectors & $\vec{p}, \vec{q}, \vec{r}$ are vectors defined by the relations

$$\vec{p} = \frac{\vec{b} \times \vec{c}}{[\vec{a} \vec{b} \vec{c}]}, \vec{q} = \frac{\vec{c} \times \vec{a}}{[\vec{a} \vec{b} \vec{c}]}, \vec{r} = \frac{\vec{a} \times \vec{b}}{[\vec{a} \vec{b} \vec{c}]}.$$

Then the value of the expression; $(\vec{a} + \vec{b}) \cdot \vec{p} + (\vec{b} + \vec{c}) \cdot \vec{q} + (\vec{c} + \vec{a}) \cdot \vec{r}$ is

78. Volume of the tetrahedron whose vertices are represented by the position vectors, A (0, 1, 2); B (3, 0, 1); C (4, 3, 6) and D (2, 3, 2) is :

3-DIMENSIONAL GEOMETRY

Introduction and Concept of DC & DR

79. If a line makes angles 90° , 60° and 30° with positive direction of x, y and z-axis respectively, then its direction-cosines are :

- (a) $\langle 0, 0, 0 \rangle$ (b) $\langle 1, 1, 1 \rangle$
(c) $\langle 0, \frac{1}{2}, \frac{\sqrt{3}}{2} \rangle$ (d) $\langle \frac{\sqrt{3}}{2}, \frac{1}{2}, 0 \rangle$

80. A line makes equal angles with co-ordinate axes. Direction cosines of this line are

- (a) $\pm 1, \pm 1, \pm 1$ (b) $\pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}$
(c) $\pm \frac{1}{3}, \pm \frac{1}{3}, \pm \frac{1}{3}$ (d) $\pm \sqrt{3}, \pm \sqrt{3}, \pm \sqrt{3}$

81. If a line passes through the points $(-2, 4, -5)$ and $(1, 2, 3)$ then its direction-cosines will be :

- (a) $\langle -\frac{2}{\sqrt{77}}, \frac{8}{\sqrt{77}}, \frac{3}{\sqrt{77}} \rangle$ (b) $\langle -3, 2, +8 \rangle$
(c) $\langle +\frac{3}{\sqrt{77}}, \frac{-2}{\sqrt{77}}, \frac{8}{\sqrt{77}} \rangle$ (d) $\langle 3, -2, 8 \rangle$

82. A line makes acute angles of α , β and γ with the co-ordinate axes such that $\cos \alpha \cos \beta = \cos \beta \cos \gamma = \frac{2}{9}$

and $\cos \gamma \cos \alpha = \frac{4}{9}$, then $\cos \alpha + \cos \beta + \cos \gamma$ is equal to :

- (a) $25/9$ (b) $5/9$
(c) $5/3$ (d) $2/3$

83. If a line has direction ratios $\langle 2, -1, -2 \rangle$, then its direction-cosines will be :

- (a) $\langle \frac{2}{3}, \frac{1}{3}, \frac{2}{3} \rangle$ (b) $\langle \frac{1}{3}, \frac{1}{3}, \frac{1}{3} \rangle$
(c) $\langle \frac{-2}{3}, \frac{-1}{3}, \frac{-2}{3} \rangle$ (d) $\langle \frac{2}{3}, \frac{-1}{3}, \frac{-2}{3} \rangle$



84. Two lines, whose direction ratios are :
 $\langle a_1, b_1, c_1 \rangle$ and $\langle a_2, b_2, c_2 \rangle$ respectively are perpendicular if

(a) $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{c_1}{c_2}$ (b) $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$
 (c) $a_1a_2 + b_1b_2 + c_1c_2 = 0$ (d) $a_1a_2 + b_1b_2 + c_1c_2 = 1$

85. A line makes angles α, β, γ with the coordinate axes. If $\alpha + \beta = 90^\circ$, then $\gamma =$

(a) 0 (b) 90°
 (c) 180° (d) None of these

86. The direction cosines of the line, $x = y = z$ are :

(a) $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$ (b) $\frac{1}{3}, \frac{1}{3}, \frac{1}{3}$
 (c) $\sqrt{5}, \sqrt{13}, \sqrt{10}$ (d) $\frac{1}{\sqrt{13}}, \frac{1}{\sqrt{10}}, \frac{1}{\sqrt{5}}$

Equation of lines in 3-D

87. A line passes through a point A with p.v. $3\hat{i} + \hat{j} - \hat{k}$ and is parallel to the vector $2\hat{i} - \hat{j} + 2\hat{k}$. If P is a point on this line such that AP = 15 units, then the p.v. of the point P may be:

(a) $13\hat{i} + 4\hat{j} - 9\hat{k}$ (b) $13\hat{i} - 4\hat{j} + 9\hat{k}$
 (c) $7\hat{i} - 6\hat{j} + 11\hat{k}$ (d) $+7\hat{i} + 6\hat{j} + 11\hat{k}$

88. Image of the point P with position vector $7\hat{i} - \hat{j} + 2\hat{k}$ in the line whose vector equation is, $\vec{r} = 9\hat{i} + 5\hat{j} + 5\hat{k} + \lambda(\hat{i} + 3\hat{j} + 5\hat{k})$ has the position vector :

(a) $(-9, 5, 2)$ (b) $(9, 5, -2)$
 (c) $(9, -5, -2)$ (d) none

89. Find the angle between the two straight lines,

$\vec{r} = 3\hat{i} - 2\hat{j} + 4\hat{k} + \lambda(-2\hat{i} + \hat{j} + 2\hat{k})$ and
 $\vec{r} = \hat{i} + 3\hat{j} - 2\hat{k} + \mu(3\hat{i} - 2\hat{j} + 6\hat{k})$:

(a) $\cos^{-1}(4/21)$ (b) $\sin^{-1}(4/21)$
 (c) $\sin^{-1}(17/21)$ (d) $\cos^{-1}(17/21)$

90. The lines,

$\vec{r}_1 = \hat{i} + 2\hat{j} + 3\hat{k} + \lambda(2\hat{i} + 3\hat{j} + 4\hat{k})$

and $\vec{r}_2 = 2\hat{i} + 3\hat{j} + 4\hat{k} + \mu(3\hat{i} + 4\hat{j} + 5\hat{k})$ are :

- (a) coplanar (b) skew
 (c) such that shortest distance between them is 1
 (d) none

91. The equations of x-axis in space are

(a) $x = 0, y = 0$ (b) $x = 0, z = 0$
 (c) $x = 0$ (d) $y = 0, z = 0$

92. Two lines $\vec{r} = \vec{a}_1 + \lambda\vec{b}_1$ and $\vec{r} = \vec{a}_2 + \mu\vec{b}_2$ would be coplanar if:

(a) $[\vec{a}_1 \vec{b}_1 \vec{b}_2] = [\vec{a}_2 \vec{b}_1 \vec{b}_2]$
 (b) $(\vec{a}_1 \cdot \vec{b}_1) \vec{b}_2 = (\vec{a}_2 \cdot \vec{b}_1) \vec{b}_2$
 (c) $\vec{a}_1 (\vec{b}_1 \cdot \vec{b}_2) = \vec{a}_2 (\vec{b}_1 \cdot \vec{b}_2)$
 (d) $\vec{a}_1 \cdot \vec{b}_1 - \vec{a}_1 \cdot \vec{b}_2 = \vec{a}_2 \cdot \vec{b}_1 - \vec{a}_2 \cdot \vec{b}_2$

93. The point of intersection of lines,

$\frac{x-4}{5} = \frac{y-1}{2} = \frac{z}{1}$ & $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ is :

(a) $(-1, -1, -1)$ (b) $(-1, -1, 1)$
 (c) $(1, -1, -1)$ (d) $(-1, 1, -1)$

94. The straight lines $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3}$ and

$\frac{x-1}{2} = \frac{y-2}{2} = \frac{z-3}{-2}$ are

- (a) parallel lines (b) Intersecting at 60°
 (c) Skew lines (d) Intersecting at right angle

95. A line with direction cosines proportional to 2, 1, 2 meets each of the lines $x = y + a = z$ and $x + a = 2y = 2z$. The co-ordinates of each of the points of intersection are given by

(a) $(3a, 3a, 3a), (a, a, a)$ (b) $(3a, 2a, 3a), (a, a, a)$
 (c) $(3a, 2a, 3a), (a, a, 2a)$ (d) $(2a, 3a, 3a), (2a, a, a)$

96. The angle between the lines $2x = 3y = -z$ and $6x = -y = -4z$ is

(a) 30° (b) 45°
 (c) 90° (d) 0°



97. If the straight lines $x = 1 + s$, $y = -3 - \lambda s$, $z = 1 + \lambda s$ and $x = \frac{t}{2}$, $y = 1 + t$, $z = 2 - t$, with parameters s and t respectively, are coplanar, then λ equals
(a) -2 (b) -1
(c) $-\frac{1}{2}$ (d) 0
98. If the lines $\frac{x-2}{1} = \frac{y-3}{1} = \frac{z-4}{-k}$ and $\frac{x-1}{k} = \frac{y-4}{2} = \frac{z-5}{1}$ are coplanar, then k can have
(a) any value (b) exactly one value
(c) exactly two values (d) exactly three value
99. The shortest distance between the skew lines $\ell_1 : \vec{r} = \vec{a}_1 + \lambda \vec{b}_1$ and $\ell_2 : \vec{r} = \vec{a}_2 + \mu \vec{b}_2$ is:
(a) $\frac{(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|}$ (b) $\frac{(\vec{a}_2 - \vec{a}_1) \cdot (\vec{a}_2 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|}$
(c) $\frac{(\vec{a}_2 - \vec{b}_2) \cdot (\vec{a}_1 \times \vec{b}_1)}{|\vec{b}_1 \times \vec{b}_2|}$ (d) $\frac{(\vec{a}_1 - \vec{b}_2) \cdot (\vec{b}_1 \times \vec{a}_2)}{|\vec{b}_1 \times \vec{b}_2|}$
100. The line, $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$ is parallel to the plane :
(a) $3x + 4y + 5z = 7$ (b) $2x + y - 2z = 0$
(c) $x + y - z = 2$ (d) $2x + 3y + 4z = 0$
101. The equation of the plane passing through the points $(3, 2, 2)$ and $(1, 0, -1)$ and parallel to the line $\frac{x-1}{2} = \frac{y-1}{-2} = \frac{z-2}{3}$, is
(a) $4x - y - 2z + 6 = 0$ (b) $4x - y + 2z + 6 = 0$
(c) $4x - y - 2z - 6 = 0$ (d) $3x - 2z - 5 = 0$
102. The equation of a plane which passes through $(2, -3, 1)$ and is normal to the line joining the points $(3, 4, -1)$ and $(2, -1, 5)$ is given by :
(a) $x + 5y - 6z + 19 = 0$ (b) $x - 5y + 6z - 19 = 0$
(c) $x + 5y + 6z + 19 = 0$ (d) $x - 5y - 6z - 19 = 0$
103. A plane meets the coordinate axes in A, B, C and (α, β, γ) is the centroid of the triangle ABC . Then the equation of the plane is
(a) $\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 3$ (b) $\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 1$
(c) $\frac{3x}{\alpha} + \frac{3y}{\beta} + \frac{3z}{\gamma} = 1$ (d) $\alpha x + \beta y + \gamma z = 1$
104. The plane $ax + by + cz = 1$ meets the co-ordinate axes in A, B and C . The centroid of the triangle is :
(a) $(3a, 3b, 3c)$ (b) $\left(\frac{a}{3}, \frac{b}{3}, \frac{c}{3}\right)$
(c) $\left(\frac{3}{a}, \frac{3}{b}, \frac{3}{c}\right)$ (d) $\left(\frac{1}{3a}, \frac{1}{3b}, \frac{1}{3c}\right)$
105. The equation of the plane which is right bisector of the line joining $(2, 3, 4)$ and $(6, 7, 8)$, is :
(a) $x + y + z - 15 = 0$ (b) $x - y + z - 15 = 0$
(c) $x - y - z - 15 = 0$ (d) $x + y + z + 15 = 0$
106. The equation of the plane containing the line $\frac{x-x_1}{\ell} = \frac{y-y_1}{m} = \frac{z-z_1}{n}$ is $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$, where :
(a) $ax_1 + by_1 + cz_1 = 0$ (b) $a\ell + bm + cn = 0$
(c) $\frac{a}{\ell} = \frac{b}{m} = \frac{c}{n}$ (d) $\ell x_1 + my_1 + nz_1 = 0$
107. The vector equation of the plane passing through the origin and the line of intersection of the plane $\vec{r} \cdot \vec{a} = \lambda$ and $\vec{r} \cdot \vec{b} = \mu$ is :
(a) $\vec{r} \cdot (\lambda \vec{a} - \mu \vec{b}) = 0$ (b) $\vec{r} \cdot (\lambda \vec{b} - \mu \vec{a}) = 0$
(c) $\vec{r} \cdot (\lambda \vec{a} + \mu \vec{b}) = 0$ (d) $\vec{r} \cdot (\lambda \vec{b} + \mu \vec{a}) = 0$
108. Equation of the plane passing through the line of intersection of the planes $P : ax + by + cz + d = 0$, $P' : a'x + b'y + c'z + d' = 0$, and parallel to x -axis is :
(a) $Pa - P'a' = 0$ (b) $P/a = P'/a' = 0$
(c) $Pa + P'a' = 0$ (d) $P/a = P'/a'$

Various form of Equation of Planes



- 109.** The Plane $2x - (1 + \lambda)y + 3\lambda z = 0$ passes through the intersection of the planes
- (a) $2x - y = 0$ and $y - 3z = 0$
 (b) $2x + 3z = 0$ and $y = 0$
 (c) $2x - y + 3z = 0$ and $y - 3z = 0$
 (d) none of these
- 110.** The equation of the plane through the intersection of the planes $x + 2y + 3z = 4$ and $2x + y - z = -5$ and perpendicular to the plane $5x + 3y + 6z + 8 = 0$ is
- (a) $7x - 2y + 3z + 81 = 0$ (b) $23x + 14y - 9z + 48 = 0$
 (c) $51x - 15y - 50z + 173 = 0$ (d) none of these
- 111.** The equation of the plane containing the two lines $\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z-0}{3}$ and $\frac{x}{-2} = \frac{y-2}{11} = \frac{z+1}{-1}$ is
- (a) $8x + y - 5z - 7 = 0$ (b) $8x + y + 5z - 7 = 0$
 (c) $8x - y - 5z - 7 = 0$ (d) none of these
- 112.** The equation of the plane through the intersection of the planes $ax + by + cz + d = 0$ and $lx + my + nz + p = 0$ and parallel to the line $y = 0, z = 0$
- (a) $(b/-am)y + (c/-an)z + d/-ap = 0$
 (b) $(am - b/l)x + (mc - bn)z + md - bp = 0$
 (c) $(na - cl)x + (bn - cm)y + nd - cp = 0$
 (d) none of these
- 113.** Equation of the plane which passes through the point of intersection of lines $\frac{x-1}{3} = \frac{y-2}{1} = \frac{z-3}{2}$ and $\frac{x-3}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ and has the largest distance from the origin is:
- (a) $7x + 2y + 4z = 54$ (b) $3x + 4y + 5z = 49$
 (c) $4x + 3y + 5z = 50$ (d) $5x + 4y + 3z = 57$
- Some Formulae of Plane**
- 114.** The distance of the point, $(-1, -5, -10)$ from the point of intersection of the line, $\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12}$ and the plane, $x - y + z = 5$, is :
- (a) 10 (b) 11
 (c) 12 (d) 13
- 115.** The point at which the line joining the points $(2, -3, 1)$ & $(3, -4, -5)$ intersects the plane $2x + y + z = 7$ is :
- (a) $(1, 2, 7)$ (b) $(1, -2, 7)$
 (c) $(-1, 2, 7)$ (d) $(1, -2, -7)$
- 116.** The point of intersection of the line $\frac{x}{1} = \frac{y-1}{2} = \frac{z+2}{3}$ and the plane, $2x + 3y + z = 0$, is :
- (a) $(0, 1, -2)$ (b) $(1, 2, 3)$
 (c) $(-1, 9, -25)$ (d) $\left(\frac{-1}{11}, \frac{9}{11}, \frac{-25}{11}\right)$
- 117.** The direction ratio of normal to the plane through $(1, 0, 0), (0, 1, 0)$, which makes an angle $\frac{\pi}{4}$ with plane $x + y = 3$, are :
- (a) $1, \sqrt{2}, 1$ (b) $1, 1, \sqrt{2}$
 (c) $1, 1, 2$ (d) $\sqrt{2}, 1, 1$
- 118.** If line $\vec{r} = (\hat{i} - 2\hat{j} - \hat{k}) + \lambda(2\hat{i} + \hat{j} + 2\hat{k})$ is parallel to the plane $\vec{r} \cdot (3\hat{i} - 2\hat{j} - m\hat{k}) = 14$ then the value of m is
- (a) +2 (b) -2
 (c) 0 (d) can not be predicted with this much informations
- 119.** The distance of the point $(1, -5, 9)$ from the plane $x - y + z = 5$ measured along a straight line $x = y = z$ is
- (a) $3\sqrt{5}$ (b) $10\sqrt{3}$
 (c) $5\sqrt{3}$ (d) $3\sqrt{10}$
- 120.** The reflection of the point (α, β, γ) in the xy -plane is
- (a) $(\alpha, \beta, 0)$ (b) $(0, 0, \gamma)$
 (c) $(-\alpha, -\beta, \gamma)$ (d) $(\alpha, \beta, -\gamma)$
- 121.** The plane $2x - 3y + 6z - 11 = 0$ makes an angle $\sin^{-1}(\alpha)$ with x -axis. The value of α is equal to
- (a) $\frac{\sqrt{3}}{2}$ (b) $\frac{\sqrt{2}}{3}$
 (c) $\frac{2}{7}$ (d) $\frac{3}{7}$



122. If a plane passes through the point $(1, 1, 1)$ and is perpendicular to the line $\frac{x-1}{3} = \frac{y-1}{0} = \frac{z-1}{4}$ then its perpendicular distance from the origin is
 (a) $3/4$ (b) $4/3$
 (c) $7/5$ (d) 1
123. The angle between the planes, $2x - y + z = 6$ and $x + y + 2z = 7$ is
 (a) 30° (b) 45°
 (c) 0° (d) 60°
124. The length of the perpendicular from origin on plane $\vec{r} \cdot (3\hat{i} - 4\hat{j} + 12\hat{k}) = 5$ is
 (a) $\frac{5}{69}$ (b) $\frac{25}{69}$
 (c) $\frac{5}{13}$ (d) $\sqrt{\frac{5}{13}}$
125. The angle between the line, $\frac{x+1}{3} = \frac{y-1}{2} = \frac{z-2}{4}$ and the plane, $2x + y - 3z + 4 = 0$, is :
 (a) $\sin^{-1}\left(\frac{4}{\sqrt{406}}\right)$ (b) $\sin^{-1}\left(\frac{14}{\sqrt{406}}\right)$
 (c) $\sin^{-1}\left(\frac{4}{14\sqrt{29}}\right)$ (d) None of these
126. If the given planes, $ax + by + cz + d = 0$ and $a'x + b'y + c'z + d' = 0$ be mutually perpendicular, then:
 (a) $\frac{a}{a'} = \frac{b}{b'} = \frac{c}{c'}$ (b) $\frac{a}{a'} + \frac{b}{b'} + \frac{c}{c'} = 0$
 (c) $aa' + bb' + cc' + dd' = 0$
 (d) $aa' + bb' + cc' = 0$
127. The angle between the line $\frac{x-2}{a} = \frac{y-2}{b} = \frac{z-2}{c}$ and the plane $ax + by + cz + 6 = 0$ is
 (a) $\sin^{-1} \frac{1}{\sqrt{a^2 + b^2 + c^2}}$ (b) 45°
 (c) 60° (d) 90°
128. The image of the point P $(1, 3, 4)$ in the plane $2x - y + z + 3 = 0$ is
 (a) $(3, 5, -2)$ (b) $(-3, 5, 2)$
 (c) $(3, -5, 2)$ (d) $(-1, 4, 2)$
129. The image of the points $(-1, 3, 4)$ in the plane $x - 2y = 0$ is
 (a) $(15, 11, 4)$ (b) $\left(-\frac{17}{3}, -\frac{19}{3}, 1\right)$
 (c) $(8, 4, 4)$ (d) $\left(\frac{9}{5}, -\frac{13}{5}, 4\right)$
130. Distance between two parallel planes $2x + y + 2z = 8$ and $4x + 2y + 4z + 5 = 0$ is
 (a) $\frac{3}{2}$ (b) $\frac{5}{2}$
 (c) $\frac{7}{2}$ (d) $\frac{9}{2}$
131. If the distance between planes, $4x - 2y - 4z + 1 = 0$ and $4x - 2y - 4z + d = 0$ is 7 , then d is:
 (a) 41 or -42 (b) 42 or -43
 (c) -41 or 43 (d) -42 or 44
132. If the angle between the line $2(x+1) = y = z + 4$ and the plane $2x - y + \sqrt{\lambda}z + 4 = 0$ is $\frac{\pi}{6}$, then the value of λ is
 (a) $\frac{135}{7}$ (b) $\frac{45}{11}$
 (c) $\frac{45}{7}$ (d) $\frac{135}{11}$
133. A symmetrical form of the line of intersection of the planes $x = ay + b$ and $z = cy + d$ is:
 (a) $\frac{x-b}{a} = \frac{y-1}{1} = \frac{z-d}{c}$
 (b) $\frac{x-b-a}{a} = \frac{y-1}{1} = \frac{z-d-c}{c}$
 (c) $\frac{x-a}{b} = \frac{y-0}{1} = \frac{z-c}{d}$
 (d) $\frac{x-b-a}{b} = \frac{y-1}{0} = \frac{z-d-c}{d}$

Numerical Value Type Questions

134. If a line makes angles α, β, γ with the co-ordinate axes, then $-(\cos 2\alpha + \cos 2\beta + \cos 2\gamma)$ is :
135. If the straight lines $x = 1 + s, y = -3 - \lambda s, z = 1 + \lambda s$ and $x = t/2, y = 1 + t, z = 2 - t$, with parameter s and t respectively, are coplanar, then $-\lambda$ equals to



EXERCISE - 2 : PREVIOUS YEAR JEE MAIN QUESTIONS

VECTORS

- The length of the projection of the line segment joining the points (5, -1, 4) and (4, -1, 3) on the plane, $x + y + z = 7$ is : **(2018)**
 - $\sqrt{\frac{2}{3}}$
 - $\frac{2}{\sqrt{3}}$
 - $\frac{2}{3}$
 - $\frac{1}{3}$
- If the position vectors of the vertices A, B and C of a $\triangle ABC$ are respectively $4\hat{i} + 7\hat{j} + 8\hat{k}$, $2\hat{i} + 3\hat{j} + 4\hat{k}$ and $2\hat{i} + 5\hat{j} + 7\hat{k}$, then the position vector of the point, where the bisector of $\angle A$ meets BC is : **(2018/Online Set-2)**
 - $\frac{1}{2}(4\hat{i} + 8\hat{j} + 11\hat{k})$
 - $\frac{1}{3}(6\hat{i} + 11\hat{j} + 15\hat{k})$
 - $\frac{1}{3}(6\hat{i} + 13\hat{j} + 18\hat{k})$
 - $\frac{1}{4}(8\hat{i} + 14\hat{j} + 19\hat{k})$
- The magnitude of the projection of the vector $2\hat{i} + 3\hat{j} + \hat{k}$ on the vector perpendicular to the plane containing the vectors $\hat{i} + \hat{j} + \hat{k}$ and $\hat{i} + 2\hat{j} + 3\hat{k}$, is **(2019-04-08/Shift-1)**
 - $\frac{\sqrt{3}}{2}$
 - $\sqrt{6}$
 - $3\sqrt{6}$
 - $\sqrt{\frac{3}{2}}$
- Let $\vec{a} = 3\hat{i} + 2\hat{j} + x\hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$, for some real x. Then $|\vec{a} \times \vec{b}| = r$ is possible if: **(2019-04-08/Shift-2)**
 - $\sqrt{\frac{3}{2}} < r \leq 3\sqrt{\frac{3}{2}}$
 - $r \geq 5\sqrt{\frac{3}{2}}$
 - $0 < r \leq \sqrt{\frac{3}{2}}$
 - $3\sqrt{\frac{3}{2}} < r < 5\sqrt{\frac{3}{2}}$
- Let $\vec{\alpha} = 3\hat{i} + \hat{j}$ and $\vec{\beta} = 2\hat{i} - \hat{j} + 3\hat{k}$. If $\vec{\beta} = \vec{\beta}_1 - \vec{\beta}_2$ where $\vec{\beta}_1$ is parallel to $\vec{\alpha}$ and $\vec{\beta}_2$ is perpendicular to $\vec{\alpha}$ then $\vec{\beta}_1 \times \vec{\beta}_2$ is equal to: **(2019-04-09/Shift-1)**
 - $-3\hat{i} + 9\hat{j} + 5\hat{k}$
 - $3\hat{i} - 9\hat{j} - 5\hat{k}$
 - $\frac{1}{2}(-3\hat{i} + 9\hat{j} + 5\hat{k})$
 - $\frac{1}{2}(3\hat{i} - 9\hat{j} + 5\hat{k})$
- Let $\vec{a} = 3\hat{i} + 2\hat{j} + 2\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - 2\hat{k}$ be two vectors. If a vector perpendicular to both the vectors $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ has the magnitude 12 then one such vector is **(2019-04-12/Shift-1)**
 - $4(2\hat{i} + 2\hat{j} + 2\hat{k})$
 - $4(2\hat{i} - 2\hat{j} - \hat{k})$
 - $4(2\hat{i} + 2\hat{j} - \hat{k})$
 - $4(-2\hat{i} - 2\hat{j} + \hat{k})$
- If the volume of parallelepiped formed by the vectors $\hat{i} + \lambda\hat{j} + \hat{k}$, $\hat{j} + \lambda\hat{k}$ and $\lambda\hat{i} + \hat{k}$ is minimum, then λ is equal to **(2019-04-12/Shift-1)**
 - $-\frac{1}{\sqrt{3}}$
 - $\frac{1}{\sqrt{3}}$
 - $\sqrt{3}$
 - $-\sqrt{3}$
- Let $\alpha \in R$ and the three vectors $\vec{a} = \alpha\hat{i} + \hat{j} + 3\hat{k}$, $\vec{b} = 2\hat{i} + \hat{j} - \alpha\hat{k}$ and $\vec{c} = \alpha\hat{i} - 2\hat{j} + 3\hat{k}$. Then the set $S = \{\alpha : \vec{a}, \vec{b} \text{ and } \vec{c} \text{ are coplanar}\}$ **(2019-04-12/Shift-2)**
 - is singleton
 - is empty
 - contains exactly two positive numbers
 - contains exactly two numbers only one of which is positive



9. Let $\vec{a} = \hat{i} - \hat{j}$, $\vec{b} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{c}$ be a vector such that $\vec{a} \times \vec{c} + \vec{b} = \vec{0}$ and $\vec{a} \cdot \vec{c} = 4$, then $|\vec{c}|^2$ is equal to:

(2019-01-09/Shift-1)

- (a) $\frac{19}{2}$ (b) 9
(c) 8 (d) $\frac{17}{2}$

10. Let $\vec{a} = \hat{i} + \hat{j} + \sqrt{2}\hat{k}$, $\vec{b} = b_1\hat{i} + b_2\hat{j} + \sqrt{2}\hat{k}$ and $\vec{c} = 5\hat{i} + \hat{j} + \sqrt{2}\hat{k}$ be three vectors such that the projection vector of $\vec{b}$ on $\vec{a}$ is $|\vec{a}|$. If $\vec{a} + \vec{b}$ is perpendicular to $\vec{c}$, then $|\vec{b}|$ is equal to:

(2019-01-09/Shift-2)

- (a) $\sqrt{32}$ (b) 6
(c) $\sqrt{22}$ (d) 4

11. Let $\vec{a} = 2\hat{i} + \lambda_1\hat{j} + 3\hat{k}$, $\vec{b} = 4\hat{i} + (3 - \lambda_2)\hat{j} + 6\hat{k}$ and $\vec{c} = 3\hat{i} + 6\hat{j} + (\lambda_3 - 1)\hat{k}$ be three vectors such that $\vec{b} = 2\vec{a}$ and $\vec{a}$ is perpendicular to $\vec{c}$. Then a possible value of $(\lambda_1, \lambda_2, \lambda_3)$ is:

(2019-01-10/Shift-1)

- (a) (1, 3, 1) (b) $\left(-\frac{1}{2}, 4, 0\right)$
(c) $\left(\frac{1}{2}, 4, -2\right)$ (d) (1, 5, 1)

12. Let $\vec{\alpha} = (\lambda - 2)\vec{a} + \vec{b}$ and $\vec{\beta} = (4\lambda - 2)\vec{a} + 3\vec{b}$ be two given vectors where vectors $\vec{a}$ and $\vec{b}$ are non-collinear. The value of λ for which vectors $\vec{\alpha}$ and $\vec{\beta}$ are collinear, is:

(2019-01-10/Shift-2)

- (a) -4 (b) 3
(c) 4 (d) 3

13. Let $\vec{a} = \hat{i} + 2\hat{j} + 4\hat{k}$, $\vec{b} = \hat{i} + \lambda\hat{j} + 4\hat{k}$ and $\vec{c} = 2\hat{i} + 4\hat{j} + (\lambda^2 - 1)\hat{k}$ be coplanar vectors. Then the

non-zero vector $\vec{a} \times \vec{c}$ is: (2019-01-11/Shift-1)

- (a) $-10\hat{i} - 5\hat{j}$ (b) $-14\hat{i} - 5\hat{j}$
(c) $-14\hat{i} + 5\hat{j}$ (d) $-10\hat{i} + 5\hat{j}$

14. Let $\sqrt{3}\hat{i} + \hat{j}$, $\hat{i} + \sqrt{3}\hat{j}$ and $\beta\hat{i} + (1 - \beta)\hat{j}$ respectively be the position vectors of the points A, B and C with respect to the origin O. If the distance of C from the bisector of the acute angle between OA and OB is $\frac{3}{\sqrt{2}}$, then the sum of all possible values of β is

(2019-01-11/Shift-2)

- (a) 4 (b) 3
(c) 2 (d) 1

15. The sum of the distinct real values of μ , for which the vectors, $\mu\hat{i} + \hat{j} + \hat{k}$, $\hat{i} + \mu\hat{j} + \hat{k}$, $\hat{i} + \hat{j} + \mu\hat{k}$ are co-planar, is _.

(2019-01-12/Shift-2)

- (a) -1 (b) 0
(c) 1 (d) 2

16. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three unit vectors, out of which vectors $\vec{b}$ and $\vec{c}$ are non-parallel. If α and β are the angles which vector $\vec{a}$ makes with vectors $\vec{b}$ and $\vec{c}$ respectively and $\vec{a} \times (\vec{b} \times \vec{c}) = \frac{1}{2}\vec{b}$, then $|\alpha - \beta|$ is equal to

(2019-01-12/Shift-2)

- (a) 30° (b) 90°
(c) 60° (d) 45°

17. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three unit vectors such that $|\vec{a} - \vec{b}|^2 + |\vec{a} - \vec{c}|^2 = 8$. Then $|\vec{a} + 2\vec{b}|^2 + |\vec{a} + 2\vec{c}|^2$ is equal to:

(2020-09-02/Shift-1)

18. Let the position vectors of points 'A' and 'B' be $\hat{i} + \hat{j} + \hat{k}$ and $2\hat{i} + \hat{j} + 3\hat{k}$, respectively. A point 'P' divides the line segment AB internally in the ratio $\lambda : 1$ ($\lambda > 0$). If O is the origin and $\overline{OB} \cdot \overline{OP} - 3|\overline{OA} \times \overline{OP}|^2 = 6$, then λ is equal to

(2020-09-02/Shift-2)



19. Let $a, b, c \in \mathbb{R}$ be such that $a^2 + b^2 + c^2 = 1$. If $a \cos \theta = b \cos \left(\theta + \frac{2\pi}{3} \right) = c \cos \left(\theta + \frac{4\pi}{3} \right)$, where $\theta = \frac{\pi}{9}$, then the angle between the vectors $a\hat{i} + b\hat{j} + c\hat{k}$ and $b\hat{i} + c\hat{j} + a\hat{k}$ is (2020-09-03/Shift-2)
- (a) $\frac{\pi}{2}$ (b) $\frac{2\pi}{3}$
(c) $\frac{\pi}{9}$ (d) 0
20. Let x_0 be the point of local maxima of $f(x) = \vec{a} \cdot (\vec{b} \times \vec{c})$, where $\vec{a} = x\hat{i} - 2\hat{j} + 3\hat{k}$, $\vec{b} = -2\hat{i} + x\hat{j} - \hat{k}$ and $\vec{c} = 7\hat{i} - 2\hat{j} + x\hat{k}$. Then the value of $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$ at $x = x_0$ is : (2020-09-04/Shift-1)
- (a) -22 (b) -4
(c) -30 (d) 14
21. If $\vec{a} = 2\hat{i} + \hat{j} + 2\hat{k}$, then the value of $|\hat{i} \times (\vec{a} \times \hat{i})|^2 + |\hat{j} \times (\vec{a} \times \hat{j})|^2 + |\hat{k} \times (\vec{a} \times \hat{k})|^2$ is equal to ... (2020-09-04/Shift-2)
22. If the volume of a parallelepiped, whose coterminal edges are given by the vectors $\vec{a} = \hat{i} + \hat{j} + n\hat{k}$, $\vec{b} = 2\hat{i} + 4\hat{j} - n\hat{k}$ and $\vec{c} = \hat{i} + n\hat{j} + 3\hat{k}$ ($n \geq 0$), is 158 cu. units, then: (2020-09-05/Shift-1)
- (a) $\vec{a} \cdot \vec{c} = 17$ (b) $\vec{b} \cdot \vec{c} = 10$
(c) $n = 9$ (d) $n = 7$
23. Let the vectors $\vec{a}, \vec{b}, \vec{c}$ be such that $|\vec{a}| = 2, |\vec{b}| = 4$ and $|\vec{c}| = 4$. If the projection of $\vec{b}$ on $\vec{a}$ is equal to the projection of $\vec{c}$ on $\vec{a}$ and $\vec{b}$ is perpendicular to $\vec{c}$, then the value of $|\vec{a} + \vec{b} - \vec{c}|$ is (2020-09-05/Shift-2)
24. If $\vec{a}$ and $\vec{b}$ are unit vectors, then the greatest value of $\sqrt{3}|\vec{a} + \vec{b}| + |\vec{a} - \vec{b}|$ is (2020-09-06/Shift-1)
25. If $\vec{x}$ and $\vec{y}$ be two non-zero vectors such that $|\vec{x} + \vec{y}| = |\vec{x}|$ and $2\vec{x} + \lambda\vec{y}$ is perpendicular to $\vec{y}$, then the value of λ is (2020-09-06/Shift-2)
26. A vector $\vec{a} = \alpha\hat{i} + 2\hat{j} + \beta\hat{k}$ ($\alpha, \beta \in \mathbb{R}$) lies in the plane of the vectors, $\vec{b} = \hat{i} + \hat{j}$ and $\vec{c} = \hat{i} - \hat{j} + 4\hat{k}$. If $\vec{a}$ bisects the angle between $\vec{b}$ and $\vec{c}$, then (2020-01-07/Shift-1)
- (a) $\vec{a} \cdot \hat{i} + 3 = 0$ (b) $\vec{a} \cdot \hat{k} + 4 = 0$
(c) $\vec{a} \cdot \hat{i} + 1 = 0$ (d) $\vec{a} \cdot \hat{k} + 2 = 0$
27. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three unit vectors such that $\vec{a} + \vec{b} + \vec{c} = 0$. If $\lambda = \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$ and $\vec{d} = \vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}$, then the ordered pair $(\lambda, \vec{d})$ is equal to : (2020-01-07/Shift-2)
- (a) $\left(\frac{3}{2}, 3\vec{a} \times \vec{c} \right)$ (b) $\left(-\frac{3}{2}, 3\vec{c} \times \vec{b} \right)$
(c) $\left(-\frac{3}{2}, 3\vec{a} \times \vec{b} \right)$ (d) $\left(\frac{3}{2}, 3\vec{b} \times \vec{c} \right)$
28. If the foot of perpendicular drawn from the point $(1, 0, 3)$ on a line passing through $(\alpha, 7, 1)$ is $\left(\frac{5}{3}, \frac{7}{3}, \frac{17}{3} \right)$, then α equal to (2020-01-07/Shift-2)
29. If volume of parallelepiped whose coterminal edges are given by $\vec{u} = \hat{i} + \hat{j} + \lambda\hat{k}$, $\vec{v} = \hat{i} + \hat{j} + 3\hat{k}$ and $\vec{w} = 2\hat{i} + \hat{j} + \hat{k}$ be 1 cu. Unit. If θ be the angle between the edges $\vec{u}$ and $\vec{w}$, then, $\cos \theta$ can be: (2020-01-08/Shift-1)
- (a) $\frac{7}{6\sqrt{6}}$ (b) $\frac{5}{7}$
(c) $\frac{7}{6\sqrt{3}}$ (d) $\frac{5}{3\sqrt{3}}$



30. Let $\vec{a} = \hat{i} - 2\hat{j} + \hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$ be two vectors. If $\vec{c}$ is a vector such that $\vec{b} \times \vec{c} = \vec{b} \times \vec{a}$ and $\vec{c} \cdot \vec{a} = 0$ then $\vec{c} \cdot \vec{b}$ is equal to: (2020-01-08/Shift-2)
- (a) $\frac{1}{2}$ (b) $-\frac{3}{2}$
- (c) $-\frac{1}{2}$ (d) -1
31. If the vectors $\vec{p} = (a+1)\hat{i} + a\hat{j} + a\hat{k}$, $\vec{q} = a\hat{i} + (a+1)\hat{j} + a\hat{k}$ and $\vec{r} = a\hat{i} + a\hat{j} + (a+1)\hat{k}$ ($a \in \mathbb{R}$) are coplanar and $3(\vec{p} \cdot \vec{q})^2 - \lambda |\vec{r} \times \vec{q}|^2 = 0$, then value of λ is _____. (2020-01-09/Shift-1)
32. The projection of the line segment joining the points $(1, -1, 3)$ and $(2, -4, 11)$ on the line joining the points $(-1, 2, 3)$ and $(3, -2, 10)$ is _____. (2020-01-09/Shift-1)
33. $\vec{a}, \vec{b}$ and $\vec{c}$ be three vectors such that $|\vec{a}| = \sqrt{3}, |\vec{b}| = 5, \vec{b} \cdot \vec{c} = 10$ and the angle between $\vec{b}$ and $\vec{c}$ is $\frac{\pi}{3}$. If $\vec{a}$ is perpendicular to vector $\vec{b} \times \vec{c}$, then $|\vec{a} \times (\vec{b} \times \vec{c})|$ is equal to (2020-01-09/Shift-2)
34. Let $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{b} = \hat{i} + \hat{j}$. If $\vec{c}$ is a vector such that $\vec{a} \cdot \vec{c} = |\vec{c}|$, $|\vec{c} - \vec{a}| = 2\sqrt{2}$ and the angle between $(\vec{a} \times \vec{b})$ and $\vec{c}$ is $\frac{\pi}{6}$, then the value of $\left| \left(\vec{a} \times \vec{b} \right) \times \vec{c} \right|$ is: (2021-07-20/Shift-1)
- (a) $\frac{2}{3}$ (b) 4
- (c) 3 (d) $\frac{3}{2}$
35. Let $\vec{a}, \vec{b}, \vec{c}$ be three mutually perpendicular vectors of the same magnitude and equally inclined at an angle θ with the vector $\vec{a} + \vec{b} + \vec{c}$. Then $36\cos^2 2\theta$ is equal to _____. (2021-07-20/Shift-1)
36. For $p > 0$ a vector $\vec{v}_2 = 2\hat{i} + (p+1)\hat{j}$ is obtained by rotating the vector $\vec{v}_1 = \sqrt{3}p\hat{i} + \hat{j}$ by an angle θ about the origin in a counter clockwise direction. If $\tan \theta = \frac{(\alpha\sqrt{3}-2)}{4\sqrt{3}+3}$, then the value of α is equal to ? (2021-07-20/Shift-2)
37. In a triangle ABC, if $|\overline{BC}| = 3, |\overline{CA}| = 5$ and $|\overline{BA}| = 7$, then the projection of the vector $\overline{BA}$ on $\overline{BC}$ is equal to ? (2021-07-20/Shift-2)
- (a) $\frac{11}{2}$ (b) $\frac{13}{2}$
- (c) $\frac{19}{2}$ (d) $\frac{15}{2}$
38. Let the vectors $(2+a+b)\hat{i} + (a+2b+c)\hat{j} - (b+c)\hat{k}$ $(1+b)\hat{i} + 2b\hat{j} - b\hat{k}$ and $(2+b)\hat{i} + 2b\hat{j} + (1-b)\hat{k}$ $a, b, c \in \mathbb{R}$ be co-planar. Then which of the following is true ? (2021-07-25/Shift-1)
- (a) $2a = b + c$ (b) $2b = a + c$
- (c) $3c = a + b$ (d) $a = b + 2c$
39. Let $\vec{p} = 2\hat{i} + 3\hat{j} + \hat{k}$ and $\vec{q} = \hat{i} + 2\hat{j} + \hat{k}$ be two vectors. If a vector $\vec{r} = (\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k})$ is perpendicular to each of the vectors $(\vec{p} + \vec{q})$ and $(\vec{p} - \vec{q})$ and $|\vec{r}| = \sqrt{3}$, then $|\alpha| + |\beta| + |\gamma|$ is equal to _____. (2021-07-25/Shift-1)



40. Let $\vec{a} = \hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = -\hat{i} + 2\hat{j} + 3\hat{k}$. Then the vector product $(\vec{a} + \vec{b}) \times (\vec{a} \times ((\vec{a} - \vec{b}) \times \vec{b})) \times \vec{b}$ is equal to:

(2021-07-27/Shift-1)

- (a) $5(30\hat{i} - 5\hat{j} + 7\hat{k})$ (b) $5(34\hat{i} - 5\hat{j} + 3\hat{k})$
(c) $7(30\hat{i} - 5\hat{j} + 7\hat{k})$ (d) $7(34\hat{i} - 5\hat{j} + 3\hat{k})$

41. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b}$ and $\vec{c} = \hat{j} - \hat{k}$ be three vectors such that $\vec{a} \times \vec{b} = \vec{c}$ and $\vec{a} \cdot \vec{b} = 1$. If the length of projection vector of the vector $\vec{b}$ on the vector $\vec{a} \times \vec{c}$ is l , then the value of $3l^2$ is equal to _____.

(2021-07-27/Shift-1)

42. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three vectors such that $\vec{a} = \vec{b} \times (\vec{b} \times \vec{c})$. If magnitudes of the vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ are $\sqrt{2}$, 1 and 2 respectively and the angle between $\vec{b}$ and $\vec{c}$ is θ ($0 < \theta < \frac{\pi}{2}$), then the value of θ ($0 < \theta < \frac{\pi}{2}$) is equal to:

(2021-07-27/Shift-2)

- (a) $\frac{\sqrt{3}+1}{\sqrt{3}}$ (b) 2
(c) $\sqrt{3}+1$ (d) 1

43. Let $\vec{a} = \hat{i} - \alpha\hat{j} + \beta\hat{k}$, $\vec{b} = 3\hat{i} + \beta\hat{j} - \alpha\hat{k}$ and $\vec{c} = -\alpha\hat{i} - 2\hat{j} + \hat{k}$, where α and β are integers. If $\vec{a} \cdot \vec{b} = -1$ and $\vec{b} \cdot \vec{c} = 10$, then $(\vec{a} \times \vec{b}) \cdot \vec{c}$ is equal to _____.

(2021-07-27/Shift-2)

44. Let three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ be such that $\vec{a} \times \vec{b} = \vec{c}$, $\vec{b} \times \vec{c} = \vec{a}$ and $|\vec{a}| = 2$. Then which one of the following is not true?

(2021-07-22/Shift-2)

- (a) Projection of $\vec{a}$ on $(\vec{b} \times \vec{c})$ is 2
(b) $|\vec{3a} + \vec{b} - 2\vec{c}|^2 = 51$

$$(c) \begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix} + \begin{bmatrix} \vec{c} & \vec{a} & \vec{b} \end{bmatrix} = 8$$

$$(d) \vec{a} \times \left(\left(\frac{\vec{b} + \vec{c}}{2} \right) \times \left(\frac{\vec{b} - \vec{c}}{2} \right) \right) = \vec{0}$$

45. Let a vector $\vec{a}$ be coplanar with vectors $\vec{b} = 2\hat{i} + \hat{j} + \hat{k}$ and $\vec{c} = \hat{i} - \hat{j} + \hat{k}$. If $\vec{a}$ is perpendicular to $\vec{d} = 3\hat{i} + 2\hat{j} + 6\hat{k}$, and $|\vec{a}| = \sqrt{10}$. Then a possible value of

$$\begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix} + \begin{bmatrix} \vec{a} & \vec{d} & \vec{b} \end{bmatrix} + \begin{bmatrix} \vec{a} & \vec{c} & \vec{d} \end{bmatrix}$$
 is equal to :

(2021-07-22/Shift-2)

- (a) -40 (b) -42
(c) -29 (d) -38

46. Let a , b and c be distinct positive numbers. If the vectors $a\hat{i} + a\hat{j} + c\hat{k}$, $\hat{i} + \hat{k}$ and $c\hat{i} + c\hat{j} + b\hat{k}$ are co-planar, then c is equal to:

(2021-07-25/Shift-2)

- (a) $\sqrt{ab}$ (b) $\frac{a+b}{2}$
(c) $\frac{1}{a} + \frac{1}{b}$ (d) $\frac{2}{\frac{1}{a} + \frac{1}{b}}$

47. If $|\vec{a}| = 2$, $|\vec{b}| = 5$ and $|\vec{a} \times \vec{b}| = 8$, then $|\vec{a} \cdot \vec{b}|$ is equal to:

(2021-07-25/Shift-2)

- (a) 5 (b) 4
(c) 6 (d) 3

48. If $(\vec{a} + 3\vec{b})$ is perpendicular to $(7\vec{a} - 5\vec{b})$ and $(\vec{a} - 4\vec{b})$

is perpendicular to $(7\vec{a} - 2\vec{b})$, then the angle between

$\vec{a}$ and $\vec{b}$ (in degrees) is _____ (2021-07-25/Shift-2)

49. Let $\vec{a} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$. Let a vector $\vec{u}$ in the plane containing $\vec{a}$ and $\vec{b}$. If $\vec{u}$ is perpendicular to the vector $3\vec{i} + 2\vec{j} - \hat{k}$ and its projection on $\vec{a}$ is 19 units, then $|2\vec{u}|^2$ is equal to _____ (2021-09-01/Shift-2)



50. If the projection of the vector $\hat{i} + 2\hat{j} + \hat{k}$ on the sum of the two vectors $2\hat{i} + 4\hat{j} - 5\hat{k}$ and $-\lambda\hat{i} + 2\hat{j} + 3\hat{k}$ is 1, then λ is equal to _____. (2021-08-26/Shift-2)
51. Let $\vec{a} = \hat{i} + 5\hat{j} + \alpha\hat{k}$, $\vec{b} = \hat{i} + 3\hat{j} + \beta\hat{k}$ and $\vec{c} = -\hat{i} + 2\hat{j} - 3\hat{k}$ be three vectors such that, $|\vec{b} \times \vec{c}| = 5\sqrt{3}$ and $\vec{a}$ is perpendicular to $\vec{b}$. Then the greatest amongst the values of $|\vec{a}|^2$ is _____. (2021-08-27/Shift-1)
52. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{j} - \hat{k}$. If $\vec{c}$ is a vector such that $\vec{a} \times \vec{c} = \vec{b}$ and $\vec{a} \cdot \vec{c} = 3$, then $\vec{a} \cdot (\vec{b} \times \vec{c})$ is equal to : (2021-08-26/Shift-1)
- (a) -2 (b) 6
(c) 2 (d) -6
53. Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|2\vec{a} + 3\vec{b}| = |3\vec{a} + \vec{b}|$ and the angle between $\vec{a}$ and $\vec{b}$ is 60° . If $\frac{1}{8}\vec{a}$ is a unit vector, then $|\vec{b}|$ is equal to ? (2021-08-31/Shift-1)
- (a) 5 (b) 8
(c) 4 (d) 6
54. Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors mutually perpendicular to each other and have same magnitude. If a vector $\vec{r}$ satisfies $\vec{a} \times \{(\vec{r} - \vec{b}) \times \vec{a}\} + \vec{b} \times \{(\vec{r} - \vec{c}) \times \vec{b}\} + \vec{c} \times \{(\vec{r} - \vec{a}) \times \vec{c}\} = \vec{0}$, then $\vec{r}$ is equal to: (2021-08-31/Shift-2)
- (a) $\frac{1}{2}(\vec{a} + \vec{b} + 2\vec{c})$ (b) $\frac{1}{2}(2\vec{a} + \vec{b} - \vec{c})$
(c) $\frac{1}{3}(\vec{a} + \vec{b} + \vec{c})$ (d) $\frac{1}{2}(\vec{a} + \vec{b} + \vec{c})$
55. Let $\vec{a} = \hat{i} + 2\hat{j} - 3\hat{k}$ and $\vec{b} = 2\hat{i} - 3\hat{j} + 5\hat{k}$. If $\vec{r} \times \vec{a} = \vec{b} \times \vec{r}$, $\vec{r} \cdot (\alpha\hat{i} + 2\hat{j} + \hat{k}) = 3$ and $\vec{r} \cdot (2\hat{i} + 5\hat{j} - \alpha\hat{k}) = -1$, $\alpha \in \mathbb{R}$, then the value of $\alpha + |\vec{r}|^2$ is equal to (2021-03-16/Shift-2)
- (a) 13 (b) 15
(c) 9 (d) 11
56. Let $\vec{c}$ be a vector perpendicular to the vectors $\vec{a} = \hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$. If $\vec{c} \cdot (\hat{i} + \hat{j} + 3\hat{k}) = 8$ then the value of $\vec{c} \cdot (\vec{a} \times \vec{b})$ is equal to _____. (2021-03-16/Shift-2)
57. Let a vector $\alpha\hat{i} + \beta\hat{j}$ be obtained by rotating the vector $\sqrt{3}\hat{i} + \hat{j}$ by an angle 45° about the origin in counter clockwise direction in the first quadrant. Then the area of triangle having vertices (α, β) , $(0, \beta)$ and $(0, 0)$ is equal to (2021-03-16/Shift-1)
- (a) 1 (b) $\frac{1}{\sqrt{2}}$
(c) $2\sqrt{2}$ (d) $\frac{1}{2}$
58. Let the position vectors of two points P and Q be $3\hat{i} - \hat{j} + 2\hat{k}$ and $\hat{i} + 2\hat{j} - 4\hat{k}$, respectively. Let R and S be two points such that the direction ratios of lines PR and QS are $(4, -1, 2)$ and $(-2, 1, -2)$, respectively. Let lines PR and QS intersect at T. If the vector $\vec{TA}$ is perpendicular to both $\vec{PR}$ and $\vec{QS}$ and the length of vector $\vec{TA}$ is $\sqrt{5}$ units, then the modulus of a position vector of A is (2021-03-16/Shift-1)
- (a) $\sqrt{5}$ (b) $\sqrt{171}$
(c) $\sqrt{227}$ (d) $\sqrt{482}$
59. Let O be the origin. Let $\vec{OP} = x\hat{i} + y\hat{j} - \hat{k}$ and $\vec{OQ} = -\hat{i} + 2\hat{j} + 3x\hat{k}$, $x, y \in \mathbb{R}$, $x > 0$, be such that $|\vec{PQ}| = \sqrt{20}$ and the vector $\vec{OP}$ is perpendicular to $\vec{OQ}$. If $\vec{OR} = 3\hat{i} + z\hat{j} - 7\hat{k}$, $z \in \mathbb{R}$, is coplanar with $\vec{OP}$ and $\vec{OQ}$, then the value of $x^2 + y^2 + z^2$ is equal to : (2021-03-17/Shift-2)
- (a) 1 (b) 9
(c) 2 (d) 7
60. Let $\vec{x}$ be a vector in the plane containing vectors $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$. If the vector $\vec{x}$ is perpendicular to $(3\hat{i} + 2\hat{j} - \hat{k})$ and its projection on $\vec{a}$ is $\frac{17\sqrt{6}}{2}$, then the value of $|\vec{x}|^2$ is equal to (2021-03-17/Shift-2)



61. Let $\vec{a} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ and $\vec{b} = 7\hat{i} + \hat{j} - 6\hat{k}$.
If $\vec{r} \times \vec{a} = \vec{r} \times \vec{b}$, $\vec{r} \cdot (\hat{i} + 2\hat{j} + \hat{k}) = -3$, then $\vec{r} \cdot (2\hat{i} - 3\hat{j} + \hat{k})$ is equal to : **(2021-03-17/Shift-1)**
(a) 12 (b) 13
(c) 10 (d) 8
62. Let $\vec{a}$ and $\vec{b}$ be two non-zero vectors perpendicular to each other and $|\vec{a}| = |\vec{b}|$. If $|\vec{a} \times \vec{b}| = |\vec{a}|$, then the angle between the vectors $(\vec{a} + \vec{b} + (\vec{a} \times \vec{b}))$ and $\vec{a}$ is equal to : **(2021-03-18/Shift-2)**
(a) $\sin^{-1}\left(\frac{1}{\sqrt{6}}\right)$ (b) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$
(c) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (d) $\cos^{-1}\left(\frac{1}{\sqrt{2}}\right)$
63. A vector $\vec{a}$ has components $3p$ and 1 with respect to a rectangular cartesian system. This system is rotated through a certain angle about the origin in the counter clockwise sense. If, with respect to new system, $\vec{a}$ has components $p + 1$ and $\sqrt{10}$, then a value of p is equal to : **(2021-03-18/Shift-1)**
(a) 1 (b) $\frac{4}{5}$
(c) $-\frac{5}{4}$ (d) -1
64. Let three vectors $\vec{a}, \vec{b}$ and $\vec{c}$ be such that $\vec{c}$ is coplanar with $\vec{a}$ and $\vec{b}$, $\vec{a} \cdot \vec{c} = 7$ and $\vec{b}$ is perpendicular to $\vec{c}$ where $\vec{a} = -\hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} + \hat{k}$, then the value of $2|\vec{a} + \vec{b} + \vec{c}|^2$ is _____. **(2021-02-24/Shift-1)**
65. Let $\vec{a} = \hat{i} + \alpha\hat{j} + 3\hat{k}$ and $\vec{b} = 3\hat{i} - \alpha\hat{j} + \hat{k}$. If the area of the parallelogram whose adjacent sides are represented by the vectors $\vec{a}$ and $\vec{b}$ is $8\sqrt{3}$ square units, then $\vec{a} \cdot \vec{b}$ is equal to: **(2021-02-25/Shift-2)**
66. Let $\vec{a} = \hat{i} + 2\hat{j} - \hat{k}$, $\vec{b} = \hat{i} - \hat{j}$ and $\vec{c} = \hat{i} - \hat{j} - \hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{a} = \vec{c} \times \vec{a}$ and $\vec{r} \cdot \vec{b} = 0$, then $\vec{r} \cdot \vec{a}$ is equal to _____. **(2021-02-25/Shift-1)**

67. If vectors $\vec{a}_1 = x\hat{i} - \hat{j} + \hat{k}$ and $\vec{a}_2 = \hat{i} + y\hat{j} + z\hat{k}$ are collinear, then a possible unit vector parallel to the vector $x\hat{i} + y\hat{j} + z\hat{k}$ is **(2021-02-26/Shift-2)**
(a) $\frac{1}{\sqrt{3}}(\hat{i} + \hat{j} - \hat{k})$ (b) $\frac{1}{\sqrt{2}}(\hat{i} - \hat{j})$
(c) $\frac{1}{\sqrt{3}}(\hat{i} - \hat{j} + \hat{k})$ (d) $\frac{1}{\sqrt{2}}(-\hat{j} + \hat{k})$
68. If $\vec{a}$ and $\vec{b}$ are perpendicular, then $\vec{a} \times (\vec{a} \times (\vec{a} \times (\vec{a} \times \vec{b})))$ is equal to **(2021-02-26/Shift-1)**
(a) $\vec{0}$ (b) $\frac{1}{2}|\vec{a}|^4 \vec{b}$
(c) $\vec{a} \times \vec{b}$ (d) $|\vec{a}|^4 \vec{b}$

3-DIMENSIONAL GEOMETRY

69. The shortest distance between the z-axis and the line $x + y + 2z - 3 = 0 = 2x + 3y + 4z - 4$, is **(2015/Online Set-2)**
(a) 1 (b) 2
(c) 4 (d) 3
70. A plane containing the point $(3, 2, 0)$ and the line $\frac{x-1}{1} = \frac{y-2}{5} = \frac{z-3}{4}$ also contains the point : **(2015/Online Set-2)**
(a) $(0, 3, 1)$ (b) $(0, 7, -10)$
(c) $(0, -3, 1)$ (d) $(0, 7, 10)$
71. If the line, $\frac{x-3}{2} = \frac{y+2}{-1} = \frac{z+4}{3}$ lies in the plane, $lx + my - z = 9$, then $l^2 + m^2$ is equal to : **(2016)**
(a) 18 (b) 5
(c) 2 (d) 26
72. The distance of the point $(1, -5, 9)$ from the plane $x - y + z = 5$ measured along the line $x = y = z$ is **(2016)**
(a) $10\sqrt{3}$ (b) $\frac{10}{\sqrt{3}}$
(c) $\frac{20}{3}$ (d) $3\sqrt{10}$



73. The shortest distance between the lines lies $\frac{x}{2} = \frac{y}{2} = \frac{z}{1}$ and $\frac{x+2}{-1} = \frac{y-4}{8} = \frac{z-5}{4}$ in the interval :
(2016/Online Set-1)
- (a) [0, 1) (b) [1, 2)
(c) (2, 3] (d) (3, 4]
74. The distance of the point (1, -2, 4) from the plane passing through the point (1, 2, 2) and perpendicular to the planes $x-y+2z=3$ and $2x-2y+z+12=0$, is : (2016/Online Set-1)
- (a) $2\sqrt{2}$ (b) 2
(c) $\sqrt{2}$ (d) $\frac{1}{\sqrt{2}}$
75. The number of distinct real values of λ for which the lines $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z+3}{\lambda^2}$ and $\frac{x-3}{1} = \frac{y-2}{\lambda^2} = \frac{z-1}{2}$ are coplanar is : (2016/Online Set-2)
- (a) 4 (b) 1
(c) 2 (d) 3
76. If the image of the point P(1, -2, 3) in the plane, $2x + 3y - 4z + 22 = 0$ measured parallel to the line, $\frac{x}{1} = \frac{y}{4} = \frac{z}{5}$ is Q then PQ is equal to : (2017)
- (a) $3\sqrt{5}$ (b) $2\sqrt{42}$
(c) $\sqrt{42}$ (d) $6\sqrt{5}$
77. The coordinates of the foot of the perpendicular from the point (1, -2, 1) on the plane containing the lines, $\frac{x+1}{6} = \frac{y-1}{7} = \frac{z-3}{8}$ and $\frac{x-1}{3} = \frac{y-2}{5} = \frac{z-3}{7}$ is : (2017/Online Set-1)
- (a) (2, -4, 2) (b) (-1, 2, -1)
(c) (0, 0, 0) (d) (1, 1, 1)
78. The line of intersection of the planes $\vec{r} \cdot (3\hat{i} - \hat{j} + \hat{k})$ and $\vec{r} \cdot (\hat{i} + 4\hat{j} - 2\hat{k}) = 2$, is: (2017/Online Set-1)
- (a) $\frac{x-\frac{4}{7}}{-2} = \frac{y}{7} = \frac{z-\frac{5}{7}}{13}$
(b) $\frac{x-\frac{4}{7}}{2} = \frac{y}{-7} = \frac{z+\frac{5}{7}}{13}$
(c) $\frac{x-\frac{6}{13}}{2} = \frac{y+\frac{5}{13}}{-7} = \frac{z}{-13}$
(d) $\frac{x-\frac{6}{13}}{2} = \frac{y-\frac{5}{13}}{-7} = \frac{z}{-13}$
79. If the line, $\frac{x-3}{1} = \frac{y+2}{-1} = \frac{z+\lambda}{-2}$ lies in the plane, $2x - 4y + 3z = 2$, then the shortest distance between this line and the line, $\frac{x-1}{12} = \frac{y}{9} = \frac{z}{4}$ is : (2017/Online Set-2)
- (a) 2 (b) 1
(c) 0 (d) 3
80. If a variable plane, at a distance of 3 units from the origin, intersects the coordinate axes at A, B and C, then the locus of the centroid of $\triangle ABC$ is : (2017/Online Set-2)
- (a) $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = 1$ (b) $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = 3$
(c) $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{1}{9}$ (d) $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = 9$



81. If L_1 is the line of intersection of the planes $2x - 2y + 3z - 2 = 0$, $x - y + z + 1 = 0$ and L_2 is the line of intersection of the planes $x + 2y - z - 3 = 0$, $3x - y + 2z - 1 = 0$, then the distance of the origin from the plane, containing the lines L_1 and L_2 , is : (2018)
- (a) $\frac{1}{\sqrt{2}}$ (b) $\frac{1}{4\sqrt{2}}$
(c) $\frac{1}{3\sqrt{2}}$ (d) $\frac{1}{2\sqrt{2}}$
82. A variable plane passes through a fixed point (3, 2, 1) and meets x, y and z axes at A, B and C respectively. A plane is drawn parallel to yz-plane through A, a second plane is drawn parallel to zx-plane through B and a third plane is drawn parallel to xy-plane through C. Then the locus of the point of intersection of these three planes, is: (2018/Online Set-1)
- (a) $\frac{x}{3} + \frac{y}{2} + \frac{z}{1} = 1$ (b) $x + y + z = 6$
(c) $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \frac{11}{6}$ (d) $\frac{3}{x} + \frac{2}{y} + \frac{1}{z} = 1$
83. An angle between the plane, $x + y + z = 5$ and the line of intersection of the planes, $3x + 4y + z - 1 = 0$ and $5x + 8y + 2z + 14 = 0$, is: (2018/Online Set-1)
- (a) $\sin^{-1}\left(\sqrt{\frac{3}{17}}\right)$ (b) $\cos^{-1}\left(\sqrt{\frac{3}{17}}\right)$
(c) $\cos^{-1}\left(\frac{3}{\sqrt{17}}\right)$ (d) $\sin^{-1}\left(\frac{3}{\sqrt{17}}\right)$
84. An angle between the lines whose direction cosines are given by the equations, $l + 3m + 5n = 0$ and $5lm - 2mn + 6nl = 0$, is : (2018/Online Set-2)
- (a) $\cos^{-1}\left(\frac{1}{3}\right)$ (b) $\cos^{-1}\left(\frac{1}{4}\right)$
(c) $\cos^{-1}\left(-\frac{1}{6}\right)$ (d) $\cos^{-1}\left(\frac{1}{8}\right)$
85. A plane bisects the line segment joining the points (1, 2, 3) and (-3, 4, 5) at right angles. Then this plane also passes through the point : (2018/Online Set-2)
- (a) (-3, 2, 1) (b) (3, 2, 1)
(c) (-1, 2, 3) (d) (1, 2, -3)
86. The sum of the intercepts on the coordinate axes of the plane passing through the point (-2, -2, 2) and containing the line joining the points (1, -1, 2) and (1, 1, 1), is : (2018/Online Set-3)
- (a) 4 (b) -4
(c) -8 (d) 12
87. If the angle between the lines, $\frac{x}{2} = \frac{y}{2} = \frac{z}{1}$ and $\frac{5-x}{-2} = \frac{7y-14}{p} = \frac{z-3}{4}$ is $\cos^{-1}\left(\frac{2}{3}\right)$, then p is equal to : (2018/Online Set-3)
- (a) $\frac{7}{2}$ (b) $\frac{2}{7}$
(c) $-\frac{7}{4}$ (d) $-\frac{4}{7}$
88. The equation of a plane containing the line of intersection of the planes $2x - y - 4 = 0$ and $y + 2z - 4 = 0$ and passing through the point (1, 1, 0) is : (2019-04-08/Shift-1)
- (a) $x - 3y - 2z = -2$ (b) $2x - z = 2$
(c) $x - y - z = 0$ (d) $x + 3y + z = 4$
89. The length of the perpendicular from the point (2, -1, 4) on the straight line, $\frac{x+3}{10} = \frac{y-2}{-7} = \frac{z}{1}$ is: (2019-04-08Shift-1)
- (a) greater than 3 but less than 4
(b) less than 2
(c) greater than 2 but less than 3
(d) greater than 4



90. The vector equation of the plane through the line of intersection of the planes $x + y + z = 1$ and $2x + 3y + 4z = 5$ which is perpendicular to the plane $x - y + z = 0$ is : (2019-04-08/Shift-2)
- (a) $\vec{r} \times (\hat{i} - \hat{k}) + 2 = 0$ (b) $\vec{r} \cdot (\hat{i} - \hat{k}) - 2 = 0$
- (c) $\vec{r} \times (\hat{i} + \hat{k}) + 2 = 0$ (d) $\vec{r} \cdot (\hat{i} - \hat{k}) + 2 = 0$
91. If a point $R(4, y, z)$ lies on the line segment joining the points $P(2, -3, 4)$ and $Q(8, 0, 10)$, then distance of R from the origin is: (2019-04-08/Shift-2)
- (a) $2\sqrt{14}$ (b) $2\sqrt{21}$
- (c) 6 (d) $\sqrt{53}$
92. If the line, $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-2}{4}$ meets the plane, $x + 2y + 3z = 15$ at a point P , then the distance of P from the origin is: (2019-04-09/Shift-1)
- (a) $\frac{\sqrt{5}}{2}$ (b) $2\sqrt{5}$
- (c) $\frac{9}{2}$ (d) $\frac{7}{2}$
93. A plane passing through the points $(0, -1, 0)$ and $(0, 0, 1)$ and making an angle $\frac{\pi}{4}$ with the plane $y - z + 5 = 0$, also passes through the point: (2019-04-09/Shift-1)
- (a) $(-\sqrt{2}, 1, -4)$ (b) $(\sqrt{2}, -1, 4)$
- (c) $(-\sqrt{2}, -1, -4)$ (d) $(\sqrt{2}, 1, 4)$
94. The vertices B and C of a ΔABC lie on the line, $\frac{x+2}{3} = \frac{y-1}{0} = \frac{z}{4}$ such that $BC = 5$ units. Then the area (in sq. units) of this triangle, given that the point $A(1, -1, 2)$, is: (2019-04-09/Shift-2)
- (a) $5\sqrt{17}$ (b) $2\sqrt{34}$
- (c) 6 (d) $\sqrt{34}$
95. Let P be the plane, which contains the line of intersection of the plane, $x + y + z - 6 = 0$ and $2x + 3y + z + 5 = 0$ and it is perpendicular to the xy -plane. Then the distance of the point $(0, 0, 256)$ from P is equal to: (2019-04-09/Shift-2)
- (a) $\frac{17}{\sqrt{5}}$ (b) $63\sqrt{5}$
- (c) $205\sqrt{5}$ (d) $\frac{11}{\sqrt{5}}$
96. If a unit vector $\vec{a}$ makes angles $\frac{\pi}{3}$ with $\hat{i}$, $\frac{\pi}{4}$ with $\hat{j}$ and $\theta \in (0, \pi)$ with $\hat{k}$ then a value of θ is: (2019-04-09/Shift-2)
- (a) $\frac{5\pi}{6}$ (b) $\frac{\pi}{4}$
- (c) $\frac{5\pi}{12}$ (d) $\frac{2\pi}{3}$
97. If $Q(0, -1, -3)$ is the image of the point P in the plane $3x - y + 4z = 2$ and R is the point $(3, -1, -2)$, then the area (in sq. units) of ΔPQR is : (2019-04-10/Shift-1)
- (a) $2\sqrt{13}$ (b) $\frac{\sqrt{91}}{4}$
- (c) $\frac{\sqrt{91}}{2}$ (d) $\frac{\sqrt{65}}{2}$
98. If the length of the perpendicular from the point $(\beta, 0, \beta)$ ($\beta \neq 0$) to the line, $\frac{x}{1} = \frac{y-1}{0} = \frac{z+1}{-1}$ is $\sqrt{\frac{3}{2}}$, then β is equal to: (2019-04-10/Shift-1)
- (a) 1 (b) 2
- (c) -1 (d) -2



99. The distance of the point having position vector $-\hat{i} + 2\hat{j} + 6\hat{k}$ from the straight line passing through the point $(2, 3, -4)$ and parallel to the vector, $6\hat{i} + 3\hat{j} - 4\hat{k}$ is:
- (2019-04-10/Shift-2)
- (a) 7 (b) $4\sqrt{3}$
(c) 6 (d) $2\sqrt{13}$
100. A perpendicular is drawn from a point on the line $\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z}{1}$ to the plane $x + y + z = 3$ such that the foot the perpendicular Q also lies on the plane $x - y + z = 3$. Then the co-ordinates of Q are:
- (2019-04-10/Shift-2)
- (a) (1, 0, 2) (b) (2, 0, 1)
(c) (-1, 0, 4) (d) (4, 0, -1)
101. If the plane $2x - y + 2z + 3 = 0$ has the distances $\frac{1}{3}$ and $\frac{2}{3}$ units from the planes $4x - 2y + 4z + \lambda = 0$ and $2x - y + 2z + \mu = 0$ respectively, then the maximum value of $\lambda + \mu$ is equal to :
- (2019-04-10/Shift-2)
- (a) 9 (b) 15
(c) 5 (d) 13
102. If the line $\frac{x-2}{3} = \frac{y+1}{2} = \frac{z-1}{-1}$ intersects the plane $2x + 3y - z + 13 = 0$ at a point P and the plane $3x + y + 4z = 16$ at a point Q, then PQ is equal to
- (2019-04-12/Shift-1)
- (a) 14 (b) $\sqrt{14}$
(c) $2\sqrt{7}$ (d) $2\sqrt{14}$
103. A plane which bisects the angle between the two given planes $2x - y + 2z - 4 = 0$ and $x + 2y + 2z - 2 = 0$ passes through the point:
- (2019-04-12/Shift-2)
- (a) (1, -4, 1) (b) (1, 4, -1)
(c) (2, 4, 1) (d) (2, -4, 1)
104. The length of the perpendicular drawn from the point $(2, 1, 4)$ to the plane containing the lines $\vec{r} = (\hat{i} + \hat{j}) + \lambda(\hat{i} + 2\hat{j} - \hat{k})$ and $\vec{r} = (\hat{i} + \hat{j}) + \mu(-\hat{i} + \hat{j} - 2\hat{k})$ is _____.
- (2019-04-12/Shift-2)
- (a) 3 (b) $\frac{1}{3}$
(c) $\sqrt{3}$ (d) $\frac{1}{\sqrt{3}}$
105. The equation of the line passing through $(-4, 3, 1)$ parallel to the plane $x + 2y - z - 5 = 0$ and intersecting the line $\frac{x+1}{-3} = \frac{y-3}{2} = \frac{z-2}{-1}$ is:
- (2019-01-09/Shift-1)
- (a) $\frac{x-4}{2} = \frac{y+3}{1} = \frac{z+1}{4}$ (b) $\frac{x+4}{1} = \frac{y-3}{1} = \frac{z-1}{3}$
(c) $\frac{x+4}{3} = \frac{y-3}{-1} = \frac{z-1}{1}$ (d) $\frac{x+4}{-1} = \frac{y-3}{1} = \frac{z-1}{1}$
106. The equation of the plane containing the straight line $\frac{x}{2} = \frac{y}{3} = \frac{z}{4}$ and perpendicular to the plane containing the straight lines $\frac{x}{3} = \frac{y}{4} = \frac{z}{2}$ and $\frac{x}{4} = \frac{y}{2} = \frac{z}{3}$ is:
- (2019-01-09/Shift-2)
- (a) $x - 2y + z = 0$ (b) $3x + 2y - 3z = 0$
(c) $x + 2y - 2z = 0$ (d) $5x + 2y - 4z = 0$
107. If the lines $x = ay + b$, $z = cy + d$ and $x = a'z + b'$, $y = c'z + d'$ are perpendicular, then:
- (2019-01-09/Shift-2)
- (a) $ab' + bc' + 1 = 0$ (b) $cc' + a + a' = 0$
(c) $bb' + cc' + 1 = 0$ (d) $aa' + c + c' = 0$



108. The plane passing through the point (4, -1, 2) and parallel to the lines $\frac{x+2}{3} = \frac{y-2}{-1} = \frac{z+1}{2}$ and $\frac{x-2}{1} = \frac{y-3}{2} = \frac{z-4}{3}$ also passes through the point:
- (2019-01-10/Shift-1)
- (a) (1, 1, -1) (b) (1, 1, 1)
(c) (-1, -1, -1) (d) (-1, -1, 1)
109. Let A be a point on the line $\vec{r} = (1-3\mu)\hat{i} + (\mu-1)\hat{j} + (2+5\mu)\hat{k}$ and B(3, 2, 6) be a point in the space. Then the value of μ for which the vector $\overrightarrow{AB}$ is parallel to the plane $x-4y+3z=1$ is
- (2019-01-10/Shift-1)
- (a) $\frac{1}{4}$ (b) $\frac{1}{8}$
(c) $\frac{1}{2}$ (d) $-\frac{1}{4}$
110. The plane which bisects the line segment joining the points (-3, -3, 4) and (3, 7, 6) at right angles, passes through which one of the following points?
- (2019-01-10/Shift-2)
- (a) (-2, 3, 5) (b) (4, -1, 7)
(c) (2, 1, 3) (d) (4, 1, -2)
111. On which of the following lines lies the point of intersection of the line $\frac{x-4}{2} = \frac{y-5}{2} = \frac{z-3}{1}$ and the plane, $x+y+z=2$?
- (2019-01-10/Shift-2)
- (a) $\frac{x+3}{3} = \frac{4-y}{3} = \frac{z+1}{-2}$ (b) $\frac{x-4}{1} = \frac{y-5}{1} = \frac{z-5}{-1}$
(c) $\frac{x-1}{1} = \frac{y-3}{2} = \frac{z+4}{-5}$ (d) $\frac{x-2}{2} = \frac{y-3}{2} = \frac{z+3}{3}$
112. The plane containing the line $\frac{x-3}{2} = \frac{y+2}{-1} = \frac{z-1}{3}$ and also containing its projection on the plane $2x+3y-z=5$, contains which one of the following points?
- (2019-01-11/Shift-1)
- (a) (2, 2, 0) (b) (-2, 2, 2)
(c) (0, -2, 2) (d) (2, 0, -2)
113. Two lines $\frac{x-3}{1} = \frac{y+1}{3} = \frac{z-6}{-1}$ and $\frac{x+5}{7} = \frac{y-2}{-6} = \frac{z-3}{4}$ intersect at the point R. The reflection of R in the xy-plane has coordinates :
- (2019-01-11/Shift-2)
- (a) (2, -4, -7) (b) (2, -4, 7)
(c) (2, 4, 7) (d) (-2, 4, 7)
114. If the point (2, α , β) lies on the plane which passes through the points (3, 4, 2) and (7, 0, 6) and is perpendicular to the plane $2x-5y=15$, then $2\alpha-3\beta$ is equal to :
- (2019-01-11/Shift-2)
- (a) 12 (b) 7
(c) 5 (d) 17
115. A tetrahedron has vertices P(1, 2, 1), Q(2, 1, 3), R(-1, 1, 2) and O(0, 0, 0). The angle between the faces OPQ and PQR is:
- (2019-01-12/Shift-1)
- (a) $\cos^{-1}\left(\frac{-17}{31}\right)$ (b) $\cos^{-1}\left(\frac{-19}{35}\right)$
(c) $\cos^{-1}\left(\frac{-9}{35}\right)$ (d) $\cos^{-1}\left(\frac{-7}{31}\right)$
116. The perpendicular distance from the origin to the plane containing the two lines, $\frac{x+2}{3} = \frac{y-2}{5} = \frac{z+5}{7}$ and $\frac{x-1}{1} = \frac{y-4}{4} = \frac{z+4}{7}$, is
- (2019-01-12/Shift-1)
- (a) $11\sqrt{6}$ (b) $\frac{11}{\sqrt{6}}$
(c) 11 (d) $6\sqrt{11}$
117. If an angle between the line, $\frac{x+1}{2} = \frac{y-2}{1} = \frac{z-3}{-2}$ and the plane, $x-2y-Kz=3$ is $\cos^{-1}\left(\frac{2\sqrt{2}}{3}\right)$ then a value of K is
- (2019-01-12/Shift-2)
- (a) $\sqrt{\frac{5}{3}}$ (b) $\sqrt{\frac{3}{5}}$
(c) $-\frac{3}{5}$ (d) $-\frac{5}{3}$



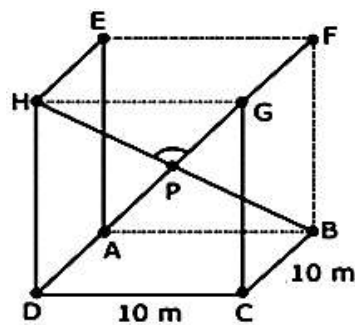
118. Let S be the set of all real values of λ such that a plane passing through the points $(-\lambda^2, 1, 1)$, $(1, -\lambda^2, 1)$ and $(1, 1, -\lambda^2)$ also passes through the point $(-1, -1, 1)$. Then S is equal to (2019-01-12/Shift-2)
- (a) $\{\sqrt{3}\}$ (b) $\{\sqrt{3}, -\sqrt{3}\}$
 (c) $\{1, -1\}$ (d) $\{3, -3\}$
119. The plane passing through the points $(1, 2, 1)$, $(2, 1, 2)$ and parallel to the line, $2x = 3y$, $z = 1$ also passes through the point : (2020-09-02/Shift-1)
- (a) $(0, -6, 2)$ (b) $(0, 6, -2)$
 (c) $(-2, 0, 1)$ (d) $(2, 0, -1)$
120. A plane passing through the point $(3, 1, 1)$ contains two lines whose direction ratios are $1, -2, 2$ and $2, 3, -1$ respectively. If this plane also passes through the point $(\alpha, -3, 5)$, then α is equal to : (2020-09-02/Shift-2)
- (a) -5 (b) 10
 (c) 5 (d) -10
121. The foot of the perpendicular drawn from the point $(4, 2, 3)$ to the line joining the points $(1, -2, 3)$ and $(1, 1, 0)$ lies on the plane : (2020-09-03/Shift-1)
- (a) $x - y - 2z = 1$ (b) $x - 2y + z = 1$
 (c) $2x + y - z = 1$ (d) $x + 2y - z = 1$
122. The lines $\vec{r} = (\hat{i} - \hat{j}) + l(2\hat{i} + \hat{k})$ and $\vec{r} = (2\hat{i} - \hat{j}) + m(\hat{i} + \hat{j} - \hat{k})$ (2020-09-03/Shift-1)
- (a) do not intersect for any values of l and m
 (b) intersect when $l = 1$ and $m = 2$
 (c) intersect when $l = 2$ and $m = \frac{1}{2}$
 (d) intersect for all values of l and m
123. The plane which bisects the line joining the points $(4, -2, 3)$ and $(2, 4, -1)$ at right angles also passes through the point : (2020-09-03/Shift-2)
- (a) $(0, -1, 1)$ (b) $(4, 0, 1)$
 (c) $(4, 0, -1)$ (d) $(0, 1, -1)$
124. Let a plane P contain two lines $\vec{r} = \hat{i} + \lambda(\hat{i} + \hat{j})$, $\lambda \in R$ and $\vec{r} = -\hat{j} + \mu(\hat{j} - \hat{k})$, $\mu \in R$. If $Q(\alpha, \beta, \gamma)$ is the foot of the perpendicular drawn from the point $M(1, 0, 1)$ to P , then $3(\alpha + \beta + \gamma)$ equals (2020-09-03/Shift-2)
125. If the equation of a plane P , passing through the intersection of the planes, $x + 4y - z + 7 = 0$ and $3x + y + 5z = 8$ is $ax + by + 6z = 15$ for some $a, b \in R$, then the distance of the point $(3, 2, -1)$ from the plane P is units. (2020-09-04/Shift-1)
126. The distance of the point $(1, -2, 3)$ from the plane $x - y + z = 5$ measured parallel to the line $\frac{x}{2} = \frac{y}{3} = \frac{z}{-6}$ is: (2020-09-04/Shift-2)
- (a) $\frac{1}{7}$ (b) 7
 (c) $\frac{7}{5}$ (d) 1
127. If (a, b, c) is the image of the point $(1, 2, -3)$ in the line, $\frac{x+1}{2} = \frac{y-3}{-2} = \frac{z}{-1}$, then $a+b+c$ is (2020-09-05/Shift-1)
- (a) 2 (b) 3
 (c) -1 (d) 1
128. If for some $\alpha \in R$, the lines $L_1 : \frac{x+1}{2} = \frac{y-2}{-1} = \frac{z-1}{1}$ and $L_2 : \frac{x+2}{\alpha} = \frac{y+1}{5-\alpha} = \frac{z+1}{1}$ are coplanar, then the line L_2 passes through the point: (2020-09-05/Shift-2)
- (a) $(2, -10, -2)$ (b) $(10, -2, -2)$
 (c) $(10, 2, 2)$ (d) $(-2, 10, 2)$
129. The shortest distance between the lines $\frac{x-1}{0} = \frac{y+1}{-1} = \frac{z}{1}$ and $x + y + z + 1 = 0$, $2x - y + z + 3 = 0$ is: (2020-09-06/Shift-1)
- (a) 1 (b) $\frac{1}{\sqrt{2}}$
 (c) $\frac{1}{\sqrt{3}}$ (d) $\frac{1}{2}$



- 130.** A plane P meets the coordinate axes at A , B and C respectively. The centroid of $\triangle ABC$ is given to be $(1, 1, 2)$. Then the equation of the line through this centroid and perpendicular to the plane P is:
(2020-09-06/Shift-2)
- (a) $\frac{x-1}{2} = \frac{y-1}{1} = \frac{z-2}{1}$ (b) $\frac{x-1}{2} = \frac{y-1}{2} = \frac{z-2}{1}$
(c) $\frac{x-1}{1} = \frac{y-1}{2} = \frac{z-2}{2}$ (d) $\frac{x-1}{1} = \frac{y-1}{1} = \frac{z-2}{2}$
- 131.** Let P be a plane passing through the points $(2, 1, 0)$, $(4, 1, 1)$ and $(5, 0, 1)$ and R be any point $(2, 1, 6)$. Then the image of R in the plane P is:
(2020-01-07/Shift-1)
- (a) $(6, 5, 2)$ (b) $(6, 5, -2)$
(c) $(4, 3, 2)$ (d) $(3, 4, -2)$
- 132.** The shortest distance between the lines $\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1}$ and $\frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4}$ is:
(2020-01-08/Shift-1)
- (a) $2\sqrt{30}$ (b) $\frac{7}{2}\sqrt{300}$
(c) 3 (d) $3\sqrt{30}$
- 133.** The mirror image of the point $(1, 2, 3)$ in a plane is $\left(-\frac{7}{3}, -\frac{4}{3}, -\frac{1}{3}\right)$. Which of the following points lies on this plane?
(2020-01-08/Shift-2)
- (a) $(1, -1, 1)$ (b) $(-1, -1, 1)$
(c) $(1, 1, 1)$ (d) $(-1, -1, -1)$
- 134.** If the distance between the plane, $23x - 10y - 2z + 48 = 0$ and the plane containing the lines $\frac{x+1}{2} = \frac{y-3}{4} = \frac{z+1}{3}$ and $\frac{x+3}{2} = \frac{y+2}{6} = \frac{z-1}{\lambda}$, $(\lambda \in \mathbb{R})$, is equal to $\frac{k}{\sqrt{633}}$, then k is equal to _____
(2020-01-09/Shift-2)
- 135.** Let P be a plane passing through the points $(1, 0, 1)$, $(1, -2, 1)$ and $(0, 1, -2)$. Let a vector $\vec{a} = \alpha \hat{i} + \beta \hat{j} + \gamma \hat{k}$ be such that $\vec{a}$ is parallel to the plane P , perpendicular to $\left(\hat{i} + 2\hat{j} + 3\hat{k}\right)$ and $\vec{a} \cdot \left(\hat{i} + \hat{j} + 2\hat{k}\right) = 2$, then $(\alpha - \beta + \gamma)^2$ equals _____.
(2021-07-20/Shift-1)
- 136.** If the shortest distance between the lines $\vec{r}_1 = \alpha \hat{i} + 2\hat{j} + 2\hat{k} + \lambda \left(\hat{i} - 2\hat{j} + 2\hat{k}\right)$, $\lambda \in \mathbb{R}$, $\alpha > 0$ and $\vec{r}_2 = -4\hat{i} - \hat{k} + \mu \left(3\hat{i} - 2\hat{j} - 2\hat{k}\right)$, $\mu \in \mathbb{R}$ is 9, then α is equal to _____.
(2021-07-20/Shift-1)
- 137.** The lines $x = ay - 1 = z - 2$ and $x = 3y - 2 = bz - 2$, $(ab \neq 0)$ are coplanar, if?
(2021-07-20/Shift-2)
- (a) $b = 1, a \in \mathbb{R} - \{0\}$ (b) $a = 2, b = 3$
(c) $a = 2, b = 2$ (d) $a = 1, b \in \mathbb{R} - \{0\}$
- 138.** Consider the line L given by the equation $\frac{x-3}{2} = \frac{y-1}{1} = \frac{z-2}{1}$. Let Q be the mirror image of the point $(2, 3, -1)$ with respect to L . Let a plane P be such that it passes through Q , and the line L is perpendicular to P . Then which of the following points is on the plane P ?
(2021-07-20/Shift-2)
- (a) $(1, 2, 2)$ (b) $(-1, 1, 2)$
(c) $(1, 1, 1)$ (d) $(1, 1, 2)$
- 139.** Let the foot of perpendicular from a point $P(1, 2, -1)$ to the straight line $L = \frac{x}{1} = \frac{y}{0} = \frac{z}{-1}$ be N . Let a line be drawn from P parallel to the plane $x + y + 2z = 0$ which meets L at point Q . If α is the acute angle between the lines PN and PQ , then $\cos \alpha$ is equal to _____.
(2021-07-25/Shift-1)
- (a) $\frac{1}{2\sqrt{3}}$ (b) $\frac{1}{\sqrt{5}}$
(c) $\frac{\sqrt{3}}{2}$ (d) $\frac{1}{\sqrt{3}}$
- 140.** Let the plane passing through the point $(-1, 0, -2)$ and perpendicular to each of the planes $2x + y - z = 2$ and $x - y - z = 3$ be $ax + by + cz + 8 = 0$, then the value of $a + b + c$ is equal to:
(2021-07-27/Shift-1)
- (a) 8 (b) 4
(c) 3 (d) 5
- 141.** Let a plane P pass through the point $(3, 7, -7)$ and contain the line, $\frac{x-2}{-3} = \frac{y-3}{2} = \frac{z+2}{1}$. If distance of the plane P from the origin is d , then d^2 is equal to _____.
(2021-07-27/Shift-1)



142. For real numbers α and $\beta \neq 0$, if the point of intersection of the straight lines $\frac{x-\alpha}{1} = \frac{y-1}{2} = \frac{z-1}{3}$ and $\frac{x-4}{\beta} = \frac{y-6}{3} = \frac{z-7}{3}$, lies on the plane $x+2y-z=8$, then $\alpha-\beta$ is equal to: (2021-07-27/Shift-2)
- (a) 5 (b) 3
(c) 7 (d) 9
143. The distance of the point $P(3, 4, 4)$ from the point of intersection of the line joining the points $Q(3, -4, -5)$ and $R(2, -3, 1)$ and the plane $2x+y+z=7$, is equal to _____. (2021-07-27/Shift-2)
144. Let L be the line of intersection of planes $\vec{r} \cdot (\hat{i} - \hat{j} + 2\hat{k}) = 2$ and $\vec{r} \cdot (2\hat{i} + \hat{j} - \hat{k}) = 2$. If $P(\alpha, \beta, \gamma)$ is the foot of perpendicular on L from the point $(1, 2, 0)$, then the value of $35(\alpha + \beta + \gamma)$ is equal to : (2021-07-22/Shift-2)
- (a) 134 (b) 119
(c) 143 (d) 101
145. If the shortest distance between the straight lines $3(x-1) = 6(y-2) = 2(z-1)$ and $4(x-2) = 2(y-\lambda) = (z-3)$, $\lambda \in \mathbb{R}$ is $\frac{1}{\sqrt{38}}$, then the integral value of λ is equal to : (2021-07-22/Shift-2)
- (a) -1 (b) 2
(c) 3 (d) 5
146. If the lines $\frac{x-k}{1} = \frac{y-2}{2} = \frac{z-3}{3}$ and $\frac{x+1}{3} = \frac{y+2}{2} = \frac{z+3}{1}$ are co-planar, then the value of k is _____. (2021-07-25/Shift-2)
147. The distance of line $3y-2z-1=0=3x-z+4$ from the point $(2, -1, 6)$ is (2021-09-01/Shift-2)
- (a) $2\sqrt{6}$ (b) $4\sqrt{2}$
(c) $2\sqrt{5}$ (d) $\sqrt{26}$
148. Let the acute angle bisector of the two planes $x-2y-2z+1=0$ and $2x-3y-6z+1=0$ be the plane P . Then which of the following points lies on P ? (2021-09-01/Shift-2)
- (a) $(4, 0, -2)$ (b) $(-2, 0, -\frac{1}{2})$
(c) $(3, 1, -\frac{1}{2})$ (d) $(0, 2, -4)$
149. Let Q be the foot of the perpendicular from the point $P(7, -2, 13)$ on the plane containing the lines, $\frac{x+1}{6} = \frac{y-1}{7} = \frac{z-3}{8}$ and $\frac{x-1}{3} = \frac{y-2}{5} = \frac{z-3}{7}$. Then $(PQ)^2$, is equal to _____. (2021-08-26/Shift-2)
150. Let P be the plane passing through the point $(1, 2, 3)$ and the line of intersection of the planes $\vec{r} \cdot (\hat{i} + \hat{j} + 4\hat{k}) = 16$ and $\vec{r} \cdot (-\hat{i} + \hat{j} + \hat{k}) = 6$. Then which of the following points lie on P ? (2021-08-26/Shift-2)
- (a) $(3, 3, 2)$ (b) $(-8, 8, 6)$
(c) $(4, 2, 2)$ (d) $(6, 6, 2)$
151. A hall has a square floor of dimension $10\text{m} \times 10\text{m}$ (see the figure) and vertical walls. If the angle GPH between the diagonals AG and BH is $\cos^{-1} \frac{1}{5}$, then the height of the hall (in meters) is: (2021-08-26/Shift-2)



- (a) $5\sqrt{3}$ (b) 5
(c) $2\sqrt{10}$ (d) $5\sqrt{2}$



152. Equation of a plane at a distance $\sqrt{\frac{2}{21}}$ from the origin, which contains the line of intersection of the planes $x - y - z - 1 = 0$ and $2x + y - 3z + 4 = 0$, is :

(2021-08-27/Shift-1)

- (a) $-x + 2y + 2z - 3 = 0$ (b) $3x - 4z + 3 = 0$
(c) $3x - y - 5z + 2 = 0$ (d) $4x - y - 5z + 2 = 0$

153. The distance of the point $(1, -2, 3)$ from the plane $x - y + z = 5$ measured parallel to a line, whose direction ratios are $2, 3, -6$ is:

(2021-08-27/Shift-1)

- (a) 2 (b) 5
(c) 3 (d) 1

154. Let the line L be the projection of the line :

$$\frac{x-1}{2} = \frac{y-3}{1} = \frac{z-4}{2}$$

In the plane $x - 2y - z = 3$. If d is the distance of the point $(0, 0, 6)$ from L, then d^2 is equal to _____.

(2021-08-26/Shift-1)

155. A plane P contains the line $x + 2y + 3z + 1 = 0$ = $x - y - z - 6$, and is perpendicular to the plane $-2x + y + z + 8 = 0$. Then which of the following points lies on P?

(2021-08-26/Shift-1)

- (a) $(1, 0, 1)$ (b) $(2, -1, 1)$
(c) $(-1, 1, 2)$ (d) $(0, 1, 1)$

156. The angle between the straight lines, whose direction cosines are given by the equation $2l + 2m - n = 0$ and $mn + nl + lm = 0$, is :

(2021-08-27/Shift-2)

- (a) $\frac{\pi}{3}$ (b) $\pi - \cos^{-1}\left(\frac{4}{9}\right)$
(c) $\cos^{-1}\left(\frac{8}{9}\right)$ (d) $\frac{\pi}{2}$

157. Let S be the mirror image of the point Q $(1, 3, 4)$ with respect to the plane $2x - y + z + 3 = 0$ and let R $(3, 5, \gamma)$ be a point on this plane. Then the square of the length of the line segment SR is _____.

(2021-08-27/Shift-2)

158. The equation of the plane passing through the line of intersection of the planes $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 1$ and

$$\vec{r} \cdot (2\hat{i} + 3\hat{j} - \hat{k}) + 4 = 0 \text{ and parallel to the x-axis is :}$$

(2021-08-27/Shift-2)

(a) $\vec{r} \cdot (\hat{i} + 3\hat{k}) + 6 = 0$ (b) $\vec{r} \cdot (\hat{j} - 3\hat{k}) - 6 = 0$

(c) $\vec{r} \cdot (\hat{i} - 3\hat{k}) + 6 = 0$ (d) $\vec{r} \cdot (\hat{j} - 3\hat{k}) + 6 = 0$

159. The square of the distance of the point of intersection of the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+1}{6}$ and the plane $2x - y + z = 6$ from the point $(-1, -1, 2)$ is _____ ?

(2021-08-31/Shift-1)

160. Let the equation of the plane, that passes through the point $(1, 4, -3)$ and contains the line of intersection of the planes $3x - 2y + 4z - 7 = 0$ and $x + 5y - 2z + 9 = 0$, be

$$\alpha x + \beta y + \gamma z + 3 = 0 \text{ then } \alpha + \beta + \gamma \text{ is equal to ?}$$

(2021-08-31/Shift-1)

- (a) -15 (b) 15
(c) -23 (d) 23

161. The distance of the point $(-1, 2, -2)$ from the line of intersection of the planes $2x + 3y + 2z = 0$ and $x - 2y + z = 0$ is:

(2021-08-31/Shift-2)

- (a) $\frac{\sqrt{42}}{2}$ (b) $\frac{5}{2}$
(c) $\frac{1}{\sqrt{2}}$ (d) $\frac{\sqrt{34}}{2}$

162. Suppose the line $\frac{x-2}{\alpha} = \frac{y-2}{-5} = \frac{z+2}{2}$ lies on the plane $x + 3y - 2z + \beta = 0$, then $(\alpha + \beta)$ is equal to _____

(2021-08-31/Shift-2)



- 163.** If the foot of the perpendicular from point $(4, 3, 8)$ on the line $L_1 : \frac{x-a}{1} = \frac{y-2}{3} = \frac{z-b}{4}$, $l \neq 0$ is $(3, 5, 7)$, then the shortest distance between the line L_1 and line $L_2 : \frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$ is equal to
(2021-03-16/Shift-2)
- (a) $\frac{1}{\sqrt{3}}$ (b) $\frac{1}{2}$
(c) $\frac{1}{\sqrt{6}}$ (d) $\sqrt{\frac{2}{3}}$
- 164.** If (x, y, z) be an arbitrary point lying on a plane P which passes through the points $(42, 0, 0)$, $(0, 42, 0)$ and $(0, 0, 42)$, then the value of the expression
$$3 + \frac{x-11}{(y-19)^2(z-12)^2} + \frac{y-19}{(x-11)^2(z-12)^2} + \frac{z-12}{(x-11)^2(y-19)^2} - \frac{x+y+z}{14(x-11)(y-19)(z-12)}$$
is equal to (2021-03-16/Shift-2)
- (a) 3 (b) 0
(c) -45 (d) 39
- 165.** If the distance of the point $(1, -2, 3)$ from the plane $x + 2y - 3z + 10 = 0$ measured parallel to the line $\frac{x-1}{3} = \frac{2-y}{m} = \frac{z+3}{1}$ is $\sqrt{\frac{7}{2}}$, then the value of $|m|$ is equal to _____. (2021-03-16/Shift-2)
- 166.** Let P be plane $lx + my + nz = 0$ containing the line, $\frac{1-x}{1} = \frac{y+4}{2} = \frac{z+2}{3}$. If plane P divides the line segment AB joining points $A(-3, -6, 1)$ and $B(2, 4, -3)$ in ratio $k : 1$ then the value of k is equal to (2021-03-16/Shift-1)
- (a) 4 (b) 3
(c) 1.5 (d) 2
- 167.** If for $a > 0$, the feet of perpendiculars from the points $A(a, -2a, 3)$ and $B(0, 4, 5)$ on the plane $lx + my + nz = 0$ are points $C(0, -a, -1)$ and D respectively, then the length of line segment CD is equal to (2021-03-16/Shift-1)
- (a) $\sqrt{66}$ (b) $\sqrt{41}$
(c) $\sqrt{55}$ (d) $\sqrt{31}$
- 168.** If the equation of plane passing through the mirror image of a point $(2, 3, 1)$ with respect to line $\frac{x+1}{2} = \frac{y-3}{1} = \frac{z+2}{-1}$ and containing the line $\frac{x-2}{3} = \frac{1-y}{2} = \frac{z+1}{1}$ is $\alpha x + \beta y + \gamma z = 24$, then $\alpha + \beta + \gamma$ is equal to : (2021-03-17/Shift-2)
- (a) 19 (b) 18
(c) 21 (d) 20
- 169.** Let P be an arbitrary point having sum of the squares of the distances from the planes $x + y + z = 0$, $lx - nz = 0$ and $x - 2y + z = 0$, equal to 9. If the locus of the point P is $x^2 + y^2 + z^2 = 9$, then the value of $l - n$ is equal to (2021-03-17/Shift-2)
- 170.** The equation of the plane which contains the y -axis and passes through the point $(1, 2, 3)$ is : (2021-03-17/Shift-1)
- (a) $3x + z = 6$ (b) $x + 3z = 0$
(c) $x + 3z = 10$ (d) $3x - z = 0$
- 171.** If the equation of the plane passing through the line of intersection of the planes $2x - 7y + 4z - 3 = 0$, $3x - 5y + 4z + 11 = 0$ and the point $(-2, 1, 3)$ is $ax + by + cz - 7 = 0$, then the value of $2a + b + c - 7$ is (2021-03-17/Shift-1)
- 172.** If $\vec{a} = \alpha\hat{i} + \beta\hat{j} + 3\hat{k}$, $\vec{b} = -\beta\hat{i} - \alpha\hat{j} - \hat{k}$ and $\vec{c} = \hat{i} - 2\hat{j} - \hat{k}$ Such that $\vec{a} \cdot \vec{b} = 1$ and $\vec{b} \cdot \vec{c} = -3$, then $\frac{1}{3}(\vec{a} \times \vec{b}) \cdot \vec{c}$ is equal to (2021-03-17/Shift-1)
- 173.** Let the mirror image of the point $(1, 3, a)$ with respect to the plane $\vec{r} \cdot (2\hat{i} - \hat{j} + \hat{k}) - b = 0$ be $(-3, 5, 2)$. Then, the value of $|a + b|$ is equal to _____. (2021-03-18/Shift-2)



174. Let P be a plane containing the line $\frac{x-1}{3} = \frac{y+6}{4} = \frac{z+5}{2}$ and parallel to the line $\frac{x-3}{4} = \frac{y-2}{-3} = \frac{z+5}{7}$. If the point $(1, -1, \alpha)$ lies on the plane, P, then the value of $|5\alpha|$ is equal to _____. (2021-03-18/Shift-2)
175. The equation of the planes parallel to the plane $x - 2y + 2z - 3 = 0$ which are at unit distance from the point $(1, 2, 3)$ is $ax + by + cz + d = 0$. If $(b-d) = K(c-a)$, then the positive value of K is (2021-03-18/Shift-1)
176. Let the plane $ax + by + cz + d = 0$ bisect the line joining the points $(4, -3, 1)$ and $(2, 3, -5)$ at the right angles. If a, b, c, d are integers, then the minimum value of $(a^2 + b^2 + c^2 + d^2)$ is (2021-03-18/Shift-1)
177. Let $a, b \in \mathbb{R}$. If the mirror image of the point $P(a, 6, 9)$ with respect to the line $\frac{x-3}{7} = \frac{y-2}{5} = \frac{z-1}{-9}$ is $(20, b, -a-9)$, then $|a+b|$ is equal to _____ (2021-02-24/Shift-2)
- (a) 88 (b) 84
(c) 90 (d) 86
178. The vector equation of the plane passing through the intersection of the planes $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 1$ and $\vec{r} \cdot (\hat{i} - 2\hat{j}) = -2$, and the point $(1, 0, 2)$ is: (2021-02-24/Shift-2)
- (a) $\vec{r} \cdot (\hat{i} + 7\hat{j} + 3\hat{k}) = 7$ (b) $\vec{r} \cdot (3\hat{i} + 7\hat{j} + 3\hat{k}) = 7$
(c) $\vec{r} \cdot (\hat{i} - 7\hat{j} + 3\hat{k}) = \frac{7}{3}$ (d) $\vec{r} \cdot (\hat{i} + 7\hat{j} + 3\hat{k}) = \frac{7}{3}$
179. Let λ be an integer. If the shortest distance between the lines $x - \lambda = 2y - 1 = -2z$ and $x = y + 2\lambda = z - \lambda$ is $\frac{\sqrt{7}}{2\sqrt{2}}$, then the value of $|\lambda|$ is _____. (2021-02-24/Shift-2)
180. The distance of the point $(1, 1, 9)$ from the point of intersection of the line $\frac{x-3}{1} = \frac{y-4}{2} = \frac{z-5}{2}$ and the plane $x + y + z = 17$ is _____ (2021-02-24/Shift-1)
- (a) $2\sqrt{19}$ (b) 38
(c) $19\sqrt{2}$ (d) $\sqrt{38}$
181. The equation of the plane passing through the point $(1, 2, -3)$ and perpendicular to the planes $3x + y - 2z = 5$ and $2x - 5y - z = 7$, is _____ (2021-02-24/Shift-1)
- (a) $6x - 5y + 2z + 10 = 0$ (b) $6x - 5y + 2z - 2 = 0$
(c) $3x - 10y + 2z + 11 = 0$ (d) $11x + y + 17z + 38 = 0$
182. A plane passes through the points $A(1, 2, 3)$, $B(2, 3, 1)$ and $C(2, 4, 2)$. If O is the origin and P is $(2, -1, 1)$, then the projection of $\vec{OP}$ on this plane is of length: _____ (2021-02-25/Shift-2)
- (a) $\sqrt{\frac{2}{3}}$ (b) $\sqrt{\frac{2}{5}}$
(c) $\sqrt{\frac{2}{7}}$ (d) $\sqrt{\frac{2}{11}}$
183. A line 'l' passing through origin is perpendicular to the lines
 $l_1 : \vec{r} = (3+t)\hat{i} + (-1+2t)\hat{j} + (4+2t)\hat{k}$
 $l_2 : \vec{r} = (3+2s)\hat{i} + (3+2s)\hat{j} + (2+s)\hat{k}$
If the co-ordinates of the point in the first octant on 'l' at a distance of $\sqrt{17}$ from the point of intersection of 'l' and 'l₁' are (a, b, c) , then $18(a+b+c)$ is equal to: _____ (2021-02-25/Shift-2)
184. Let α be the angle between the lines whose direction cosines satisfy the equations $l + m - n = 0$ and $l^2 + m^2 - n^2 = 0$. Then the value of $\sin^4 \alpha + \cos^4 \alpha$ is: _____ (2021-02-25/Shift-1)
- (a) $\frac{5}{8}$ (b) $\frac{3}{4}$
(c) $\frac{1}{2}$ (d) $\frac{3}{8}$



185. The equation of the line through the point (0, 1, 2) and perpendicular to the line $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{-2}$ is:
- (2021-02-25/Shift-1)
- (a) $\frac{x}{-3} = \frac{y-1}{4} = \frac{z-2}{3}$ (b) $\frac{x}{3} = \frac{y-1}{-4} = \frac{z-2}{3}$
 (c) $\frac{x}{3} = \frac{y-1}{4} = \frac{z-2}{3}$ (d) $\frac{x}{3} = \frac{y-1}{4} = \frac{z-2}{-3}$
186. If the mirror image of the point (1, 3, 5) with respect to the plane $4x - 5y + 2z = 8$ is (α, β, γ) , then $5(\alpha + \beta + \gamma)$ equals :
- (2021-02-26/Shift-2)
- (a) 39 (b) 41
 (c) 47 (d) 43
187. Let L be a line obtained from the intersection of two planes $x + 2y + z = 6$ and $y + 2z = 4$. If point $P(\alpha, \beta, \gamma)$ is the foot of perpendicular from (3, 2, 1) on L, then the value of $21(\alpha + \beta + \gamma)$ equals:
- (2021-02-26/Shift-2)
- (a) 68 (b) 102
 (c) 142 (d) 136
188. If $(1, 5, 35), (7, 5, 5), (1, \lambda, 7)$ and $(2\lambda, 1, 2)$ are coplanar, then the sum of all possible values of λ is :
- (2021-02-26/Shift-1)
- (a) $\frac{44}{5}$ (b) $\frac{39}{5}$
 (c) $-\frac{44}{5}$ (d) $-\frac{39}{5}$
189. Consider the three planes
 $P_1 : 3x + 15y + 21z = 9$
 $P_2 : x - 3y - z = 5$, and
 $P_3 : 2x + 10y + 14z = 5$
 Then, which one of the following is true?
- (2021-02-26/Shift-1)
- (a) P_1 and P_2 are parallel
 (b) P_1, P_2 and P_3 are parallel
 (c) P_1 and P_3 are parallel
 (d) P_2 and P_3 are parallel
190. Let $(\lambda, 2, 1)$ be a point on the plane which passes through the point (4, -2, 2). If the plane is perpendicular to the line joining the points (-2, -21, 29) and (-1, -16, 23), then $\left(\frac{\lambda}{11}\right)^2 - \frac{4\lambda}{11} - 4$ is equal to ____.
- (2021-02-26/Shift-1)



EXERCISE - 3 : ADVANCED OBJECTIVE QUESTIONS

VECTORS

Objective Questions I [Only one correct option]

- If $\vec{a}$ & $\vec{b}$ lie on a plane normal to the plane containing $\vec{c}$ & $\vec{d}$ then, $(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d})$ is equal to
(a) -1 (b) 1
(c) 0 (d) none
- 'P' is a point inside the triangle ABC such that $BC(\vec{PA}) + CA(\vec{PB}) + AB(\vec{PC}) = 0$ then for the triangle ABC the point P is its
(a) incentre (b) circumcentre
(c) centroid (d) orthocentre
- Let $\vec{p}$ is the p.v. of the orthocentre & $\vec{g}$ is the p.v. of the centroid of the triangle ABC where circumcentre is the origin. If $\vec{p} = K\vec{g}$ then K =
(a) 3 (b) 2
(c) 1/3 (d) 2/3
- If the unit vectors $\vec{e}_1$ and $\vec{e}_2$ are inclined at an angle 2θ and $|\vec{e}_1 - \vec{e}_2| < 1$, then for $\theta \in [0, \pi]$, θ may lie in the interval :
(a) $\left[0, \frac{\pi}{6}\right]$ (b) $\left[\frac{\pi}{6}, \frac{\pi}{2}\right]$
(c) $\left[\frac{5\pi}{6}, \pi\right]$ (d) $\left[\frac{\pi}{2}, \frac{5\pi}{6}\right]$
- Two given points P and Q in the rectangular cartesian co-ordinates lie on $y = 2^{x+2}$ such that $\vec{OP} \cdot \hat{i} = -1$ and $\vec{OQ} \cdot \hat{i} = +2$ where $\hat{i}$ is a unit vector along the x-axis. The magnitude of $\vec{OQ} - 4\vec{OP}$ will be :
(a) 10 (b) 20
(c) 30 (d) none
- A, B, C and D are four points in a plane with p.v.'s $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$ respectively such that
 $(\vec{a} - \vec{d}) \cdot (\vec{b} - \vec{c}) = (\vec{b} - \vec{d}) \cdot (\vec{c} - \vec{a}) = 0$.
Then for the triangle ABC, D is its :
(a) incentre (b) circumcentre
(c) orthocentre (d) centroid
- Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ and $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$ be three non-zero vectors such that $\vec{c}$ is a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$. If the angle between $\vec{a}$ and $\vec{b}$ is $\pi/6$, then $\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}^2$ is equal to :
(a) 0
(b) 1
(c) $\frac{1}{4}(a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2)$
(d) none of these
- The vectors $\vec{p}$ & $\vec{q}$ satisfy the system of equations $2\vec{p} + \vec{q} = \vec{a}$, $\vec{p} + 2\vec{q} = \vec{b}$ and the angle between $\vec{p}$ & $\vec{q}$ is θ . If it is known that in the rectangular system of co-ordinates the vectors $\vec{a}$ & $\vec{b}$ have the forms $\vec{a} = (1, 1)$ & $\vec{b} = (1, -1)$ then $\cos \theta =$
(a) $\frac{4}{5}$ (b) $-\frac{4}{5}$
(c) $-\frac{3}{5}$ (d) none



9. Let A ($\vec{a}$) and B ($\vec{b}$) be points on two skew lines $\vec{r} = \vec{a} + \lambda \vec{p}$ and $\vec{r} = \vec{b} + \mu \vec{q}$ and the shortest distance between the skew lines is 1, where $\vec{p}$ and $\vec{q}$ are unit vectors forming adjacent sides of a parallelogram enclosing an area of $\frac{1}{2}$ units. If an angle between AB and the line of shortest distance is 60° , then AB =
- (a) $\frac{1}{2}$ (b) 2
(c) 1 (d) $\lambda \in \mathbb{R} - \{0\}$
10. If $\vec{b}$ & $\vec{c}$ are any two perpendicular unit vectors and $\vec{a}$ is any vector, then, $(\vec{a} \cdot \vec{b}) \vec{b} + (\vec{a} \cdot \vec{c}) \vec{c} + \frac{\vec{a} \cdot (\vec{b} \times \vec{c})}{|\vec{b} \times \vec{c}|^2} (\vec{b} \times \vec{c})$ is equal to :
- (a) $\vec{a}$ (b) $\vec{b}$
(c) $\vec{c}$ (d) none of these
11. If $A_1, A_2, A_3, \dots, A_n$ are the vertices of a regular plane polygon with n sides & O is its centre then
- $$\sum_{i=1}^{n-1} (\vec{OA}_i \times \vec{OA}_{i+1}) =$$
- (a) $(1-n) \vec{OA}_2 \times \vec{OA}_1$ (b) $(n-1) \vec{OA}_2 \times \vec{OA}_1$
(c) $n \vec{OA}_2 \times \vec{OA}_1$ (d) none
12. If the vector product of a constant vector $\vec{OA}$ with a variable vector $\vec{OB}$ in a fixed plane OAB be a constant vector, then locus of B is :
- (a) a straight line perpendicular to $\vec{OA}$
(b) a circle with centre O radius equal to $|\vec{OA}|$
(c) a straight line parallel to $\vec{OA}$
(d) none of these
13. Let $\vec{A}, \vec{B}, \vec{C}$ be vectors of length 3, 4 and 5 respectively. Let $\vec{A}$ be perpendicular to $\vec{B} + \vec{C}$, $\vec{B}$ to $\vec{C} + \vec{A}$ and $\vec{C}$ to $\vec{A} + \vec{B}$. Then length of the vector, $\vec{A} + \vec{B} + \vec{C}$ is :
- (a) $-5\sqrt{2}$ (b) $\sqrt{2}$
(c) $5\sqrt{2}$ (d) none of these
14. If $\vec{a}$ & $\vec{b}$ are unit vectors such that $(\vec{a} + \vec{b}) \cdot (2\vec{a} + 3\vec{b}) \times (3\vec{a} - 2\vec{b}) = 0$, then angle between $\vec{a}$ and $\vec{b}$
- (a) 0 (b) $\pi/2$
(c) π (d) indeterminate
15. If a plane cuts off intercepts $OA = a, OB = b, OC = c$ from the coordinate axes, then the area of the triangle ABC =
- (a) $\frac{1}{2} \sqrt{b^2 c^2 + c^2 a^2 + a^2 b^2}$
(b) $\frac{1}{2} (bc + ca + ab)$
(c) $\frac{1}{2} abc$
(d) $\frac{1}{2} \sqrt{(b-c)^2 + (c-a)^2 + (a-b)^2}$
16. Let $\vec{a} = \hat{i} - \hat{k}, \vec{b} = x\hat{i} + \hat{j} + (1-x)\hat{k}$ and $\vec{c} = y\hat{i} + x\hat{j} + (1+x-y)\hat{k}$. Then $[\vec{a} \vec{b} \vec{c}]$ depends on
- (a) only x (b) only y
(c) neither x nor y (d) both x and y
17. The value of a so that the volume of the parallelepiped formed by the vectors $\hat{i} + a\hat{j} - \hat{k}, \hat{j} + a\hat{k}, \hat{i} + \hat{j} + \hat{k}$ becomes minimum is
- (a) $\sqrt{3}$ (b) $\frac{1}{2}$
(c) $\frac{1}{\sqrt{3}}$ (d) $\frac{7}{4}$



18. If the vectors $a\hat{i} + \hat{j} + \hat{k}$, $\hat{i} + b\hat{j} + \hat{k}$ and $\hat{i} + \hat{j} + c\hat{k}$ ($a \neq b \neq c \neq 1$) are coplanar then the value of
- $$\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} =$$
- (a) 1 (b) -1
(c) 0 (d) none
19. $[(\vec{a} \times \vec{b}) \times (\vec{b} \times \vec{c}) (\vec{b} \times \vec{c}) \times (\vec{c} \times \vec{a}) (\vec{c} \times \vec{a}) \times (\vec{a} \times \vec{b})] =$
- (a) $[\vec{a} \vec{b} \vec{c}]^2$ (b) $[\vec{a} \vec{b} \vec{c}]^3$
(c) $[\vec{a} \vec{b} \vec{c}]^4$ (d) none
20. The triple product $(\vec{d} + \vec{a}) \cdot [\vec{a} \times (\vec{b} \times (\vec{c} \times \vec{d}))]$ simplifies to
- (a) $(\vec{b} \cdot \vec{d}) [\vec{a} \vec{c} \vec{d}]$ (b) $(\vec{b} \cdot \vec{c}) [\vec{a} \vec{b} \vec{d}]$
(c) $(\vec{b} \cdot \vec{a}) [\vec{a} \vec{b} \vec{d}]$ (d) none
21. Let $\vec{r}, \vec{a}, \vec{b}$ and $\vec{c}$ be four non-zero vectors such that $\vec{r} \cdot \vec{a} = 0$, $|\vec{r} \times \vec{b}| = |\vec{r}| |\vec{b}|$, $|\vec{r} \times \vec{c}| = |\vec{r}| |\vec{c}|$, then $[a \ b \ c] =$
- (a) $|a| |b| |c|$ (b) $-|a| |b| |c|$
(c) 0 (d) none of these
22. If $\vec{a}, \vec{b}, \vec{c}$ are such that $[\vec{a} \ \vec{b} \ \vec{c}] = 1$, $\vec{c} = \lambda \vec{a} \times \vec{b}$, $\vec{a} \wedge \vec{b} < \frac{2\pi}{3}$ and $|\vec{a}| = \sqrt{2}$, $|\vec{b}| = \sqrt{3}$, $|\vec{c}| = \frac{1}{\sqrt{3}}$, then the angle between $\vec{a}$ and $\vec{b}$ is
- (a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$
(c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$
23. If $\vec{a} + 2\vec{b} + 3\vec{c} = \vec{0}$, then $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}$ is equal to
- (a) $6(\vec{b} \times \vec{c})$ (b) $6(\vec{c} \times \vec{a})$
(c) $6(\vec{a} \times \vec{b})$ (d) none of these
24. If $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) \cdot (\vec{a} \times \vec{d}) = 0$, then which of the following is always true
- (a) $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are necessarily coplanar
(b) either $\vec{a}$ or $\vec{d}$ must lie in the plane of $\vec{b}$ and $\vec{c}$
(c) either $\vec{b}$ or $\vec{c}$ must lie in plane of $\vec{a}$ and $\vec{d}$
(d) either $\vec{a}$ or $\vec{b}$ must lie in plane of $\vec{c}$ and $\vec{d}$
25. If $P_1: \vec{r} \cdot \vec{n}_1 - d_1 = 0$, $P_2: \vec{r} \cdot \vec{n}_2 - d_2 = 0$ and $P_3: \vec{r} \cdot \vec{n}_3 - d_3 = 0$ are three planes and $\vec{n}_1, \vec{n}_2$ and $\vec{n}_3$ are three non-coplanar vectors then, the three lines $P_1=0$, $P_2=0$ and $P_3=0$ are
- (a) parallel lines (b) coplanar lines
(c) coincident lines (d) concurrent lines
26. If $\vec{a}, \vec{b}$ and $\vec{c}$ are three unit vectors equally inclined to each other at an angle α . Then the angle between $\vec{a}$ and plane of $\vec{b}$ and $\vec{c}$ is
- (a) $\theta = \cos^{-1} \left(\frac{\cos \alpha}{\cos \frac{\alpha}{2}} \right)$ (b) $\theta = \sin^{-1} \left(\frac{\cos \alpha}{\cos \frac{\alpha}{2}} \right)$
(c) $\theta = \cos^{-1} \left(\frac{\sin \frac{\alpha}{2}}{\sin \alpha} \right)$ (d) $\theta = \sin^{-1} \left(\frac{\sin \frac{\alpha}{2}}{\sin \alpha} \right)$
27. For any vector $\vec{A}$, $\hat{i} \times (\hat{i} \times \vec{A}) + \hat{j} \times (\hat{j} \times \vec{A}) + \hat{k} \times (\hat{k} \times \vec{A})$ simplifies to
- (a) $3\vec{A}$ (b) $\vec{A}$
(c) $-\vec{A}$ (d) $-2\vec{A}$
28. If $\vec{a}, \vec{b}, \vec{c}$ be the unit vectors such that $\vec{b}$ is not parallel to $\vec{c}$ and $\vec{a} \times (2\vec{b} \times \vec{c}) = \vec{b}$ then the angle that $\vec{a}$ makes with $\vec{b}$ & $\vec{c}$ are respectively
- (a) $\frac{\pi}{3}$ & $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$ & $\frac{2\pi}{3}$
(c) $\frac{\pi}{2}$ & $\frac{2\pi}{3}$ (d) $\frac{\pi}{2}$ & $\frac{\pi}{3}$



29. If a vector $\vec{a}$ is expressed as the sum of two vectors $\vec{a}'$ and $\vec{a}''$ along and perpendicular to a given vector $\vec{b}$ then $\vec{a}''$ is

(a) $\frac{(\vec{a} \times \vec{b}) \times \vec{b}}{|\vec{b}|}$ (b) $\frac{\vec{b} \times (\vec{a} \times \vec{b})}{|\vec{b}|^2}$
 (c) $\frac{\vec{b} \times (\vec{a} \times \vec{b})}{|\vec{b}|}$ (d) $\frac{\vec{a} \times \vec{b}}{|\vec{b}|^2} \vec{b}$

30. If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are any three vectors, then

$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \times \vec{c}$ is true if :

- (a) $\vec{b}$ & $\vec{c}$ are collinear (b) $\vec{a}$ & $\vec{c}$ are collinear
 (c) $\vec{a}$ & $\vec{b}$ are collinear (d) none of these

31. $(\vec{r} \cdot \hat{i})(\hat{i} \times \vec{r}) + (\vec{r} \cdot \hat{j})(\hat{j} \times \vec{r}) + (\vec{r} \cdot \hat{k})(\hat{k} \times \vec{r}) =$

- (a) 0 (b) $\vec{r}$
 (c) $2\vec{r}$ (d) $3\vec{r}$

32. For a non zero vector $\vec{A}$ if the equations $\vec{A} \cdot \vec{B} = \vec{A} \cdot \vec{C}$ and $\vec{A} \times \vec{B} = \vec{A} \times \vec{C}$ hold simultaneously, then

- (a) $\vec{A}$ is perpendicular to $\vec{B} - \vec{C}$
 (b) $\vec{A} = \vec{B}$
 (c) $\vec{B} = \vec{C}$
 (d) $\vec{C} = \vec{A}$

33. Let $\vec{a} = \hat{i} + \hat{j}$ and $\vec{b} = 2\hat{i} - \hat{k}$. The point of intersection of the lines $\vec{r} \times \vec{a} = \vec{b} \times \vec{a}$ and $\vec{r} \times \vec{b} = \vec{a} \times \vec{b}$ is :

- (a) $-\hat{i} + \hat{j} + \hat{k}$ (b) $3\hat{i} - \hat{j} + \hat{k}$
 (c) $3\hat{i} + \hat{j} - \hat{k}$ (d) $\hat{i} - \hat{j} - \hat{k}$

Objective Questions II [One or more than one correct option]

34. The vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ are of the same length & pairwise form equal angles. If $\vec{a} = \hat{i} + \hat{j}$ & $\vec{b} = \hat{j} + \hat{k}$ then $\vec{c}$ can be

(a) (1, 0, 1) (b) $\left(-\frac{4}{3}, \frac{1}{3}, -\frac{4}{3}\right)$
 (c) $\left(\frac{1}{3}, -\frac{4}{3}, \frac{1}{3}\right)$ (d) $\left(-\frac{1}{3}, \frac{4}{3}, -\frac{1}{3}\right)$

35. The vector $\vec{a} \times (\vec{b} \times \vec{a})$ is

- (a) perpendicular to $\vec{a}$
 (b) perpendicular to $\vec{b}$
 (c) coplanar with $\vec{a}$ & $\vec{b}$
 (d) perpendicular to $\vec{a} \times \vec{b}$

36. $(\vec{u} \times \vec{v}) \cdot (\vec{u} \times \vec{v})$ is equal to

(a) $\left| \begin{matrix} \vec{u} \cdot \vec{u} & \vec{u} \cdot \vec{v} \\ \vec{u} \cdot \vec{v} & \vec{v} \cdot \vec{v} \end{matrix} \right|$ (b) $(\vec{u} \cdot \vec{v})^2 - \vec{u}^2 \cdot \vec{v}^2$
 (c) $|\vec{u}|^2 |\vec{v}|^2 - (\vec{u} \cdot \vec{v})^2$ (d) none

37. If $\vec{z}_1 = a\hat{i} + b\hat{j}$ & $\vec{z}_2 = c\hat{i} + d\hat{j}$ are two vectors in $\hat{i}$ & $\hat{j}$ system where $|\vec{z}_1| = |\vec{z}_2| = r$ & $\vec{z}_1 \cdot \vec{z}_2 = 0$ then $\vec{w}_1 = a\hat{i} + c\hat{j}$ and $\vec{w}_2 = b\hat{i} + d\hat{j}$ satisfy

- (a) $|\vec{w}_1| = r$ (b) $|\vec{w}_2| = r$
 (c) $\vec{w}_1 \cdot \vec{w}_2 = 0$ (d) none of these

38. If $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$ are the pv's of the point A, B, C and D respectively in three dimensional space and satisfy the relation $3\vec{a} - 2\vec{b} + \vec{c} - 2\vec{d} = 0$, then :

- (a) A, B, C and D are coplanar
 (b) the line joining the points B and D divides the line joining the point A and C in the ratio 2 : 1.
 (c) the line joining the points A and C divides the line joining the points B and D in the ratio 1 : 1.
 (d) the four vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ & $\vec{d}$ are linearly dependents.



39. $\vec{a}$ and $\vec{b}$ are two non collinear unit vectors. Then $\vec{a}, \vec{b}, x\vec{a} - y\vec{b}$ form a triangle, if:
- (a) $x = -1; y = 1$ and $|\vec{a} + \vec{b}| = 2\cos\left(\frac{\vec{a} \wedge \vec{b}}{2}\right)$
- (b) $x = -1; y = 1$ and $\cos(\vec{a} \wedge \vec{b}) + |\vec{a} + \vec{b}| \cos[\vec{a} \wedge -(\vec{a} + \vec{b})] = -1$
- (c) $|\vec{a} + \vec{b}| = -2\cot\left(\frac{\vec{a} \wedge \vec{b}}{2}\right)\cos\left(\frac{\vec{a} \wedge \vec{b}}{2}\right)$ and $x = -1, y = 1$
- (d) none of these
40. If $\vec{OA} = \vec{a}; \vec{OB} = \vec{b}; \vec{OC} = 2\vec{a} + 3\vec{b}; \vec{OD} = \vec{a} - 2\vec{b}$, the length of $\vec{OA}$ is three times the length of $\vec{OB}$ and $\vec{OA}$ is perpendicular to $\vec{DB}$ then $(\vec{BD} \times \vec{AC}) \cdot (\vec{OD} \times \vec{OC})$ is :
- (a) $7|\vec{a} \times \vec{b}|^2$ (b) $42|\vec{a} \times \vec{b}|^2$
- (c) 0 (d) none of these
41. $\hat{a}$ & $\hat{b}$ are two given unit vectors at right angle. The unit vector equally inclined with $\hat{a}$ & $\hat{b}$ and $\hat{a} \times \hat{b}$ will be :
- (a) $-\frac{1}{\sqrt{3}}(\hat{a} + \hat{b} + \hat{a} \times \hat{b})$ (b) $\frac{1}{\sqrt{3}}(\hat{a} + \hat{b} + \hat{a} \times \hat{b})$
- (c) $\frac{1}{\sqrt{3}}(\hat{a} + \hat{b} - \hat{a} \times \hat{b})$ (d) $-\frac{1}{\sqrt{3}}(\hat{a} + \hat{b} - \hat{a} \times \hat{b})$
42. If $\vec{a}$ and $\vec{b}$ unequal unit vectors such that $(\vec{a} - \vec{b}) \times [(\vec{b} + \vec{a}) \times (2\vec{a} + \vec{b})] = \vec{a} + \vec{b}$, then angle θ between $\vec{a}$ and $\vec{b}$ is
- (a) $\frac{\pi}{2}$ (b) 0
- (c) π (d) $\frac{\pi}{4}$
43. A vector $(\vec{d})$ is equally inclined to three vectors $\vec{a} = \hat{i} - \hat{j} + \hat{k}$, $\vec{b} = 2\hat{i} + \hat{j}$ and $\vec{c} = 3\hat{j} - 2\hat{k}$. Let $\vec{x}, \vec{y}, \vec{z}$ be three vector in the plane of $\vec{a}, \vec{b}; \vec{b}, \vec{c}; \vec{c}, \vec{a}$ respectively then
- (a) $\vec{x} \cdot \vec{d} = 14$
- (b) $\vec{y} \cdot \vec{d} = 3$
- (c) $\vec{z} \cdot \vec{d} = 0$
- (d) $\vec{r} \cdot \vec{d} = 0$ where $\vec{r} = \lambda\vec{x} + \mu\vec{y} + \delta\vec{z}$
44. Identify the statement(s) which is/are INCORRECT ?
- (a) $\vec{a} \times [\vec{a} \times (\vec{a} \times \vec{b})] = (\vec{a} \times \vec{b})(\vec{a}^2)$
- (b) If $\vec{a}, \vec{b}, \vec{c}$ are non-zero, non coplanar vectors, and $\vec{v} \cdot \vec{a} = \vec{v} \cdot \vec{b} = \vec{v} \cdot \vec{c} = 0$ then $\vec{v}$ must be a null vector
- (c) If $\vec{a}$ and $\vec{b}$ lie in a plane normal to the plane containing the vectors $\vec{a} - \vec{b}, \vec{c} - \vec{d}$, where $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are non-zero vectors, then $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = \vec{0}$
- (d) If $\vec{a}, \vec{b}, \vec{c}$ and $\vec{a}', \vec{b}', \vec{c}'$ are reciprocal system of vectors then $\vec{a} \cdot \vec{b}' + \vec{b} \cdot \vec{c}' + \vec{c} \cdot \vec{a}' = 3$

Numerical Value Type Questions

45. If $|\vec{a}| = 3, |\vec{b}| = 4, |\vec{c}| = 5$ and $\vec{a} \perp (\vec{b} + \vec{c}), \vec{b} \perp (\vec{c} + \vec{a}), \vec{c} \perp (\vec{a} + \vec{b})$ then $|\vec{a} + \vec{b} + \vec{c}|^2$ is
46. $\vec{a}, \vec{b}, \vec{c}$ are three unit vectors and every two are inclined to each other at an angle $\cos^{-1}(3/5)$. If $\vec{a} \times \vec{b} = p\vec{a} + q\vec{b} + r\vec{c}$, where p, q, r are scalars, then $55q^2$ is equal to



Assertion & Reason

Use the following codes to answer the questions

- (A) If both assertion and reason are correct and reason is the correct explanation of assertion.
 (B) If both assertion and reason are true but and reason is not the correct explanation of assertion.
 (C) If assertion is true but reason is false.
 (D) If assertion is false but reason is true.
 (E) If assertion and reason are both false.

47. **Assertion :** Let $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$, $\vec{b} = 4\hat{i} + 6\hat{j} - 2\hat{k}$, then $\vec{a} \times \vec{b} = \vec{0}$

Reason : If $\vec{a} \neq \vec{0}$, $\vec{b} \neq \vec{0}$ and $\vec{a}$ and $\vec{b}$ are non-collinear vectors, then $\vec{a} \times \vec{b} = ab \sin \theta \hat{n}$, where θ is the smaller angle between the vectors $\vec{a}$ and $\vec{b}$ and $\hat{n}$ is unit vector such that $\vec{a}, \vec{b}, \hat{n}$ taken in this order form right handed orientation

- (a) A (b) B
 (c) C (d) D
 (e) E

48. **Assertion :** Let $\vec{a} = 3\hat{i} - \hat{j}$, $\vec{b} = 2\hat{i} + \hat{j} - 3\hat{k}$. If $\vec{b} = \vec{b}_1 + \vec{b}_2$ such that $\vec{b}_1$ is collinear with $\vec{a}$ and $\vec{b}_2$ is perpendicular to $\vec{a}$ is possible, then $\vec{b}_2 = \hat{i} + 3\hat{j} - 3\hat{k}$.

Reason : If $\vec{a}$ and $\vec{b}$ are non-zero, non-collinear vectors, then $\vec{b}$ can be expressed as $\vec{b} = \vec{b}_1 + \vec{b}_2$, where $\vec{b}_1$ is collinear with $\vec{a}$ and $\vec{b}_2$ is perpendicular to $\vec{a}$.

- (a) A (b) B
 (c) C (d) D
 (e) E

Match the Following

Each question has two columns. Four options are given representing matching of elements from Column-I and Column-II. Only one of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.

49. **Column-I** **Column-II**

(A) If $\vec{a} + \vec{b} = \hat{j}$ and

(P) 1

$$2\vec{a} - \vec{b} = 3\hat{i} + \frac{\hat{j}}{2}, \text{ then}$$

cosine of the angle between

$\vec{a}$ and $\vec{b}$ is

(B) If $|\vec{a}| = |\vec{b}| = |\vec{c}|$, angle between

(Q) $5\sqrt{3}$

each pair of vectors is $\frac{\pi}{3}$ and

$$|\vec{a} + \vec{b} + \vec{c}| = \sqrt{6}, \text{ then } |\vec{a}| =$$

(C) Area of the parallelogram whose diagonals represent the vectors $3\hat{i} + \hat{j} - 2\hat{k}$ and $\hat{i} - 3\hat{j} + 4\hat{k}$ is

(R) 7

(D) If $\vec{a}$ is perpendicular to

(S) $-\frac{3}{5}$

$\vec{b} + \vec{c}$, $\vec{b}$ is perpendicular to

$\vec{c} + \vec{a}$, $\vec{c}$ is perpendicular to

$$\vec{a} + \vec{b}, |\vec{a}| = 2, |\vec{b}| = 3 \text{ and}$$

$$|\vec{c}| = 6, \text{ then } |\vec{a} + \vec{b} + \vec{c}| =$$

The correct matching is

- (a) A-R; B-P; C-Q; D-S
 (b) A-P; B-R; C-Q; D-S
 (c) A-Q; B-R; C-R; D-P
 (d) A-S; B-P; C-Q; D-R



50. Match the following

Column-I

Column-II

(A) If the vectors $\vec{a}, \vec{b}, \vec{c}$ form sides $\overrightarrow{BA}, \overrightarrow{CA}, \overrightarrow{AB}$ of $\triangle ABC$, then

(P) $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a}$

(B) If $\vec{a}, \vec{b}, \vec{c}$ are forming three adjacent sides of regular tetrahedron, then

(Q) $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 0$

(C) If $\vec{a} \times \vec{b} = \vec{c}$, $\vec{b} \times \vec{c} = \vec{a}$ then

(R) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$

(D) $\vec{a}, \vec{b}, \vec{c}$ are unit vectors

(S) $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = -\frac{3}{2}$

and $\vec{a} + \vec{b} + \vec{c} = 0$ then

The correct matching is

(a) $A \rightarrow R$; $B \rightarrow P$; $C \rightarrow P, Q$; $D \rightarrow S$

(b) $A \rightarrow R, S$; $B \rightarrow P$; $C \rightarrow S$; $D \rightarrow Q$

(c) $A \rightarrow P$; $B \rightarrow R$; $C \rightarrow Q$; $D \rightarrow P, S$

(d) $A \rightarrow R$; $B \rightarrow Q$; $C \rightarrow P$; $D \rightarrow S$

Paragraph Type Questions

Using the following passage, solve Q.51 to Q.54

PASSAGE

Three vectors $\hat{a}$, $\hat{b}$ and $\hat{c}$ are such that

$$\hat{a} \times \hat{b} = \hat{c}, \hat{b} \times \hat{c} = \hat{a}, \hat{c} \times \hat{a} = \hat{b}.$$

Answer the following questions :

51. If vector $3\hat{a} - 2\hat{b} + 2\hat{c}$ and $-\hat{a} - 2\hat{c}$ are adjacent sides of a parallelogram, then an angle between the diagonals is

(a) $\frac{\pi}{4}$

(b) $\frac{\pi}{3}$

(c) $\frac{\pi}{2}$

(d) $\frac{2\pi}{3}$

52. Vectors $2\hat{a} - 3\hat{b} + 4\hat{c}$, $\hat{a} + 2\hat{b} - \hat{c}$ and $x\hat{a} - \hat{b} + 2\hat{c}$ are coplanar, then x =

(a) $\frac{8}{5}$

(b) $\frac{5}{8}$

(c) 0

(d) 1

53. Let $\vec{x} = \hat{a} + \hat{b}$, $\vec{y} = 2\hat{a} - \hat{b}$, then the point of intersection of straight lines $\vec{r} \times \vec{x} = \vec{y} \times \vec{x}$, $\vec{r} \times \vec{y} = \vec{x} \times \vec{y}$ is

(a) $2\hat{b}$

(c) $3\hat{b}$

(c) $3\hat{a}$

(d) $2\hat{a}$

54. $\hat{a} \cdot (\hat{b} \times \hat{c}) + \hat{b} \cdot (\hat{c} \times \hat{a}) + \hat{c} \cdot (\hat{a} \times \hat{b})$ is equal to

(a) 1

(b) 3

(c) 0

(d) -12

3-DIMENSIONAL GEOMETRY

Objective Questions I [Only one correct option]

55. Consider a tetrahedron with faces f_1, f_2, f_3, f_4 . Let $\vec{a}_1, \vec{a}_2, \vec{a}_3, \vec{a}_4$ be the vectors whose magnitudes are respectively equal to the areas of f_1, f_2, f_3, f_4 & whose directions are perpendicular to these faces in the outward direction. Then

(a) $\vec{a}_1 + \vec{a}_2 + \vec{a}_3 + \vec{a}_4 = \vec{0}$

(b) $\vec{a}_1 + \vec{a}_3 = \vec{a}_2 + \vec{a}_4$

(c) $\vec{a}_1 + \vec{a}_2 = \vec{a}_3 + \vec{a}_4$

(d) none

56. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be non-coplanar unit vectors equally inclined to one another at an acute angle θ . Then $[\vec{a} \vec{b} \vec{c}]$ in terms of θ is equal to :

(a) $(1 + \cos \theta) \sqrt{\cos 2\theta}$

(b) $(1 + \cos \theta) \sqrt{1 - 2 \cos 2\theta}$

(c) $(1 - \cos \theta) \sqrt{1 + 2 \cos \theta}$

(d) none of these

57. If line makes angle $\alpha, \beta, \gamma, \delta$ with the four diagonals of a cube, then the value of $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta =$

(a) 1

(b) $4/3$

(c) $2/3$

(d) Variable

58. If the sum of the squares of the distance of a point from the three coordinate axes be 36, then its distance from the origin is

(a) 6

(b) $3\sqrt{2}$

(c) $2\sqrt{3}$

(d) $6\sqrt{2}$



59. The direction ratio's of the line $x - y + z - 5 = 0 = x - 3y - 6$ are
 (a) 3, 1, -2 (b) 2, -4, 1
 (c) $\frac{3}{\sqrt{14}}, \frac{1}{\sqrt{14}}, \frac{-2}{\sqrt{14}}$ (d) $\frac{2}{\sqrt{21}}, \frac{-4}{\sqrt{21}}, \frac{1}{\sqrt{21}}$
60. A variable plane passes through a fixed point (1, 2, 3). The locus of the foot of the perpendicular drawn from origin to this plane is :
 (a) $x^2 + y^2 + z^2 - x - 2y - 3z = 0$
 (b) $x^2 + 2y^2 + 3z^2 - x - 2y - 3z = 0$
 (c) $x^2 + 4y^2 + 9z^2 + x + 2y + 3 = 0$
 (d) $x^2 + y^2 + z^2 + x + 2y + 3z = 0$
61. The straight lines $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3}$ and $\frac{x-1}{2} = \frac{y-2}{2} = \frac{z-3}{-2}$ are
 (a) parallel lines (b) Intersecting at 60°
 (c) Skew lines (d) Intersecting at right angle
62. The equation of the plane which bisects the angle between the planes $3x - 6y + 2z + 5 = 0$ and $4x - 12y + 3z - 3 = 0$ which contains the origin is
 (a) $33x - 13y + 32z + 45 = 0$
 (b) $x - 3y + z - 5 = 0$
 (c) $33x + 13y + 32z + 45 = 0$
 (d) None of these
63. P is fixed point (a, a, a) on a line through the origin equally inclined to the axes, then any plane through P perpendicular to OP, makes intercepts on the axes, the sum of whose reciprocals is equal to
 (a) a (b) a/2
 (c) 3a/2 (d) 1/a
64. If $\vec{a}_1, \vec{a}_2$ and $\vec{a}_3$ are non-coplanar vectors and $(x + y - 3) \vec{a}_1 + (2x - y + 2) \vec{a}_2 + (2x + y + \lambda) \vec{a}_3 = \vec{0}$ holds for some 'x' and 'y' then 'λ' is
 (a) $\frac{7}{3}$ (b) 2
 (c) $-\frac{10}{3}$ (d) $\frac{5}{3}$
65. If the foot of the perpendicular from the origin to a plane is P (a, b, c), the equation of the plane is
 (a) $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 3$
 (b) $ax + by + cz = 3$
 (c) $ax + by + cz = a^2 + b^2 + c^2$
 (d) $ax + by + cz = a + b + c$
66. Equation of line in the plane P : $2x - y + z - 4 = 0$ which is perpendicular to the line l whose equation is $\frac{x-2}{1} = \frac{y-2}{-1} = \frac{z-3}{-2}$ and which passes through the point of intersection of l and P is
 (a) $\frac{x-2}{3} = \frac{y-1}{5} = \frac{z-1}{-1}$ (b) $\frac{x-1}{3} = \frac{y-3}{5} = \frac{z-5}{-1}$
 (c) $\frac{x+2}{2} = \frac{y+1}{-1} = \frac{z+1}{1}$ (d) $\frac{x-2}{2} = \frac{y-1}{-1} = \frac{z-1}{1}$
67. Equation of plane which passes through the point of intersection of lines $\frac{x-1}{3} = \frac{y-2}{1} = \frac{z-3}{2}$ and $\frac{x-3}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ and at greatest distance from the point (0, 0, 0) is :
 (a) $4x + 3y + 5z = 25$ (b) $4x + 3y + 5z = 50$
 (c) $3x + 4y + 5z = 49$ (d) $x + 7y - 5z = 2$



68. Let $A(1, 1, 1)$, $B(2, 3, 5)$, $C(-1, 0, 2)$ be three points, then equation of a plane parallel to the plane ABC which is at distance 2 from plane ABC

- (a) $2x - 3y + z + 2\sqrt{14} = 0$
 (b) $2x - 3y + z - \sqrt{14} = 0$
 (c) $2x - 3y + z + 2 = 0$
 (d) $2x - 3y + z - 2 = 0$

Numerical Value Type Questions

69. If equation of the plane through the straight line

$$\frac{x-1}{2} = \frac{y+2}{-3} = \frac{z}{5} \text{ and perpendicular to the plane}$$

$x - y + z + 2 = 0$ is $ax - by + cz + 4 = 0$, then find the value of $10^3a + 10^2b + 10c$.

70. In a regular tetrahedron let θ be the angle between any

edge and a face not containing the edge. If $\cos^2\theta = \frac{a}{b}$

where $a, b \in \mathbb{I}^+$ also a and b are coprime, then find the value of $(10a + b)$

71. If the planes $x - cy - bz = 0$, $cx - y + az = 0$ and $bx + ay - z = 0$ pass through a straight line, then find the value of $a^2 + b^2 + c^2 + 2abc$.

72. If the reflection of the point $P(1, 0, 0)$ in the line

$$\frac{x-1}{2} = \frac{y+1}{-3} = \frac{z+10}{8} \text{ is } (\alpha, \beta, \gamma). \text{ Find } -(\alpha + \beta + \gamma)$$

Assertion & Reason

Use the following codes to answer the questions

- (A) If both assertion and reason are correct and reason is the correct explanation of assertion.
 (B) If both assertion and reason are true but reason is not the correct explanation of assertion.
 (C) If assertion is true but reason is false.
 (D) If assertion is false but reason is true.
 (E) If assertion and reason are both false.

73. **Assertion :** If $\vec{a} = 3\hat{i} + \hat{k}$, $\vec{b} = -\hat{i} + 2\hat{j} + \hat{k}$, $\vec{c} = \hat{i} + \hat{j} + \hat{k}$ and

$\vec{d} = 2\hat{i} - \hat{j}$, then there exist real numbers α, β, γ such that

$$\vec{a} = \alpha\vec{b} + \beta\vec{c} + \gamma\vec{d}$$

Reason : $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are four vectors in a 3-dimensional

space. If $\vec{b}, \vec{c}, \vec{d}$ are non-coplanar, then there exist real

numbers α, β, γ such that $\vec{a} = \alpha\vec{b} + \beta\vec{c} + \gamma\vec{d}$

- (a) A (b) B
 (c) C (d) D
 (e) E

74. **Assertion :** Let $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ are position vectors of four

points A, B, C and D and $3\vec{a} - 2\vec{b} + 5\vec{c} - 6\vec{d} = \vec{0}$, then points

A, B, C and D are coplanar.

Reason : Three non zero, linearly dependent co-initial

vectors $(\vec{PQ}, \vec{PR} \text{ and } \vec{PS})$ are coplanar.

- (a) A (b) B
 (c) C (d) D
 (e) E

75. **Assertion :** A point on the straight line $2x + 3y - 4z = 5$,

$3x - 2y + 4z = 7$ can be determined by taking $x = k$ and then solving the two equations for y and z , where k is any real number except $12/5$.

Reason : If $c' \neq kc$, then the straight line $ax + by + cz + d = 0$, $kax + kby + c'z + d' = 0$, does not intersect the plane $z = \alpha$, where α is any real number except

$$\frac{d' - kd}{kc - c'}$$

- (a) A (b) B
 (c) C (d) D
 (e) E



Match the Following

Each question has two columns. Four options are given representing matching of elements from Column-I and Column-II. Only one of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.

76. **Column-I** **Column-II**
- (A) Foot of perp. drawn for point (1, 2, 3) to the line $\frac{x-2}{2} = \frac{y-1}{3} = \frac{z-2}{4}$ is
- (B) Image of line point (1, 2, 3) in the line $\frac{x-2}{2} = \frac{y-1}{3} = \frac{z-2}{4}$ is
- (C) Foot of perpendicular from the point (2, 3, 5) to the plane $2x + 3y - 4z + 17 = 0$ is
- (D) Image of the point (2, 5, 1) in the plane $3x - 2y + 4z - 5 = 0$ is
- (P) $\left(\frac{107}{29}, \frac{30}{29}, \frac{69}{29}\right)$
- (Q) $\left(\frac{88}{29}, \frac{125}{29}, \frac{69}{29}\right)$
- (R) $\left(\frac{68}{29}, \frac{44}{29}, \frac{78}{29}\right)$
- (S) $\left(\frac{38}{29}, \frac{57}{29}, \frac{185}{29}\right)$

The correct matching is

- (a) A-R; B-P; C-S; D-Q
- (b) A-R; B-S; C-P; D-Q
- (c) A-R; B-P; C-Q; D-S
- (d) A-P; B-R; C-S; D-Q

77. Consider three planes

$$P_1: 2x + y + z = 1$$

$$P_2: x - y + z = 2$$

$$P_3: \alpha x + y + 3z = 5$$

The three planes intersect each other at point P on XOY plane and at point Q on YOZ plane. O is the origin.

- | Column-I | Column-II |
|---|-----------|
| (A) The value of α is | (P) 1 |
| (B) The length of projection of PQ on x-axis is | (Q) 2 |
| (C) If the coordinates of point R on the line PQ situated at minimum distance from point 'O' are (a, b, c), then the value of $7a + 14b + 14c$ is | (R) 4 |
| (D) If the area of ΔPOQ is $\sqrt{\frac{a}{b}}$ | (S) 3 |
- Then the value of $a - b$ is where a & b are co prime numbers

The correct matching is

- (a) A $\rightarrow$ S ; B $\rightarrow$ P ; C $\rightarrow$ Q ; D $\rightarrow$ R
- (b) A $\rightarrow$ P ; B $\rightarrow$ R ; C $\rightarrow$ Q ; D $\rightarrow$ S
- (c) A $\rightarrow$ R ; B $\rightarrow$ Q ; C $\rightarrow$ P ; D $\rightarrow$ S
- (d) A $\rightarrow$ R ; B $\rightarrow$ P ; C $\rightarrow$ Q ; D $\rightarrow$ S



Paragraph Type Questions

Using the following passage, solve Q. 78 to Q. 80

Passage

Let $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ be two planes, where $d_1, d_2 > 0$. Then origin lies in acute angle if $a_1a_2 + b_1b_2 + c_1c_2 < 0$ and origin lies in obtuse angle if $a_1a_2 + b_1b_2 + c_1c_2 > 0$.

Further point (x_1, y_1, z_1) and origin both lie either in acute angle or in obtuse angle, if $(a_1x_1 + b_1y_1 + c_1z_1 + d_1)(a_2x_1 + b_2y_1 + c_2z_1 + d_2) > 0$. One of (x_1, y_1, z_1) and origin lie in acute angle and the other in obtuse angle, if

$$(a_1x_1 + b_1y_1 + c_1z_1 + d_1)(a_2x_1 + b_2y_1 + c_2z_1 + d_2) < 0$$

78. Given the planes $2x + 3y - 4z + 7 = 0$ and $x - 2y + 3z - 5 = 0$, if a point P is $(1, -2, 3)$, then
- (a) O and P both lie in acute angle between the planes
 - (b) O and P both lie in obtuse angle
 - (c) O lies in acute angle, P lies in obtuse angle
 - (d) O lies in obtuse angle, P lies in acute angle.
79. Given the planes $x + 2y - 3z + 5 = 0$ and $2x + y + 3z + 1 = 0$. If a point P is $(2, -1, 2)$, then
- (a) O and P both lie in acute angle between the planes
 - (b) O and P both lie in obtuse angle
 - (c) O lies in acute angle, P lies in obtuse angle
 - (d) O lies in obtuse angle, P lies in acute angle.
80. Given the planes $x + 2y - 3z + 2 = 0$ and $x - 2y + 3z + 7 = 0$, if the point P is $(1, 2, 2)$, then
- (a) O and P both lie in acute angle between the planes
 - (b) O and P both lie in obtuse angle
 - (c) O lies in acute angle, P lies in obtuse angle
 - (d) O lies in obtuse angle, P lies in acute angle.



EXERCISE - 4 : PREVIOUS YEAR JEE ADVANCED QUESTIONS

VECTORS

Objective Question I [Only one correct option]

1. If the vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ form the sides BC, CA and AB respectively of a triangle ABC, then : (2000)
- (a) $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = 0$ (b) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$
- (c) $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a}$ (d) $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a} = 0$
2. If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are unit coplanar vectors, then the scalar triple product $\left[(2\vec{a} - \vec{b})(2\vec{b} - \vec{c})(2\vec{c} - \vec{a}) \right]$ is equal to : (2000)
- (a) 0 (b) 1
- (c) $-\sqrt{3}$ (d) $\sqrt{3}$
3. If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are unit vectors, then $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2$ does not exceed : (2001)
- (a) 4 (b) 9
- (c) 8 (d) 6
4. Let $\vec{a} = \hat{i} - \hat{k}$, $\vec{b} = x\hat{i} + \hat{j} + (1-x)\hat{k}$ and $\vec{c} = y\hat{i} + x\hat{j} + (1+x-y)\hat{k}$. Then $[\vec{a} \vec{b} \vec{c}]$ depends on : (2001)
- (a) only x (b) only y
- (c) neither x nor y (d) both x and y
5. If $\vec{a}$ and $\vec{b}$ are two unit vectors such that $\vec{a} + 2\vec{b}$ and $5\vec{a} - 4\vec{b}$ are perpendicular to each other, then the angle between $\vec{a}$ and $\vec{b}$ is : (2002)
- (a) 45° (b) 60°
- (c) $\cos^{-1}\left(\frac{1}{3}\right)$ (d) $\cos^{-1}\left(\frac{2}{7}\right)$
6. Let $\vec{V} = 2\hat{i} + \hat{j} - \hat{k}$ and $\vec{W} = \hat{i} + 3\hat{k}$. If $\vec{U}$ is a unit vector, then the maximum value of the scalar triple product $[\vec{U} \vec{V} \vec{W}]$ is : (2002)
- (a) -1 (b) $\sqrt{10} + \sqrt{6}$
- (c) $\sqrt{59}$ (d) $\sqrt{60}$
7. If $\vec{a} = (\hat{i} + \hat{j} + \hat{k})$, $\vec{a} \cdot \vec{b} = 1$ and $\vec{a} \times \vec{b} = \hat{j} - \hat{k}$, then $\vec{b}$ is : (2003)
- (a) $\hat{i} - \hat{j} + \hat{k}$ (b) $2\hat{j} - \hat{k}$
- (c) $\hat{i}$ (d) $2\hat{i}$
8. If $\vec{a}$, $\vec{b}$, $\vec{c}$ are three non-zero, non-coplanar vectors and
- $$\vec{b}_1 = \vec{b} - \frac{\vec{b} \cdot \vec{a}}{|\vec{a}|^2} \vec{a}, \quad \vec{b}_2 = \vec{b} + \frac{\vec{b} \cdot \vec{a}}{|\vec{a}|^2} \vec{a},$$
- $$\vec{c}_1 = \vec{c} - \frac{\vec{c} \cdot \vec{a}}{|\vec{a}|^2} \vec{a} - \frac{\vec{c} \cdot \vec{b}}{|\vec{b}|^2} \vec{b}, \quad \vec{c}_2 = \vec{c} - \frac{\vec{c} \cdot \vec{a}}{|\vec{a}|^2} \vec{a} - \frac{\vec{c} \cdot \vec{b}_1}{|\vec{b}_1|^2} \vec{b}_1,$$
- $$\vec{c}_3 = \vec{c} - \frac{\vec{c} \cdot \vec{a}}{|\vec{a}|^2} \vec{a} - \frac{\vec{c} \cdot \vec{b}_2}{|\vec{b}_2|^2} \vec{b}_2, \quad \vec{c}_4 = \vec{a} - \frac{\vec{c} \cdot \vec{a}}{|\vec{a}|^2} \vec{a}.$$
- Then which of the following is a set of mutually orthogonal vectors ? (2005)
- (a) $\{\vec{a}, \vec{b}_1, \vec{c}_1\}$ (b) $\{\vec{a}, \vec{b}_1, \vec{c}_2\}$
- (c) $\{\vec{a}, \vec{b}_2, \vec{c}_3\}$ (d) $\{\vec{a}, \vec{b}_2, \vec{c}_4\}$
9. Let $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$, $\vec{c} = \hat{i} + \hat{j} - \hat{k}$. A vector coplanar to $\vec{a}$ and $\vec{b}$ has a projection along $\vec{c}$ of magnitude $\frac{1}{\sqrt{3}}$, then the vector is : (2006)
- (a) $4\hat{i} - \hat{j} + 4\hat{k}$ (b) $4\hat{i} + \hat{j} - 4\hat{k}$
- (c) $2\hat{i} + \hat{j} + \hat{k}$ (d) none of these



10. Let $\vec{a}, \vec{b}, \vec{c}$ be unit vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$. Which one of the following is correct ? (2007)
- (a) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a} = \vec{0}$
- (b) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a} \neq \vec{0}$
- (c) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{a} \times \vec{c} \neq \vec{0}$
- (d) $\vec{a} \times \vec{b}, \vec{b} \times \vec{c}, \vec{c} \times \vec{a}$ are mutually perpendicular
11. Let two non-collinear unit vectors $\hat{a}$ and $\hat{b}$ form an acute angle. A point P moves so that at any time t the position vector $\vec{OP}$ (where O is the origin) is given by $\hat{a} \cos t + \hat{b} \sin t$. When P is farthest from origin O, let M be the length of $\vec{OP}$ and $\hat{u}$ be the unit vector along $\vec{OP}$. Then, (2008)
- (a) $\hat{u} = \frac{\hat{a} + \hat{b}}{|\hat{a} + \hat{b}|}$ and $M = (1 + \hat{a} \cdot \hat{b})^{1/2}$
- (b) $\hat{u} = \frac{\hat{a} - \hat{b}}{|\hat{a} - \hat{b}|}$ and $M = (1 + \hat{a} \cdot \hat{b})^{1/2}$
- (c) $\hat{u} = \frac{\hat{a} + \hat{b}}{|\hat{a} + \hat{b}|}$ and $M = (1 + 2\hat{a} \cdot \hat{b})^{1/2}$
- (d) $\hat{u} = \frac{\hat{a} - \hat{b}}{|\hat{a} - \hat{b}|}$ and $M = (1 + 2\hat{a} \cdot \hat{b})^{1/2}$
12. If $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ are the unit vectors such that $(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = 1$ and $\vec{a} \cdot \vec{c} = \frac{1}{2}$, then (2009)
- (a) $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar
- (b) $\vec{a}, \vec{b}, \vec{d}$ are non-coplanar
- (c) $\vec{b}, \vec{d}$ are non-parallel
- (d) $\vec{a}, \vec{d}$ are parallel and $\vec{b}, \vec{c}$ are parallel
13. Let P, Q, R and S be the points on the plane with position vectors $-2\hat{i} - \hat{j}$, $4\hat{i}$, $3\hat{i} + 3\hat{j}$ and $-3\hat{i} + 2\hat{j}$ respectively. The quadrilateral PQRS must be a (2010)
- (a) parallelogram, which is neither a rhombus nor a rectangle
- (b) square
- (c) rectangle, but not a square
- (d) rhombus, but not a square
14. Two adjacent sides of a parallelogram ABCD are given by $\vec{AB} = 2\hat{i} + 10\hat{j} + 11\hat{k}$ and $\vec{AD} = -\hat{i} + 2\hat{j} + 2\hat{k}$. The side AD is rotated by an acute angle α in the plane of the parallelogram so that AD becomes AD'. If AD' makes a right angle with the side AB, then the cosine of the angle α is given by (2010)
- (a) $\frac{8}{9}$ (b) $\frac{\sqrt{17}}{9}$
- (c) $\frac{1}{9}$ (d) $\frac{4\sqrt{5}}{9}$
15. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$ and $\vec{c} = \hat{i} - \hat{j} - \hat{k}$ be three vectors. A vector $\vec{v}$ in the plane of $\vec{a}$ and $\vec{b}$, whose projection on $\vec{c}$ is $\frac{1}{\sqrt{3}}$, is given by (2011)
- (a) $\hat{i} - 3\hat{j} + 3\hat{k}$ (b) $-3\hat{i} - 3\hat{j} - \hat{k}$
- (c) $3\hat{i} - \hat{j} + 3\hat{k}$ (d) $\hat{i} + 3\hat{j} - 3\hat{k}$
16. If $\vec{a}$ and $\vec{b}$ are vectors such that $|\vec{a} + \vec{b}| = \sqrt{29}$ and $\vec{a} \times (2\hat{i} + 3\hat{j} + 4\hat{k}) = (2\hat{i} + 3\hat{j} + 4\hat{k}) \times \vec{b}$, then a possible value of $(\vec{a} + \vec{b}) \cdot (-7\hat{i} + 2\hat{j} + 3\hat{k})$ is (2012)
- (a) 0 (b) 3
- (c) 4 (d) 8
17. The equation of the plane passing through the point (1, 1, 1) and perpendicular to the planes $2x + y - 2z = 5$ and $3x - 6y - 2z = 7$, is (2017)
- (a) $-14x + 2y + 15z = 3$ (b) $14x - 2y + 15z = 27$
- (c) $14x + 2y - 15z = 1$ (d) $14x + 2y + 15z = 31$



18. Let O be the origin and let PQR be an arbitrary triangle. The point S is such that
 $\overrightarrow{OP} \cdot \overrightarrow{OQ} + \overrightarrow{OR} \cdot \overrightarrow{OS} = \overrightarrow{OR} \cdot \overrightarrow{OP} + \overrightarrow{OQ} \cdot \overrightarrow{OS} = \overrightarrow{OQ} \cdot \overrightarrow{OR} + \overrightarrow{OP} \cdot \overrightarrow{OS}$
 Then the triangle PQR has S as its (2017)
 (a) incentre (b) circumcentre
 (c) orthocenter (d) centroid
19. Let O be the origin, and $\overrightarrow{OX}, \overrightarrow{OY}, \overrightarrow{OZ}$ be three unit vectors in the directions of the sides $\overrightarrow{QR}, \overrightarrow{RP}, \overrightarrow{PQ}$, respectively, of a triangle PQR. Then (2017)
 $|\overrightarrow{OX} \times \overrightarrow{OY}| =$
 (a) $\sin(P+R)$ (b) $\sin 2R$
 (c) $\sin(P+Q)$ (d) $\sin(Q+R)$
20. If the triangle PQR varies, then the minimum value of $\cos(P+Q) + \cos(Q+R) + \cos(R+P)$ (2017)
 (a) $-\frac{3}{2}$ (b) $\frac{3}{2}$
 (c) $\frac{5}{3}$ (d) $-\frac{5}{3}$
23. A line l passing through the origin is perpendicular to the lines
 $l_1: (3+t)\hat{i} + (-1+2t)\hat{j} + (4+2t)\hat{k}, -\infty < t < \infty$
 $l_2: (3+2s)\hat{i} + (3+2s)\hat{j} + (2+s)\hat{k}, -\infty < s < \infty$
 Then, the coordinate(s) of the point(s) on l_2 at a distance of $\sqrt{17}$ from the point of intersection of l and l_1 are (2013)
 (a) $\left(\frac{7}{3}, \frac{7}{3}, \frac{5}{3}\right)$ (b) $(-1, -1, 0)$
 (c) $(1, 1, 1)$ (d) $\left(\frac{7}{9}, \frac{7}{9}, \frac{8}{9}\right)$
24. Let $\vec{x}, \vec{y}$ and $\vec{z}$ be three vectors each of magnitude $\sqrt{2}$ and the angle between each pair of them is $\frac{\pi}{3}$. If $\vec{a}$ is a non-zero vector perpendicular to $\vec{x}$ and $\vec{y} \times \vec{z}$ and $\vec{b}$ is a non-zero vector perpendicular to $\vec{y}$ and $\vec{z} \times \vec{x}$, then (2014)
 (a) $\vec{b} = (\vec{b} \cdot \vec{z})(\vec{z} - \vec{x})$
 (b) $\vec{a} = (\vec{a} \cdot \vec{y})(\vec{y} - \vec{z})$
 (c) $\vec{a} \cdot \vec{b} = -(\vec{a} \cdot \vec{y})(\vec{b} \cdot \vec{z})$
 (d) $\vec{a} = (\vec{a} \cdot \vec{y})(\vec{z} - \vec{y})$

Objective Questions II [One or more than one correct option]

21. Let $\vec{A}$ be a vector parallel to line of intersection of planes P_1 and P_2 through the origin. P_1 is parallel to the vectors $2\hat{j} + 3\hat{k}$ and $4\hat{j} - 3\hat{k}$ and P_2 is parallel to $\hat{j} - \hat{k}$ and $3\hat{i} + 3\hat{j}$, then the angle between vector $\vec{A}$ and $2\hat{i} + \hat{j} - 2\hat{k}$ is (2006)
 (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$
 (c) $\frac{\pi}{6}$ (d) $\frac{3\pi}{4}$
22. The vector(s) which is/are coplanar with vectors $\hat{i} + \hat{j} + 2\hat{k}$ and $\hat{i} + 2\hat{j} + \hat{k}$, and are perpendicular to the vector $\hat{i} + \hat{j} + \hat{k}$ is/are (2011)
 (a) $\hat{j} - \hat{k}$ (b) $-\hat{i} + \hat{j}$
 (c) $\hat{i} - \hat{j}$ (d) $-\hat{j} + \hat{k}$
25. Let ΔPQR be a triangle. Let $\vec{a} = \overrightarrow{QR}$, $\vec{b} = \overrightarrow{RP}$ and $\vec{c} = \overrightarrow{PQ}$. If $|\vec{a}| = 12$, $|\vec{b}| = 4\sqrt{3}$, $\vec{b} \cdot \vec{c} = 24$, then which of the following is (are) true? (2015)
 (a) $\frac{|\vec{c}|^2}{2} - |\vec{a}| = 12$ (b) $\frac{|\vec{c}|^2}{2} + |\vec{a}| = 30$
 (c) $|\vec{a} \times \vec{b} + \vec{c} \times \vec{a}| = 48\sqrt{3}$ (d) $\vec{a} \cdot \vec{b} = -72$



26. Let $\hat{u} = u_1\hat{i} + u_2\hat{j} + u_3\hat{k}$ be a unit vector in R^3 and

$$\hat{w} = \frac{1}{\sqrt{6}}(\hat{i} + \hat{j} + 2\hat{k}). \text{ Given that there exists a vector } \vec{v} \text{ in}$$

R^3 such that $|\hat{u} \times \vec{v}| = 1$ and $\hat{w} \cdot |\hat{u} \times \vec{v}| = 1$. Which of the following statement(s) is(are) correct? (2016)

- (a) There is exactly one choice for such $\vec{v}$
- (b) There are infinitely many choices for such
- (c) If $\hat{u}$ lies in the xy-plane then $|u_1| = |u_2|$
- (d) If $\hat{u}$ lies in the xz-plane then $2|u_1| = |u_3|$

27. Let L_1 and L_2 be the following straight line.

$$L_1 : \frac{x-1}{1} = \frac{y}{-1} = \frac{z-1}{3} \text{ and } L_2 : \frac{x-1}{-3} = \frac{y}{-1} = \frac{z-1}{1}$$

Suppose the straight line

$$L : \frac{x-\alpha}{l} = \frac{y-1}{m} = \frac{z-\gamma}{-2}$$

lies in the plane containing L_1 and L_2 , and passes through the point of intersection of L_1 and L_2 . If the line L bisects the acute angle between the lines L_1 and L_2 , then which of the following statements is/are TRUE? (2020)

- (a) $\alpha - \gamma = 3$
- (b) $l + m = 2$
- (c) $\alpha - \gamma = 1$
- (d) $l + m = 0$

28. Let a and b be positive real numbers. Suppose

$\overrightarrow{PQ} = a\hat{i} + b\hat{j}$ and $\overrightarrow{PS} = a\hat{i} - b\hat{j}$ are adjacent sides of a parallelogram PQRS. Let $\vec{u}$ and $\vec{v}$ be the projection vectors of $\vec{w} = \hat{i} + \hat{j}$ along $\overrightarrow{PQ}$ and $\overrightarrow{PS}$, respectively. If

$|\vec{u}| + |\vec{v}| = |\vec{w}|$ and if the area of the parallelogram PQRS is 8, then which of the following statements is/are TRUE? (2020)

- (a) $a + b = 4$
- (b) $a - b = 2$
- (c) The length of the diagonal PQ of the parallelogram PQRS is 4
- (d) $\vec{w}$ is an angle bisector of the vectors $\overrightarrow{PQ}$ and $\overrightarrow{PS}$

29. Let O be the origin and $\overrightarrow{OA} = 2\hat{i} + 2\hat{j} + \hat{k}$,

$$\overrightarrow{OB} = \hat{i} - 2\hat{j} + 2\hat{k} \text{ and } \overrightarrow{OC} = \frac{1}{2}(\overrightarrow{OB} - \lambda\overrightarrow{OA}) \text{ for some}$$

$\lambda > 0$. If $|\overrightarrow{OB} \times \overrightarrow{OC}| = \frac{9}{2}$, then which of the following statements is (are) TRUE? (2021)

(a) Projection of $\overrightarrow{OC}$ on $\overrightarrow{OA}$ is $-\frac{3}{2}$

(b) Area of the triangle OAB is $\frac{9}{2}$

(c) Area of triangle ABC is $\frac{9}{2}$

(d) The acute angle between the diagonals of the parallelogram with adjacent sides $\overrightarrow{OA}$ and $\overrightarrow{OC}$ is $\frac{\pi}{3}$

Numerical Value Type Questions

30. If $\vec{a}$ and $\vec{b}$ are vectors in space given by $\vec{a} = \frac{\hat{i} - 2\hat{j}}{\sqrt{5}}$ and

$$\vec{b} = \frac{2\hat{i} + \hat{j} + 3\hat{k}}{\sqrt{14}}, \text{ then the value of}$$

$$\left(2\vec{a} + \vec{b} \right) \cdot \left[\left(\vec{a} \times \vec{b} \right) \times \left(\vec{a} - 2\vec{b} \right) \right] \text{ is.... (2010)}$$

31. Let $\vec{a} = -\hat{i} - \hat{k}$, $\vec{b} = -\hat{i} + \hat{j}$ and $\vec{c} = \hat{i} + 2\hat{j} + 3\hat{k}$ be three given

vectors. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$ and

$$\vec{r} \cdot \vec{a} = 0, \text{ then the value of } \vec{r} \cdot \vec{b} \text{ is.... (2011)}$$

32. If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are unit vectors satisfying

$$|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 = 9, \text{ then}$$

$$|2\vec{a} + 5\vec{b} + 5\vec{c}| \text{ is (2012)}$$



33. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non-coplanar unit vectors such

that the angle between every pair of them is $\frac{\pi}{3}$. If

$\vec{a} \times \vec{b} + \vec{b} \times \vec{c} = p\vec{a} + q\vec{b} + r\vec{c}$, where p, q and r are scalars,

then the value of $\frac{p^2 + 2q^2 + r^2}{q^2}$ is (2014)

34. Let $\vec{a}$ and $\vec{b}$ be two unit vectors such that $\vec{a} \cdot \vec{b} = 0$. For

some $x, y \in \mathbb{R}$, let $\vec{c} = x\vec{a} + y\vec{b} + (\vec{a} \times \vec{b})$. If $|\vec{c}| = 2$ and the

vector $\vec{c}$ is inclined at the same angle α to both $\vec{a}$ and $\vec{b}$

and then the value of $8\cos^2\alpha$ is (2018)

35. Let P be a point in the first octant, whose image Q in the plane $x + y = 3$ lies on the z -axis (that is, the line segment PQ is perpendicular to the plane $x + y = 3$ and the mid-point of PQ lies in the plane $x + y = 3$). Let the distance of P from the x -axis be 5. If R is the image of P in the xy -plane, then the length of PR is (2018)

36. Let $\vec{a} = 2\hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$ be two vectors,

consider a vector $\vec{c} = \alpha\vec{a} + \beta\vec{b}$, $\alpha, \beta \in \mathbb{R}$. If the projection

of $\vec{c}$ on the vector $(\vec{a} + \vec{b})$ is $3\sqrt{2}$, then the minimum value

of $(\vec{c} - (\vec{a} \times \vec{b})) \cdot \vec{c}$ equals (2019)

37. In a triangle PQR , let $\vec{a} = \overrightarrow{QR}$, $\vec{b} = \overrightarrow{RP}$ and $\vec{c} = \overrightarrow{PQ}$. If

$|\vec{a}| = 3, |\vec{b}| = 4$ and $\frac{\vec{a} \cdot (\vec{c} - \vec{b})}{\vec{c} \cdot (\vec{a} - \vec{b})} = \frac{|\vec{a}|}{|\vec{a}| + |\vec{b}|}$, then the value

of $|\vec{a} \times \vec{b}|^2$ is (2020)

38. Let α, β and γ be real numbers such that the system of linear equations

$$x + 2y + 3z = \alpha$$

$$4x + 5y + 6z = \beta$$

$$7x + 8y + 9z = \gamma - 1$$

is consistent. Let $|M|$ represent the determinant of the matrix

$$M = \begin{bmatrix} \alpha & 2 & \gamma \\ \beta & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

Let P be the plane containing all those (α, β, γ) for which the above system of linear equations is consistent, and D be the square of the distance of the point $(0, 1, 0)$ from the plane P . The value of D is (2021)

39. Let $\vec{u}, \vec{v}$ and $\vec{w}$ be vectors in three-dimensional space, where $\vec{u}$ and $\vec{v}$ are unit vectors which are not perpendicular to each other and $\vec{u} \cdot \vec{w} = 1, \vec{v} \cdot \vec{w} = 1, \vec{w} \cdot \vec{w} = 4$. If the volume of the parallelepiped, whose adjacent sides are represented by the vectors $\vec{u}, \vec{v}$ and $\vec{w}$, is $\sqrt{2}$, then the value of $|3\vec{u} + 5\vec{v}|$ is (2021)

Assertion & Reason

For the following questions choose the correct answer from the codes (a), (b), (c) and (d) defined as follows.

- (a) Statement I is true, Statement II is also true; Statement II is the correct explanation of Statement I.
(b) Statement I is true, Statement II is also true; Statement II is not the correct explanation of Statement I.
(c) Statement I is true, Statement II is false.
(d) Statement I is false, Statement II is true.

40. Let the vectors $\overrightarrow{PQ}, \overrightarrow{QR}, \overrightarrow{RS}, \overrightarrow{ST}, \overrightarrow{TU}$ and $\overrightarrow{UP}$ represent the sides of a regular hexagon.

Assertion : $\overrightarrow{PQ} \times (\overrightarrow{RS} + \overrightarrow{ST}) \neq \vec{0}$.

Reason : $\overrightarrow{PQ} \times \overrightarrow{RS} = \vec{0}$ and $\overrightarrow{PQ} \times \overrightarrow{ST} \neq \vec{0}$. (2007)

- (a) A (b) B
(c) C (d) D



Match the Following

Each question has two columns. Four options are given representing matching of elements from Column-I and Column-II. Only one of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.

41. Match List-I with List-II and select the correct answer using the code given below the lists.

List-I

List-II

P. Volume of parallelopiped determined by

1. 100

vectors $\vec{a}$, $\vec{b}$ & $\vec{c}$ is 2. Then, the volume of the parallelopiped determined by vectors

$2(\vec{a} \times \vec{b})$, $3(\vec{b} \times \vec{c})$ and $(\vec{c} \times \vec{a})$ is

Q. Volume of parallelopiped determined by

2. 30

vectors $\vec{a}$, $\vec{b}$ & $\vec{c}$ is 5. Then, the volume of the parallelopiped determined by vectors

$3(\vec{a} + \vec{b})$, $(\vec{b} + \vec{c})$ and $2(\vec{c} + \vec{a})$ is

R. Area of a triangle with adjacent sides

3. 24

determined by vector $\vec{a}$ and $\vec{b}$ is 20. Then, the area of the triangle with adjacent sides determined by vectors

$(2\vec{a} + 3\vec{b})$ and $(\vec{a} - \vec{b})$ is

S. Area of a parallelogram with adjacent

4. 60

sides determined by vectors $\vec{a}$ and $\vec{b}$ is

30. Then, the area of the parallelogram with adjacent sides determined by

vectors $(\vec{a} + \vec{b})$ and $\vec{a}$ is

(2015)

P Q R S P Q R S

(a) 4 2 3 1 (b) 2 3 1 4

(c) 3 4 1 2 (d) 1 4 3 2

Text

42. If $\vec{u}, \vec{v}, \vec{w}$ are three non-coplanar unit vectors and α, β, γ are the angles between $\vec{u}$ and $\vec{v}, \vec{v}$ and $\vec{w}, \vec{w}$ and $\vec{u}$ respectively and $\vec{x}, \vec{y}, \vec{z}$ are unit vectors along the bisectors of the angles α, β, γ respectively. Prove that

$$[\vec{x} \times \vec{y} \quad \vec{y} \times \vec{z} \quad \vec{z} \times \vec{x}] = \frac{1}{16} [\vec{u} \quad \vec{v} \quad \vec{w}]^2 \sec^2 \frac{\alpha}{2} \sec^2 \frac{\beta}{2} \sec^2 \frac{\gamma}{2}. \quad (2003)$$

43. A plane is parallel to two lines whose direction ratios are $(1, 0, -1)$ & $(-1, 1, 0)$ and it contains the point $(1, 1, 1)$. If it cuts coordinate axes at A, B, C. Then find the volume of the tetrahedron OABC. (2004)

44. If $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are four distinct vectors satisfying the conditions $\vec{a} \times \vec{b} = \vec{c} \times \vec{d}$ and $\vec{a} \times \vec{c} = \vec{b} \times \vec{d}$, then prove that $\vec{a} \cdot \vec{b} + \vec{c} \cdot \vec{d} \neq \vec{a} \cdot \vec{c} + \vec{b} \cdot \vec{d}$. (2004)

45. Consider the cube in the first octant with sides OP, OQ and OR of length 1, along the x-axis, y-axis and z-axis, respectively, where O(0, 0, 0) is the origin. Let

$S\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$ be the centre of the cube and T be the vertex

of the cube opposite to the origin O such that S lies on the diagonal OT. If $\vec{p} = \vec{SP}, \vec{q} = \vec{SQ}, \vec{r} = \vec{SR}$ and $\vec{t} = \vec{ST}$, then

the value of $\left| (\vec{p} \times \vec{q}) \times (\vec{r} \times \vec{t}) \right|$ is ____ (2018)

3-DIMENSIONAL GEOMETRY

Objective Question I [Only one correct option]

46. Let the vectors $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ be such that $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = 0$. Let P_1 and P_2 be planes determined by the pairs of vectors $\vec{a}, \vec{b}$ and $\vec{c}, \vec{d}$ respectively, then the angle between P_1 and P_2 is: (2000)

- (a) 0 (b) $\pi/4$
(c) $\pi/3$ (d) $\pi/2$



47. The positive value of 'a' so that the volume of the parallelepiped formed by $\hat{i} + a\hat{j} + \hat{k}$, $\hat{j} + a\hat{k}$ and $a\hat{i} + \hat{k}$ becomes minimum is : (2003)
 (a) 4 (b) 3
 (c) $1/\sqrt{3}$ (d) $\sqrt{3}$
48. The value of k such that $\frac{x-4}{1} = \frac{y-2}{1} = \frac{z-k}{2}$ lies in the plane $2x - 4y + z = 7$, is (2003)
 (a) 7 (b) -7
 (c) No real value (d) 4
49. The unit vector which is orthogonal to the vector $3\hat{i} + 2\hat{j} + 6\hat{k}$ and is coplanar with the vectors $2\hat{i} + \hat{j} + \hat{k}$ and $\hat{i} - \hat{j} + \hat{k}$ is : (2004)
 (a) $\frac{2\hat{i} - 6\hat{j} + \hat{k}}{\sqrt{41}}$ (b) $\frac{2\hat{i} - 3\hat{j}}{\sqrt{13}}$
 (c) $\frac{3\hat{j} - \hat{k}}{\sqrt{10}}$ (d) $\frac{4\hat{i} + 3\hat{j} - 3\hat{k}}{\sqrt{34}}$
50. If the lines $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4}$ and $\frac{x-3}{1} = \frac{y-k}{2} = \frac{z}{1}$ intersect, then the value of k is (2004)
 (a) $\frac{3}{2}$ (b) $\frac{9}{2}$
 (c) $-\frac{2}{9}$ (d) $-\frac{3}{2}$
51. A variable plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ at a unit distance from origin cuts the coordinate axes at A, B and C. Centroid (x, y, z) satisfies the equation $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = K$. Then value of K is (2005)
 (a) 9 (b) 3
 (c) $1/9$ (d) $1/3$
52. A plane passes through (1, -2, 1) and is perpendicular to two planes $2x - 2y + z = 0$ and $x - y + 2z = 4$, then the distance of the plane from the point (1, 2, 2) is (2006)
 (a) 0 (b) 1
 (c) $\sqrt{2}$ (d) $2\sqrt{2}$
53. The number of distinct real values of λ , for which the vectors $-\lambda^2\hat{i} + \hat{j} + \hat{k}$, $\hat{i} - \lambda^2\hat{j} + \hat{k}$ and $\hat{i} + \hat{j} - \lambda^2\hat{k}$ are coplanar, is : (2007)
 (a) zero (b) one
 (c) two (d) three
54. The edges of a parallelepiped are of unit length and are parallel to non-coplanar unit vectors $\hat{a}, \hat{b}, \hat{c}$ such that $\hat{a} \cdot \hat{b} = \hat{b} \cdot \hat{c} = \hat{c} \cdot \hat{a} = \frac{1}{2}$. Then the volume of the parallelepiped is (2008)
 (a) $\frac{1}{\sqrt{2}}$ cu unit (b) $\frac{1}{2\sqrt{2}}$ cu unit
 (c) $\frac{\sqrt{3}}{2}$ cu unit (d) $\frac{1}{\sqrt{3}}$ cu unit
55. Let P (3, 2, 6) be a point in space and Q be a point on the line $\vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \mu(-3\hat{i} + \hat{j} + 5\hat{k})$. Then the value of μ for which the vector $\overrightarrow{PQ}$ is parallel to the plane $x - 4y + 3z = 1$ is (2009)
 (a) $\frac{1}{4}$ (b) $-\frac{1}{4}$
 (c) $\frac{1}{8}$ (d) $-\frac{1}{8}$
56. The point P is the intersection of the straight line joining the points Q(2, 3, 5) and R(1, -1, 4) with the plane $5x - 4y - z = 1$. If S is the foot of the perpendicular drawn from the point T (2, 1, 4) to QR, then the length of the line segment PS is (2012)
 (a) $\frac{1}{\sqrt{2}}$ (b) $\sqrt{2}$
 (c) 2 (d) $2\sqrt{2}$
57. The equation of a plane passing through the line of intersection of the planes $x + 2y + 3z = 2$ & $x - y + z = 3$ & at a distance $\frac{2}{\sqrt{3}}$ from the point (3, 1, -1) is (2012)
 (a) $5x - 11y + z = 17$ (b) $\sqrt{2}x + y = 3\sqrt{2} - 1$
 (c) $x + y + z = \sqrt{3}$ (d) $x - \sqrt{2}y = 1 - \sqrt{2}$



58. Let $\overrightarrow{PR} = 3\hat{i} + \hat{j} - 2\hat{k}$ and $\overrightarrow{SQ} = \hat{i} - 3\hat{j} - 4\hat{k}$ determine diagonals of a parallelogram PQRS and $\overrightarrow{PT} = \hat{i} + 2\hat{j} + 3\hat{k}$ be another vector. Then, the volume of the parallelepiped determined by the vectors $\overrightarrow{PT}$, $\overrightarrow{PQ}$ and $\overrightarrow{PS}$ is (2013)
- (a) 5 (b) 20
(c) 10 (d) 30
59. Perpendiculars are drawn from points on the line $\frac{x+2}{2} = \frac{y+1}{-1} = \frac{z}{3}$ to the plane $x + y + z = 3$. The feet of perpendiculars lie on the line. (2013)
- (a) $\frac{x}{5} = \frac{y-1}{8} = \frac{z-2}{-13}$ (b) $\frac{x}{2} = \frac{y-1}{3} = \frac{z-2}{-5}$
(c) $\frac{x}{4} = \frac{y-1}{3} = \frac{z-2}{-7}$ (d) $\frac{x}{2} = \frac{y-1}{-7} = \frac{z-2}{5}$
60. From a point $P(\lambda, \lambda, \lambda)$, perpendiculars PQ and PR are drawn respectively on the lines $y = x, z = 1$ and $y = -x, z = -1$. If P is such that $\angle QPR$ is a right angle, then the possible value(s) of λ is(are) (2014)
- (a) $\sqrt{2}$ (b) 1
(c) -1 (d) $-\sqrt{2}$
61. Let P be the image of the point (3, 1, 7) with respect to the plane $x - y + z = 3$. Then the equation of the plane passing through P and containing the straight line $\frac{x}{1} = \frac{y}{2} = \frac{z}{1}$ is (2016)
- (a) $x + y - 3z = 0$ (b) $3x + z = 0$
(c) $x - 4y + 7z = 0$ (d) $2x - y = 0$
62. If the straight lines $\frac{x-1}{2} = \frac{y+1}{K} = \frac{z}{2}$ and $\frac{x+1}{5} = \frac{y+1}{2} = \frac{z}{K}$ are coplanar, then the plane(s) containing these two lines is/are (2012)
- (a) $y + 2z = -1$ (b) $y + z = -1$
(c) $y - z = -1$ (d) $y - 2z = -1$
63. Two lines $L_1 : x = 5, \frac{y}{3-\alpha} = \frac{z}{-2}$ and $L_2 : x = \alpha, \frac{y}{-1} = \frac{z}{2-\alpha}$ are coplanar. Then, α can take value(s). (2013)
- (a) 1 (b) 2
(c) 3 (d) 4
64. In R^3 , consider the planes $P_1 : y = 0$ and $P_2 : x + z = 1$. Let P_3 be the plane, different from P_1 and P_2 , which passes through the intersection of P_1 and P_2 . If the distance of the point (0, 1, 0) from P_3 is 1 and the distance of a point (α, β, γ) from P_3 is 2, then which of the following relations is (are) true? (2015)
- (a) $2\alpha + \beta + 2\gamma + 2 = 0$ (b) $2\alpha - \beta + 2\gamma + 4 = 0$
(c) $2\alpha + \beta - 2\gamma - 10 = 0$ (d) $2\alpha - \beta + 2\gamma - 8 = 0$
65. In R^3 , let L be a straight line passing through the origin. Suppose that all the points on L at a constant distance from the two planes $P_1 : x + 2y - z + 1 = 0$ & $P_2 : 2x - y + z - 1 = 0$. Let M be the locus of the foot of the perpendiculars drawn from the points on L to plane P_1 . Which of the following point(s) lie(s) on M. (2015)
- (a) $\left(0, -\frac{5}{6}, -\frac{2}{3}\right)$ (b) $\left(-\frac{1}{6}, -\frac{1}{3}, \frac{1}{6}\right)$
(c) $\left(-\frac{5}{6}, 0, \frac{1}{6}\right)$ (d) $\left(-\frac{1}{3}, 0, \frac{2}{3}\right)$
66. Consider a pyramid OPQRS located in the first octant ($x \geq 0, y \geq 0, z \geq 0$) with O as origin, and OP and OR along the x-axis and the y-axis, respectively. The base OPQR of the pyramid is a square with $OP = 3$. The point S is directly above the mid-point T of diagonal OQ such that $TS = 3$. Then (2016)
- (a) the acute angle between OQ and OS is $\frac{\pi}{3}$
(b) the equation of the plane containing the triangle OQS is $x - y = 0$
(c) the length of the perpendicular from P to the plane containing the triangle OQS is $\frac{3}{\sqrt{2}}$
(d) the perpendicular distance from O to the straight line containing RS is $\sqrt{\frac{15}{2}}$

Objective Questions II [One or more than one correct option]

62. If the straight lines $\frac{x-1}{2} = \frac{y+1}{K} = \frac{z}{2}$ and $\frac{x+1}{5} = \frac{y+1}{2} = \frac{z}{K}$ are coplanar, then the plane(s) containing these two lines is/are (2012)
- (a) $y + 2z = -1$ (b) $y + z = -1$
(c) $y - z = -1$ (d) $y - 2z = -1$



67. Let $P_1 : 2x + y - z = 3$ and $P_2 : x + 2y + z = 2$ be two planes. Then, which of the following statements(s) is (are) TRUE?

(2018)

- (a) The line of intersection of P_1 and P_2 has direction ratios 1, 2, -1

- (b) The line $\frac{3x-4}{9} = \frac{1-3y}{9} = \frac{z}{3}$ is perpendicular to the

line of intersection of P_1 and P_2

- (c) The acute angle between P_1 and P_2 is 60°

- (d) If P_3 is the plane passing through the point (4, 2, -2) and perpendicular to the line of intersection of P_1 and P_2 , then the distance of the point (2, 1, 1)

from the plane P_3 is $\frac{2}{\sqrt{3}}$.

68. Let L_1 and L_2 denote the lines

$$\vec{r} = \hat{i} + \lambda(-\hat{i} + 2\hat{j} + 2\hat{k}), \lambda \in R \text{ and}$$

$$\vec{r} = \mu(2\hat{i} - \hat{j} + 2\hat{k}), \mu \in R$$

respectively. If L_3 is a line which is perpendicular to both L_1 and L_2 and cuts both of them, then which of the following options describe (s) L_3 ?

(2019)

(a) $\vec{r} = \frac{1}{3}(2\hat{i} + \hat{k}) + t(2\hat{i} + 2\hat{j} - \hat{k}), t \in R$

(b) $\vec{r} = \frac{2}{9}(4\hat{i} + \hat{j} + \hat{k}) + t(2\hat{i} + 2\hat{j} - \hat{k}), t \in R$

(c) $\vec{r} = \frac{2}{9}(2\hat{i} - \hat{j} + 2\hat{k}) + t(2\hat{i} + 2\hat{j} - \hat{k}), t \in R$

(d) $\vec{r} = t(2\hat{i} + 2\hat{j} - \hat{k}), t \in R$

69. Three lines

$$L_1 : \vec{r} = \lambda \hat{i}, \lambda \in R$$

$$L_2 : \vec{r} = \hat{k} + \mu \hat{j}, \mu \in R \text{ and}$$

$$L_3 : \vec{r} = \hat{i} + \hat{j} + \gamma \hat{k}, \gamma \in R.$$

are given. For which point(s) Q on L_2 can we find a point P on L_1 and a point R on L_3 so that P, Q and R are collinear.

(2019)

(a) $\hat{k} + \frac{1}{2}\hat{j}$

(b) $\hat{k} + \hat{j}$

(c) $\hat{k}$

(d) $\hat{k} - \frac{1}{2}\hat{j}$

70. Let $\alpha, \beta, \gamma, \delta$ be real numbers such that

$$\alpha^2 + \beta^2 + \gamma^2 \neq 0 \text{ and } \alpha + \gamma = 1. \text{ Suppose the point}$$

(3, 2, -1) is the mirror image of the point (1, 0, -1) with

respect to the plane $\alpha x + \beta y + \gamma z = \delta$. Then which of

the following statements is/are TRUE?

(2020)

(a) $\alpha + \beta = 2$

(b) $\delta - \gamma = 3$

(c) $\delta + \beta = 4$

(d) $\alpha + \beta + \gamma = \delta$

Numerical Value Type Questions

71. Consider the set of eight vectors

$$V = [\hat{a}\hat{i} + \hat{b}\hat{j} + \hat{c}\hat{k} : \hat{a}, \hat{b}, \hat{c} \in \{-1, 1\}]. \text{ Three non-coplanar}$$

vectors can be chosen from V in 2^p ways. Then, p is

(2013)

72. Three lines are given by

$$\vec{r} = \lambda \hat{i}, \lambda \in R$$

$$\vec{r} = \mu(\hat{i} + \hat{j}), \mu \in R \text{ and } \vec{r} = \gamma(\hat{i} + \hat{j} + \hat{k}), \gamma \in R$$

Let the lines cut the plane $x + y + z = 1$ at points A, B and C respectively. If the area of the triangle ABC is Δ then

the value of $(6\Delta)^2$ equals

(2019)



Assertion & Reason

For the following questions choose the correct answer from the codes (a), (b), (c) and (d) defined as follows.

- (a) Statement I is true, Statement II is also true; Statement II is the correct explanation of Statement I.
 (b) Statement I is true, Statement II is also true; Statement II is not the correct explanation of Statement I.
 (c) Statement I is true, Statement II is false.
 (d) Statement I is false, Statement II is true.

73. Consider the planes
 $3x - 6y - 2z = 15$ and $2x + y - 2z = 5$.
Statement-I : The parametric equations of the line of intersection of the given planes are $x = 3 + 14t$, $y = 1 - 2t$, $z = 15t$.

Statement-II : The vector $14\hat{i} + 2\hat{j} + 15\hat{k}$ is parallel to the line of intersection of the given planes. (2007)

- (a) A (b) B
 (c) C (d) D

74. Consider three planes
 $P_1 : x - y + z = 1$
 $P_2 : x + y - z = 1$ and $P_3 : x - 3y + 3z = 2$
 Let L_1, L_2, L_3 be the lines of intersection of the planes P_2 and P_3, P_3 and P_1, P_1 and P_2 , respectively.
Statement-I : At least two of the lines L_1, L_2 and L_3 are non-parallel.
Statement-II : The three planes do not have a common point. (2008)

- (a) A (b) B
 (c) C (d) D

Match the Following

Each question has two columns. Four options are given representing matching of elements from Column-I and Column-II. Only one of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.

75. Consider the following linear equations

$$\begin{aligned} ax + by + cz &= 0, \\ bx + cy + az &= 0, \\ cx + ay + bz &= 0 \end{aligned}$$

(2007)

Column-I

(A) $a + b + c \neq 0$ and
 $a^2 + b^2 + c^2$
 $= ab + bc + ca$

(B) $a + b + c = 0$ and
 $a^2 + b^2 + c^2$
 $\neq ab + bc + ca$

(C) $a + b + c \neq 0$ and
 $a^2 + b^2 + c^2$
 $\neq ab + bc + ca$

(D) $a + b + c = 0$ and
 $a^2 + b^2 + c^2$
 $= ab + bc + ca$

Column-II

(P) the equations represent planes meeting only at a single point

(Q) the equations represent the lines $x = y = z$

(R) the equations represent identical planes

(S) the equations represent the whole of the three dimensional space.

The correct matching is

- (a) A-R, B-Q; C-P; D-S
 (b) A-Q, B-R; C-P; D-S
 (c) A-S, B-Q; C-P; D-R
 (d) A-R, B-Q; C-S; D-P

76. Consider the lines

$$L_1: \frac{x-1}{2} = \frac{y}{-1} = \frac{z+3}{1}, \quad L_2: \frac{x-4}{1} = \frac{y+3}{1} = \frac{z+3}{2} \text{ and the}$$

planes $P_1 : 7x + y + 2z = 3$, $P_2 : 3x + 5y - 6z = 4$. Let $ax + by + cz = d$ the equation of the plane passing through the point of intersection of lines L_1 and L_2 and perpendicular to planes P_1 and P_2 .

Match List-I with List-II and select the correct answer using the code give below the lists. (2013)

List-I

- P. $a =$
 Q. $b =$
 R. $c =$
 S. $d =$

List-II

1. 13
 2. -3
 3. 1
 4. -2

P Q R S

- (a) 3 2 4 1
 (c) 3 2 1 4

P Q R S

- (b) 1 3 4 2
 (d) 2 4 1 3



77. Match the following.

(2015) Passage Based Problem

Column-I

Column-II

(A) In a triangle ΔXYZ , let a , b , and c be the lengths of the sides opposite to the angles X , Y and Z , respectively.

(P) 1

If $2(a^2 - b^2) = c^2$ and $\lambda = \frac{\sin(X-Y)}{\sin Z}$,

then possible value of

n for which $\cos(n\pi\lambda) = 0$ is

(B) In a triangle ΔXYZ , let a , b and c be the lengths of the sides opposite to the angles X , Y and Z respectively. If

(Q) 2

$1 + \cos 2X - 2\cos 2Y = 2 \sin X \sin Y$,

then possible value(s) of $\frac{a}{b}$ is are

(C) In R^2 , let

(R) 3

$\sqrt{3}\hat{i} + \hat{j}$, $\hat{i} + \sqrt{3}\hat{j}$ and $\beta\hat{i} + (1-\beta)\hat{j}$

be the position vectors X , Y and Z with respect to the origin O , respectively. If the distance of Z from the bisector of the acute

angle of $\overline{OX}$ with $\overline{OY}$ is $\frac{3}{\sqrt{2}}$, the

possible value(s) of $|\beta|$ is(are)

(D) Suppose that $F(\alpha)$ denotes the area of the region bounded

(S) 5

by $x = 0$, $x = 2$, $y^2 = 4x$ and

$y = |ax - 1| + |\alpha x - 2| + ax$,

where $\alpha \in \{0, 1\}$. Then the

value(s) of $F(\alpha) + \frac{8}{3}\sqrt{2}$,

when $\alpha = 0$ and $\alpha = 1$, is (are)

(T) 6

The correct matching is

- (a) A-P,R,S; B-P; C-P,Q; D-S,T
- (b) A-P,R,S; B-P; C-T,Q; D-S
- (c) A-P,S; B-Q; C-P; D-S,T
- (d) A-S,T; B-P; C-P,Q; D-P,R,S

Using the following passage, solve Q.78 to Q.80

Consider the lines

$$L_1: \frac{x+1}{3} = \frac{y+2}{1} = \frac{z+1}{2}, \quad L_2: \frac{x-2}{1} = \frac{y+2}{2} = \frac{z-3}{3} \quad (2008)$$

78. The unit vector perpendicular to both L_1 and L_2 is

$$(a) \frac{-\hat{i} + 7\hat{j} + 7\hat{k}}{\sqrt{99}}$$

$$(b) \frac{-\hat{i} - 7\hat{j} + 5\hat{k}}{5\sqrt{3}}$$

$$(c) \frac{-\hat{i} + 7\hat{j} + 5\hat{k}}{5\sqrt{3}}$$

$$(d) \frac{7\hat{i} - 7\hat{j} - \hat{k}}{\sqrt{99}}$$

79. The shortest distance between L_1 and L_2 is

(a) 0 unit

(b) $17/\sqrt{3}$ unit

(c) $41/5\sqrt{3}$ unit

(d) $17/5\sqrt{3}$ unit

80. The distance of the point $(1, 1, 1)$ from the plane passing through the point $(-1, -2, -1)$ and whose normal is perpendicular to both the lines L_1 and L_2 is

(a) $2/\sqrt{75}$ unit

(b) $7/\sqrt{75}$ unit

(c) $13/\sqrt{75}$ unit

(d) $23/\sqrt{75}$ unit

Text

81. Find 3-dimensional vectors $\vec{v}_1, \vec{v}_2, \vec{v}_3$ satisfying

$$\vec{v}_1 \cdot \vec{v}_1 = 4, \vec{v}_1 \cdot \vec{v}_2 = -2, \vec{v}_1 \cdot \vec{v}_3 = 6,$$

$$\vec{v}_2 \cdot \vec{v}_2 = 2, \vec{v}_2 \cdot \vec{v}_3 = -5, \vec{v}_3 \cdot \vec{v}_3 = 29 \quad (2001)$$

82. (a) Find the equation of the plane passing through the points $(2, 1, 0)$, $(5, 0, 1)$ and $(4, 1, 1)$.

(b) If P is the point $(2, 1, 6)$, then the point Q such that PQ is perpendicular to the plane in (a) and the mid point of PQ lies on it. (2003)



83. T is a parallelopiped in which A, B, C and D are vertices of one face and the face just above it has corresponding vertices A', B', C', D', T is now compressed to S with face ABCD remaining same and A', B', C', D' shifted to A'', B'', C'', D'' in S. The volume of parallelopiped S is reduced to 90% of T. Prove that locus of A'' is a plane. (2004)
84. Find the equation of the plane containing the lines $2x - y + z - 3 = 0$, $3x + y + z = 5$ and at a distance of $\frac{1}{\sqrt{6}}$ from the point $(2, 1, -1)$. (2005)
85. Incident ray is along the unit vector $\hat{v}$ and the reflected ray is along the unit vector $\hat{w}$. The normal is along unit vector $\hat{a}$ outwards. Express $\hat{w}$ in terms of $\hat{a}$ and $\hat{v}$. (2005)

Find Answer Key and Detailed Solutions at the end of this book



VECTORS AND 3-DIMENSIONAL GEOMETRY



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PROBABILITY

PROBABILITY



1. INTRODUCTION

1.1 Factorial notation :

If $n \in \mathbb{N}$, then the product $1 \times 2 \times 3 \times \dots \times n$ is defined as factorial n which is denoted by $n!$ or $\underline{n}$

i.e., $n! = 1 \times 2 \times 3 \times \dots \times n$

We also define $0! = 1$

NOTES :

$$n! = n(n-1)!$$

1.2 Permutation :

If n objects are given and we have to arrange $r (r \leq n)$ out of them such that the order in which we are arranging the objects is important, then such an arrangement is called permutation of n objects taking r at a time. This is denoted by

$${}^n P_r = \frac{n!}{(n-r)!}$$

1.3 Combination :

If n objects are given and we have to choose $r (r \leq n)$ out of them such that the order in which we are choosing the objects is not important, then such a choice is called combination of n objects taking r at a time. This is denoted by

$${}^n C_r = \frac{n!}{r!(n-r)!}$$

1.4 Fundamental Principle of Counting :

If an event can occur in 'm' different ways following which another event can occur in 'n' different ways following which another event can occur in 'p' different ways then the total number of ways of simultaneous occurrence of all these events in a definite order is $m \times n \times p$.

(where the events are independent of each other)

2. DEFINITIONS AND TYPES OF EVENTS

2.1 Random experiment :

If an act or an experiment has more than one possible results which are known in advance and it is not possible to predict which one is going to occur, then such an experiment is called a **random experiment**.

The following are some random experiments :

- Tossing of a coin
- Throwing a six-faced die
- Drawing a card from a well-shuffled pack of cards
- Ten horses run a race
- Two persons are selected out of 10 persons to form a committee.
- A ball is drawn from a bag containing 7 balls.

2.2 Outcome :

The result of a random experiment is called an outcome.

2.3 Sample space :

The set of all possible outcomes of a random experiment is called a **sample space** and its elements are called **sample points**. A sample space is usually denoted by S .

Illustrations ;

- When a fair coin is tossed, then either head or tail will turn up.
Hence $S = \{H, T\}$ S contains 2 sample points.
- When a six-faced die is thrown, then only one of 1, 2, 3, 4, 5, 6 will turn up.
Hence $S = \{1, 2, 3, 4, 5, 6\}$. S contains 6 sample points.
- Suppose a bag contains 7 balls.

Consider the sample points.

- The experiment is : one ball is drawn. We can draw one ball out of the 7 balls in ${}^7 C_1 = 7$ ways.
 $\therefore$ The sample space for this experiment contains 7 sample points.



(b) The experiment is : two balls are drawn. We can draw

$$2 \text{ balls out of the } 7 \text{ balls in } {}^7C_2 = \frac{7 \times 6}{1 \times 2} = 21 \text{ ways}$$

$\therefore$ the sample space for this experiment contains 21 sample points.

(c) The experiment is : three balls are drawn.

The sample space for this experiment contains

$${}^7C_3 = \frac{7 \times 6 \times 5}{1 \times 2 \times 3} = 35 \text{ sample points}$$

2.4 Event :

Any subset of a sample space is called an event.

Example :

In a single throw of a die, the event of getting a prime number is $E \equiv \{2, 3, 5\}$ The sample space

$$S \equiv \{1, 2, 3, 4, 5, 6\}$$

$\therefore E \subseteq S \Rightarrow E$ is an event

2.5 Complementary event :

Let A be an event in a sample space S . Then A is a subset of S . We can hence think of the complement of A in S , i.e., $S - A$. This is also a subset of S and hence an event in S . This event is called the complementary event of A and is denoted by $\bar{A}$ or A' .

Now, suppose S contains n sample points, A contains m sample points, Then A' will contain

$n - m$ sample points.

2.6 Impossible event :

Let S be a sample space. Since $\phi \subseteq S$, So ϕ is an event, called an impossible event.

NOTES :

The event E and E' are such that only one of them can occur in a trial and at least one of them must occur.

2.7 Union of events :

If A and B are two events of the sample space S then $A \cup B$ or $A + B$ is the event that either A or B (or both) take place.

2.8 Intersection of events :

If A and B are two events of the sample space S then $A \cap B$ or $A \cdot B$ is the event that both A and B take place.

2.9 Mutually Exclusive events :

Two events A and B of the sample space S are said to be mutually exclusive if they cannot occur simultaneously. In such case $A \cap B$ is a null set.

2.10 Exhaustive events :

Two events A and B of the sample space S are said to be exhaustive if $A \cup B = S$ i.e. $A \cup B$ contains all sample points.

2.11 Probability of an event :

Let A be an event in a sample space S . Then the probability of the event A denoted by $P(A)$ is defined as,

$$P(A) = \frac{\text{number of sample points in } A}{\text{number of sample points in } S} = \frac{n(A)}{n(S)}$$

Theorem :

If E is an event of a sample space S , prove that $0 \leq P(E) \leq 1$ and $P(E') = 1 - P(E)$, where E' is the complementary event of E .

Proof :

Suppose the sample space S contains n sample points and

the event E contains m sample points. Then $P(E) = \frac{m}{n}$

$$\text{Now, } 0 \leq m \leq n \therefore 0 \leq \frac{m}{n} \leq 1$$

$$\therefore 0 \leq P(E) \leq 1$$

Further E' contains $n - m$ sample points,

$$\therefore P(E') = 1 - P(E)$$

2.12 Equally likely event :

The events are said to be equally likely if none of them is expected to occur in preference to the other.

Ex :- When a die is thrown, then all the side faces are equally likely to come.



2.13 Odds in favour of A : It is defined as $P(A) : P(\bar{A})$

Odds against A : It is defined as $P(\bar{A}) : P(A)$

If $P(A) : P(\bar{A}) = x : y$ then

$$P(A) = \frac{x}{x+y} \text{ and } P(\bar{A}) = \frac{y}{x+y}$$

3. EXPERIMENT NO 1 : TOSSING COINS

Tossing one coin :

Let S be the sample space $S \equiv \{H, T\}$, $n(S) = 2$

Tossing two coin :

Let S be the sample space $S \equiv \{HH, HT, TH, TT\} \Rightarrow n(S) = 4$

Tossing three coin :

Let S be the sample space

$S \equiv \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$
 $\Rightarrow n(S) = 8$

NOTES :

1. No head means all tails.
2. At least one head means one head or two heads or three heads
3. At most two heads means two heads or one head or no head (all tail).

4. EXPERIMENT NO 2 : THROWING DIE / DICE

One six faced die is thrown :

$S \equiv \{1, 2, 3, 4, 5, 6\} \Rightarrow n(S) = 6$

Two dice are thrown :

$$S \equiv \left\{ \begin{array}{l} (1,1), (2,1), (3,1), (4,1), (5,1), (6,1) \\ (1,2), (2,2), (3,2), (4,2), (5,2), (6,2) \\ (1,3), (2,3), (3,3), (4,3), (5,3), (6,3) \\ (1,4), (2,4), (3,4), (4,4), (5,4), (6,4) \\ (1,5), (2,5), (3,5), (4,5), (5,5), (6,5) \\ (1,6), (2,6), (3,6), (4,6), (5,6), (6,6) \end{array} \right\}$$

$\Rightarrow n(S) = 36$

NOTES :

If one coin is tossed n times or n coins are tossed once the sample space consists of same number of sample points.

i.e. $n(S) = (2)^n$

Prime numbers are 2, 3, 5, 7, 11, ...

1 card is picked $n(S) = {}^{52}C_1 = 52$

2 cards are drawn. $n(S) = {}^{52}C_2 = 1326$

5. EXPERIMENT NO 3: PACK OF CARDS

- 5.1 There are 4 suits (spade, heart, diamond and club) each having 13 cards.
- 5.2 There are two colours red (heart and diamond) and black (spade and club) each having 26 cards.
- 5.3 Jack, Queen and King are face cards. Therefore face cards are 12 in pack of cards. Face card is also called a picture card.
- 5.4 There are four aces. Ace is not a picture card.
- 5.5 Face cards and Ace cards are known Coloured Cards.

Theorem 1

If E is an event of sample space S then $0 \leq P(E) \leq 1$

Proof :

As $E \subseteq S$

$$\therefore 0 \leq n(E) \leq n(S)$$

$$\therefore \frac{0}{n(S)} \leq \frac{n(E)}{n(S)} \leq \frac{n(S)}{n(S)} \quad (\because n(S) \neq 0)$$

$$\therefore 0 \leq P(E) \leq 1$$

NOTES :

$P(E) = 0$ if and only if E is an impossible event and

$P(E) = 1$ if and only if E is a certain event

Theorem 2

If E is an event of sample space S and E' is the event that E does not happen then

$$P(E') = 1 - P(E)$$

Proof :

E' is the event that E does not happen.

$\therefore$ E and E' are complements of each other.



$\therefore n(E) + n(E') = n(S)$ Dividing by $n(S)$, ($\because n(S) \neq 0$.)

$$\therefore \frac{n(E)}{n(S)} + \frac{n(E')}{n(S)} = \frac{n(S)}{n(S)}$$

$$\therefore P(E) + P(E') = 1$$

$$\therefore P(E') = 1 - P(E)$$

Theorem 3

- (i) If A and B are two events of sample space S, prove that $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- (ii) $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$

Theorem 4

Show that if A and B are independent events defined on S then

(i) A & B' (ii) A' & B (iii) A' & B' are

independent events where A and B are mutually exclusive events of A' and B' respectively.

Ans. Since A and B are known to be independent events.

We have $P(A \cap B) = P(A) \cdot P(B)$

$$\begin{aligned} \text{(i)} \quad P(A \cap B') &= P(A) - P(A \cap B) = P(A) - P(A) \cdot P(B) \\ &= P(A) (1 - P(B)) \\ &= P(A) \cdot P(B') \end{aligned}$$

$\therefore$ A and B' are independent events

$$\begin{aligned} \text{(ii)} \quad P(A' \cap B) &= P(B) - P(A \cap B) = P(B) - P(A) \cdot P(B) \\ &= P(B) (1 - P(A)) = P(B) \cdot P(A') \end{aligned}$$

$\therefore$ A' and B are independent events

$$\begin{aligned} \text{(iii)} \quad P(A' \cap B') &= P(A \cup B)' = 1 - P(A \cup B) \\ &= 1 - [P(A) + P(B) - P(A \cap B)] \\ &= 1 - [P(A) + P(B) - P(A) \cdot P(B)] \\ &= 1 - P(A) - P(B) + P(A) \cdot P(B) \\ &= [1 - P(A)] [1 - P(B)] = P(A') \cdot P(B') \end{aligned}$$

$\therefore$ A' and B' are independent events.

6. IMPORTANT RESULTS

1. A and B are mutually exclusive if $P(A \cap B) = 0$.
2. They are independent if $P(A \cap B) = P(A) \cdot P(B)$
3. Two independent events with non-zero probabilities cannot be mutually exclusive.

4. If A, B and C are independent events with non-zero probabilities then $P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$
5. If $A_1, A_2, A_3, \dots, A_n$ are independent events with non-zero probabilities, then $P(A_1 \cap A_2 \cap A_3 \cap \dots \cap A_n) = P(A_1) \cdot P(A_2) \dots P(A_n)$
6. $P[\text{exactly one of A or B occurs}] = P(A \cap B') + P(A' \cap B) = P(A \cup B) - P(A \cap B) = P(A) + P(B) - 2P(A \cap B)$
7. $P(A' \cap B') = P(A \cup B)' = 1 - P(A \cup B)$
8. $P(A' \cup B') = P(A \cap B)' = 1 - P(A \cap B)$
9. $P(A) = P(A \cap B) + P(A \cap B')$
10. $P(B) = P(A \cap B) + P(A' \cap B)$
11. $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$
12. $P(A \cup B \cup C) = P(A) + P(B) + P(C)$ if A, B and C are mutually exclusive.
13. If A and B are mutually exclusive and exhaustive events then $P(A) + P(B) = 1$
14. If A, B and C are mutually exclusive and exhaustive events then $P(A) + P(B) + P(C) = 1$

7. CONDITIONAL PROBABILITY

Let A and B be two events associated with a random experiment. Then, the probability of occurrence of event A under the condition that B has already occurred and $P(B) \neq 0$, is called the conditional probability of event A and it is denoted by $P(A/B)$. Thus, we have

$P(A/B)$ = Probability of occurrence of A given that B has already occurred.

$$P(A/B) = \frac{\text{Number of elementary events favourable to } A \cap B}{\text{Number of elementary events favourable to } B}$$

$$\Rightarrow P(A/B) = \frac{n(A \cap B)}{n(B)}$$

$$\Rightarrow P(A/B) = \frac{P(A \cap B)}{P(B)}$$

Theorem

If A and B are two events associated with a random experiment, then

$$P(A \cap B) = P(A) P(B/A), \text{ if } P(A) \neq 0$$

NOTES:

1. $0 \leq P(A/B) \leq 1$
2. $P(A/A) = 1$

8. INDEPENDENT EVENTS

Definition :

Events are said to be independent, if the occurrence or non-occurrence of one does not affect the probability of the occurrence or non-occurrence of the other.

Theorem 1 :

If A and B are independent events associated with a random experiment, then $P(A \cap B) = P(A) P(B)$.

Theorem 2 :

If $A_1, A_2, \dots, A_n$ are independent events associated with a random experiment, then

$$P(A_1 \cap A_2 \cap A_3 \dots \cap A_n) = P(A_1) P(A_2) \dots P(A_n)$$

Theorem 3 :

If A and B are independent events associated with a random experiment, then

- (i) $\bar{A}$ and B are independent events
- (ii) A and $\bar{B}$ are independent events
- (iii) $\bar{A}$ and $\bar{B}$ are also independent events.

9. MULTIPLICATION THEOREM (OR) PRODUCT THEOREM (OR) THEROREM OF COMPOUND PROBABILITY

If A and B are two events in a sample space S such that $P(A) \neq 0$ & $P(B) \neq 0$, then the probability of simultaneous occurrence of the two events A and B is given by

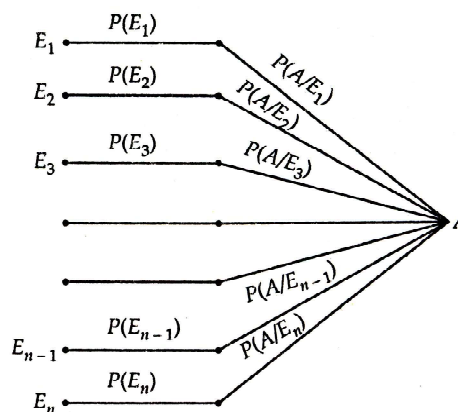
$$P(A \cap B) = P(A) \cdot P(B/A) = P(B) \cdot P(A/B).$$

- * For three events A, B and C $P(A \cap B \cap C) = P(A) \cdot P(B/A) \cdot P(C/A \cap B)$
- * For four events A, B, C and D $P(A \cap B \cap C \cap D) = P(A) \cdot P(B/A) \cdot P(C/A \cap B) \cdot P(D/A \cap B \cap C)$

10. THE LAW OF TOTAL PROBABILITY

Theorem : (Law of Total Probability) Let S be the sample space and let $E_1, E_2, \dots, E_n$ be n mutually exclusive and exhaustive events associated with a random experiment. If A is any event which occurs with E_1 or E_2 or ... or E_n , then

$$P(A) = P(E_1) P(A/E_1) + P(E_2) P(A/E_2) + \dots + P(E_n) P(A/E_n)$$

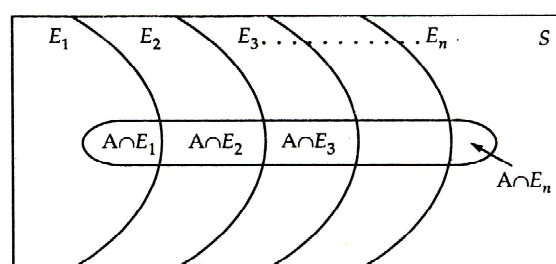


11. STATEMENT OF BAYES' THEOREM

If $B_1, B_2, B_3, \dots, B_n$ are mutually exclusive and exhaustive events & if A is an event consequent to these B_i 's then for each i, where $i = 1, 2, 3, \dots, n$

$$P(B_i/A) = \frac{P(B_i) P(A/B_i)}{P(B_1) P(A/B_1) + P(B_2) P(A/B_2) + \dots + P(B_n) P(A/B_n)}$$

$$P(B_i/A) = \frac{P(B_i) P(A/B_i)}{\sum_{i=1}^n P(B_i) P(A/B_i)}$$



12. RANDOM VARIABLE AND PROBABILITY DISTRIBUTION

12.1 Random Variable : Let S be a sample space associated with a random experiment. Then a real valued function $X: S \rightarrow R$ is



called a random function or random variable.

12.2 Discrete Random Variable : A real valued function defined on discrete sample space S is called a discrete random variable.

12.3 Continuous Random Variable : A random variable X defined on continuous sample space S , which can take all real values in an interval (a, b) is called a continuous random variable.

12.4 Discrete Random Variable : If a discrete variable X can assume values $X_1, X_2, X_3, \dots, X_n$ with respective probabilities $P(X_1), P(X_2), P(X_3) \dots P(X_n)$ such that $P(X_i) \geq 0, \forall i$ and $\sum P(X_i) = 1$, then X is said to be a discrete random variable.

12.5 Probability Mass Functions : If X is a discrete random variable which can assume values $X_i; i = 1, 2, 3, \dots$. With respective probabilities $P_i; i = 1, 2, 3, \dots$ such that

$$\sum_{i=1}^{\infty} P_i = 1, \text{ then } P(X=x_i) = P_i; i = 1, 2, 3, \dots \text{ is called probability}$$

mass function of a discrete random variable X . (OR) If any function $P(X=x)$ gives the probabilities of various values of a discrete random variable X in its range, then that function is called probability mass function.

12.6 Probability Distribution : The set of ordered pairs $\{x_i, P(x_i)\}$ is called the probability distribution of a discrete random variable X .

13. MEANS AND VARIANCE

If $\{x_i, P(x_i)\}$ is the probability distribution of a discrete random variable X , then its:

13.1 Mean or Average : ($\bar{x}$ or μ) : Expected value of x or mathematical expectation of x : $E(x)$ or First moment about origin: $\mu_1'(0)$ is defined as $\bar{x} = \mu = \mu_1'(0) = \sum x_i P(x_i)$.

13.2 Variance (σ^2) or second moment about mean or 2nd central moment $= \mu_2$.

$$\begin{aligned} \sigma^2 = \mu_2 &= E\{x - E(x)\}^2 \\ &= E(x^2) - \{E(x)\}^2 \\ &= \mu_2'(0) - \mu_1'(0)^2 \\ &= \sum x_i^2 P(x_i) - \{\sum x_i P(x_i)\}^2 \end{aligned}$$

14. BINOMIAL DISTRIBUTION

14.1 Bernoulli Trials : Trials of a random experiment are called Bernoulli trials if they satisfy the following conditions :

- They are finite in number.
- They are independent of each other.
- Each trial has exactly two outcomes : success or failure.
- The probability of success or failure remains same in each trial.
- The probability of success is p and failure is q such that $p + q = 1$
- The probability of r successes in n trials in any order is given by ${}^nC_r p^r q^{n-r}$.

14.2 Binomial Distribution : Let X denote the random variable which associates every outcome to the number of successes in it. Then, X assumes values $0, 1, 2, \dots, n$ such that

$$P(X=r) = {}^nC_r p^r q^{n-r}, r = 0, 1, 2, \dots, n.$$

The probability distribution of the random variable X is therefore given by

$$\begin{array}{ccccccccccc} X: & 0 & 1 & 2 & \dots & r & \dots & n \\ P(X): & {}^nC_0 p^0 q^{n-0} & {}^nC_1 p^1 q^{n-1} & {}^nC_2 p^2 q^{n-2} & \dots & {}^nC_r p^r q^{n-r} & \dots & {}^nC_n p^n q^{n-n} \end{array}$$

14.3 Mean & Variance :

$$\text{Mean} = np$$

$$\text{Variance} = npq$$



SOLVED EXAMPLES

Example – 1

5 letters are to be posted in 5 post boxes. If any number of letters can be posted in 5 post boxes, what is the probability that each box contains only one letter?

Sol. Since any number of letters can be posted in all 5 post boxes, each letter can be posted in 5 different ways.

$$\therefore n(S) = 5 \times 5 \times 5 \times 5 \times 5 = 5^5$$

Let $A \equiv$ the event that each box contains only one letter.

The first letter can be posted in 5 post boxes in 5 different ways. Since each box contains only one letter, the second letter can be posted in the remaining 4 post boxes in 4 different ways.

Similarly, the third letter can be posted in 3 different ways, the fourth letter can be posted in 2 different ways and the fifth letter can be posted in 1 way.

$$\therefore n(A) = 5 \times 4 \times 3 \times 2 \times 1 = 5!$$

$$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{5!}{5^5}$$

$$= \frac{5 \times 4 \times 3 \times 2 \times 1}{5 \times 5^4} = \frac{24}{625}$$

Example – 2

A doctor is called to see a sick child. The doctor has prior information that 80% of sick children in that area have the flu, while the other 20% are sick with measles. Assume that there is no other disease in that area. A well-known symptom of measles is a rash. From the past records it is known that, chances of having rashes given that sick child is suffering from measles is 0.95. However, occasionally children with flu also develop rash, whose chances are 0.08. Upon examining the child, the doctor finds a rash. What is the probability that the child has measles?

Sol. Let $A \equiv$ event that the child is sick with flu

$B \equiv$ event that the child is sick with measles

$C \equiv$ event that the child has rash

$$\therefore P(A) = 80\% = \frac{80}{100} = \frac{4}{5}$$

$$P(B) = 20\% = \frac{20}{100} = \frac{1}{5}$$

Since the chances of having rashes, if the child is suffering from measles is 0.95 and the chances of having rashes, if the child has flu is 0.08,

$$P(C/B) = 0.95 = \frac{95}{100}$$

$$\text{and } P(C/A) = 0.08 = \frac{8}{100}$$

By Baye's Theorem, probability that the child has measles provided he has the rashes is given by

$$P(B/C) = \frac{P(B) \cdot P(C/B)}{P(A) \cdot P(C/A) + P(B) \cdot P(C/B)}$$

$$= \frac{\left(\frac{1}{5}\right) \left(\frac{95}{100}\right)}{\left(\frac{4}{5}\right) \left(\frac{8}{100}\right) + \left(\frac{1}{5}\right) \left(\frac{95}{100}\right)}$$

$$= \frac{95}{32 + 95} = \frac{95}{127}$$

$$= 0.748$$

Example – 3

Suppose you have a large barrel containing a number of plastic eggs. Some eggs contain pearls, the rest contain nothing. Some eggs are painted blue, the rest are painted

red. Suppose that 40% of the eggs are painted blue, $\frac{5}{13}$ of

the eggs painted blue contain pearls and 20% of the red eggs are empty. What is the probability that an egg containing pearl is painted blue ?

Sol. Let event $A =$ An egg is painted blue.

event $B =$ An egg is painted red.

The barrel contains egg with blue paint as 40% and red paint as 60%.



$$\therefore P(A) = 40\% = \frac{40}{100} = \frac{2}{5}$$

$$\therefore P(B) = 60\% = \frac{60}{100} = \frac{3}{5}$$

Let event C = an egg selected contains a pearl.

Then C/A = A blue painted egg contains pearl. given that $P(C/A) = 5/13$.

$P(C/B)$ = A red painted egg contains pearl. given that 20% of red eggs are empty.

i.e. 80% of red eggs contain pearls.

$$\therefore P(C/B) = 80\% = \frac{80}{100} = \frac{4}{5}$$

$\therefore$ Required probability, that an egg containing pearl is painted blue is

$$P(A/C) = \frac{P(A).P(C/A)}{P(A).P(C/A) + P(B).P(C/B)}$$

$$= \frac{\frac{2}{5} \times \frac{5}{13}}{\frac{2}{5} \times \frac{5}{13} + \frac{3}{5} \times \frac{4}{5}}$$

$$= \frac{\frac{2}{13}}{\frac{2}{13} + \frac{12}{25}} = \frac{50}{206}$$

$$= \frac{25}{103} = 0.243$$

Example - 4

The mean and variance of a binomial distribution are 4 and $4/3$ respectively, find $P(X \geq 1)$.

Sol. Let X be a binomial variate with parameters n and p. Then,

Mean = np and Variance = npq

$$\Rightarrow np = 4 \text{ and } npq = \frac{4}{3}$$

$$[\because \text{Mean} = 4, \text{Var}(X) = \frac{4}{3} \text{ (Given)}]$$

$$\Rightarrow \frac{npq}{np} = \frac{\frac{4}{3}}{4} \Rightarrow q = \frac{1}{3} \Rightarrow p = 1 - \frac{1}{3} = \frac{2}{3}$$

$$[p = 1 - q]$$

Putting $p = 2/3$ in $np = 4$, we get

$$n \times \frac{2}{3} = 4 \Rightarrow n = 6$$

Thus, we have

$$n = 6, p = \frac{2}{3} \text{ and } q = \frac{1}{3}$$

$$\therefore P(X = r) = {}^nC_r p^r q^{n-r} \Rightarrow P(X = r) = {}^6C_r \left(\frac{2}{3}\right)^r \left(\frac{1}{3}\right)^{6-r},$$

$$r = 0, 1, 2, \dots, 6$$

Now, $P(X \geq 1) = 1 - P(X < 1)$

$$\Rightarrow P(X \geq 1) = 1 - P(X = 0)$$

$$\Rightarrow P(X \geq 1) = 1 - {}^6C_0 \left(\frac{2}{3}\right)^0 \left(\frac{1}{3}\right)^6 = 1 - \left(\frac{1}{3}\right)^6 = 1 - \frac{1}{729} = \frac{728}{729}$$

Example - 5

Events A, B, C are mutually exclusive events such that

$$P(A) = \frac{3x+1}{3}, P(B) = \frac{1-x}{4} \text{ and } P(C) = \frac{1-2x}{2}. \text{ Then}$$

set of possible values of x are in the interval

$$(a) \left[\frac{1}{3}, \frac{2}{3}\right]$$

$$(b) \left[\frac{1}{3}, \frac{13}{3}\right]$$

$$(c) [0, 1]$$

$$(d) \left[\frac{1}{3}, \frac{1}{2}\right]$$

Ans. (d)

Sol. A, B, C are mutually exclusive

$$\therefore P(A) + P(B) + P(C) \leq 1 \quad \dots (i)$$

$$0 \leq P(A), P(B), P(C) \leq 1 \quad \dots (ii)$$

Now on solving (i) and (ii), we get

$$\frac{3x+1}{3} + \frac{1-x}{4} + \frac{1-2x}{2} \leq 1$$

$$12x + 4 + 3 - 3x + 6 - 13 \times \leq 12$$

$$13 - 3 \times \leq 12$$

$$\Rightarrow \boxed{x \geq \frac{1}{3}}$$

Also, $P(C) \geq 0$



$$\Rightarrow 1 - 2x \geq 0$$

$$2x \leq 1$$

$$\Rightarrow x \leq \frac{1}{2}$$

$$\Rightarrow \frac{1}{3} \leq x \leq \frac{1}{2}$$

Example – 6

Five horses are in a race. Mr. A selects two of the horses at random and bets on them. The probability that Mr. A selected the winning horse, is

- (a) $3/5$ (b) $1/5$
(c) $2/5$ (d) $4/5$

Ans. (c)

Sol. Let two horses selected are A and B.

No. of horses = 5

$\therefore$ Probability that A can't win the race

$$\text{Where } p(A) = \frac{1}{5} \text{ and } P(B) = \frac{1}{4}$$

$$= p(\bar{A}) \cdot p(\bar{B}) = \frac{4}{5} \times \frac{3}{4}$$

Probability that 'A' must win the race = $1 - P(\bar{A})P(\bar{B})$

$$= 1 - \frac{12}{20} = \frac{2}{5}$$

Example – 7

The mean and variance of a random variable X having a binomial distribution are 4 and 2 respectively, then $P(X=1)$ is

- (a) $1/16$ (b) $1/8$
(c) $1/4$ (d) $1/32$

Ans. (d)

Sol. Given mean $np = 4$, $npq = 2$

$$\Rightarrow \frac{npq}{np} = \frac{2}{4} \Rightarrow q = \frac{1}{2} \therefore q = p = \frac{1}{2} \text{ and } n = 8$$

$$\text{Now } P(X=r) = {}^8C_r p^r q^{8-r}$$

$$(\text{Use } P(X=r) = {}^nC_r p^r q^{n-r})$$

$$\therefore P(X=1) = {}^8C_1 \left(\frac{1}{2}\right)^8 = \frac{8}{16 \times 16} = \frac{1}{32}$$

Example – 8

The mean and the variance of a binomial distribution are 4 and 2 respectively. Then the probability of 2 successes is

- (a) $128/256$ (b) $219/256$
(c) $37/256$ (d) $28/256$

Ans. (d)

Sol. Given $np = 4$ and $npq = 2$

$$q = \frac{npq}{np} = \frac{2}{4} = \frac{1}{2} \Rightarrow p = 1 - \frac{1}{2} = \frac{1}{2}$$

$$\text{Now } npq = 2 \therefore n = 8$$

$\therefore$ Binomial Distribution is given by

$$P(X=r) = {}^8C_r p^r q^{8-r}$$

$$\therefore P(X=r=2) = {}^8C_2 \left(\frac{1}{2}\right)^8 = \frac{28}{256}$$

Example – 9

A pair of fair dice is thrown independently three times. The probability of getting a score of exactly 9 twice, is

- (a) $8/729$ (b) $8/243$
(c) $1/729$ (d) $8/9$

Ans. (b)

Sol. Possibility of getting 9 are (5, 4), (4, 5), (6, 3), (3, 6)

Probability of getting score 9 in a single throw

$$= p = \frac{4}{36} = \frac{1}{9}$$

so, probability of not getting 9 in a single throw

$$= 1 - \frac{1}{9} = \frac{8}{9}$$

Required probability = probability of getting score 9

$$\text{exactly twice} = {}^3C_2 \left(\frac{1}{9}\right)^2 \times \left(\frac{8}{9}\right) = \frac{8}{243}$$



Example – 10

A die is thrown. Let A be the event that the number obtained is greater than 3, Let B be the event that the number obtained is less than 5. Then $P(A \cup B)$ is

- (a) $\frac{2}{5}$ (b) $\frac{3}{5}$
(c) 0 (d) 1

Ans. (d)

Sol. $A = \{4, 5, 6\}$

Also $B = \{1, 2, 3, 4\}$

We have $A \cup B = \{1, 2, 3, 4, 5, 6\} = S$

Where S is the sample space of the experiment of throwing a die. $P(S) = 1$, for it is a sure event.

Hence $P(A \cup B) = 1$

Example – 11

A set S contains 7 elements. A non-empty subset A of S and an element x of S are chosen at random. Then the probability that $x \in A$ is

- (a) $\frac{1}{2}$ (b) $\frac{64}{127}$
(c) $\frac{63}{128}$ (d) $\frac{31}{128}$

Ans. (b)

Sol. $S = \{a, b, c, d, e, f, g\}$

A can be selected in $(2^7 - 1)$ ways

Hence, the number of non-empty subsets of S can be chosen in

$${}^7C_1 + {}^7C_2 + {}^7C_3 + \dots + {}^7C_7 = 128 - 1 \text{ ways} \\ = 127$$

Now, any x if not included in A can happen in $2^6 - 1$ ways

$\therefore$ Number of sets which include x are

$$= (2^7 - 1) - (2^6 - 1) \\ = 2^6(2 - 1) = 64 \text{ ways}$$

Hence, probability that $x \in A$ is

$$\frac{64}{2^7 - 1} = \frac{64}{127}$$

Example – 12

One hundred identical coins, each with probability p, of showing up heads are tossed once. If $0 < p < 1$ and the probability of heads showing on 50 coins is equal to that of heads showing on 51 coins, then the value of p is

- (a) $1/2$ (b) $49/101$
(c) $50/101$ (d) $51/101$

Ans. (d)

Sol. Let X be the number of coins showing heads. Let X be a binomial variate with parameters $n = 100$ and p .

Since, $P(X = 50) = P(X = 51)$

$$\Rightarrow {}^{100}C_{50} p^{50} (1-p)^{50} = {}^{100}C_{51} (p)^{51} (1-p)^{49}$$

$$\Rightarrow \frac{(100)!}{(50!)(50!)} \cdot \frac{(51!) \times (49!)}{100!} = \frac{p}{1-p}$$

$$\Rightarrow \frac{p}{1-p} = \frac{51}{50} \Rightarrow p = \frac{51}{101}$$

Example – 13

Three of the six vertices of a regular hexagon are chosen at random. The probability that the triangle with three vertices is equilateral, equals

- (a) $1/2$ (b) $1/5$
(c) $1/10$ (d) $1/20$

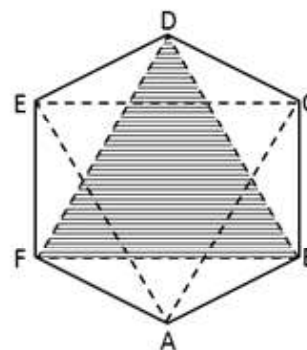
Ans. (c)

Sol. Three vertices out of 6 can be chosen in 6C_3 ways.

So, total ways = ${}^6C_3 = 20$

Only two equilateral triangles can be formed

$\triangle AEC$ and $\triangle BFD$



$\therefore$ Favourable ways = 2

$$\text{So, required probability} = \frac{2}{20} = \frac{1}{10}$$



Example – 14

If the integers m and n are chosen at random between 1 and 100, then the probability that a number of the form $7^m + 7^n$ is divisible by 5, equals

- (a) $\frac{1}{4}$ (b) $\frac{1}{7}$
(c) $\frac{1}{8}$ (d) $\frac{1}{49}$

Ans. (a)

Sol. $7^1 = 7, 7^2 = 49, 7^3 = 343, 7^4 = 2401, \dots$

There fore, for $7^r, r \in \mathbb{N}$ then number ends at unit Place 7, 9, 3, 1, 7, ...

$\therefore 7^m + 7^n$ will be divisible by 5 if it end at 5 or 0.

But it cannot end at 5.

Also for end at 0.

For this m and n should be as follows

	m	n
1	$4r$	$4r - 2$
2	$4r - 1$	$4r - 3$
3	$4r - 2$	$4r$
4	$4r - 3$	$4r - 1$

For any given value of m , there will be 25 values of n .

Hence, the probability of the required event is

$$= \frac{100 \times 25}{100 \times 100} = \frac{1}{4}$$

Example – 15

If M and N are any two events, then the probability that exactly one of them occurs is

- (a) $P(M) + P(N) - 2P(M \cap N)$
(b) $P(M) + P(N) - P(\overline{M \cup N})$
(c) $P(\overline{M}) + P(\overline{N}) - 2P(\overline{M} \cap \overline{N})$
(d) $P(M \cap \overline{N}) - P(\overline{M} \cap N)$

Ans. (a,c)

Sol. $P(\text{exactly one of } M, N \text{ occurs})$

$$= P\{(M \cap \overline{N}) \cup (\overline{M} \cap N)\} = P(M \cap \overline{N}) + P(\overline{M} \cap N)$$

$$= P(M) - P(M \cap N) + P(N) - P(M \cap N)$$

$$= P(M) + P(N) - 2P(M \cap N)$$

Also, $P(\text{exactly one of them occurs})$

$$= \{1 - P(\overline{M} \cap \overline{N})\} \{1 - P(\overline{M} \cup \overline{N})\}$$

$$= P(\overline{M} \cup \overline{N}) - P(\overline{M} \cap \overline{N}) = P(\overline{M}) + P(\overline{N}) - 2P(\overline{M} \cap \overline{N})$$

Hence, (a) and (c) are correct answers.

Example – 16

For two given events A and B , $P(A \cap B)$ is

- (a) not less than $P(A) + P(B) - 1$
(b) not greater than $P(A) + P(B)$
(c) equal to $P(A) + P(B) - P(A \cup B)$
(d) equal to $P(A) + P(B) + P(A \cup B)$

Ans. (a,b,c)

Sol. We know that,

$$P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

Also, $P(A \cup B) \leq 1$

$$\therefore P(A \cap B)_{\min}, \text{ when } P(A \cup B)_{\max} = 1$$

$$\Rightarrow P(A \cap B) \geq P(A) + P(B) - 1$$

$\therefore$ Option (a) is true.

Again, $P(A \cup B) \geq 0$

$$\therefore P(A \cap B)_{\max}, \text{ when } P(A \cup B)_{\min} = 0$$

$$\Rightarrow P(A \cap B) \leq P(A) + P(B)$$

$\therefore$ Option (b) is true.

Also, $P(A \cap B) = P(A) + P(B) - P(A \cup B)$, Thus, (c) is also correct.

Hence, (a), (b), (c) are correct options.

Example – 17

If E and F are independent events such that $0 < P(E) < 1$ and $0 < P(F) < 1$, then

- (a) E and F are mutually exclusive
(b) E and F^c (the complement of the event F) are independent
(c) E^c and F^c are independent
(d) $P(E/F) + P(E^c/F) = 1$

Ans. (b,c,d)

Sol. Since, E and F are independent events. Therefore, $P(E \cap F) = P(E) \cdot P(F) \neq 0$, so E and F are not mutually exclusive events.



$$\text{Now, } P(E \cap \bar{F}) = P(E) - P(E \cap F) = P(E) - P(E) \cdot P(F)$$

$$= P(E) [1 - P(F)] = P(E) \cdot P(\bar{F})$$

$$\text{and, } P(\bar{E} \cap \bar{F}) = P(\overline{E \cup F}) = 1 - P(E \cup F)$$

$$= 1 - [1 - P(\bar{E}) \cdot P(\bar{F})]$$

$$[\because E \text{ and } F \text{ are independent}]$$

$$= P(\bar{E}) \cdot P(\bar{F})$$

So, E and $\bar{F}$ as well as $\bar{E}$ and $\bar{F}$ are independent events.

$$\text{Now, } P(E/F) + P(\bar{E}/F) = \frac{P(E \cap F) + P(\bar{E} \cap F)}{P(F)}$$

$$= \frac{P(F)}{P(F)} = 1$$

Example – 18

For any two events A and B in a sample space

$$(a) P\left(\frac{A}{B}\right) \geq \frac{P(A) + P(B) - 1}{P(B)}, P(B) \neq 0 \text{ is always true}$$

$$(b) P(A \cap \bar{B}) = P(A) - P(A \cap B) \text{ does not hold}$$

$$(c) P(A \cup B) = 1 - P(\bar{A}) \cdot P(\bar{B}), \text{ if } A \text{ and } B \text{ are independent}$$

$$(d) P(A \cup B) = 1 - P(\bar{A}) \cdot P(\bar{B}), \text{ if } A \text{ and } B \text{ are disjoint}$$

Ans. (a,c)

Sol. We know that,

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{P(A) + P(B) - P(A \cup B)}{P(B)}$$

$$\text{Since, } P(A \cup B) \leq 1$$

$$\Rightarrow -P(A \cup B) \geq -1$$

$$\Rightarrow P(A) + P(B) - P(A \cup B) \geq P(A) + P(B) - 1$$

$$\Rightarrow \frac{P(A) + P(B) - P(A \cup B)}{P(B)} \geq \frac{P(A) + P(B) - 1}{P(B)}$$

$$\Rightarrow P\left(\frac{A}{B}\right) \geq \frac{P(A) + P(B) - 1}{P(B)}$$

Hence, option (a) is correct.

The choice (b) holds only for disjoint i.e. $P(A \cap B) = 0$

$$\text{Finally, } P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= P(A) + P(B) - P(A) \cdot P(B),$$

if A, B are independent

$$= 1 - \{1 - P(A)\} \{1 - P(B)\} = 1 - P(\bar{A}) \cdot P(\bar{B})$$

Hence, option (c) is correct, but option (d) is not correct.

Example – 19

Let E and F be two independent events. The probability that both E and F happen is $1/12$ and the probability that neither E nor F happen is $1/2$. Then,

$$(a) P(E) = 1/3, P(F) = 1/4$$

$$(b) P(E) = 1/2, P(F) = 1/6$$

$$(c) P(E) = 1/6, P(F) = 1/2$$

$$(d) P(E) = 1/4, P(F) = 1/3$$

Ans. (a,d)

$$\text{Sol. Both } E \text{ and } F \text{ happen} \Rightarrow P(E \cap F) = \frac{1}{12}$$

$$\text{and neither } E \text{ nor } F \text{ happens} \Rightarrow P(\bar{E} \cap \bar{F}) = \frac{1}{2}$$

But for independent events, we have

$$P(E \cap F) = P(E)P(F) = \frac{1}{12} \quad \dots(i)$$

$$\text{and } P(\bar{E} \cap \bar{F}) = P(\bar{E})P(\bar{F})$$

$$= \{1 - P(E)\} \cdot \{1 - P(F)\}$$

$$= 1 - P(E) - P(F) + P(E)P(F)$$

$$\Rightarrow \frac{1}{2} = 1 - \{P(E) + P(F)\} + \frac{1}{12}$$

$$\Rightarrow P(E) + P(F) = 1 - \frac{1}{2} + \frac{1}{12} = \frac{7}{12} \quad \dots(ii)$$

On solving Eqs. (i) and (ii), we get

$$\text{either } P(E) = \frac{1}{4} \text{ and } P(F) = \frac{1}{3}.$$

$$\text{or, } P(E) = \frac{1}{3} \text{ and } P(F) = \frac{1}{4}$$



Example – 20

The probabilities that a student passes in Mathematics, Physics and Chemistry are m , p and c respectively. Of these subjects, the student has a 75% chance of passing in at least one, a 50% chance of passing in at least two, and a 40% chance of passing in exactly two. Which of the following relations are true?

(a) $p + m + c = \frac{19}{20}$ (b) $p + m + c = \frac{27}{20}$

(c) $pmc = \frac{1}{10}$ (d) $pmc = \frac{1}{4}$

Ans. (b,c)

Sol. Let A, B and C respectively denote the events that the student passes in Maths, Physics and Chemistry.

It is given.

$$P(A) = m, P(B) = p \text{ and } P(C) = c \text{ and}$$

$$P(\text{passing atleast in one subject}) = P(A \cup B \cup C) = 0.75$$

$$\Rightarrow 1 - P(A' \cap B' \cap C') = 0.75$$

$$\therefore [P(A) = 1 - P(\bar{A})]$$

$$\text{and } [P(\overline{A \cup B \cup C}) = P(A' \cap B' \cap C')]$$

$$\Rightarrow 1 - P(A') \cdot P(B') \cdot P(C') = 0.75$$

($\because$ A, B and C are independent events, therefore A' , B' and C' are independent events)

$$\Rightarrow 0.75 = 1 - (1 - m)(1 - p)(1 - c)$$

$$\Rightarrow 0.25 = (1 - m)(1 - p)(1 - c) \quad \dots(i)$$

$$\text{Also, } P(\text{passing exactly in two subjects}) = 0.4$$

$$\Rightarrow P(A \cap B \cap \bar{C}) + P(A \cap \bar{B} \cap C) + P(\bar{A} \cap B \cap C) = 0.4$$

$$\Rightarrow P(A \cap B \cap \bar{C}) + P(A \cap \bar{B} \cap C) + P(\bar{A} \cap B \cap C) = 0.4$$

$$\Rightarrow P(A) P(B) P(\bar{C}) + P(A) P(\bar{B}) P(C)$$

$$+ P(\bar{A}) P(B) P(C) = 0.4$$

$$\Rightarrow pm(1 - c) + p(1 - m)c + (1 - p)mc = 0.4$$

$$\Rightarrow pm - pmc + pc - pmc + mc - pmc = 0.4 \quad \dots(iii)$$

$$\text{Again, } P(\text{passing atleast in two subjects}) = 0.5$$

$$\Rightarrow P(A \cap B \cap \bar{C}) + P(A \cap \bar{B} \cap C) + P(\bar{A} \cap B \cap C)$$

$$+ P(A \cap B \cap C) = 0.5$$

$$\Rightarrow pm(1 - c) + pc(1 - m) + cm(1 - p) + pcm = 0.5$$

$$\Rightarrow pm - pmc + pc - pmc + cm - pmc + pcm = 0.5$$

$$\Rightarrow (pm + pc + mc) - 2pmc = 0.5 \quad \dots(ii)$$

From Eq. (ii),

$$pm + pc + mc - 3pmc = 0.4 \quad \dots(iv)$$

From Eq.(i),

$$0.25 = 1 - (m + p + c) + (pm + pc + cm) - pmc \quad \dots(v)$$

On solving Eqs. (iii), (iv) and (v), we get

$$p+m+c = 1.35 = \frac{27}{20}$$

Therefore, option (b) is correct.

$$\text{Also, from Eqs. (ii) and (iii), we get } pmc = \frac{1}{10}$$

Hence, option (c) is correct.

Example – 21

If p & q are chosen randomly from the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ with replacement. Determine the probability that the roots of the equation $x^2 + px + q = 0$ are real.

Ans. (31/50)

Sol. The required probability = 1 - (probability of the event that the roots of $x^2 + px + q = 0$ are non-real).

The roots of $x^2 + px + q = 0$ will be non-real if and only if $p^2 - 4q < 0$, i.e. if $p^2 < 4q$

The possible values of p and q can be possible according to the following table.

Value of q	Value of p	Number of pairs of p, q
1	1	1
2	1, 2	2
3	1, 2, 3	3
4	1, 2, 3	3
5	1, 2, 3, 4	4
6	1, 2, 3, 4	4
7	1, 2, 3, 4, 5	5
8	1, 2, 3, 4, 5	5
9	1, 2, 3, 4, 5	5
10	1, 2, 3, 4, 5, 6	6

Therefore, the number of possible pairs = 38

Also, the total number of possible pairs is $10 \times 10 = 100$

$$\therefore \text{The required probability} = 1 - \frac{38}{100} = 1 - 0.38 = 0.62$$



Example – 22

There is 30% chance that it rains on any particular day. What is the probability that there is at least one rainy day within a period of 7 – days ? Given that there is at least one rainy day, what is the probability that there are at least two rainy days ?

Ans. $1 - (0.7)^7, \frac{[1 - (0.7)^7 - {}^7C_1(0.3)(0.7)^6]}{1 - (0.7)^7}$

Sol. Probability of rain on any particular day = 0.3

Probability of no rain on any particular day
= $1 - 0.3 = 0.7$

Probability of no rain for 7 days = $(0.7)^7$

probability for atleast one rainy day = $1 - (0.7)^7$

Now let A = there are atleast two rainy days

B = there is atleast one rainy day.

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{[1 - (0.7)^7 - {}^7C_1(0.3)(0.7)^6]}{1 - (0.7)^7}$$

Example – 23

3 players A, B & C toss a coin cyclically in that order (that is A, B, C, A, B, C, A, B,) till a head shows. Let p be the probability that the coin shows a head. Let α , β & γ be respectively the probabilities that A, B and C gets the first head. Prove that

$\beta = (1 - p)\alpha$. Determine α , β & γ (in terms of p).

Ans. $\alpha = \frac{p}{1 - (1 - p)^3}, \beta = \frac{(1 - p)p}{1 - (1 - p)^3}, \gamma = \frac{(1 - p)^2 p}{1 - (1 - p)^3}$

Sol. Let $q = 1 - p$ = probability of getting the tail. We have

α = probability of A getting the head on tossing firstly

$$= P(H_1 \text{ or } T_1 T_2 T_3 H_4 \text{ or } T_1 T_2 T_3 T_4 T_5 T_6 H_7 \text{ or } \dots)$$

$$= P(H) + P(H)P(T)^3 + P(H)P(T)^6 + \dots$$

$$= \frac{P(H)}{1 - P(T)^3} = \frac{p}{1 - q^3}$$

Also,

β = probability of B getting the head on tossing secondly

$$= P(T_1 H_2 \text{ or } T_1 T_2 T_3 H_4 \text{ or } T_1 T_2 T_3 T_4 T_5 T_6 T_7 H_8 \text{ or } \dots)$$

$$= P(H)[P(T) + P(H)P(T)^4 + P(H)P(T)^7 + \dots]$$

$$= P(T)[P(H) + P(H)P(T)^3 + P(H)P(T)^6 + \dots]$$

$$= q\alpha = (1 - p)\alpha = \frac{p(1 - p)}{1 - q^3}$$

Again, we have

$$\alpha + \beta + \gamma = 1$$

$$\Rightarrow \gamma = 1 - (\alpha + \beta) = 1 - \frac{p + p(1 - p)}{1 - q^3}$$

$$= 1 - \frac{p + p(1 - p)}{1 - (1 - p)^3}$$

$$= \frac{1 - (1 - p)^3 - p - p(1 - p)}{1 - (1 - p)^3}$$

$$\gamma = \frac{1 - (1 - p)^3 - 2p + p^2}{1 - (1 - p)^3} = \frac{p - 2p^2 + p^3}{1 - (1 - p)^3}$$

$$\text{Also, } \alpha = \frac{p}{1 - (1 - p)^3}, \beta = \frac{p(1 - p)}{1 - (1 - p)^3}$$

Example – 24

Eight players $P_1, P_2, P_3, \dots, P_8$ play a knock-out tournament. It is known that whenever the players P_i and P_j play, the player P_i will win if $i < j$. Assuming that the players are paired at random in each round, what is the probability that the player P_4 reaches the final.

Ans. (4/35)

Sol. The number of ways in which $P_1, P_2, \dots, P_8$ can be paired in four pairs

$$= \frac{1}{4!} [({}^8C_2)({}^6C_2)({}^4C_2)({}^2C_2)]$$

$$= \frac{1}{4!} \times \frac{8!}{2!6!} \times \frac{6!}{2!4!} \times \frac{4!}{2!2!} \times 1$$

$$= \frac{1}{4!} \times \frac{8 \times 7}{2! \times 1} \times \frac{6 \times 5}{2! \times 1} \times \frac{4 \times 3}{2! \times 1} = \frac{8 \times 7 \times 6 \times 5}{2.2.2.2} = 105$$

Now, atleast two players certainly reach the second round between P_1, P_2 and P_3 and P_4 can reach in final if exactly two players play against each other between P_1, P_2, P_3 and remaining player will play against one of the remaining three from $P_5 \dots P_8$.

This can be possible in



$${}^3C_2 \times {}^4C_1 \times {}^3C_1 = 3.4.3 = 36 \text{ ways}$$

$\therefore$ Probability that P_4 and exactly one of $P_5 \dots P_8$ reach second round

$$= \frac{36}{105} = \frac{12}{35}$$

If P_i, P_j, P_4 and P_j , where $i=2$ or 3 and $j=5$ or 6 or 7 reach the second round, then they

can be paired in 2 pairs in $\frac{1}{2!}({}^4C_2)({}^2C_2) = 3$ ways.

But P_4 will reach the final, if P_i plays

Against P_i and P_4 plays against P_j ,

Hence, the probability that P_4 will reach the final round

from the second = $\frac{1}{3}$

$\therefore$ Probability that P_4 will reach the final is $\frac{12}{35} \times \frac{1}{3} = \frac{4}{35}$

Example – 25

Four cards are drawn from a pack of 52 playing cards. Find the probability (correct upto two places of decimals) of drawing exactly one pair.

Ans. (0.3)

Sol. Total number of outcomes = ${}^{52}C_4$

Now there are exactly 13 group of cards each group containing card having equal number. Choose any one group and in that group out of 4 cards choose any two cards which will make exactly one pair. Now rest of 12 groups select one card from each group. Hence

$$\text{Favourable outcomes} = {}^{13}C_1 \times {}^4C_2 \times {}^{12}C_2 \times {}^4C_1 \times {}^4C_1$$

Required probability

$$= \frac{{}^{13}C_1 \times {}^4C_2 \times {}^{12}C_2 \times {}^4C_1 \times {}^4C_1}{{}^{52}C_4} = 0.3$$

Example – 26

A box contains 2 black, 4 white and 3 red balls. One ball is drawn at random from the box and kept aside. From the remaining balls in the box, another ball is drawn at random and kept beside the first. This process is repeated till all the balls are drawn from the box. Find the probability that the balls drawn are in the sequence of 2 black, 4 white and 3 red.

Ans. $\frac{1}{1260}$

Sol. Since, the drawn balls are in the sequence black, black, white, white, white, white, red, red and red.

Let the corresponding probabilities be $p_1, p_2, \dots, p_9$

$$\text{Then, } p_1 = \frac{2}{9}, p_2 = \frac{1}{8}, p_3 = \frac{4}{7}, p_4 = \frac{3}{6}, p_5 = \frac{2}{5},$$

$$p_6 = \frac{1}{4}, p_7 = \frac{3}{3}, p_8 = \frac{2}{2}, p_9 = 1$$

$\therefore$ Required probability

$$p_1 \cdot p_2 \cdot p_3 \dots p_9 = \left(\frac{2}{9}\right)\left(\frac{1}{8}\right)\left(\frac{4}{7}\right)\left(\frac{3}{6}\right)\left(\frac{2}{5}\right)\left(\frac{1}{4}\right)\left(\frac{3}{3}\right)\left(\frac{2}{2}\right)(1) = \frac{1}{1260}$$

Example – 27

An anti-aircraft gun can take a maximum of four shots at an enemy plane moving away from it. The probabilities of hitting the plane at the first, second, third and fourth shot are 0.4, 0.3, 0.2, and 0.1 respectively. What is the probability that the gun hits the plane?

Ans. (0.6976)

Sol. Let $P(H_1) = 0.4, P(H_2) = 0.3, P(H_3) = 0.2, P(H_4) = 0.1$

P (gun hits the plane)

$$= 1 - P(\bar{H}_1) \cdot P(\bar{H}_2) \cdot P(\bar{H}_3) \cdot P(\bar{H}_4)$$

$$= 1 - (0.6)(0.7)(0.8)(0.9) = 1 - 0.3024 = 0.6976$$



Example – 28

Cards are drawn one by one at random from a well shuffled full pack of 52 playing cards until 2 aces are obtained for the first time. If N is the number of cards required to be drawn, then show that

$$P_r \{N = n\} = \frac{(n-1)(52-n)(51-n)}{50 \times 49 \times 17 \times 13}$$

where $2 < n \leq 50$.

Sol. $P(N^{\text{th}}$ draw gives 2nd ace)
 $= P\{1 \text{ ace and } (n-2) \text{ other cards are drawn in } (N-1) \text{ draws}\}$
 $\times P\{N^{\text{th}}$ draw is 2nd ace}

$$= \frac{{}^4C_1 \times 48 {}^{C_{n-2}} \times 3 {}^{C_1}}{52 {}^{C_{n-1}} \times (53-n)}$$

$$= \frac{4 \cdot (48)! \cdot (n-1)! \cdot (53-n)!}{(52)! \cdot (n-2)! \cdot (50-n)!} \cdot \frac{3}{(53-n)}$$

$$= \frac{4(n-1)(52-n)(51-n) \cdot 3}{52 \cdot 51 \cdot 50 \cdot 49}$$

$$= \frac{(n-1)(52-n)(51-n)}{50 \cdot 49 \cdot 17 \cdot 13}$$

Example – 29

A, B, C are events such that

$$P_r(A) = 0.3, P_r(B) = 0.4, P_r(C) = 0.8,$$

$$P_r(AB) = 0.08, P_r(AC) = 0.28, P_r(ABC) = 0.09.$$

If $P_r(A \cup B \cup C) \geq 0.75$, then show that $P_r(BC)$ lies in the interval $0.23 \leq x \leq 0.48$.

Sol. We know that,
 $P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - (C \cap A)$
 $+ P(A \cap B \cap C) = P(A \cup B \cup C)$
 $\Rightarrow 0.3 + 0.4 + 0.8 - \{0.08 + 0.28 + P(BC)\} + 0.09$
 $= P(A \cup B \cup C)$
 $\Rightarrow 1.23 - P(BC) = P(A \cup B \cup C)$
 where, $0.75 \leq P(A \cup B \cup C) \leq 1$
 $\Rightarrow 0.75 \leq 1.23 - P(BC) \leq 1$
 $\Rightarrow -0.48 \leq -P(BC) \leq -0.23$
 $\Rightarrow 0.23 \leq P(BC) \leq 0.48$
 $\Rightarrow 0.23 \leq x \leq 0.48$

Example – 30

In a certain city only two newspapers A and B are published, it is known that 25% of the city population reads A and 20% reads B , while 8% reads both A and B . It is also known that 30% of those who read A but not B look into advertisements and 40% of those who read B but not A look into advertisements while 50% of those who read both A and B look into advertisements. What is the percentage of the population reads an advertisement?

Ans. (13.9%)

Sol. Let $P(A)$ and $P(B)$ denote respectively the percentage of city population that reads newspapers A and B . Then,

$$P(A) = \frac{25}{100} = \frac{1}{4}, P(B) = \frac{20}{100} = \frac{1}{5},$$

$$P(A \cap B) = \frac{8}{100} = \frac{2}{25},$$

$$P(A \cap B) = P(A) - P(A \cap B) = \frac{1}{4} - \frac{2}{25} = \frac{17}{100},$$

$$P(A \cap B) = P(B) - P(A \cap B) = \frac{1}{5} - \frac{2}{25} = \frac{3}{25}$$

Let $P(C)$ be the probability that the population who reads advertisements.

$$\therefore P(C) = 30\% \text{ of } P(A \cap \bar{B}) + 40\% \text{ of } P(\bar{A} \cap B)$$

$$+ 50\% \text{ of } P(A \cap B)$$

[since, $A \cap \bar{B}$, $\bar{A} \cap B$ and $A \cap B$ are all mutually exclusive]

$$\Rightarrow P(C) = \frac{3}{10} \times \frac{17}{100} + \frac{2}{5} \times \frac{3}{25} + \frac{1}{2} \times \frac{2}{25} = \frac{139}{1000} = 13.9\%$$

Example – 31

In a multiple-choice question there are four alternative answers, of which one or more are correct. A candidate will get marks in the question only if he ticks the correct answers. The candidate decides to tick the answers at random. If he is allowed upto three chances to answer the questions, find the probability that he will get marks in the question.

Ans. $\frac{1}{5}$

Sol. The total number of ways to answer the question



$$= {}^4C_1 + {}^4C_2 + {}^4C_3 + {}^4C_4 = 2^4 - 1 = 15$$

P(getting marks) = P(correct answer in I chance)
+ P(correct answer in II chance)
+ P(correct answer in III chance)

$$= \frac{1}{15} + \left(\frac{14}{15} \cdot \frac{1}{14} \right) + \left(\frac{14}{15} \cdot \frac{13}{14} \cdot \frac{1}{13} \right) = \frac{3}{15} = \frac{1}{5}$$

Example – 32

A lot contains 20 articles. The probability that the lot contains exactly 2 defective articles is 0.4 and the probability that the lot contains exactly 3 defective articles is 0.6. Articles are drawn from the lot at random one by one without replacement and are tested till all defective articles are found. What is the probability that the testing procedure ends at the twelfth testing ?

Ans. $\frac{99}{1900}$

Sol. The testing procedure may terminate at the twelfth testing in two mutually exclusive ways.

I : When lot contains 2 defective articles.

II : When lot contains 3 defective articles.

Let A = testing procedure ends at twelfth testing

A_1 = lot contains 2 defective articles

A_2 = lot contains 3 defective articles

$\therefore$ Required probability = $P(A_1) \cdot P(A/A_1) + P(A_2) \cdot P(A/A_2)$

Here, $P(A/A_1)$ = probability that first 11 draws contains 10 non-defective and one-defective and twelfth draw contains a defective article.

$$= \frac{{}^{18}C_{10} \times {}^2C_1}{{}^{20}C_{11}} \times \frac{1}{9} \dots (i)$$

$P(A/A_2)$ = probability that first 11 draws contains 9 non-defective and 2-defective articles and twelfth draw

$$\text{contains defective} = \frac{{}^{17}C_9 \times {}^3C_2}{{}^{20}C_{11}} \times \frac{1}{9} \dots (ii)$$

$\therefore$ Required probability = $(0.4)P(A/A_1) + 0.6P(A/A_2)$

$$= \frac{0.4 \times {}^{18}C_{10} \times {}^2C_1}{{}^{20}C_{11}} \times \frac{1}{9} + \frac{0.6 \times {}^{17}C_9 \times {}^3C_2}{{}^{20}C_{11}} \times \frac{1}{9} = \frac{99}{1900}$$

Example – 33

A man takes a step forward with probability 0.4 and backwards with probability 0.6. Find the probability that at the end of eleven steps he is one step away from the starting point.

Ans. $({}^{11}C_6 (0.24)^5)$

Sol. The man will be one step away from the starting point, if (i) either he is one step ahead or (ii) one step behind the starting point

The man will be one step ahead at the end of eleven steps, if he moves six steps forward and five steps backward. The probability of this event is ${}^{11}C_6 (0.4)^6 (0.6)^5$.

The man will be one step behind at the end of eleven steps, if he moves six steps backward and five steps forward. The probability of this event is ${}^{11}C_6 (0.6)^6 (0.4)^5$.

$\therefore$ Required probability = ${}^{11}C_6 (0.4)^6 (0.6)^5$

$$+ {}^{11}C_6 (0.6)^6 (0.4)^5 = {}^{11}C_6 (0.24)^5$$

Example – 34

An urn contains 2 white and 2 blacks balls. A ball is drawn at random. If it is white it is not replaced into the urn. Otherwise it is replaced along with another ball of the same colour. The process is repeated. Find the probability that the third ball drawn is black.

Ans. $\frac{23}{30}$

Sol. Let B_i = i th ball drawn is black.

W_i = i th ball drawn is white, where $i = 1, 2$

and A = third ball drawn is black.

We observe that the black ball can be drawn in the third draw in one of the following mutually exclusive ways.

i) Both first and second balls drawn are white and third ball drawn is black.

i.e. $(W_1 \cap W_2) \cap A$

ii) Both first and second balls are black and third ball drawn is black.

i.e. $(B_1 \cap B_2) \cap A$

iii) The first ball drawn is white, the second ball drawn is black and the third ball drawn is black

i.e. $(W_1 \cap B_2) \cap A$

iv) The first ball drawn is black, the second ball drawn is



white and the third ball drawn is black

i.e. $(B_1 \cap W_2) \cap A$

$\therefore P(A) = P[\{(W_1 \cap W_2) \cap A\} \cup \{(B_1 \cap B_2) \cap A\}$

$\cup \{(W_1 \cap B_2) \cap A\} \cup \{(B_1 \cap W_2) \cap A\}$

$= P[\{(W_1 \cap W_2) \cap A\} + P\{(B_1 \cap B_2) \cap A\}$

$+ P\{(W_1 \cap B_2) \cap A\} + P\{(B_1 \cap W_2) \cap A\}$

$= P\{(W_1 \cap W_2)\} \cdot P(A/(W_1 \cap W_2)) + P(B_1 \cap B_2)$

$\therefore P(A/(B_1 \cap B_2)) + P(W_1 \cap B_2) \cdot P(A/(W_1 \cap B_2))$

$+ P(B_1 \cap W_2) \cdot P(A/(B_1 \cap W_2))$

$$= \left(\frac{2}{4} \times \frac{1}{3}\right) \times 1 + \left(\frac{2}{4} \times \frac{3}{5}\right) \times \frac{4}{6} + \left(\frac{2}{4} \times \frac{2}{3}\right) \times \frac{3}{4} + \left(\frac{2}{4} \times \frac{2}{5}\right) \times \frac{3}{4}$$

$$= \frac{1}{6} + \frac{1}{5} + \frac{1}{4} + \frac{3}{20} = \frac{23}{30}$$

Example – 35

A box contains 2 fifty paise coins, 5 twenty five paise coins and a certain fixed number n (≥ 2) of ten and five paise coins. Five coins are taken out of the box at random. Find the probability that the total value of these 5 coins is less than one rupee and fifty paise.

Ans. $1 - \frac{10(n+2)}{n+7} C_5$

Sol. There are $(n+7)$ coins in the box out of which five coin can be taken out in ${}^{n+7}C_5$ ways.

The total value of 5 coins can be equal to or more than one rupee and fifty paise in the following ways.

- When one 50 paise coin and four 25 paise coins are chosen.
- When two 50 paise coins and three 25 paise coins are chosen.
- When two 50 paise coins, 2 twenty five paise coins and one from n coins of ten and five paise

$\therefore$ The total number of ways of selecting the coins

so that the total value of the coins is not less than one rupee and fifty paise is

$$({}^2C_1 \cdot {}^5C_4 \cdot {}^nC_0) + ({}^2C_2 \cdot {}^5C_3 \cdot {}^nC_0) + ({}^2C_2 \cdot {}^5C_2 \cdot {}^nC_1) = 10 + 10 + 10n = 10(n+2)$$

So, the number of ways of selecting five coins, so that the total value of the coins is less than one rupee and fifty paise is ${}^{n+7}C_5 - 10(n+2)$

$$\therefore \text{Required probability} = \frac{{}^{n+7}C_5 - 10(n+2)}{{}^{n+7}C_5}$$

$$= 1 - \frac{10(n+2)}{{}^{n+7}C_5}$$

Example – 36

Suppose the probability for A to win a game against B is 0.4. If A has an option of playing either a “best of 3 games” or a “best of 5 games” match against B, which option should choose so that the probability of his winning the match is higher? (No game ends in a draw).

Ans. (best of 3 games)

Sol. **Case I:** When A plays 3 games against B.

In this case, we have $n = 3$, $p = 0.4$ and $q = 0.6$

Let X denote the number of wins. Then,

$$P(X=r) = {}^nC_r p^r q^{n-r} = {}^3C_r (0.4)^r (0.6)^{3-r}; r = 0, 1, 2, 3$$

$\therefore P_1$ = probability of winning the best of 3 games

$$= P(X \geq 2)$$

$$= P(X=2) + P(X=3)$$

$$= {}^3C_2 (0.4)^2 (0.6)^1 + {}^3C_3 (0.4)^3 (0.6)^0$$

$$= 0.288 + 0.064 = 0.352$$

Case II: When A plays 5 games against B.

In this case, we have $n = 5$, $p = 0.4$ and $q = 0.6$

Let X denote the number of wins in 5 games.

Then,

$$P(X=r) = {}^5C_r (0.4)^r (0.6)^{5-r}; \text{ where } r = 0, 1, 2, \dots, 5$$

$\therefore P_2$ = probability of winning the best of 5 games

$$= P(X \geq 3)$$

$$= P(X=3) + P(X=4) + P(X=5)$$

$$= {}^5C_3 (0.4)^3 (0.6)^2 + {}^5C_4 (0.4)^4 (0.6) + {}^5C_5 (0.4)^5 (0.6)^0$$

$$= 0.2304 + 0.0768 + 0.1024 = 0.31744$$

Clearly, $P_1 > P_2$. Therefore, first option i.e. ‘best of 3 games’ has higher probability of winning the match.



Example – 37

An unbiased coin is tossed. If the result is a head, a pair of unbiased dice is rolled and the number obtained by adding the numbers on the two faces is noted. If the result is a tail, a card from a well shuffled pack of eleven cards numbered 2, 3, 4, ..., 12 is picked and the number on the card is noted. What is the probability that the noted number is either 7 or 8 ?

Ans. $\frac{193}{792}$

Sol. Let, E_1 = the event that the noted number is 7

E_2 = the event that the noted number is 8

H = getting head on coin

T = getting tail on coin

∴ By law of total probability,

$$P(E_1) = P(H) \cdot P(E_1/H) + P(T) \cdot P(E_1/T)$$

$$\text{and } P(E_2) = P(H) \cdot P(E_2/H) + P(T) \cdot P(E_2/T)$$

Where, $P(H) = 1/2 = P(T)$

$P(E_1/H)$ = probability of getting a sum of 7 on two dice.

Here, favourable cases are

$$\{(1, 6), (6, 1), (2, 5), (5, 2), (3, 4), (4, 3)\}.$$

$$\therefore P(E_1/H) = \frac{6}{36} = \frac{1}{6}$$

Also, $P(E_1/T)$ = probability of getting 7 numbered card out of 11 cards

$$= \frac{1}{11}$$

$P(E_2/H)$ = probability of getting a sum of 8 on two dice

Here, favourable cases are

$$\{(2, 6), (6, 2), (4, 4), (5, 3), (3, 5)\}$$

$$\therefore P(E_2/H) = \frac{5}{36}$$

$P(E_2/T)$ = probability of getting '8' numbered card out of 11 cards

$$= 1/11$$

$$\therefore P(E_1) = \left(\frac{1}{2} \times \frac{1}{6}\right) + \left(\frac{1}{2} \times \frac{1}{11}\right) = \frac{1}{12} + \frac{1}{22} = \frac{17}{132}$$

$$\text{and } P(E_2) = \left(\frac{1}{2} \times \frac{5}{36}\right) + \left(\frac{1}{2} \times \frac{1}{11}\right) = \frac{1}{2} \left(\frac{91}{396}\right) = \frac{91}{792}$$

Now, E_1 and E_2 are mutually exclusive events.

Therefore,

$$P(E_1 \text{ or } E_2) = P(E_1) + P(E_2) = \frac{17}{132} + \frac{91}{792} = \frac{193}{792}$$

Example – 38

In how many ways three girls and nine boys can be seated in two vans, each having numbered seats, 3 in the front and 4 at the back ? How many seating arrangements are possible if 3 girls should sit together in a back row on adjacent seats ? Now, if all the seating arrangements are equally likely, what is the probability of 3 girls sitting together in a back row on adjacent seats ?

Ans. $\frac{14!}{2!}, 2 \times 2 \times 3! \times {}^{11}P_9, \frac{1}{91}$

Sol. We have 14 seats in two vans and there are 9 boys and 3 girls. The number of ways of arranging 12 people on 14 seats without restriction is

$${}^{14}P_{12} = \frac{14!}{2!} = 7(13!)$$

Now, the number of ways of choosing back seats is 2. and the number of ways of arranging 3 girls on adjacent seats is $2(3!)$ and the number of ways of arranging 9 boys on the remaining 11 seats is ${}^{11}P_9$ ways.

Therefore, the required number of ways.

$$= 2 \cdot (2 \cdot 3!) \cdot {}^{11}P_9 = \frac{4 \cdot 3! \cdot 11!}{2!} = 12!$$

Hence, the probability of the required event

$$= \frac{12!}{7 \cdot 13!} = \frac{1}{91}$$



Example – 39

Sixteen players $S_1, S_2, \dots, S_{16}$ play in a tournament. They are divided into eight pairs at random. From each pair a winner is decided on the basis of a game played between the two players of the pair. Assume that all the players are of equal strength.

- Find the probability that the players S_1 is among the eight winners.
- Find the probability that exactly one of the two players S_1 and S_2 is among the eight winners.

Ans. (a) $\frac{1}{2}$ (b) $\frac{8}{15}$

Sol. i) Probability of S_1 to be among the eight winners
 = (Probability of S_1 being a pair) $\times$ (Probability of S_1 winning in the group)

$$= 1 \times \frac{1}{2} = \frac{1}{2} \text{ [since, } S_1 \text{ is definitely in a group]}$$

ii) If S_1 and S_2 are in the same pair, then exactly one wins.

If S_1 and S_2 are in two pairs separately, then exactly one of S_1 and S_2 will be among the eight winners. If S_1 wins and S_2 loses or S_1 loses and S_2 wins.

Now, the probability of S_1, S_2 being in the same pair and one wins

$$= (\text{Probability of } S_1, S_2 \text{ being in the same pair}) \times (\text{Probability of any one winning in the pair})$$

$$\text{and the probability of } S_1, S_2 \text{ being the same pair} = \frac{n(E)}{n(S)}$$

where, $n(E)$ = the number of ways in which 16 persons can be divided in 8 pairs.

$$\therefore n(E) = \frac{(14)!}{(2!)^7 \cdot 7!} \text{ and } n(S) = \frac{(16)!}{(2!)^8 \cdot 8!}$$

$\therefore$ Probability of S_1, S_2 being in the same pair

$$= \frac{(14)! \cdot (2!)^8 \cdot 8!}{(2!)^7 \cdot 7! \cdot (16)!} = \frac{1}{15}$$

The probability of any one winning in the pairs of $S_1, S_2 = P(\text{certain event}) = 1$

$\therefore$ The pairs of S_1, S_2 being in two pairs separately and S_1 wins, S_2 loses + The probability of S_1, S_2 being in two pairs separately and S_1 loses, S_2 wins.

$$= \left[1 - \frac{(14)!}{(2!)^7 \cdot 7!} \cdot \frac{1}{(16)!} \right] \times \frac{1}{2} \times \frac{1}{2} + \left[1 - \frac{(14)!}{(2!)^7 \cdot 7!} \cdot \frac{1}{(16)!} \right] \times \frac{1}{2} \times \frac{1}{2}$$

$$= \frac{1}{2} \times \frac{14 \times (14)!}{15 \times (14)!} = \frac{7}{15}$$

$$\therefore \text{Required probability} = \frac{1}{15} + \frac{7}{15} = \frac{8}{15}$$

Example – 40

A determinant is chosen at random from the set of all determinants of order 2 with elements 0 or 1 only. The probability that the value of the determinant chosen is positive, is....

Ans. $\frac{3}{16}$

Sol. Since, determinat is of order 2×2 and each element is 0 or 1 only.

$$\therefore n(S) = 2^4 = 16$$

$$\text{and the determinat is positive are } \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}, \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix}, \begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix}$$

$$\therefore n(E) = 3$$

$$\text{Thus, the required probability} = \frac{3}{16}$$



EXERCISE - 1 : BASIC OBJECTIVE QUESTIONS

Classical Probability

- Three houses are available in a locality. Three persons apply for the houses. Each applies for one house without consulting others. The probability that all the three apply for the same house, is
 (a) $\frac{1}{9}$ (b) $\frac{2}{9}$
 (c) $\frac{7}{9}$ (d) $\frac{8}{9}$
- Fifteen coupons are numbered 1, 2, ..., 15, respectively. Seven coupons are selected at random one at a time with replacement. The probability that the largest number appearing on a selected coupon is 9, is
 (a) $\left(\frac{9}{16}\right)^6$ (b) $\left(\frac{8}{15}\right)^7$
 (c) $\left(\frac{3}{5}\right)^7$ (d) None of these
- Seven white balls and three black balls are randomly placed in a row. The probability that no two black balls are placed adjacently, equals
 (a) $\frac{1}{2}$ (b) $\frac{7}{15}$
 (c) $\frac{2}{15}$ (d) $\frac{1}{3}$
- An Urn contains 'm' white and 'n' black balls. All the balls except for one ball, are drawn from it. The probability that the last ball remaining in the Urn is white, is
 (a) $\frac{m}{m+n}$ (b) $\frac{n}{m+n}$
 (c) $\frac{1}{(m+n)!}$ (d) $\frac{mn}{(m+n)!}$
- Three of the six vertices of a regular hexagon are chosen at random. The probability that the triangle with three vertices is equilateral equals
 (a) $\frac{1}{2}$ (b) $\frac{1}{5}$
 (c) $\frac{1}{10}$ (d) $\frac{1}{20}$
- $x_1, x_2, x_3, \dots, x_{50}$ are fifty numbers such that $x_r < x_{r+1}$ for $r = 1, 2, 3, \dots, 49$. Five numbers out of these are picked up at random. The probability that the five numbers have x_{20} as the middle number is
 (a) $\frac{{}^{20}C_2 \times {}^{30}C_2}{{}^{50}C_5}$ (b) $\frac{{}^{30}C_2 \times {}^{19}C_2}{{}^{50}C_5}$
 (c) $\frac{{}^{19}C_2 \times {}^{31}C_3}{{}^{50}C_5}$ (d) none of these
- The probability that the birthdays of six different people will fall in exactly two calendar months is
 (a) $\frac{1}{6}$ (b) ${}^{12}C_2 \times \frac{2^6}{12^6}$
 (c) ${}^{12}C_2 \times \frac{2^6 - 1}{12^6}$ (d) $\frac{341}{12^5}$
- A, B, C are three events for which $P(A) = 0.6$, $P(B) = 0.4$, $P(C) = 0.5$, $P(A \cup B) = 0.8$, $P(A \cap C) = 0.3$ and $P(A \cap B \cap C) = 0.2$. If $P(A \cup B \cup C) \geq 0.85$ then the interval of values of $P(B \cap C)$ is
 (a) [0.2, 0.35] (b) [0.55, 0.7]
 (c) [0.2, 0.55] (d) none of these
- A and B are two events. Odds against A are 2 : 1. Odds in favour of $A \cup B$ are 3 : 1. If $x \leq P(B) \leq y$, then the ordered pair (x, y) is
 (a) $\left(\frac{5}{12}, \frac{3}{4}\right)$ (b) $\left(\frac{2}{3}, \frac{3}{4}\right)$
 (c) $\left(\frac{1}{3}, \frac{3}{4}\right)$ (d) none of these



10. If $P(B) = \frac{3}{5}$, $P(A/B) = \frac{1}{2}$ and $P(A \cup B) = \frac{4}{5}$, then

$$P((A \cup B)') + P(A' \cup B) =$$

- (a) $\frac{1}{5}$ (b) $\frac{4}{5}$
(c) $\frac{1}{2}$ (d) 1

Multiplication Theorem & Independent Events

11. A problem in mathematics is given to three student A, B, C and their respective probability of solving the problem is $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$. Probability that the problem is solved, is

- (a) $\frac{3}{4}$ (b) $\frac{1}{2}$
(c) $\frac{2}{3}$ (d) $\frac{1}{3}$

12. The probability that A speaks truth is $\frac{4}{5}$, while this probability for B is $\frac{3}{4}$. The probability that they contradict each other when asked to speak on a fact, is

- (a) $\frac{7}{20}$ (b) $\frac{1}{5}$
(c) $\frac{3}{20}$ (d) $\frac{4}{5}$

13. If A and B are independent events such that $0 < P(A) < 1$ and $0 < P(B) < 1$, then which of the following is not correct ?

- (a) A and B are mutually exclusive
(b) A and B' are independent
(c) A' and B are independent
(d) A' and B' are independent

14. Let A and B be two events such that

$$P(\overline{A \cup B}) = \frac{1}{6}, P(A \cap B) = \frac{1}{4} \text{ and}$$

$$P(\overline{A}) = \frac{1}{4}, \text{ where } \overline{A} \text{ stands for the complement of the}$$

event A. Then the events A and B are :

- (a) independent and equally likely
(b) mutually exclusive and independent
(c) equally likely but not independent
(d) independent but not equally likely

15. If from each of the three boxes containing 3 white and 1 black, 2 white and 2 black, 1 white and 3 black balls, one ball is drawn at random, then the probability that 2 white and 1 black balls will be drawn, is

- (a) $\frac{13}{32}$ (b) $\frac{1}{4}$
(c) $\frac{1}{32}$ (d) $\frac{3}{16}$

16. Three persons A, B and C fire at a target in turn, starting with A. Their probabilities of hitting the target are 0.4, 0.3 and 0.2 respectively. The probability of two hits is

- (a) 0.024 (b) 0.188
(c) 0.336 (d) 0.452

17. A box contains 3 orange balls, 3 green balls and 2 blue balls. Three balls are drawn at random from the box without replacement. The probability of drawing 2 green balls and one blue ball is

- (a) $\frac{2}{21}$ (b) $\frac{3}{28}$
(c) $\frac{1}{28}$ (d) $\frac{167}{168}$

18. The probability that certain electronic component fails when first used is 0.10. If it does not fail immediately, the probability that it lasts for one year is 0.99. The probability that a new component will last for one year is

- (a) 0.891 (b) 0.692
(c) 0.92 (d) none of these

19. The probability that an event A happens in one trial of an experiment is 0.4. Three independent trials of the experiments are performed. The probability that the event A happens at least once is

- (a) 0.936 (b) 0.784
(c) 0.904 (d) None of these

20. India plays two matches each with West Indies and Australia. In any match the probabilities of India getting points 0, 1 and 2 are 0.45, 0.05 and 0.50 respectively. Assuming that the outcomes are independent, the probability of India getting at least 7 points, is

- (a) 0.8750 (b) 0.0875
(c) 0.0625 (d) 0.0250



21. An unbiased die with faces marked 1, 2, 3, 4, 5 and 6 is rolled four times. Out of four face values obtained, the probability that the minimum face value is not less than 2 and the maximum face value is not greater than 5, is
(a) $\frac{16}{81}$ (b) $\frac{1}{81}$
(c) $\frac{80}{81}$ (d) $\frac{65}{81}$
22. A coin is tossed n times. The probability of getting at least one head is greater than that of getting at least two tails by $\frac{5}{32}$. Then n is
(a) 5 (b) 10
(c) 15 (d) none of these
23. An unbiased cubic die marked with 1, 2, 2, 3, 3, 3 is rolled 3 times. The probability of getting a total score of 4 or 6 is
(a) $\frac{16}{216}$ (b) $\frac{50}{216}$
(c) $\frac{60}{216}$ (d) none
24. A Urn contains 'm' white and 'n' black balls. Balls are drawn one by one till all the balls are drawn. Probability that the second drawn ball is white, is
(a) $\frac{m}{m+n}$ (b) $\frac{n(m+n-1)}{(m+n)(m+n-1)}$
(c) $\frac{m(m-1)}{(m+n)(m+n-1)}$ (d) $\frac{mn}{(m+n)(m+n-1)}$
25. The probabilities of four cricketers A, B, C and D scoring more than 50 runs in a match are $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$ and $\frac{1}{10}$. It is known that exactly two of the players scored more than 50 runs in a particular match. The probability that players were A and B is
(a) $\frac{27}{65}$ (b) $\frac{5}{6}$
(c) $\frac{1}{6}$ (d) none of these
26. Assume that in a family, each child is equally likely to be a boy or a girl. A family with three children is chosen at random. The probability that the eldest child is a girl given that the family has atleast one girl is
(a) $\frac{1}{2}$ (b) $\frac{1}{3}$
(c) $\frac{2}{3}$ (d) $\frac{4}{7}$
27. Let $A = \{2, 3, 4, \dots, 20, 21\}$. A number is chosen at random from the set A and it is found to be a prime number. The probability that it is more than 10 is
(a) $\frac{9}{10}$ (b) $\frac{1}{10}$
(c) $\frac{1}{2}$ (d) none of these
28. Three distinguishable balls are distributed in three cells. The probability that all three occupy the same cell, given that atleast two of them are in the same cell, is
(a) $\frac{1}{7}$ (b) $\frac{1}{9}$
(c) $\frac{1}{6}$ (d) none of these
29. In a certain town, 40% of the people have brown hair, 25% have brown eyes and 15% have both brown hair and brown eyes. If a person selected at random from the town, has brown hair, the probability that he also has brown eyes, is
(a) $\frac{1}{5}$ (b) $\frac{3}{8}$
(c) $\frac{1}{3}$ (d) $\frac{2}{3}$

Conditional Probability

25. The probabilities of four cricketers A, B, C and D scoring more than 50 runs in a match are $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$ and $\frac{1}{10}$. It is known that exactly two of the players scored more than 50 runs in a particular match. The probability that players were A and B is
(a) $\frac{27}{65}$ (b) $\frac{5}{6}$
(c) $\frac{1}{6}$ (d) none of these
26. Assume that in a family, each child is equally likely to be a boy or a girl. A family with three children is chosen at random. The probability that the eldest child is a girl given that the family has atleast one girl is
(a) $\frac{1}{2}$ (b) $\frac{1}{3}$
(c) $\frac{2}{3}$ (d) $\frac{4}{7}$
27. Let $A = \{2, 3, 4, \dots, 20, 21\}$. A number is chosen at random from the set A and it is found to be a prime number. The probability that it is more than 10 is
(a) $\frac{9}{10}$ (b) $\frac{1}{10}$
(c) $\frac{1}{2}$ (d) none of these
28. Three distinguishable balls are distributed in three cells. The probability that all three occupy the same cell, given that atleast two of them are in the same cell, is
(a) $\frac{1}{7}$ (b) $\frac{1}{9}$
(c) $\frac{1}{6}$ (d) none of these
29. In a certain town, 40% of the people have brown hair, 25% have brown eyes and 15% have both brown hair and brown eyes. If a person selected at random from the town, has brown hair, the probability that he also has brown eyes, is
(a) $\frac{1}{5}$ (b) $\frac{3}{8}$
(c) $\frac{1}{3}$ (d) $\frac{2}{3}$



30. A pair of numbers is picked up randomly (without replacement) from the set $\{1, 2, 3, 5, 7, 11, 12, 13, 17, 19\}$. The probability that the number 11 was picked given that the sum of the numbers was even, is nearly :

(a) 0.1 (b) 0.125
(c) 0.24 (d) 0.18

31. For a biased die the probabilities for the different faces to turn up are given below :

Faces :	1	2	3	4	5	6
Probabilities :	0.10	0.32	0.21	0.15	0.05	0.17

The die is tossed & you are told that either face one or face two has turned up. Then the probability that it is face one is :

(a) $1/6$ (b) $1/10$
(c) $5/49$ (d) $5/21$

32. The probability that an automobile will be stolen and found within one week is 0.0006. The probability that an automobile will be stolen is 0.0015. The probability that a stolen automobile will be found in one week is

(a) 0.3 (b) 0.4
(c) 0.5 (d) 0.6

Total Probability Law and Baye's Theorem

33. A letter is known to have come either from LONDON or CLIFTON; on the postmark only the two consecutive letters ON are legible. The probability that it came from LONDON is

(a) $\frac{5}{17}$ (b) $\frac{12}{17}$
(c) $\frac{17}{30}$ (d) $\frac{3}{5}$

34. For $k = 1, 2, 3$ the box B_k contains k red balls and $(k + 1)$ white balls. Let $P(B_1) = \frac{1}{2}$, $P(B_2) = \frac{1}{3}$ and $P(B_3) = \frac{1}{6}$. A box is selected at random and a ball is drawn from it. If a red ball is drawn, then the probability that it has come from box B_2 , is

(a) $\frac{35}{78}$ (b) $\frac{14}{39}$
(c) $\frac{10}{13}$ (d) $\frac{12}{13}$

35. In an entrance test there are multiple choice questions. There are four possible answers to each question of which one is correct. The probability that a student knows the answer to a question is 90%. If he gets the correct answer to a question, then the probability that he was guessing, is

(a) $\frac{37}{40}$ (b) $\frac{1}{37}$
(c) $\frac{36}{37}$ (d) $\frac{1}{9}$

36. Two coins are available, one fair and the other two headed. Choose a coin and toss it once assume that the

unbiased coin is chosen with probability $\frac{3}{4}$. Given that the outcome is head, the probability that the two-headed coin was chosen is

(a) $\frac{3}{5}$ (b) $\frac{2}{5}$
(c) $\frac{1}{5}$ (d) $\frac{2}{7}$

37. A man is known to speak truth 3 out of 4 times. He throws a dice and reports that it is six. The probability that it is actually six is

(a) $\frac{3}{8}$ (b) $\frac{1}{5}$
(c) $\frac{3}{5}$ (d) none of these

38. One bag contains 5 white and 4 black balls. Another bag contains 7 white and 9 black balls. A ball is transferred from the first bag to the second and then a ball is drawn from second. The probability that the ball is white, is

(a) $8/17$ (b) $40/153$
(c) $5/9$ (d) $4/9$

39. Three groups A, B, C are competing for positions on the Board of Directors of a company. The probabilities of their winning are 0.5, 0.3, 0.2 respectively. If the group A wins, the probability of introducing a new product is 0.7 and the corresponding probabilities for group B and C are 0.6 and 0.5 respectively. The probability that the new product will be introduced, is

(a) 0.18 (b) 0.35
(c) 0.10 (d) 0.63



40. A survey of people in a given region showed that 20% were smokers. The probability of death due to lung cancer, given that a person smoked, was 10 times the probability of death due to lung cancer, given that a person did not smoke. If the probability of death due to lung cancer in the region is 0.006, what is the probability of death due to lung cancer given that a person is a smoker
- (a) $\frac{1}{140}$ (b) $\frac{1}{70}$
(c) $\frac{3}{140}$ (d) $\frac{1}{10}$
41. A, B and C are contesting the election for the post of secretary of a club which does not allow ladies to become members. The probabilities of A, B and C winning the election are $\frac{1}{3}$, $\frac{2}{9}$ and $\frac{4}{9}$ respectively. The probabilities of introducing the clause of admitting lady members to the club by A, B and C are 0.6, 0.7 and 0.5 respectively. The probability that ladies will be taken as members in the club after the election is
- (a) $\frac{26}{45}$ (b) $\frac{5}{9}$
(c) $\frac{19}{45}$ (d) none of these
42. A certain player, say X, is known to win with probability 0.3 if the track is fast and 0.4 if the track is slow. For Monday, there is a 0.7 probability of a fast-track and 0.3 probability of slow track. The probability that player X will win on Monday, is
- (a) 0.22 (b) 0.11
(c) 0.33 (d) none of these
43. A coin is tossed 7 times. Each time a man calls head. The probability that he wins the toss on more occasions is
- (a) $\frac{1}{4}$ (b) $\frac{5}{8}$
(c) $\frac{1}{2}$ (d) none of these
44. If X and Y are the independent random variables $B\left(5, \frac{1}{2}\right)$ and $B\left(7, \frac{1}{2}\right)$, then $P(X + Y \geq 1) =$
- (a) $\frac{4095}{4096}$ (b) $\frac{309}{4096}$
(c) $\frac{4032}{4096}$ (d) none of these
45. The mean and variance of a random variable X having a binomial distribution are 4 and 2 respectively, then $P(X = 1)$ is
- (a) $\frac{1}{4}$ (b) $\frac{1}{32}$
(c) $\frac{1}{16}$ (d) $\frac{1}{8}$
46. 6 ordinary dice are rolled. The probability that at least half of them will show at least 3 is
- (a) $41 \times \frac{2^4}{3^6}$ (b) $\frac{2^4}{3^6}$
(c) $20 \times \frac{2^4}{3^6}$ (d) none of these
47. The probability of guessing correctly at least 8 out of 10 answers on a true-false type examination is
- (a) $\frac{7}{64}$ (b) $\frac{7}{128}$
(c) $\frac{45}{1024}$ (d) $\frac{7}{41}$
48. Suppose that a random variable X follows Binomial distribution with parameters n and p, where $0 < p < 1$. If $P(X = r)/P(X = n - r)$ is independent of n and r, then p is equal to
- (a) $\frac{1}{2}$ (b) $\frac{1}{3}$
(c) $\frac{1}{5}$ (d) $\frac{1}{7}$
49. How many times must a man toss a fair coin so that the probability of having at least one head is more than 90%?
- (a) 2 (b) 3
(c) 4 (d) 5

Bernoulli trials and binomial distribution

43. A coin is tossed 7 times. Each time a man calls head. The probability that he wins the toss on more occasions is
- (a) $\frac{1}{4}$ (b) $\frac{5}{8}$
(c) $\frac{1}{2}$ (d) none of these



50. One hundred identical coins, each with probability, p , of showing up heads are tossed once. If $0 < p < 1$ and the probability of heads showing on fifty coins is equal to that of heads showing on 51 coins, then the value of p is:

- (a) $\frac{1}{2}$ (b) $\frac{49}{101}$
(c) $\frac{50}{101}$ (d) $\frac{51}{101}$

Misc Examples-Probability

51. An unbiased die is tossed until a number greater than 4 appears. The probability that an even number of tosses is needed is

- (a) $\frac{1}{2}$ (b) $\frac{2}{5}$
(c) $\frac{1}{5}$ (d) $\frac{2}{3}$

52. Ram and Shyam throw with one dice for a prize of Rs 88 which is to be won by the player who throws 1 first. If Ram starts, then mathematical expectation for Shyam is

- (a) Rs 32 (b) Rs 40
(c) Rs 48 (d) none of these

53. A square is inscribed in a circle. If p_1 is the probability that a randomly chosen point of the circle lies within the square and p_2 is the probability that the point lies outside the square then

- (a) $p_1 = p_2$
(b) $p_1 > p_2$ and $p_1^2 - p_2^2 < \frac{1}{3}$
(c) $p_1 < p_2$
(d) none of these

54. A pair of dice is rolled again and again till a total of 5 or a total of 7 is obtained. The chance that a total of 5 comes before a total of 7 is

- (a) $\frac{2}{5}$ (b) $\frac{3}{7}$
(c) $\frac{3}{13}$ (d) none of these

55. A determinant is chosen at random from the set of all determinants of order 2 with elements 0 or 1 only. The probability that value of the determinant chosen is positive is

- (a) $\frac{16}{81}$ (b) $\frac{7}{16}$
(c) $\frac{3}{16}$ (d) none of these

56. If the integers m and n belongs to set of first hundred natural numbers then the probability that a number of the form $7^m + 7^n$ is divisible by 5 is

- (a) $\frac{1}{5}$ (b) $\frac{1}{7}$
(c) $\frac{1}{4}$ (d) $\frac{1}{49}$

57. The number 'a' is randomly selected from the set $\{0, 1, 2, 3, \dots, 98, 99\}$. The number 'b' is selected from the same set. Probability that the number $3^a + 7^b$ has a digit equal to 8 at the units place, is

- (a) $\frac{1}{16}$ (b) $\frac{2}{16}$
(c) $\frac{4}{16}$ (d) $\frac{3}{16}$

58. Four tickets marked 00, 01, 10 and 11 respectively are placed in bag. A ticket is drawn at random five times, being replaced each time. The probability that the sum of the numbers on the ticket is 15, is

- (a) $\frac{3}{1024}$ (b) $\frac{5}{1024}$
(c) $\frac{7}{1024}$ (d) none of these

59. $2n$ boys are randomly divided into two subgroups containing n boys each. The probability that the two tallest boys are in different groups is

- (a) $\frac{n}{2n-1}$ (b) $\frac{n-1}{2n-1}$
(c) $\frac{2n-1}{4n^2}$ (d) none of these

60. 5 girls and 10 boys sit at random in a row having 15 chairs numbered as 1 to 15. Find the probability that end seats are occupied by the girls and between any two girls odd number of boys sit, is

- (a) $\frac{20 \times 10 \times 5!}{15!}$ (b) $\frac{20 \times 10!}{15!}$
(c) $\frac{20 \times 5!}{15!}$ (d) none of these



EXERCISE - 2 : PREVIOUS YEAR JEE MAIN QUESTIONS

1. If A and B are any two events such that $P(A) = \frac{2}{5}$ and $P(A \cap B) = \frac{3}{20}$, then the conditional probability, $P(A | (A' \cup B'))$, where A' denotes the complement of A, is equal to : **(2016/Online Set-1)**
 - (a) $\frac{1}{4}$
 - (b) $\frac{5}{17}$
 - (c) $\frac{8}{17}$
 - (d) $\frac{11}{20}$
2. An experiment succeeds twice as often as it fails. The probability of at least 5 successes in the six trials of this experiment is : **(2016/Online Set-2)**
 - (a) $\frac{240}{729}$
 - (b) $\frac{192}{729}$
 - (c) $\frac{256}{729}$
 - (d) $\frac{496}{729}$
3. For three events A, B and C, $P(\text{Exactly one of A or B occurs}) = P(\text{Exactly one of B or C occurs}) = P(\text{Exactly one of C or A occurs}) = \frac{1}{4}$ and $P(\text{All the three events occur simultaneously}) = \frac{1}{16}$. Then the probability that at least one of the events occurs, is: **(2017)**
 - (a) $\frac{7}{32}$
 - (b) $\frac{7}{16}$
 - (c) $\frac{7}{64}$
 - (d) $\frac{3}{16}$
4. If two different numbers are taken from the set $\{0, 1, 2, 3, \dots, 10\}$; then the probability that their sum as well as absolute difference are both multiple of 4, **(2017)**
 - (a) $\frac{6}{55}$
 - (b) $\frac{12}{55}$
 - (c) $\frac{14}{45}$
 - (d) $\frac{7}{55}$
5. Three persons P, Q and R independently try to hit a target. If the probabilities of their hitting the target are $\frac{3}{4}$, $\frac{1}{2}$ and $\frac{5}{8}$ respectively, then the probability that the target is hit by P or Q but not by R is : **(2017)**
 - (a) $\frac{21}{64}$
 - (b) $\frac{9}{64}$
 - (c) $\frac{15}{64}$
 - (d) $\frac{39}{64}$
6. An unbiased coin is tossed eight times. The probability of obtaining at least one head and at least one tail is : **(2017)**
 - (a) $\frac{255}{256}$
 - (b) $\frac{127}{128}$
 - (c) $\frac{63}{64}$
 - (d) $\frac{1}{2}$
7. From a group of 10 men and 5 women, four member committees are to be formed each of which must contain at least one woman. Then the probability for these committees to have more women than men, is **(2017)**
 - (a) $\frac{21}{220}$
 - (b) $\frac{3}{11}$
 - (c) $\frac{1}{11}$
 - (d) $\frac{2}{23}$



8. Let E and F be two independent events. The probability that both E and F happen is $\frac{1}{12}$ and the probability that neither E nor F happens is $\frac{1}{2}$, then a value of $\frac{P(E)}{P(F)}$ is: (2017)
- (a) $\frac{4}{3}$ (b) $\frac{3}{2}$
(c) $\frac{1}{3}$ (d) $\frac{5}{12}$
9. A bag contains 4 red and 6 black balls. A ball is drawn at random from the bag, its colour is observed and this ball along with two additional balls of the same colour are returned to the bag. If now a ball is drawn at random from the bag, then the probability that this drawn ball is red, is: (2018)
- (a) $\frac{3}{4}$ (b) $\frac{3}{10}$
(c) $\frac{2}{5}$ (d) $\frac{1}{5}$
10. A box 'A' contains 2 white, 3 red and 2 black balls. Another box 'B' contains 4 white, 2 red and 3 black balls. If two balls are drawn at random, without replacement, from a randomly selected box and one ball turns out to be white while the other ball turns out to be red, then the probability that both balls are drawn from box 'B' is: (2018/Online Set-1)
- (a) $\frac{9}{16}$ (b) $\frac{7}{16}$
(c) $\frac{9}{32}$ (d) $\frac{7}{8}$
11. A player X has a biased coin whose probability of showing heads is p and a player Y has a fair coin. They start playing a game with their own coins and play alternately. The player who throws a head first is a winner. If X starts the game, and the probability of winning the game by both the players is equal, then the value of 'p' is: (2018/Online Set-2)
- (a) $\frac{1}{5}$ (b) $\frac{1}{3}$
(c) $\frac{2}{5}$ (d) $\frac{1}{4}$
12. Two different families A and B are blessed with equal number of children. There are 3 tickets to be distributed amongst the children of these families so that no child gets more than one ticket. If the probability that all the tickets go to the children of the family B is $\frac{1}{12}$, then the number of children in each family is: (2018/Online Set-3)
- (a) 3 (b) 4
(c) 5 (d) 6
13. The minimum number of times one has to toss a fair coin so that the probability of observing at least one head is at least 90% is: (2019-04-08/Shift-2)
- (a) 5 (b) 3
(c) 4 (d) 2
14. Four persons can hit a target correctly with probabilities $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$ and $\frac{1}{8}$ respectively. If all hit at the target independently, then the probability that the target would be hit, is: (2019-04-09/Shift-1)
- (a) $\frac{25}{192}$ (b) $\frac{7}{32}$
(c) $\frac{1}{192}$ (d) $\frac{25}{32}$
15. Minimum number of times a fair coin must be tossed so that the probability of getting at least one head is more than 99% is: (2019-04-10/Shift-2)
- (a) 5 (b) 6
(c) 8 (d) 7
16. If three of the six vertices of a regular hexagon are chosen at random, then the probability that the triangle formed with these chosen vertices is equilateral is: (2019-04-12/Shift-1)
- (a) $\frac{1}{10}$ (b) $\frac{1}{5}$
(c) $\frac{3}{10}$ (d) $\frac{3}{20}$



17. Let a random variable X have a binomial distribution with mean 8 and variance 4. If $P(X \leq 2) = \frac{k}{2^{16}}$, then k is equal to _____. (2019-04-12/Shift-1)
18. For an initial screening of an admission test, a candidate is given fifty problems to solve. If the probability that the candidate can solve any problem is $\frac{4}{5}$, then the probability that he is unable to solve less than two problems is : (2019-04-12/Shift-2)
- (a) $\frac{201}{5} \left(\frac{1}{5}\right)^{49}$ (b) $\frac{316}{25} \left(\frac{4}{5}\right)^{48}$
(c) $\frac{54}{5} \left(\frac{4}{5}\right)^{49}$ (d) $\frac{164}{25} \left(\frac{1}{5}\right)^{48}$
19. A person throws two fair dice. He wins Rs. 15 for throwing a doublet (same numbers on the two dice), wins Rs. 12 when the throw results in the sum of 9, and loses Rs. 6 for any other outcome on the throw. Then the expected gain/loss (in Rs.) of the person is _____. (2019-04-12/Shift-2)
- (a) $\frac{1}{2}$ gain (b) $\frac{1}{4}$ loss
(c) $\frac{1}{2}$ loss (d) 2 gain
20. Two cards are drawn successively with replacement from a well-shuffled deck of 52 cards. Let X denote the random variable of number of aces obtained in the two drawn cards. Then $P(X = 1) + P(X = 2)$ equals: (2019-01-09/Shift-1)
- (a) 49/169 (b) 52/169
(c) 24/169 (d) 25/169
21. An urn contains 5 red and 2 green balls. A ball is drawn at random from the urn. If the drawn ball is green, then a red ball is added to the urn and if the drawn ball is red, then a green ball is added to the urn; the original ball is not returned to the urn. Now, a second ball is drawn at random from it. The probability that the second ball is red is: (2019-01-09/Shift-2)
- (a) $\frac{21}{49}$ (b) $\frac{27}{49}$
(c) $\frac{26}{49}$ (d) $\frac{32}{49}$
22. An unbiased coin is tossed. If the outcome is a head then a pair of unbiased dice is rolled and the sum of the numbers obtained on them is noted. If the toss of the coin results in tail then a card from a well-shuffled pack of nine cards numbered 1, 2, 3, ..., 9 is randomly picked and the number on the card is noted. The probability that the noted number is either 7 or 8 is: (2019-01-10/Shift-1)
- (a) $\frac{13}{36}$ (b) $\frac{15}{72}$
(c) $\frac{19}{72}$ (d) $\frac{19}{36}$
23. If the probability of hitting a target by a shooter, in any shot, is $\frac{1}{3}$, then the minimum number of independent shots at the target required by him so that the probability of hitting the target at least once is greater than $\frac{5}{6}$, is (2019-01-10/Shift-2)
- (a) 3 (b) 6
(c) 5 (d) 4
24. Two integers are selected at random from the set $\{1, 2, \dots, 11\}$. Given that the sum of selected numbers is even, the conditional probability that both the numbers are even is: (2019-01-11/Shift-1)
- (a) $\frac{7}{10}$ (b) $\frac{1}{2}$
(c) $\frac{2}{5}$ (d) $\frac{3}{5}$
25. A bag contains 30 white balls and 10 red balls. 16 balls are drawn one by one randomly from the bag with replacement. If X be the number of white balls drawn, the $\left(\frac{\text{mean of } X}{\text{standard deviation of } X} \right)$ is equal to: (2019-01-11/Shift-2)
- (a) 4 (b) $4\sqrt{3}$
(c) $3\sqrt{2}$ (d) $\frac{4\sqrt{3}}{3}$



26. Let $S = \{1, 2, \dots, 20\}$. A subset B of S is said to be "nice", if the sum of the elements of B is 203. Then the probability that a randomly chosen subset of S is "nice" is :
(2019-01-11/Shift-2)
- (a) $\frac{7}{2^{20}}$ (b) $\frac{5}{2^{20}}$
(c) $\frac{4}{2^{20}}$ (d) $\frac{6}{2^{20}}$
27. In a game, a man wins Rs100 if he gets 5 or 6 on a throw of a fair die and loses Rs 50 forgetting any other number on the die. If he decides to throw the die either till he gets a five or a six or to a maximum of three throws, then his expected gain/loss (in rupees) is:
(2019-01-12/Shift-2)
- (a) $\frac{400}{9}$ loss (b) 0
(c) $\frac{400}{3}$ gain (d) $\frac{400}{3}$ loss
28. In a class of 60 students, 40 opted for NCC, 30 opted for NSS and 20 opted for both NCC and NSS. If one of these students is selected at random, then the probability that the student selected has opted neither for NCC nor for NSS is :
(2019-01-12/Shift-2)
- (a) $\frac{1}{6}$ (b) $\frac{1}{3}$
(c) $\frac{2}{3}$ (d) $\frac{5}{6}$
29. Box I contains 30 cards numbered 1 to 30 and Box II contains 20 cards numbered 31 to 50. A box is selected at random and a card is drawn from it. The number on the card is found to be a non-prime number. The probability that the card was drawn from Box I is :
(2020-09-02/Shift-1)
- (a) $\frac{4}{17}$ (b) $\frac{8}{17}$
(c) $\frac{2}{5}$ (d) $\frac{2}{3}$
30. Let E^C denote the complement of an event E . Let E_1, E_2 and E_3 be any pairwise independent events with $P(E_1) > 0$ and $P(E_1 \cap E_2 \cap E_3) = 0$. Then is equal to :
(2020-09-02/Shift-2)
- (a) $P(E_3^C) - P(E_2^C)$ (b) $P(E_3) - P(E_2^C)$
(c) $P(E_3^C) - P(E_2)$ (d) $P(E_2^C) + P(E_3)$
31. A die is thrown two times and the sum of the scores appearing on the die is observed to be a multiple of 4. Then the conditional probability that the score 4 has appeared at least once is :
(2020-09-03/Shift-1)
- (a) $\frac{1}{3}$ (b) $\frac{1}{4}$
(c) $\frac{1}{8}$ (d) $\frac{1}{9}$
32. The probability that a randomly chosen 5-digit number is made from exactly two digits is :
(2020-09-03/Shift-2)
- (a) $\frac{134}{10^4}$ (b) $\frac{121}{10^4}$
(c) $\frac{135}{10^4}$ (d) $\frac{150}{10^4}$
33. The probability of a man hitting a target is $\frac{1}{10}$. The least number of shots required, so that the probability of his hitting the target at least once is greater than $\frac{1}{4}$, is....
(2020-09-04/Shift-1)
34. In a game two players A and B take turns in throwing a pair of fair dice starting with player A and total of scores on the two dice, in each throw is noted. A wins the game if he throws a total of 6 before B throws a total of 7 and B wins the game if he throws a total of 7 before A throws a total of 6. The game stops as soon as either of the players wins. The probability of A winning the game is :
(2020-09-04/Shift-2)
- (a) $\frac{5}{31}$ (b) $\frac{31}{61}$
(c) $\frac{30}{61}$ (d) $\frac{5}{6}$



35. Four fair dice are thrown independently 27 times. Then the expected number of times, at least two dice shown up a three or a five, is
(2020-09-05/Shift-1)
36. In a bombing attack, there is 50% chance that a bomb will hit the target. At least two independent hits are required to destroy the target completely. Then the minimum number of bombs, that must be dropped to ensure that there is at least 99% chance of completely destroying the target, is _____
(2020-09-05/Shift-2)
37. Out of 11 consecutive natural numbers if three numbers are selected at random (without repetition), then the probability that they are in A.P. with positive common difference, is: (2020-09-06/Shift-1)
- (a) $\frac{10}{99}$ (b) $\frac{5}{33}$
(c) $\frac{15}{101}$ (d) $\frac{5}{101}$
38. The probabilities of three events A, B and C are given by $P(A)=0.6$, $P(B)=0.4$ and $P(C)=0.5$. If $P(A \cup B)=0.8$, $P(A \cap C)=0.3$, $P(A \cap B \cap C)=0.2$, $P(B \cap C)=\beta$ and $P(A \cup B \cup C)=\alpha$, where $0.85 \leq \alpha \leq 0.95$, then β lies in the interval:
(2020-09-06/Shift-2)
- (a) [0.36, 0.40] (b) [0.25, 0.35]
(c) [0.35, 0.36] (d) [0.20, 0.25]
39. An unbiased coin is tossed 5 times. Suppose that a variable x is assigned the value k when k consecutive heads are obtained for $k=3, 4, 5$, otherwise x takes the value -1. The expected value of x, is
(2020-01-07/Shift-3)
- (a) $\frac{1}{8}$ (b) $\frac{3}{16}$
(c) $-\frac{1}{8}$ (d) $-\frac{3}{16}$
40. In a workshop, there are five machines and the probability of any one of them to be out of service on a day is $\frac{1}{4}$. If the probability that at most two machines will be out of service on the same day is $\left(\frac{3}{4}\right)^3 k$, then k is equal to : (2020-01-07/Shift-2)
- (a) $\frac{17}{2}$ (b) 4
(c) $\frac{17}{4}$ (d) $\frac{17}{8}$
41. Let A and B be two independent events such that $P(A)=\frac{1}{3}$ and $P(B)=\frac{1}{6}$. Then, which of the following is TRUE? (2020-01-08/Shift-1)
- (a) $P(A/(A \cup B))=\frac{1}{4}$ (b) $P(A/B')=\frac{1}{3}$
(c) $P(A/B)=\frac{2}{3}$ (d) $P(A'/B')=\frac{1}{3}$
42. Let A and B be two events such that the probability that exactly one of them occurs is $\frac{2}{5}$ and the probability that A or B occurs is $\frac{1}{2}$, then the probability of both of them occur together is (2020-01-08/Shift-2)
- (a) 0.10 (b) 0.20
(c) 0.01 (d) 0.02
43. In a box, there are 20 cards out of which 10 are labelled as A and remaining 10 are labelled as B. Cards are drawn at random, one after the other and with replacement, till a second A-card is obtained. The probability that the second A-card appears before the third B-card is: (2020-01-09/Shift-1)
- (a) 15/16 (b) 9/16
(c) 13/16 (d) 11/16



44. If 10 different balls has to be placed in 4 distinct boxes at random, then the probability that two of these boxes contain exactly 2 and 3 balls is : **(2020-01-09/Shift-2)**

(a) $\frac{965}{2^{10}}$ (b) $\frac{945}{2^{10}}$
(c) $\frac{945}{2^{11}}$ (d) $\frac{965}{2^{11}}$

45. A random variable X has the following probability distribution:

X	1	2	3	4	5
P(X)	K^2	$2K$	K	$2K$	$5K^2$

Then $P(X > 2)$ is equal to **(2020-01-09/Shift-2)**

(a) $\frac{7}{12}$ (b) $\frac{23}{36}$
(c) $\frac{1}{36}$ (d) $\frac{1}{6}$

46. Words with or without meaning are to be formed using all the letters of the word EXAMINATION. The probability that the letter M appears at the fourth position in any such word is: **(2021-07-20/Shift-1)**

(a) $\frac{1}{9}$ (b) $\frac{1}{66}$
(c) $\frac{2}{11}$ (d) $\frac{1}{11}$

47. The probability of selecting integers $a \in [-5, 30]$ such that $x^2 + 2(a+4)x - 5a + 64 > 0$, for all $x \in \mathbb{R}$, is:

(2021-07-20/Shift-1)

(a) $\frac{1}{4}$ (b) $\frac{7}{36}$
(c) $\frac{2}{9}$ (d) $\frac{1}{6}$

48.

Let A, B and C be three events such that the probability that exactly one of A and B occurs is $(1-k)$, the probability that exactly one of B and C occurs is $(1-2k)$, the probability that exactly one of C and A occurs is $(1-k)$ and the probability of all A, B and C occur simultaneously is k^2 , where $0 < k < 1$. Then the probability that at least one of A, B and C occur is ?

(2021-07-20/Shift-2)

- (a) Greater than $\frac{1}{2}$
(b) Greater than $\frac{1}{4}$ but less than $\frac{1}{2}$
(c) Exactly equal to $\frac{1}{2}$
(d) Greater than $\frac{1}{8}$ but less than $\frac{1}{4}$

49.

Let 9 distinct balls be distributed among 4 boxes, B_1, B_2, B_3 and B_4 . If the probability that B_3 contains exactly 3

balls is $k\left(\frac{3}{4}\right)^9$ then k lies in the set ?

(2021-07-25/Shift-1)

- (a) $\{x \in \mathbb{R} : |x-5| \leq 1\}$ (b) $\{x \in \mathbb{R} : |x-2| \leq 1\}$
(c) $\{x \in \mathbb{R} : |x-3| < 1\}$ (d) $\{x \in \mathbb{R} : |x-1| < 1\}$

50.

A student appeared in an examination consisting of 8 true – false type questions. The student guesses the answers with equal probability. The smallest value of n, so that the probability of guessing at least 'n' correct

answers is less than $\frac{1}{2}$, is: **(2021-07-27/Shift-2)**

- (a) 5 (b) 3
(c) 6 (d) 4

51.

Four dice are thrown simultaneously and the numbers shown on these dice are recorded in 2×2 matrices. The probability that such formed matrices have all different entries and are non-singular, is:

(2021-07-22/Shift-2)

(a) $\frac{23}{81}$ (b) $\frac{22}{81}$
(c) $\frac{45}{162}$ (d) $\frac{43}{162}$



52. Let X be a random variable such that the probability function of a distribution is given by $P(X=0) = \frac{1}{2}$,
 $P(X=j) = \frac{1}{3^j} (j=1,2,3,\dots,\infty)$. Then the mean of the distribution and $P(X \text{ is positive and even})$ respectively are: **(2021-07-25/Shift-2)**
- (a) $\frac{3}{4}$ and $\frac{1}{9}$ (b) $\frac{3}{4}$ and $\frac{1}{16}$
 (c) $\frac{3}{8}$ and $\frac{1}{8}$ (d) $\frac{3}{4}$ and $\frac{1}{8}$
53. A fair coin is tossed n -times such that the probability of getting at least one head is at least 0.9. Then the minimum value of n is _____. **(2021-07-25/Shift-2)**
54. A fair die is tossed until six is obtained on it. Let X be the number of required tosses, then the conditional probability $P(x \geq 5 | x > 2)$ is: **(2021-08-26/Shift-2)**
- (a) $\frac{25}{36}$ (b) $\frac{11}{36}$
 (c) $\frac{125}{216}$ (d) $\frac{5}{6}$
55. When a certain biased die is rolled, a particular face occurs with probability $\frac{1}{6} - x$ and its opposite face occurs with probability $\frac{1}{6} + x$. All other faces occur with probability $\frac{1}{6}$. Note that opposite faces sum to in any die. If $0 < x < \frac{1}{6}$, and the probability of obtaining total sum = 7, when such a die is rolled twice, is $\frac{13}{96}$, then the value of x is: **(2021-08-27/Shift-1)**
- (a) $\frac{1}{9}$ (b) $\frac{1}{12}$
 (c) $\frac{1}{8}$ (d) $\frac{1}{16}$
56. Let A and B be independent events such that $P(A) = p$, $P(B) = 2p$. The largest value of p , for which $P(\text{exactly one of } A, B \text{ occurs}) = \frac{5}{9}$, is: **(2021-08-26/Shift-1)**
- (a) $\frac{4}{9}$ (b) $\frac{2}{9}$
 (c) $\frac{1}{3}$ (d) $\frac{5}{12}$
57. Each of the persons A and B independently tosses three fair coins. The probability that both of them get the same number of heads is: **(2021-08-27/Shift-2)**
- (a) $\frac{5}{8}$ (b) $\frac{5}{16}$
 (c) $\frac{1}{8}$ (d) 1
58. An electric instrument consists of two units. Each unit must function independently for the instrument to operate. The probability that the first unit functions is 0.9 and that of the second unit is 0.8. The instrument is switched on and it fails to operate. If the probability that only the first unit failed and second unit is functioning is p , then $98p$ is equal to _____. **(2021-08-31/Shift-1)**
59. Let $S = \{1, 2, 3, 4, 5, 6\}$. Then the probability that a randomly chosen onto function g from S to S satisfies $g(3) = 2g(1)$ is: **(2021-08-31/Shift-2)**
- (a) $\frac{1}{5}$ (b) $\frac{1}{30}$
 (c) $\frac{1}{15}$ (d) $\frac{1}{10}$
60. The probability distribution of random variable X is given by :
- | | | | | | |
|--------|-----|------|------|------|-----|
| X | 1 | 2 | 3 | 4 | 5 |
| $P(X)$ | K | $2K$ | $2K$ | $3K$ | K |
- Let $p = P\left(\frac{1 < X < 4}{X < 3}\right)$, If $5p = \lambda K$, then λ is equal to _____. **(2021-08-27/Shift-2)**



61. Let A denote the event that a 6-digit integer formed by 0, 1, 2, 3, 4, 5, 6 without repetitions, be divisible by 3. Then probability of event A is equal to

(2021-03-16/Shift-2)

- (a) $\frac{9}{56}$ (b) $\frac{11}{27}$
(c) $\frac{3}{7}$ (d) $\frac{4}{9}$

62. A pack of cards has one card missing. Two cards are drawn randomly and are found to be spades. The probability that the missing card is not a spade, is

(2021-03-16/Shift-1)

- (a) $\frac{39}{50}$ (b) $\frac{52}{867}$
(c) $\frac{22}{425}$ (d) $\frac{3}{4}$

63. Let a computer program generate only the digits 0 and 1 to form a string of binary numbers with probability of occurrence of 0 at even places be $\frac{1}{2}$ and probability of occurrence of 0 at the odd place be $\frac{1}{3}$. Then the probability that '10' is followed by '01' is equal to :

(2021-03-17/Shift-2)

- (a) $\frac{1}{3}$ (b) $\frac{1}{9}$
(c) $\frac{1}{18}$ (d) $\frac{1}{6}$

64. Two dices are rolled. If both dices have six faces numbered 1, 2, 3, 5, 7 and 11, then the probability that the sum of the numbers on the top faces is less than or equal to 8 is :

(2021-03-17/Shift-1)

- (a) $\frac{5}{12}$ (b) $\frac{1}{2}$
(c) $\frac{17}{36}$ (d) $\frac{4}{9}$

65. Let there be three independent events E_1 , E_2 and E_3 . The probability that only E_1 occurs is α , only E_2 occurs is β and only E_3 occurs is γ . Let 'p' denote the probability of none of events occurs that satisfies the equation $(\alpha - 2\beta)p = \alpha\beta$ and $(\beta - 3\gamma)p = 2\beta\gamma$. All the given probabilities are assumed to lie in the interval (0, 1).

Then, $\frac{\text{Probability of occurrence of } E_1}{\text{Probability of occurrence of } E_3}$ is equal to

(2021-03-17/Shift-1)

66. Let in a Binomial distribution, consisting of 5 independent trials, probabilities of exactly 1 and 2 successes be 0.4096 and 0.2048 respectively. Then the probability of getting exactly 3 successes is equal to :

(2021-03-18/Shift-2)

- (a) $\frac{40}{243}$ (b) $\frac{80}{243}$
(c) $\frac{32}{625}$ (d) $\frac{128}{625}$

67. The probability that two randomly selected subsets of the set $\{1, 2, 3, 4, 5\}$ have exactly two elements in their intersection, is :

(2021-02-24/Shift-2)

- (a) $\frac{65}{2^8}$ (b) $\frac{35}{2^7}$
(c) $\frac{135}{2^9}$ (d) $\frac{65}{2^7}$

68. An ordinary dice is rolled for a certain number of times. If the probability of getting an odd number 2 times is equal to the probability of getting an even number 3 times, then the probability of getting an odd number for odd number of times is

(2021-02-24/Shift-1)

- (a) $\frac{1}{32}$ (b) $\frac{5}{16}$
(c) $\frac{3}{16}$ (d) $\frac{1}{2}$



69. Let B_i ($i=1,2,3$) be three independent events in a sample space. The probability that only B_1 occurs is α , only B_2 occurs is β and only B_3 occurs is γ . Let p be the probability that none of the events B_i occurs and these 4 probabilities satisfy the equations $(\alpha - 2\beta)p = \alpha\beta$ and $(\beta - 3\gamma)p = 2\beta\gamma$ (All the probabilities are assumed to lie in the interval $(0, 1)$). Then $\frac{P(B_1)}{P(B_3)}$ is equal to _____.
(2021-02-24/Shift-1)
70. Let A be a set of all 4-digit natural numbers whose exactly one digit is 7. Then the probability that a randomly chosen element of A leaves remainder 2 when divided by 5 is :
(2021-02-25/Shift-2)
- (a) $\frac{97}{297}$ (b) $\frac{122}{297}$
(c) $\frac{2}{9}$ (d) $\frac{1}{5}$
71. In a group of 400 people, 160 are smokers and non-vegetarian; 100 are smokers and vegetarian and the remaining 140 are non-smokers and vegetarian. Their chances of getting a particular chest disorder are 35%, 20% and 10% respectively. A person is chosen from the group at random and is found to be suffering from the chest disorder. The probability that the selected person is a smoker and non-vegetarian is:
(2021-02-25/Shift-2)
- (a) $\frac{14}{45}$ (b) $\frac{7}{45}$
(c) $\frac{8}{45}$ (d) $\frac{28}{45}$
72. The coefficients a , b and c of the quadratic equation, $ax^2 + bx + c = 0$ are obtained by throwing a dice three times. The probability that this equation has equal roots is:
(2021-02-25/Shift-1)
- (a) $\frac{5}{216}$ (b) $\frac{1}{36}$
(c) $\frac{1}{54}$ (d) $\frac{1}{72}$
73. When a missile is fired from a ship, the probability that it is intercepted is $\frac{1}{3}$ and the probability that the missile hits the target, given that it is not intercepted, is $\frac{3}{4}$. If three missiles are fired independently from the ship, then the probability that all three hit the target, is:
(2021-02-25/Shift-1)
- (a) $\frac{3}{4}$ (b) $\frac{3}{8}$
(c) $\frac{1}{27}$ (d) $\frac{1}{8}$
74. A seven digit number is formed using digits 3, 3, 4, 4, 4, 5, 5. The probability, that number so formed is divisible by 2, is
(2021-02-26/Shift-2)
- (a) $\frac{1}{7}$ (b) $\frac{6}{7}$
(c) $\frac{4}{7}$ (d) $\frac{3}{7}$
75. A fair coin is tossed a fixed number of times. If the probability of getting 7 heads is equal to probability of getting 9 heads, then the probability of getting 2 heads is :
(2021-02-26/Shift-1)
- (a) $\frac{15}{2^{13}}$ (b) $\frac{15}{2^8}$
(c) $\frac{15}{2^{14}}$ (d) $\frac{15}{2^{12}}$



EXERCISE - 3 : ADVANCED OBJECTIVE QUESTIONS

Objective Questions I [Only one correct option]

- A purse contains 2 six sided dice. One is a normal fair die, while the other has 2 ones, 2 threes, and 2 fives. A die is picked up and rolled. Because of some secret magnetic attraction of the unfair die, there is 75% chance of picking the unfair die and a 25% chance of picking a fair die. The die is rolled and shows up the face 3. The probability that a fair die was picked up, is
 (a) $\frac{1}{7}$ (b) $\frac{1}{4}$
 (c) $\frac{1}{6}$ (d) $\frac{1}{24}$
- An instrument consists of two units. Each unit must function for the instrument to operate. The reliability of the first unit is 0.9 & that of the second unit is 0.8. The instrument is tested & fails. The probability that "only the first unit failed & the second unit is sound" is :
 (a) $\frac{1}{7}$ (b) $\frac{2}{7}$
 (c) $\frac{3}{7}$ (d) $\frac{4}{7}$
- A box has four dice in it. Three of them are fair dice but the fourth one has the number five on all of its faces. A die is chosen at random from the box and is rolled three times and shows up the face five on all the three occasions. The chance that the die chosen was a rigged die, is
 (a) $\frac{216}{217}$ (b) $\frac{215}{219}$
 (c) $\frac{216}{219}$ (d) none
- On a Saturday night 20% of all drivers in U.S.A. are under the influence of alcohol. The probability that a driver under the influence of alcohol will have an accident is 0.001. The probability that a sober driver will have an accident is 0.0001. If a car on a Saturday night smashed into a tree, the probability that the driver was under the influence of alcohol, is
 (a) $\frac{3}{7}$ (b) $\frac{4}{7}$
 (c) $\frac{5}{7}$ (d) $\frac{6}{7}$
- Mr. Dupont is a professional wine taster. When given a French wine, he will identify it with probability 0.9 correctly as French, and will mistake it for a Californian wine with probability 0.1. When given a Californian wine, he will identify it with probability 0.8 correctly as Californian, and will mistake it for a French wine with probability 0.2. Suppose that Mr. Dupont is given ten unlabelled glasses of wine, three with French and seven with Californian wines. He randomly picks a glass, tries the wine, and solemnly says : "French". The probability that the wine he tasted was Californian, is nearly equal to
 (a) 0.14 (b) 0.24
 (c) 0.34 (d) 0.44
- A box contains a normal coin and a double headed coin. A coin selected at random and tossed twice, fell headwise on both the occasions. The probability that the drawn coin is a double headed coin is
 (a) $\frac{2}{3}$ (b) $\frac{5}{8}$
 (c) $\frac{3}{4}$ (d) $\frac{4}{5}$
- A box contains 5 red and 4 white marbles. Two marbles are drawn successively from the box without replacement and the second drawn marble drawn is found to be white. Probability that the first marble is also white is
 (a) $\frac{3}{8}$ (b) $\frac{1}{2}$
 (c) $\frac{1}{3}$ (d) $\frac{1}{4}$
- Events A and C are independent. If the probabilities relating A, B and C are $P(A) = \frac{1}{5}$;
 $P(B) = \frac{1}{6}$; $P(A \cap C) = \frac{1}{20}$; $P(B \cup C) = \frac{3}{8}$ then
 (a) events B and C are independent
 (b) events B and C are mutually exclusive
 (c) events B and C are neither independent nor mutually exclusive
 (d) events B and C are equiprobable



9. Assume that the birth of a boy or girl to a couple to be equally likely, mutually exclusive, exhaustive and independent of the other children in the family. For a couple having 6 children, the probability that their "three oldest are boys" is
- (a) $\frac{20}{64}$ (b) $\frac{1}{64}$
(c) $\frac{2}{64}$ (d) $\frac{8}{64}$
10. A and B are two events such that $P(A) = 0.2$ and $P(A \cup B) = 0.7$. If A and B are independent events then $P(B)$ equals
- (a) $\frac{2}{7}$ (b) $\frac{7}{9}$
(c) $\frac{5}{8}$ (d) none of these
11. Box A contains 3 red and 2 blue marbles while box B contains 2 red and 8 blue marbles. A fair coin is tossed. If the coin turns up heads, a marble is drawn from A, if it turns up tails, a marble is drawn from bag B. The probability that a red marble is chosen, is
- (a) $\frac{1}{5}$ (b) $\frac{2}{5}$
(c) $\frac{3}{5}$ (d) $\frac{1}{2}$
12. Lot A consists of 3G and 2D articles. Lot B consists of 4G and 1D article. A new lot C is formed by taking 3 articles from A and 2 from B. The probability that an article chosen at random from C is defective, is
- (a) $\frac{1}{3}$ (b) $\frac{2}{5}$
(c) $\frac{8}{25}$ (d) none
13. A bowl has 6 red marbles and 3 green marbles. The probability that a blind folded person will draw a red marble on the second draw from the bowl without replacing the marble from the first draw, is
- (a) $\frac{2}{3}$ (b) $\frac{1}{4}$
(c) $\frac{1}{2}$ (d) $\frac{5}{8}$
14. A number x is chosen at random from the set $\{1, 2, 3, 4, \dots, 100\}$. Define the event: A = the chosen number x satisfies $\frac{(x-10)(x-50)}{(x-30)} \geq 0$. Then $P(A)$ is:
- (a) 0.71 (b) 0.70
(c) 0.51 (d) 0.20
15. Three identical dice are rolled. The probability that the same number will appear on each of them, is
- (a) $\frac{1}{6}$ (b) $\frac{1}{36}$
(c) $\frac{1}{18}$ (d) $\frac{3}{28}$
16. A die is tossed 5 times. Getting an odd number is considered a success. Then the variance of distribution of success is
- (a) $\frac{8}{3}$ (b) $\frac{3}{8}$
(c) $\frac{4}{5}$ (d) $\frac{5}{4}$
17. A fair die is tossed repeatedly. A wins if it is 1 or 2 on two consecutive tosses and B wins if it is 3, 4, 5 or 6 on two consecutive tosses. The probability that A wins if the die is tossed indefinitely, is
- (a) $\frac{1}{3}$ (b) $\frac{5}{21}$
(c) $\frac{1}{4}$ (d) $\frac{2}{5}$
18. A and B play a game of tennis. The situation of the game is as follows; if one scores two consecutive points after a deuce he wins; if loss of a point is followed by win of a point, it is deuce. The chance of a server to win a point is $\frac{2}{3}$. The game is at deuce and A is serving. Probability that A will win the match is, (serves are changed after each pt)
- (a) $\frac{3}{5}$ (b) $\frac{2}{5}$
(c) $\frac{1}{2}$ (d) $\frac{4}{5}$
19. Two aeroplanes I and II bomb a target in succession. The probabilities of I and II scoring a hit correctly are 0.3 and 0.2, respectively. The second plane will bomb if the first misses the target. The probability that the target is hit by the second plane, is
- (a) 0.2 (b) 0.7
(c) 0.06 (d) 0.14



20. It is given that the event A and B are such that $P(A) = \frac{1}{4}$, $P\left(\frac{A}{B}\right) = \frac{1}{2}$ and $P\left(\frac{B}{A}\right) = \frac{2}{3}$. Then P (B) is
- (a) $\frac{1}{2}$ (b) $\frac{1}{6}$
(c) $\frac{1}{3}$ (d) $\frac{2}{3}$
21. Let A and E by any two events with positive probabilities:
Statement 1 : $P(E/A) \geq P(A/E)P(E)$
Statement 2 : $P(A/E) \geq P(A \cap E)$
(a) Both the statements are false
(b) Both statements are True
(c) Statement–1 is true, Statement–2 is false
(d) Statement–1 is false, Statement–2 is true
22. If A and B are two independent events such that $P(A) > 0$, and $P(B) \neq 1$, then $P(\bar{A} / \bar{B})$ is equal to
- (a) $1 - P(\bar{A} / B)$ (b) $1 - P(A / \bar{B})$
(c) $\frac{1 - P(A \cup B)}{P(B)}$ (d) $\frac{P(\bar{A})}{P(\bar{B})}$
23. Indicate the correct order sequence in respect of the following :
- I.** If the probability that a computer will fail during the first hour of operation is 0.01, then if we turn on 100 computers, exactly one will fail in the first hour of operation.
- II.** A man has ten keys only one of which fits the lock. He tries them in a door one by one discarding the one he has tried. The probability that fifth key fits the lock is 1/10.
- III.** Given the events A and B in a sample space. If $P(A) = 1$, then A and B are independent.
- IV.** When a fair six sided die is tossed on a table top, the bottom face can not be seen. The probability that the product of the numbers on the five faces that can be seen is divisible by 6 is one.
- (a) FTFT (b) FTTF
(c) TFTF (d) TFFF
24. A child throws 2 fair dice. If the numbers showing are unequal, he adds them together to get his final score. On the other hand, if the numbers showing are equal, he throws 2 more dice & adds all 4 numbers showing to get his final score. The probability that his final score is 6 is:
- (a) $\frac{145}{1296}$ (b) $\frac{146}{1296}$
(c) $\frac{147}{1296}$ (d) $\frac{148}{1296}$
25. A examination consists of 8 questions in each of which one of the 5 alternatives is the correct one. On the assumption that a candidate who has done no preparatory work chooses for each question any one of the five alternatives with equal probability, the probability that he gets more than one correct answer is equal to :
- (a) $(0.8)^8$ (b) $3(0.8)^8$
(c) $1 - (0.8)^8$ (d) $1 - 3(0.8)^8$
26. A number is chosen at random from the numbers 10 to 99. By seeing the number a man will laugh if product of the digits is 12. If he choose three numbers with replacement then the probability that he will laugh at least once is
- (a) $1 - \left(\frac{3}{5}\right)^3$ (b) $\left(\frac{43}{45}\right)^3$
(c) $1 - \left(\frac{4}{25}\right)^3$ (d) $1 - \left(\frac{43}{45}\right)^3$
27. A fair die is tossed eight times. Probability that on the eighth throw a third six is observed is,
- (a) ${}^8C_3 \frac{5^5}{6^8}$
(b) $\frac{{}^7C_2 \cdot 5^5}{6^8}$
(c) $\frac{{}^7C_2 \cdot 5^5}{6^7}$
(d) none of these



28. Two cards are drawn from a well shuffled pack of 52 playing cards one by one. If

A : the event that the second card drawn is an ace and

B : the event that the first card drawn is an ace card.

then which of the following is true?

(a) $P(A) = \frac{4}{17}$; $P(B) = \frac{1}{13}$

(b) $P(A) = \frac{1}{13}$; $P(B) = \frac{1}{13}$

(c) $P(A) = \frac{1}{13}$; $P(B) = \frac{1}{17}$

(d) $P(A) = \frac{16}{221}$; $P(B) = \frac{4}{51}$

29. There are n different gift coupons, each of which can occupy N ($N > n$) different envelopes, with the same probability $1/N$

P_1 : The probability that there will be one gift coupon in each of n definite envelopes out of N given envelopes

P_2 : The probability that there will be one gift coupon in each of n arbitrary envelopes out of N given envelopes

Consider the following statements

(i) $P_1 = P_2$ (ii) $P_1 = \frac{n!}{N^n}$

(iii) $P_2 = \frac{N!}{N^n (N-n)!}$

(iv) $P_2 = \frac{n!}{N^n (N-n)!}$ (v) $P_1 = \frac{N!}{N^n}$

Now, which of the following is true

- (a) Only (i) (b) (ii) and (iii)
(c) (ii) and (iv) (d) (iii) and (v)

30. A bag contains 3 R & 3 G balls and a person draws out 3 at random. He then drops 3 blue balls into the bag & again draws out 3 at random. The chance that the 3 later balls being all of different colours is

- (a) 15% (b) 20%
(c) 27% (d) 40%

31. From an urn containing six balls, 3 white and 3 black ones, a person selects at random an even number of balls (all the different ways of drawing an even number of balls are considered equally probable, irrespective of their number). Then the probability that there will be the same number of black and white balls among them

(a) $\frac{4}{5}$ (b) $\frac{11}{15}$

(c) $\frac{11}{30}$ (d) $\frac{2}{5}$

32. One purse contains 6 copper coins and 1 silver coin ; a second purse contains 4 copper coins. Five coins are drawn from the first purse and put into the second, and then 2 coins are drawn from the second and put into the first. The probability that the silver coin is in the second purse is

(a) $\frac{1}{2}$ (b) $\frac{4}{9}$

(c) $\frac{5}{9}$ (d) $\frac{2}{3}$

33. If $\frac{(1+3p)}{3}$, $\frac{(1-p)}{4}$ & $\frac{(1-2p)}{2}$ are the probabilities of three mutually exclusive events defined on a sample space S , then the true set of all values of p is

(a) $\left[\frac{1}{3}, \frac{1}{2}\right]$ (b) $\left[\frac{1}{3}, 1\right]$

(c) $\left[\frac{1}{4}, \frac{1}{3}\right]$ (d) $\left[\frac{1}{4}, \frac{1}{2}\right]$

34. The probabilities of events, $A \cap B$, A , B & $A \cup B$ are respectively in A.P. with probability of second term equal to the common difference. Therefore the events A and B are

- (a) compatible
(b) independent
(c) such that one of them must occur
(d) such that one is twice as likely as the other



Objective Questions II [One or more than one correct option]

35. A bag initially contains one red & two blue balls. An experiment consisting of selecting a ball at random, noting its colour & replacing it together with an additional ball of the same colour. If three such trials are made, then :
- probability that atleast one blue ball is drawn is 0.9
 - probability that exactly one blue ball is drawn is 0.2
 - probability that all the drawn balls are red given that all the drawn balls are of same colour is 0.2
 - probability that atleast one red ball is drawn is 0.6.
36. If E_1 and E_2 are two events such that $P(E_1) = 1/4$, $P(E_2/E_1) = 1/2$ and $P(E_1/E_2) = 1/4$
- then E_1 and E_2 are independent
 - E_1 and E_2 are exhaustive
 - E_2 is twice as likely to occur as E_1
 - Probabilities of the events $E_1 \cap E_2$, E_1 and E_2 are in GP.
37. Let $0 < P(A) < 1$, $0 < P(B) < 1$ & $P(A \cup B) = P(A) + P(B) - P(A) \cdot P(B)$, then :
- $P(B/A) = P(B) - P(A)$
 - $P(A^c \cup B^c) = P(A^c) + P(B^c)$
 - $P((A \cup B)^c) = P(A^c) \cdot P(B^c)$
 - $P(A/B) = P(A)$
38. If M & N are independent events such that $0 < P(M) < 1$ & $0 < P(N) < 1$, then :
- M & N are mutually exclusive
 - M & $\bar{N}$ are independent
 - $\bar{M}$ & $\bar{N}$ are independent
 - $P(M/N) + P(\bar{M}/N) = 1$
39. If $\bar{E}$ and $\bar{F}$ are the complementary events of E and F respectively and if $0 < P(F) < 1$, then
- $P(E/F) + P(\bar{E}/F) = 1$
 - $P(E/F) + P(E/\bar{F}) = 1$
 - $P(\bar{E}/F) + P(E/\bar{F}) = 1$
 - $P(E/\bar{F}) + P(\bar{E}/\bar{F}) = 1$
40. Two real numbers, x & y are selected at random. Given that $0 \leq x \leq 1$; $0 \leq y \leq 1$. Let A be the event that $y^2 \leq x$; B be the event that $x^2 \leq y$, then :
- $P(A \cap B) = \frac{1}{3}$
 - A & B are exhaustive events
 - A & B are mutually exclusive
 - A & B are independent events.
41. If A & B are two events such that $P(B) \neq 1$, B^c denotes the event complementary to B , then
- $P(A/B^c) = \frac{P(A) - P(A \cap B)}{1 - P(B)}$
 - $P(A \cap B) \geq P(A) + P(B) - 1$
 - $P(A) > P(A/B)$ if $P(A/B^c) > P(A)$
 - $P(A/B^c) + P(A^c/B^c) = 1$
42. For any two events A & B defined on a sample space ,
- $P(A/B) \geq \frac{P(A) + P(B) - 1}{P(B)}$, $P(B) \neq 0$ is always true
 - $P(A \cap \bar{B}) = P(A) - P(A \cap B)$
 - $P(A \cup B) = 1 - P(A^c) \cdot P(B^c)$, if A & B are independent
 - $P(A \cup B) = 1 - P(A^c) \cdot P(B^c)$, if A & B are disjoint
43. For two given events A & B , $P(A \cap B)$ is :
- not less than $P(A) + P(B) - 1$
 - not greater than $P(A) + P(B)$
 - equal to $P(A) + P(B) - P(A \cup B)$
 - equal to $P(A) + P(B) + P(A \cup B)$



Numerical Value Type Questions

44. Two integers r and s are chosen one at a time without replacement from the numbers $1, 2, 3, \dots, 100$. Let p be the probability that $r \leq 25$ given that $s \leq 25$. Find the value of $33p$.
45. A bag contains $n + 1$ coins. It is known that one of these coins has heads on both sides, whereas the other coins are fair. One coin is selected at random and tossed. If the probability that the toss results in heads is $7/12$, find n .
46. 7 persons are stopped on the road at random and asked about their birthdays. If the probability that 3 of them are born on Wednesday, 2 on Thursday and the remaining 2

on Sunday is $\frac{K}{7^6}$, then K is equal to

Assertion & Reason

- (A) If both assertion and reason are correct and reason is the correct explanation of assertion.
- (B) If both assertion and reason are true but reason is not the correct explanation of assertion.
- (C) If assertion is true but reason is false.
- (D) If assertion is false but reason is true.

47. From an urn containing a white and b black balls, k ($k < a, b$) are drawn and laid aside, their colour unnoted. Then another ball, that is, $(k + 1)^{\text{th}}$ ball is drawn.

Assertion : Probability that $(k + 1)^{\text{th}}$ ball drawn is white

is $\frac{a}{a + b}$.

Reason : Probability that $(k + 1)^{\text{th}}$ ball drawn is black is

$\frac{a}{a + b}$

- (a) A (b) B
(c) C (d) D

48. Let A and B are two events such that $P(A) > 0$.

Assertion : If $P(A) + P(B) > 1$, then

$P(B/A) \geq 1 - P(B')/P(A)$

Reason : If $P(A/B') \geq P(A)$, then $P(A) \geq P(A/B)$.

- (a) A (b) B
(c) C (d) D

Match the Following

Each question has two columns. Four options are given representing matching of elements from Column-I and Column-II. Only one of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.

49. A determinant Δ is chosen at random from the set of all determinant of order two with elements 0 and 1 only.

Value of Δ	Probability
(A) 1	(P) $5/8$
(B) 0	(Q) $3/16$
(C) 2	(R) $3/8$
(D) non zero	(S) 0

The correct matching is

- (a) A-Q; B-P; C-S; D-R
(b) A-P; B-Q; C-S; D-R
(c) A-Q; B-S; C-P; D-R
(d) A-Q; B-P; C-R; D-S

50. A ten digit number N is formed by using the digits 0 to 9 exactly once. The probability that N is divisible by

- (A) 4 (P) 1
(B) 5 (Q) $20/81$
(C) 45 (R) $17/81$
(D) 12 (S) $2/81$

The correct matching is

- (a) A-Q; B-R; C-R; D-Q
(b) A-R; B-Q; C-R; D-Q
(c) A-Q; B-R; C-Q; D-R
(d) A-R; B-R; C-Q; D-Q



Paragraph Type Questions

Passage

Using the following passage, solve Q.51 to Q.53

Let S and T are two events defined on a sample space with probabilities

$$P(S) = 0.5, P(T) = 0.69, P(S/T) = 0.5$$

51. Events S and T are:
 (a) mutually exclusive
 (b) independent
 (c) mutually exclusive and independent
 (d) neither mutually exclusive nor independent
52. The value of $P(S \text{ and } T)$
 (a) 0.3450 (b) 0.2500
 (c) 0.6900 (d) 0.350
53. The value of $P(S \text{ or } T)$
 (a) 0.6900 (b) 1.19
 (c) 0.8450 (d) 0

Using the following passage, solve Q.54 to Q.56

A JEE aspirant estimates that she will be successful with an 80 percent chance if she studies 10 hours per day, with a 60 percent chance if she studies 7 hours per day and with a 40 percent chance if she studies 4 hours per day. She further believes that she will study 10 hours, 7 hours and 4 hours per day with probabilities 0.1, 0.2 and 0.7, respectively

54. The chance she will be successful, is
 (a) 0.28 (b) 0.38
 (c) 0.48 (d) 0.58
55. Given that she is successful, the chance she studied for 4 hours, is
 (a) $\frac{6}{12}$ (b) $\frac{7}{12}$
 (c) $\frac{8}{12}$ (d) $\frac{9}{12}$
56. Given that she does not achieve success, the chance she studied for 4 hour, is
 (a) $\frac{18}{26}$ (b) $\frac{19}{26}$
 (c) $\frac{20}{26}$ (d) $\frac{21}{26}$

Using the following passage, solve Q.57 to Q.60

Read the passage given below carefully before attempting these questions.

A standard deck of playing cards has 52 cards. There are four suit (clubs, diamonds, hearts and spades), each of which has thirteen numbered cards (2,, 9, 10, Jack, Queen, King, Ace)

In a game of card, each card is worth an amount of points. Each numbered card is worth its number (e.g. a 5 is worth 5 points); the Jack, Queen and King are each worth 10 points; and the Ace is worth your choice of either 1 point or 11 points. The object of the game is to have more points in your set of cards than your opponent without going over 21. Any set of cards with sum greater than 21 automatically loses.

Here's how the game played. You and your opponent are each dealt two cards. Usually the first card for each player is dealt face down, and the second card for each player is dealt face up. After the initial cards are dealt, the first player has the option of asking for another card or not taking any cards. The first player can keep asking for more cards until either he or she goes over 21, in which case the player loses, or stops at some number less than or equal to 21. When the first player stops at some number less than or equal to 21, the second player then can take more cards until matching or exceeding the first player's number without going over 21, in which case the second player wins, or until going over 21, in which case the first player wins.

We are going to simplify the game a little and assume that all cards are dealt face up, so that all cards are visible. Assume your opponent is dealt cards and plays first.

57. The chance that the second card will be a heart and a Jack, is
 (a) $\frac{4}{52}$ (b) $\frac{13}{52}$
 (c) $\frac{17}{52}$ (d) $\frac{1}{52}$
58. The chance that the first card will be a heart or a Jack, is
 (a) $\frac{13}{52}$ (b) $\frac{16}{52}$
 (c) $\frac{17}{52}$ (d) none



59. Given that the first card is a Jack, the chance that it will be the heart, is

- (a) $\frac{1}{13}$ (b) $\frac{4}{13}$
(c) $\frac{1}{4}$ (d) $\frac{1}{3}$

60. Your opponent is dealt a King and a 10, and you are dealt a Queen and a 9. Being smart, your opponent does not take any more cards and stays at 20. The chance that you will win if you are allowed to take as many cards as you need, is

- (a) 0.771 (b) 0.088
(c) 0.0797 (d) 0.0907

Text

61. In a test an examinee either guesses or copies or knows the answer to a multiple choice question with four choices.

The probability that he make a guess is $\frac{1}{3}$ and the

probability that he copies the answer is $\frac{1}{6}$. The probability

that his answer is correct given that he copied it, is $\frac{1}{8}$.

Find the probability that he knew the answer to the question given that he correctly answered it.

62. Six boys and six girls sit in a row at random. Find the probability that

- (a) the six girls sit together
(b) the boys and girls sit alternatively

Fill in the blanks

63. For a biased die the probabilities for the different faces to turn up are given below

Face	1	2	3	4	5	6
Probability	0.1	0.32	0.21	0.15	0.05	0.17

This die is tossed and you are told that either face 1 or face 2 has turned up. Then, the probability that it is face 1, is....

64. A box contains 100 tickets numbered 1, 2, ..., 100. Two tickets are chosen at random. It is given that the maximum number on the two chosen tickets is not more than 10. The minimum number on them is 5 with probability....

65. If two events A and B are such that $P(A^c) = 0.3$, $P(B) = 0.4$ and $P(A \cap B^c) = 0.5$ then $P[B / (A \cup B^c)] = \dots$



EXERCISE - 4 : PREVIOUS YEAR JEE ADVANCED QUESTIONS

Objective Questions I [Only one correct option]

1. If $P(B) = \frac{3}{4}$, $P(A \cap B \cap \bar{C}) = \frac{1}{3}$ and $P(\bar{A} \cap B \cap \bar{C}) = \frac{1}{3}$, then $P(B \cap C)$ is (2002)
 - (a) $\frac{1}{12}$ (b) $\frac{1}{6}$
 - (c) $\frac{1}{15}$ (d) $\frac{1}{9}$
2. Two numbers are selected randomly from the set $S = \{1, 2, 3, 4, 5, 6\}$ without replacement one by one. The probability that minimum of the two numbers is less than 4, is (2003)
 - (a) $1/15$ (b) $14/15$
 - (c) $1/5$ (d) $4/5$
3. If three distinct numbers are chosen randomly from the first 100 natural numbers, then the probability that all three of them are divisible by both 2 and 3, is (2004)
 - (a) $\frac{4}{55}$ (b) $\frac{4}{35}$
 - (c) $\frac{4}{33}$ (d) $\frac{4}{1155}$
4. A fair die is rolled. The probability that the first time 1 occurs at the even throw, is (2005)
 - (a) $1/6$ (b) $5/11$
 - (c) $6/11$ (d) $5/36$
5. One Indian and four American men and their wives are to be seated randomly around a circular table. Then, the conditional probability that the Indian man is seated adjacent to his wife given that each American man is seated adjacent to his wife, is (2007)
 - (a) $\frac{1}{2}$ (b) $\frac{1}{3}$
 - (c) $\frac{2}{5}$ (d) $\frac{1}{5}$
6. Let E^c denotes the complement of an event E . Let E, F, G be pairwise independent events with $P(G) > 0$ and $P(E \cap F \cap G) = 0$. Then, $P(E^c \cap F^c | G)$ equals (2007)
 - (a) $P(E^c) + P(F^c)$ (b) $P(E^c) - P(F^c)$
 - (c) $P(E^c) - P(F)$ (d) $P(E) - P(F^c)$
7. An experiment has 10 equally likely outcomes. Let A and B be two non-empty events of the experiment. If A consists of 4 outcomes, the number of outcomes that B must have so that A and B are independent, is (2008)
 - (a) 2, 4, or 8 (b) 3, 6 or 9
 - (c) 4 or 8 (d) 5 or 10
8. Let ω be a complex cube root of unity with $\omega \neq 1$. A fair die is thrown three times, If r_1, r_2 and r_3 are the numbers obtained on the die, then the probability that $\omega^{r_1} + \omega^{r_2} + \omega^{r_3} = 0$ is (2010)
 - (a) $1/18$ (b) $1/9$
 - (c) $2/9$ (d) $1/36$
9. A signal which can be green or red with probability $\frac{4}{5}$ and $\frac{1}{5}$ respectively, is received by station A and then transmitted to station B. The probability of each station receiving the signal correctly is $\frac{3}{4}$. If the signal received at station B is green, then the probability that the original signal green is (2010)
 - (a) $\frac{3}{5}$ (b) $\frac{6}{7}$
 - (c) $\frac{20}{23}$ (d) $\frac{9}{20}$
10. Three boys and two girls stand in a queue. The probability, that the number of boys ahead of every girl is at least one more than the number of girls ahead of her, is (2014)
 - (a) $\frac{1}{2}$ (b) $\frac{1}{3}$
 - (c) $\frac{2}{3}$ (d) $\frac{3}{4}$



11. A computer producing factory has only two plants T_1 and T_2 . Plant T_1 produces 20% and plant T_2 produces 80% of the total computers produced. 7% of computers produced in the factory turn out to be defective. It is known that $P(\text{computer turns out to be defective given that it is produced in plant } T_1) = 10P(\text{computer turns out to be defective given that it is produced in plant } T_2)$, where $P(E)$ denotes the probability of an event E . A computer produced in the factory is randomly selected and it does not turn out to be defective. Then the probability that it is produced in plants T_2 is (2016)

- (a) $\frac{36}{73}$ (b) $\frac{47}{79}$
(c) $\frac{78}{93}$ (d) $\frac{75}{83}$

12. Three randomly chosen nonnegative integers x, y and z are found to satisfy the equation $x + y + z = 10$. Then the probability that z is even, is (2017)

- (a) $\frac{1}{2}$ (b) $\frac{36}{55}$
(c) $\frac{6}{11}$ (d) $\frac{5}{11}$

13. Let C_1 and C_2 be two biased coins such that the probabilities of getting head in a single toss are $\frac{2}{3}$ and

$\frac{1}{3}$, respectively. Suppose α is the number of heads that appear when C_1 is tossed twice, independently, and suppose β is the number of heads that appear when C_2 is tossed twice, independently. Then probability that the roots of the quadratic polynomial $x^2 - \alpha x + \beta$ are real and equal, is (2020)

- (a) $\frac{40}{81}$ (b) $\frac{20}{81}$
(c) $\frac{1}{2}$ (d) $\frac{1}{4}$

14. Consider three sets $E_1 = \{1, 2, 3\}$, $F_1 = \{1, 3, 4\}$ and $G_1 = \{2, 3, 4, 5\}$. Two elements are chosen at random, without replacement, from the set E_1 , and let S_1 denote the set of these chosen elements. Let $E_2 = E_1 - S_1$ and $F_2 = F_1 \cup S_1$. Now two elements are chosen at random, without replacement, from the set F_2 and let S_2 denote the set of these chosen elements.

Let $G_2 = G_1 \cup S_2$. Finally, two elements are chosen at random, without replacement from the set G_2 and let S_3 denote the set of these chosen elements.

Let $E_3 = E_2 \cup S_3$. Given that $E_1 = E_3$, let p be the conditional probability of the event $S_1 = \{1, 2\}$. Then the value of p is (2021)

- (a) $\frac{1}{5}$ (b) $\frac{3}{5}$
(c) $\frac{1}{2}$ (d) $\frac{2}{5}$

Objective Questions II [One or more than one correct option]

15. Let E and F be two independent events. The probability that exactly one of them occurs is $\frac{11}{25}$ and the probability

of none of them occurring is $\frac{2}{25}$. If $P(T)$ denotes the probability of occurrence of the event T , then (2011)

- (a) $P(E) = \frac{4}{5}$, $P(F) = \frac{3}{5}$ (b) $P(E) = \frac{1}{5}$, $P(F) = \frac{2}{5}$
(c) $P(E) = \frac{2}{5}$, $P(F) = \frac{1}{5}$ (d) $P(E) = \frac{3}{5}$, $P(F) = \frac{4}{5}$

16. Let X and Y be two events such that $P(X) = \frac{1}{3}$,

$P(X|Y) = \frac{1}{2}$ and $P(Y|X) = \frac{2}{5}$. Then (2017)

- (a) $P(Y) = \frac{4}{15}$ (b) $P(X|Y) = \frac{1}{2}$
(c) $P(X \cup Y) = \frac{2}{5}$ (d) $P(X \cap Y) = \frac{1}{5}$



17. There are three bags B_1 , B_2 and B_3 . The bag B_1 contains 5 red and 5 green balls. B_2 contains 3 red and 5 green balls and B_3 contains 5 red and 3 green balls. Bags B_1 , B_2 and B_3 have probabilities $3/10$, $3/10$ and $4/10$ respectively of being chosen. A bag is selected at random and a ball is chosen at random from the bag. Then which of the following options is/are correct ?
(2019)
- (a) Probability that the chosen ball is green, given that the selected bag is B_3 , equals $\frac{3}{8}$
- (b) Probability that the selected bag is B_3 , given that the chosen ball is green, equals $\frac{5}{13}$
- (c) Probability that the chosen ball is green equals $\frac{39}{80}$
- (d) Probability that the selected bag is B_3 , given that the chosen ball is green, equals $\frac{3}{10}$
18. Let E , F and G be three events having probabilities $P(E) = \frac{1}{8}$, $P(F) = \frac{1}{6}$ and $P(G) = \frac{1}{4}$, and $P(E \cap F \cap G) = \frac{1}{10}$. For any event H , if H^c denotes its complement, then which of the following statements is/are TRUE?
(2021)
- (a) $P(E \cap F \cap G^c) \leq \frac{1}{40}$
- (b) $P(E^c \cap F \cap G) \leq \frac{1}{15}$
- (c) $P(E \cup F \cup G) \leq \frac{13}{24}$
- (d) $P(E^c \cap F^c \cap G^c) \leq \frac{5}{12}$
20. Let S be the sample space of all 3×3 matrices with entries from the set $\{0, 1\}$. Let the events E_1 and E_2 is given by
 $E_1 = \{A \in S : \det A = 0\}$
 $E_2 = \{A \in S : \text{Sum of Entries of } A \text{ is } 7\}$
If a matrix is chosen at random from S , then the conditional probability $P(E_1|E_2)$ equals _____.
(2019)
21. Let $|X|$ denote the number of elements in a set X . Let $S = \{1, 2, 3, 4, 5, 6\}$ be a sample space, where each element is equally likely to occur. If A and B are independent events associated with S , then the number of ordered pairs (A, B) such that $1 \leq |B| < |A|$ equals _____.
(2019)
22. The probability that a missile hits a target successfully is 0.75. In order to destroy the target completely, at least three successful hits are required. Then the minimum number of missiles that have to be fired so that the probability of completely destroying the target is NOT less than 0.95, is _____.
(2020)
23. Two fair dice, each with faces numbered 1, 2, 3, 4, 5 and 6, are rolled together and the sum of the numbers on the faces is observed. This process is repeated till the sum is either a prime number or a perfect square. Suppose the sum turns out to be a perfect square before it turns out to be a prime number. If P is the probability that this perfect square is an odd number, then the value of $14P$ is _____.
(2020)
24. A number is chosen at random from the set $\{1, 2, 3, \dots, 2000\}$. Let p be the probability that the chosen number is a multiple of 3 or a multiple of 7. Then the value of $500p$ is _____.
(2021)

Assertion & Reason

- (A) If both assertion and reason are correct and reason is the correct explanation of assertion.
- (B) If both assertion and reason are true but reason is not the correct explanation of assertion.
- (C) If assertion is true but reason is false.
- (D) If assertion is false but reason is true.

25. Let $H_1, H_2, \dots, H_n$ be mutually exclusive and exhaustive events with $P(H_i) > 0$, $i = 1, 2, \dots, n$. Let E be any other event with $0 < P(E) < 1$.
(2007)

Assertion : $P(H_i|E) > P(H_i) \cdot P(H_i)$ for $i = 1, 2, \dots, n$

Reason : $\sum_{i=1}^n P(H_i) = 1$

- (a) A (b) B
(c) C (d) D

Numerical Value Type Questions

19. The minimum number of times a fair coin needs to be tossed, so that the probability of getting at least two heads is at least 0.96, is.
(2015)



26. Consider the system of equations

$$ax + by = 0, cx + dy = 0, \text{ where } a, b, c, d \in \{0, 1\}.$$

Assertion : The probability that the system of equations has a unique solution, is $3/8$.

Reason : The probability that the system of equations has a solution, is 1. (2008)

- (a) A (b) B
(c) C (d) D

Comprehension Type

Passage – 1

Using the following passage, solve Q.27 to Q.29

There are n urns each containing $n + 1$ balls such that the i^{th} urn contains i white balls and $(n + 1 - i)$ red balls. Let u_i be the event of selecting i^{th} urn, $i = 1, 2, 3, \dots, n$ and w denotes the event of getting a white ball.

(2006)

27. If $P(u_i) \propto i$ where $i = 1, 2, 3, \dots, n$ then $\lim_{n \rightarrow \infty} P(w)$ is equal to

- (a) 1 (b) $2/3$
(c) $3/4$ (d) $1/4$

28. If $P(u_i) = c$, where c is a constant then $P(u_n/w)$ is equal to

- (a) $\frac{2}{n+1}$ (b) $\frac{1}{n+1}$
(c) $\frac{n}{n+1}$ (d) $\frac{1}{2}$

29. If n is even and E denotes the event of choosing even numbered urn $\left(P(u_i) = \frac{1}{n} \right)$, then the value of $P(w/E)$, is

- (a) $\frac{n+2}{2n+1}$ (b) $\frac{n+2}{(n+1)}$
(c) $\frac{n}{n+1}$ (d) $\frac{1}{n+1}$

Passage – 2

Using the following passage, solve Q.30 to Q.32

A fair die is tossed repeatedly until a six is obtained. Let X denote the number of tosses required. (2009)

30. The probability that $X = 3$ equals

- (a) $\frac{25}{216}$ (b) $\frac{25}{36}$
(c) $\frac{5}{36}$ (d) $\frac{125}{216}$

31. The probability that $X \geq 3$ equals

- (a) $\frac{125}{216}$ (b) $\frac{25}{36}$
(c) $\frac{5}{36}$ (d) $\frac{25}{216}$

32. The conditional probability that $X \geq 6$ given $X > 3$ equals

- (a) $\frac{125}{216}$ (b) $\frac{25}{216}$
(c) $\frac{5}{36}$ (d) $\frac{25}{36}$

Passage – 3

Using the following passage, solve Q.33 and Q.34

Box 1 contains three cards bearing numbers 1,2,3 ; box 2 contains five cards bearing numbers 1,2,3,4,5 and box 3 contains seven cards bearing numbers 1,2,3,4,5,6,7. A card is drawn from each of the boxes. Let x_i be the number on the card drawn from the i^{th} box, $i = 1, 2, 3$. (2014)

33. The probability that $x_1 + x_2 + x_3$ is odd, is

- (a) $\frac{29}{105}$ (b) $\frac{53}{105}$
(c) $\frac{57}{105}$ (d) $\frac{1}{2}$



34. The probability that x_1, x_2, x_3 are in an arithmetic progression, is

- (a) $\frac{9}{105}$ (b) $\frac{10}{105}$
(c) $\frac{11}{105}$ (d) $\frac{7}{105}$

Passage – 4

Using the following passage, solve Q.35 and Q.36

Let n_1 and n_2 be the number of red and black balls, respectively, in box I. Let n_3 and n_4 be the number of red and black balls, respectively, in box II. (2015)

35. One of the two boxes, box I and box II, was selected at random and a ball was drawn randomly out of this box. The ball was found to be red. If the probability that this

red ball was drawn from box II is $\frac{1}{3}$, then the correct option(s) with the possible values of n_1, n_2, n_3 and n_4 is (are).

- (a) $n_1 = 3, n_2 = 3, n_3 = 5, n_4 = 15$
(b) $n_1 = 3, n_2 = 6, n_3 = 10, n_4 = 50$
(c) $n_1 = 8, n_2 = 6, n_3 = 5, n_4 = 20$
(d) $n_1 = 6, n_2 = 12, n_3 = 5, n_4 = 20$

36. A ball is drawn at random from box I and transferred to box II. If the probability of drawing a red ball from box I,

after this transfer, is $\frac{1}{3}$, then the correct option(s) with the possible values of n_1 and n_2 is(are)

- (a) $n_1 = 4$ and $n_2 = 6$
(b) $n_1 = 2$ and $n_2 = 3$
(c) $n_1 = 10$ and $n_2 = 20$
(d) $n_1 = 3$ and $n_2 = 6$

Passage – 5

Using the following passage, solve Q.37 and Q.38

Football teams T_1 and T_2 have to play two games against each other. It is assumed that the outcomes of the two games are independent. The probabilities of T_1 winning, drawing and losing a game against T_2 are $\frac{1}{2}, \frac{1}{6}$ and $\frac{1}{3}$, respectively. Each team gets 3 points for a win, 1 point for a draw and 0 point for a loss in a game. Let X and Y denote the total points scored by teams T_1 and T_2 , respectively, after two games. (2016)

37. $P(X > Y)$ is

- (a) $\frac{1}{4}$ (b) $\frac{5}{12}$
(c) $\frac{1}{2}$ (d) $\frac{7}{12}$

38. $P(X = Y)$ is

- (a) $\frac{11}{36}$ (b) $\frac{1}{3}$
(c) $\frac{13}{36}$ (d) $\frac{1}{2}$

Passage – 6

Using the following passage, solve Q.39 and Q.40

There are five students S_1, S_2, S_3, S_4 and S_5 in a music class and for them there are five seats R_1, R_2, R_3, R_4 and R_5 arranged in a row, where initially the seat R_i is allotted to the student $S_i, i = 1, 2, 3, 4, 5$. But, on the examination day, the five students are randomly allotted the five seats. (2018)

39. The probability that, on the examination day, the student S_1 gets the previously allotted seat R_1 , and none of the remaining students gets the seat previously allotted to him/her is

- (a) $\frac{3}{40}$ (b) $\frac{1}{8}$
(c) $\frac{7}{40}$ (d) $\frac{1}{5}$



40. For $i = 1, 2, 3, 4$, let T_i denote the event that the students S_i and S_{i+1} do not sit adjacent to each other on the day of the examination. Then, the probability of the event $T_1 \cap T_2 \cap T_3 \cap T_4$ is

- (a) $\frac{1}{15}$ (b) $\frac{1}{10}$
(c) $\frac{7}{60}$ (d) $\frac{1}{5}$

Passage – 7

Using the following passage, solve Q.41 and Q.42

41. Three numbers are chosen at random, one after another with replacement, from the set $S = \{1, 2, 3, \dots, 100\}$. Let p_1 be the probability that the maximum of chosen numbers is at least 81 and p_2 be the probability that the minimum of chosen numbers is at most 40.

The value of $\frac{625}{4} p_1$ is _____. (2021)

42. Three numbers are chosen at random, one after another with replacement, from the set $S = \{1, 2, 3, \dots, 100\}$. Let p_1 be the probability that the maximum of chosen numbers is at least 81 and p_2 be the probability that the minimum of chosen numbers is at most 40.

The value of $\frac{125}{4} p_2$ is _____. (2021)

Text

43. A coin has probability 'p' of showing head when tossed. It is tossed 'n' times. Let p_n denote the probability that no two (or more) consecutive heads occur. Prove that, $p_1 = 1$, $p_2 = 1 - p^2$ & $p_n = (1 - p)p_{n-1} + p(1 - p)p_{n-2}$, for all $n \geq 3$. (2000)

44. A and B are two independent events. The probability that both occur simultaneously is $1/6$ and the probability that neither occurs is $1/3$. Find the probabilities of occurrence of the events A and B separately. (2000)

45. Two cards are drawn at random from a pack of playing cards. Find the probability that one card is a heart and the other is an ace. (2001)

46. (a) An urn contains 'm' white and 'n' black balls. A ball is drawn at random and is put back into the urn along with K additional balls of the same colour as that of the ball drawn. A ball is again drawn at random. What is the probability that the ball drawn now is white.

- (b) An unbiased die, with faces numbered 1, 2, 3, 4, 5, 6 is thrown n times and the list of n numbers showing up is noted. What is the probability that among the numbers 1, 2, 3, 4, 5, 6, only three numbers appear in the list. (2001)

47. A box contains N coins, m of which are fair and the rest are biased. The probability of getting a head when a fair coin is tossed is $1/2$, while it is $2/3$ when a biased coin is tossed. A coin is drawn from the box at random and is tossed twice. The first time it shows head and the second time it shows tail. What is the probability that the coin drawn is fair? (2002)

48. (a) A person takes three tests in succession. The probability of his passing the first test is p, that of his passing each successive test is p or $p/2$ according as he passes or fails in the preceding one. He gets selected provided he passes at least two tests. Determine the probability that the person is selected.

- (b) In a combat, A targets B, and both B and C target A. The probabilities of A, B, C hitting their targets are $2/3$, $1/2$ and $1/3$ respectively. They shoot simultaneously and A is hit. Find the probability that B hits his target whereas C does not. (2003)



49. (a) If A and B are independent events, prove that $P(A \cup B) \cdot P(A' \cap B') \leq P(C)$, where C is an event defined that exactly one of A or B occurs.
- (b) A bag contains 12 red balls and 6 white balls. Six balls are drawn one by one without replacement of which atleast 4 balls are white. Find the probability that in the next two draws exactly one white ball is drawn (leave the answer in terms of nC_r). (2004)
50. A person goes to office either by car, scooter, bus or train probability of which being $\frac{1}{7}, \frac{3}{7}, \frac{2}{7}$ and $\frac{1}{7}$ respectively. Probability that he reaches office late, if he takes car, scooter, bus or train is $\frac{2}{9}, \frac{1}{9}, \frac{4}{9}$ and $\frac{1}{9}$ respectively. Given that he reached office in time, then what is the probability that he travelled by a car. (2005)

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PROBABILITY



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Answer Key



CHAPTER -8 | VECTORS & 3-DIMENSIONAL GEOMETRY

EXERCISE - 1: BASIC OBJECTIVE QUESTIONS



DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | | | | | |
|----------|----------|----------|----------|----------|----------|----------|----------|
| 1. (c) | 2. (b) | 3. (d) | 4. (c) | 77. (3) | 78. (6) | 79. (c) | 80. (b) |
| 5. (b) | 6. (a) | 7. (b) | 8. (c) | 81. (c) | 82. (c) | 83. (d) | 84. (c) |
| 9. (b) | 10. (c) | 11. (d) | 12. (a) | 85. (b) | 86. (a) | 87. (b) | 88. (b) |
| 13. (a) | 14. (a) | 15. (a) | 16. (a) | 89. (a) | 90. (a) | 91. (d) | 92. (a) |
| 17. (c) | 18. (b) | 19. (b) | 20. (b) | 93. (a) | 94. (d) | 95. (b) | 96. (c) |
| 21. (d) | 22. (a) | 23. (c) | 24. (b) | 97. (a) | 98. (c) | 99. (a) | 100. (b) |
| 25. (c) | 26. (c) | 27. (a) | 28. (a) | 101. (d) | 102. (a) | 103. (a) | 104. (d) |
| 29. (a) | 30. (b) | 31. (d) | 32. (a) | 105. (a) | 106. (b) | 107. (b) | 108. (d) |
| 33. (d) | 34. (b) | 35. (c) | 36. (b) | 109. (a) | 110. (d) | 111. (a) | 112. (a) |
| 37. (d) | 38. (a) | 39. (b) | 40. (c) | 113. (c) | 114. (d) | 115. (b) | 116. (d) |
| 41. (c) | 42. (d) | 43. (a) | 44. (a) | 117. (b) | 118. (a) | 119. (b) | 120. (d) |
| 45. (d) | 46. (b) | 47. (a) | 48. (b) | 121. (c) | 122. (c) | 123. (d) | 124. (c) |
| 49. (d) | 50. (b) | 51. (c) | 52. (a) | 125. (a) | 126. (d) | 127. (d) | 128. (b) |
| 53. (c) | 54. (c) | 55. (4) | 56. (46) | 129. (d) | 130. (c) | 131. (c) | 132. (c) |
| 57. (17) | 58. (2) | 59. (2) | 60. (5) | 133. (b) | 134. (1) | 135. (2) | |
| 61. (1) | 62. (3) | 63. (2) | 64. (0) | | | | |
| 65. (7) | 66. (2) | 67. (17) | 68. (0) | | | | |
| 69. (2) | 70. (50) | 71. (70) | 72. (3) | | | | |
| 73. (2) | 74. (14) | 75. (3) | 76. (3) | | | | |

CHAPTER - 8 | VECTORS & 3-DIMENSIONAL GEOMETRY

EXERCISE - 2 :
PREVIOUS YEAR JEE MAIN QUESTIONS

DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | | | | | |
|------------|-------------|----------|-----------|-----------|-----------|-----------|-----------|
| 1. (a) | 2. (c) | 3. (d) | 4. (b) | 97. (c) | 98. (c) | 99. (a) | 100. (b) |
| 5. (c) | 6. (b) | 7. (b) | 8. (b) | 101. (d) | 102. (d) | 103. (d) | 104. (c) |
| 9. (a) | 10. (b) | 11. (b) | 12. (a) | 105. (c) | 106. (a) | 107. (d) | 108. (b) |
| 13. (d) | 14. (d) | 15. (a) | 16. (a) | 109. (a) | 110. (d) | 111. (c) | 112. (d) |
| 17. (2) | 18. (0.8) | 19. (a) | 20. (a) | 113. (a) | 114. (b) | 115. (b) | 116. (b) |
| 21. (18) | 22. (b) | 23. (6) | 24. (4) | 117. (a) | 118. (b) | 119. (c) | 120. (c) |
| 25. (1) | 26. (BONUS) | 27. (c) | 28. (4) | 121. (c) | 122. (a) | 123. (c) | 124. (5) |
| 29. (c) | 30. (c) | 31. (1) | 32. (81) | 125. (3) | 126. (d) | 127. (a) | 128. (a) |
| 33. (30) | 34. (d) | 35. (4) | 36. (6) | 129. (c) | 130. (b) | 131. (b) | 132. (d) |
| 37. (a) | 38. (b) | 39. (3) | 40. (d) | 133. (a) | 134. (3) | 135. (81) | 136. (6) |
| 41. (2) | 42. (b) | 43. (9) | 44. (b) | 137. (a) | 138. (a) | 139. (d) | 140. (b) |
| 45. (b) | 46. (a) | 47. (c) | 48. (60) | 141. (3) | 142. (c) | 143. (7) | 144. (b) |
| 49. (1494) | 50. (5) | 51. (90) | 52. (a) | 145. (c) | 146. (1) | 147. (a) | 148. (b) |
| 53. (a) | 54. (d) | 55. (b) | 56. (28) | 149. (96) | 150. (d) | 151. (d) | 152. (d) |
| 57. (d) | 58. (b) | 59. (b) | 60. (486) | 153. (d) | 154. (26) | 155. (d) | 156. (d) |
| 61. (a) | 62. (b) | 63. (d) | 64. (75) | 157. (72) | 158. (d) | 159. (61) | 160. (c) |
| 65. (2) | 66. (12) | 67. (c) | 68. (d) | 161. (d) | 162. (7) | 163. (c) | 164. (a) |
| 69. (b) | 70. (d) | 71. (c) | 72. (a) | 165. (2) | 166. (d) | 167. (a) | 168. (a) |
| 73. (c) | 74. (a) | 75. (d) | 76. (b) | 169. (0) | 170. (d) | 171. (4) | 172. (2) |
| 77. (c) | 78. (d) | 79. (c) | 80. (a) | 173. (1) | 174. (38) | 175. (4) | 176. (28) |
| 81. (c) | 82. (d) | 83. (a) | 84. (c) | 177. (a) | 178. (a) | 179. (1) | 180. (d) |
| 85. (a) | 86. (b) | 87. (a) | 88. (c) | 181. (d) | 182. (d) | 183. (44) | 184. (a) |
| 89. (a) | 90. (d) | 91. (a) | 92. (c) | 185. (a) | 186. (c) | 187. (b) | 188. (a) |
| 93. (d) | 94. (d) | 95. (d) | 96. (d) | 189. (c) | 190. (8) | | |

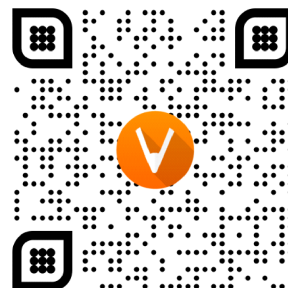
CHAPTER - 8 | VECTORS & 3-DIMENSIONAL GEOMETRY

EXERCISE - 3 :
ADVANCED OBJECTIVE QUESTIONS

DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | |
|-------------|-------------|-------------|-------------|
| 1. (c) | 2. (a) | 3. (a) | 4. (a) |
| 5. (a) | 6. (c) | 7. (c) | 8. (b) |
| 9. (b) | 10. (a) | 11. (a) | 12. (c) |
| 13. (c) | 14. (d) | 15. (a) | 16. (c) |
| 17. (b) | 18. (a) | 19. (c) | 20. (a) |
| 21. (c) | 22. (b) | 23. (a) | 24. (c) |
| 25. (d) | 26. (a) | 27. (d) | 28. (d) |
| 29. (b) | 30. (b) | 31. (a) | 32. (c) |
| 33. (c) | 34. (a,d) | 35. (a,c,d) | 36. (a,c) |
| 37. (a,b,c) | 38. (a,c,d) | 39. (a, b) | 40. (a,b,c) |
| 41. (a,b) | 42. (a,c) | 43. (c,d) | 44. (a,c,d) |
| 45. (50) | 46. (9) | 47. (b) | 48. (d) |
| 49. (d) | 50. (a) | 51. (a) | 52. (a) |
| 53. (c) | 54. (b) | 55. (a) | 56. (c) |
| 57. (b) | 58. (b) | 59. (a) | 60. (a) |
| 61. (d) | 62. (d) | 63. (d) | 64. (c) |
| 65. (c) | 66. (b) | 67. (b) | 68. (a) |
| 69. (1710) | 70. (13) | 71. (1) | 72. (7) |
| 73. (b) | 74. (b) | 75. (d) | 76. (a) |
| 77. (d) | 78. (b) | 79. (c) | 80. (b) |

EXERCISE - 4 :
PREVIOUS YEAR JEE ADVANCED QUESTIONS

DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | |
|-------------|-----------------------------------|-------------------|-------------|
| 1. (b) | 2. (a) | 3. (b) | 4. (c) |
| 5. (b) | 6. (c) | 7. (c) | 8. (b) |
| 9. (a) | 10. (b) | 11. (a) | 12. (c) |
| 13. (a) | 14. (b) | 15. (c) | 16. (c) |
| 17. (d) | 18. (c) | 19. (c) | 20. (a) |
| 21. (b,d) | 22. (a, d) | 23. (b, d) | 24. (a,b,c) |
| 25. (a,c,d) | 26. (b,c) | 27. (a,b) | 28. (a,c) |
| 29. (a,b,c) | 30. (5) | 31. (9) | 32. (3) |
| 33. (4) | 34. (3) | 35. (8) | 36. (18.00) |
| 37. (108) | 38. (1.50) | 39. (7) | 40. (c) |
| 41. (c) | 43. $\frac{9}{2} \text{ cu unit}$ | 45. $\frac{1}{2}$ | 46. (a) |
| 47. (c) | 48. (a) | 49. (c) | 50. (b) |
| 51. (a) | 52. (d) | 53. (c) | 54. (a) |
| 55. (a) | 56. (a) | 57. (a) | 58. (c) |
| 59. (d) | 60. (c) | 61. (c) | 62. (b, c) |
| 63. (a,d) | 64. (b,d) | 65. (a,b) | 66. (b,c,d) |
| 67. (c,d) | 68. (a,b,c) | 69. (a, d) | 70. (a,b,c) |
| 71. (5) | 72. (0.75) | 73. (d) | 74. (d) |
| 75. (a) | 76. (a) | 77. (a) | |
| 78. (b) | 79. (d) | 80. (c) | |
81. $(\vec{v}_1 = 2\hat{i}, \vec{v}_2 = -\hat{i} + \hat{j} \text{ and } \vec{v}_3 = 3\hat{i} - 2\hat{j} + 4\hat{k})$
82. (a) $x + y - 2z = 3$, (b) $Q(6, 5, -2)$
84. $2x - y + z - 3 = 0$ and $62x + 29y + 19z - 105 = 0$
85. $\hat{w} = \hat{v} - 2(\hat{a} \cdot \hat{v})\hat{a}$

Answer Key

CHAPTER -9 | PROBABILITY

EXERCISE - 1: BASIC OBJECTIVE QUESTIONS



DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | |
|---------|---------|---------|---------|
| 1. (a) | 2. (d) | 3. (b) | 4. (a) |
| 5. (c) | 6. (b) | 7. (d) | 8. (a) |
| 9. (a) | 10. (d) | 11. (a) | 12. (a) |
| 13. (a) | 14. (d) | 15. (a) | 16. (b) |
| 17. (b) | 18. (a) | 19. (b) | 20. (b) |
| 21. (a) | 22. (a) | 23. (b) | 24. (a) |
| 25. (a) | 26. (d) | 27. (c) | 28. (a) |
| 29. (b) | 30. (c) | 31. (d) | 32. (b) |
| 33. (b) | 34. (b) | 35. (b) | 36. (b) |
| 37. (a) | 38. (d) | 39. (d) | 40. (c) |
| 41. (a) | 42. (c) | 43. (c) | 44. (a) |
| 45. (b) | 46. (a) | 47. (b) | 48. (a) |
| 49. (c) | 50. (d) | 51. (b) | 52. (b) |
| 53. (b) | 54. (a) | 55. (c) | 56. (c) |
| 57. (d) | 58. (b) | 59. (a) | 60. (a) |

EXERCISE - 2: PREVIOUS YEAR JEE MAIN QUESTIONS



DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | |
|-----------|----------|----------|----------|
| 1. (b) | 2. (c) | 3. (b) | 4. (a) |
| 5. (a) | 6. (b) | 7. (c) | 8. (a) |
| 9. (c) | 10. (b) | 11. (b) | 12. (c) |
| 13. (c) | 14. (d) | 15. (d) | 16. (a) |
| 17. (137) | 18. (c) | 19. (c) | 20. (d) |
| 21. (d) | 22. (c) | 23. (c) | 24. (c) |
| 25. (b) | 26. (b) | 27. (b) | 28. (a) |
| 29. (b) | 30. (c) | 31. (d) | 32. (c) |
| 33. (3) | 34. (c) | 35. (11) | 36. (11) |
| 37. (b) | 38. (b) | 39. (a) | 40. (d) |
| 41. (b) | 42. (a) | 43. (d) | 44. (b) |
| 45. (b) | 46. (d) | 47. (c) | 48. (a) |
| 49. (c) | 50. (a) | 51. (d) | 52. (d) |
| 53. (4) | 54. (a) | 55. (c) | 56. (d) |
| 57. (b) | 58. (28) | 59. (d) | 60. (30) |
| 61. (d) | 62. (a) | 63. (b) | 64. (c) |
| 65. (6) | 66. (c) | 67. (c) | 68. (d) |
| 69. (6) | 70. (a) | 71. (d) | 72. (a) |
| 73. (d) | 74. (d) | 75. (a) | |

CHAPTER - 9 | PROBABILITY

EXERCISE - 3 :
ADVANCED OBJECTIVE QUESTIONS

DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | |
|---------------------|---|--------------------|-------------|
| 1. (a) | 2. (b) | 3. (c) | 4. (c) |
| 5. (c) | 6. (d) | 7. (a) | 8. (a) |
| 9. (d) | 10. (c) | 11. (b) | 12. (c) |
| 13. (a) | 14. (a) | 15. (b) | 16. (d) |
| 17. (b) | 18. (c) | 19. (d) | 20. (c) |
| 21. (b) | 22. (b) | 23. (b) | 24. (d) |
| 25. (d) | 26. (d) | 27. (b) | 28. (b) |
| 29. (b) | 30. (c) | 31. (b) | 32. (c) |
| 33. (a) | 34. (d) | 35. (a,b,c,d) | 36. (a,c,d) |
| 37. (c,d) | 38. (b,c,d) | 39. (a,d) | 40. (a,b) |
| 41. (a,b,c,d) | 42. (a,b,c) | 43. (a,b,c) | 44. (8) |
| 45. (5) | 46. (30) | 47. (c) | 48. (b) |
| 49. (a) | 50. (a) | 51. (b) | 52. (a) |
| 53. (c) | 54. (c) | 55. (b) | 56. (d) |
| 57. (d) | 58. (b) | 59. (c) | 60. (d) |
| 61. $\frac{24}{29}$ | 62. (a) $\frac{1}{132}$ (b) $\frac{1}{462}$ | 63. $\frac{5}{21}$ | |
| 64. $\frac{1}{9}$ | 65. $\frac{1}{4}$ | | |

EXERCISE - 4 :
PREVIOUS YEAR JEE ADVANCED QUESTIONS

DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | |
|---|---|--|------------|
| 1. (a) | 2. (d) | 3. (d) | 4. (b) |
| 5. (c) | 6. (c) | 7. (d) | 8. (c) |
| 9. (c) | 10. (a) | 11. (c) | 12. (c) |
| 13. (b) | 14. (a) | 15. (a,d) | 16. (a,b) |
| 17. (a,c) | 18. (a,b,c) | 19. (8) | 20. (0.50) |
| 21. (422) | 22. (6) | 23. (8) | 24. (214) |
| 25. (d) | 26. (b) | 27. (b) | 28. (a) |
| 29. (b) | 30. (a) | 31. (b) | 32. (d) |
| 33. (b) | 34. (c) | 35. (a,b) | 36. (c,d) |
| 37. (b) | 38. (c) | 39. (a) | 40. (c) |
| 41. (76.25) | 42. (24.50) | 44. $\frac{1}{2} \& \frac{1}{3}$ or $\frac{1}{3} \& \frac{1}{2}$ | |
| 45. $\frac{1}{26}$ | 46. (a) $\frac{m}{m+n}$ (b) $\frac{{}^6C_3 (3^n - 3 \cdot 2^n - 3)}{6^n}$ | | |
| 47. $\frac{9m}{m+8N}$ | 48. (a) $p^2(2-p)$ (b) $1/2$ | | |
| 49. $\frac{{}^{12}C_2 {}^6C_4 {}^{10}C_1 {}^2C_1 + {}^{12}C_1 {}^6C_5 {}^{11}C_1 {}^1C_1}{{}^{12}C_2 ({}^{12}C_2 {}^6C_4 + {}^{12}C_1 {}^6C_5 + {}^{12}C_0 {}^6C_6)}$ | | | |
| 50. $\frac{1}{7}$ | | | |

MASTER INDEX

VOLUME 1:

Determinants & Matrices

Relations , Functions & Inverse Trigonometric Functions

Continuity, Differentiability & Differentiation

Applications of Derivatives

VOLUME 2:

Indefinite Integration

Definite Integration, Area under curves

Differential Equations

VOLUME 3:

Vectors & 3-Dimensional Geometry

Probability

Creating Impact at Scale



65 Million+

Monthly
Youtube Views



49 Million+

Monthly
Web + App Visitors



40+ Countries

Where Students
Take Live Classes



25 Million+

Monthly Active
Users



26 Million+

Hours of
Live Learning



25

MILLION+

Parents
Trust
Vedantu



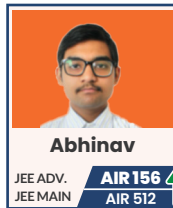
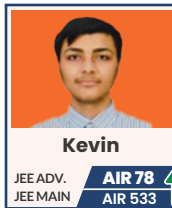
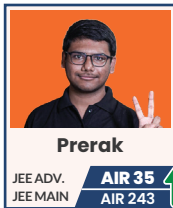
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MILLION+

Happy
Students

Our Achievers of 2021

JEE Advanced 2021



1500+

Vedantu students ace
JEE Advanced 2021



6%

of IITs' upcoming batch
will be from Vedantu

JEE Main 2021

903

students in
Top 10,000

78

students scored
99.9+ PERCENTILE

869

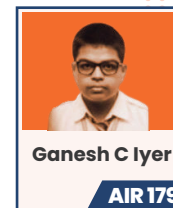
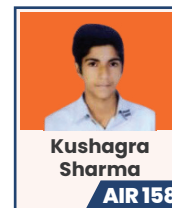
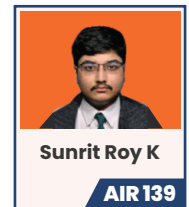
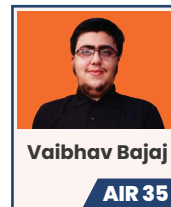
students scored
99.9+ PERCENTILE

68

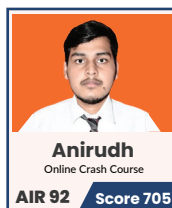
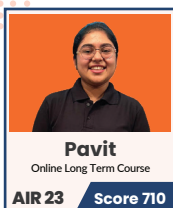
students in
Top AIR 1000

444

students in
Top AIR 5000



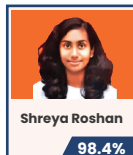
NEET (UG) 2021



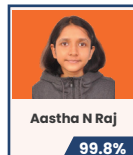
1172

NEET
Qualifiers

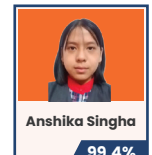
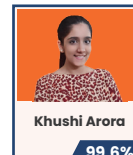
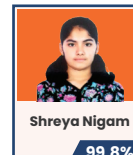
CBSE Class 12



ISC Class 12



CBSE Class 10



ICSE Class 10

